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Council of the University of Novi Sad, Faculty of Agriculture, at its fourth meeting on 18th of February 1981 made a decision that the Conference for students of agriculture and veterinary medicine with international participation, held each year at the Faculty of Agriculture will be held in the memory of academician ***Dr Petar Drezgić***, professor, and the prizes awarded at the Conference will bear his name.

PREFACE

The first Conference for students of agriculture and veterinary medicine with international participation was held in Novi Sad 44 years ago. To the present date, many papers from national and international authors were presented (Hungary – Gödölö and Debrecen, Germany – Kassel, Slovakia – Nitra, Poland – Wroclaw and Warsaw, Romania – Timisoara, Cluj and Bucharest, Bulgaria – Plovdiv, Russia – Moscow and Kemerovo, Macedonia – Skopje and Bitola, Bosnia and Herzegovina – Sarajevo, Banja Luka, Croatia – Zagreb). Since 2009, papers presented at the Conference are published in a special publication named “PROCEEDINGS OF THE 44th CONFERENCE FOR STUDENTS OF AGRICULTURE AND VETERINARY MEDICINE WITH INTERNATIONAL PARTICIPATION” where all published papers have all elements of the original scientific paper. Starting this year, we enabled students and mentors who are interested in benefitting more from this Conference to publish their papers in the journal Contemporary Agriculture published by the University of Novi Sad, Faculty of Agriculture.

Also, besides the Faculty of Agriculture from Novi Sad, co-organizer of the Conference is also Faculty of Agriculture from Timisoara and there is a prospect of joining another University center from the region. This Conference is strongly supported by the Student Parliament of the Faculty of Agriculture, Novi Sad.

As the host and the principal organizer of the Conference University of Novi Sad, Faculty of Agriculture has a special honor and a pleasure to wish a warm welcome to all participants, successful work and amusing socializing in Novi Sad.

Vice-dean for science and international cooperation
Dr Branko Ćupina, full professor

President of the organizing committee
Dr Dragana Budakov, associate professor

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OPPORTUNITIES REGARDING THE FORTIFICATION OF WHEAT FLOUR-BASED FOOD PRODUCTS

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SUMMARY

The food fortification with micronutrients, such as vitamins and minerals represents a technology by which the deficient nutrient is specifically added to food – an intervention to improve the quality of food and health in the general or segment population. The inadequate eating habits, the high calories intake and some metabolic disorders may lead to micronutrient's deficiency, affecting the health of people all over the world. The consumption of enriched food represents the most suitable method for prophylaxis of deficiencies and nutritional diseases. The paper presents several opportunities identified for wheat flour and products fortification through specific technologies, using either common fortificants, such as vitamins and minerals, or, more recently investigated, bioactive ingredients rich in dietary fibers, proteins, polyunsaturated fatty acids and antioxidant compounds. The nutritional profile of fortified products is improved, but evaluation of physical-chemical, rheological and sensory properties becomes mandatory in order to produce an acceptable fortified food.

Key words: fortification, enriched food, flour, wheat

INTRODUCTION

Nowadays, consumers have a higher level of perception of health benefits from a healthy diet and became more familiar with issues of functional foods and nutraceuticals. Thus, a lot of countries, particularly those with an advanced economy, such as Japan, Canada, USA or EU, have established national programs on healthy eating to be carefully implemented.

Based on the purpose of changing diet and lifestyle, those involved in agriculture and food manufacturing approached the huge possibility of raw materials and foods to provide bioactive ingredients with disease-preventing properties (Linseisen et al., 2007).

Beside the nutritional role of macronutrients, plant foods contain a lot of micronutrients, as bioactive molecules formed during secondary metabolism, which have been shown to possess various health benefits, such as vitamins, biominerals, polyphenols, phenolic acids, terpenes, sulfur-containing compounds, carotenoids. Foods rich in such biologically active compounds or ingredients containing these biomolecules to be used in different food

products, have been considered as functional food with positive health impact (Goldberg, 1994).

Food products which contain essential nutrients with the role of preventing their deficiencies to population have been defined as fortified or enriched foods (FAO/WHO, 1994). Fortification could be done for establishing the natural level of some nutrients that have been lost during processing, or above the natural level in case of foods with special destination (disease-preventing), for the purpose of public health and safety issues (Harris, 1968).

The most common fortified foods with various vitamins or biominerals (Berner et al., 2001; Noorfarahzilah et al., 2014) are presented in Table 1.

Table 1. Type of common fortified foods with different essential nutrients.

Type of food	Fortificants
Cereals and their products	Vitamins (A, D, E, C, B ₂ , B ₆ , B ₉ , B ₁₂) Minerals (Fe, Ca, Mg, Zn) Legumes Seeds Natural extracts
Moh Milk and milk products	Vitamins (A, D, C, B ₆) Minerals (Ca, Fe)
Margarine	β -carotene, vitamins (A, D, E)
Fruit juices and drinks	Vitamins (A, D, E, C, B ₉) Minerals (Fe, Ca, Mg)

Food fortification has been considered a long term cost-effective solution to overcome micronutrient malnutrition in a population or segment of population, but implementation of specific programs has to be considered accurately by each nation in order to achieve the targeted fortification with maintenance of food safety (Mannar et al., 2017).

FORTIFICATION OF WHEAT FLOUR BASED PRODUCTS

The deficiency of food micronutrients, especially of vitamins and minerals, led to different diseases, frequently in children and women. Unfortunately, the malnutrition is the main cause of death in developing countries. Thus, one successful solution to combat the malnutrition is the fortification of categories of foods that the population highly consumes on daily basis. The food fortification strategies have laid a special emphasis on flour fortification, the flour being one of the staple foods to meet the nutritional needs and to improve the health of vulnerable people.

Fortified flours can constitutes as important source not only of added nutrients, such as vitamins or biominerals, but also of bioactive compounds, due to the fact that the flour can be

enriched with selected ingredients rich in fiber or different phytochemicals, showing improved benefits for human health (Akhtar et al., 2011). It is well-known that vitamins and minerals are essential elements to maintain physiological functions, enzyme activity and to increase the metabolism.

1. Fortification with vitamins and minerals

Fortification of wheat flour with vitamins and minerals is performed for establishing the natural level that has been lost during processings (*e.g.* milling of grains), focusing the public health and safety issues (Harris, 1968).

The factors which influence the selection of the technology of wheat fortification are the bioavailability of micronutrients, the cost, the acceptance of sensory properties and the stability of the fortificants during storage. Most food manufacturers and researchers are involved in performing fortification of wheat flour and products with appropriate levels of vitamins, such as not to change the sensory properties (*e.g.* taste, texture, color) and at the same time, to be affordable to the targeted population – people in developing countries which have low incomes.

Considering the widely biologically active properties of vitamin D (calciferol) including antiviral, anti-inflammatory and anticancer (Zhang et al., 2010; Ilie et al., 2020), beside the accepted role in calcium absorption and mineral balance (Norman, 2012), fortification of wheat flour or products (breakfast cereals) becomes one of the most important solution of increasing the vitamin D₃ consumption and of the long-term treatment of the vitamin D deficiency (Khayyatadeh et al., 2019). Vitamin A is added to flour in the form of palmitate powder (25000-75000 UI/ g vitamin A), while other vitamins are added in crystalline forms (Banu et al., 2005). As vitamin A may be lost during storage, attempts have been done to stabilize it by addition of antioxidants and other stabilizers (Sayed Ahmad et al., 2018).

The iron deficiency can lead to anemia, this being the most widespread disease globally, so that bread fortification with iron is widely considered by different national strategies (Sadighi et al., 2008). Minerals are added in the form of different salts (fumarate, lactate, gluconate, chloride, stearate, sulfate, phosphate or carbonate) (Banu et al., 2005). The level of iron fortification is in correlation to flour consumption per day, varying from 20 to 60 ppm of ferrous sulfate or fumarate, according to WHO recommendations. Fortification is done such as to ensure vitamin and mineral bioavailability and no changes of color and taste. The form of fortificants highly influences not only the bioavailability, but also the homogeneity and stability of the final products.

Fortification technology of wheat flour is realized by using vitamin and mineral premixes, usually customized blends provided by different suppliers (Mannar et al., 2017), while technology of wheat bread fortification is done using different approaches: (1) use of yeast strains enriched with vitamins and minerals; (2) use of vitamin-mineral premixes initially dissolved in small amounts of technological water; (3) use of salt enriched with vitamins and minerals (Banu et al., 2005).

2. Fortification with bioactive ingredients (dietary fibers/ proteins/ antioxidants)

Nowadays, there is an increasing demand for enriched foods, in particular widely-consumed wheat products, that can be achieved by addition of nutritional or bioactive ingredients, which add value to bread through their nutritional profile (fibers, proteins, polyunsaturated fatty acids, antioxidant compounds) (Das et al., 2016).

The fortification of wheat flour with fibers, both soluble and insoluble, may provide many health benefits, such as maintaining the intestinal integrity and lowering the cholesterol level (Anderson et al., 2009). Scientific studies reported technology and advantages of bread fortification with seeds of pumpkin, grape, fennel, cumin and caraway, or with by-products from oil-seed pressing (Sayed Ahmad et al., 2018), ingredients which provide fibers, proteins, fatty acids, polyphenols and minerals. Incorporation at levels of 10-15% did not affect rheological properties of dough (Aghamirzaei et al., 2015), products showing an increased nutritional level and antioxidant activity.

Wheat flour has been also fortified at 1, 2 and 3% (w/w) with dried leafy vegetable powders (amaranth, African eggplant), the final product showing high antioxidant activity due to the increase of the polyphenols content (Adeola et al., 2019).

Other studies have been conducted for the wheat flour and products fortification with legume flours or isolates from chickpeas, peas, soya, grains, beans or lentils, improving their protein profiles (Noorfarahzilah et al., 2014). Legume flour was added in amounts of 0-40% during bread processing. Bread fortification with legume flours lead to a decrease of some properties (volume, lightness values of color, overall acceptability) and increase of others (density, firmness and hardness, proteins, fibers and minerals). Higher amounts of legume flours until total substitution of wheat, were added to other wheat-products such as biscuits, cakes or cooked pasta (Noor Aziah et al., 2012; Campbell et al., 2016; Laleg et al., 2017). Recent research has been conducting to optimize several process parameters (mixing speed, hydration, use of structuring agents such as gum or ovalbumins), in order to reduce the effects of legume fortificants on the product structuring during processing. However, the nutritional quality of the fortified bread or other wheat-based products has been improved.

FUTURE TRENDS

Micronutrients malnutrition determines long term health effects, being the greatest health challenge, in particular in developing countries. Thus, according to the studies that have been carried out, diverse fortification of foods, *e.g.* wheat flours and products, as main staple foods consumed all over the world, is an efficient and low-cost alternative to combat vitamins and mineral deficiencies, and to enrich with valuable bioactive compounds, such as fibers, fatty acids or polyphenols. National programs of food fortification is in strong relation to objective factors (socioeconomic situation, purchasing power of the population) and subjective ones (national food traditions) as well, that actually determines the structure of such foods.

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CHANGES IN THYROID HORMONE LEVELS IN SHEEP DURING LACTATION

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Summary: *The aim of the research was to determine the changes in the concentration of thyroid hormones (T3 and T4) in sheep during the lactation period. 20 Polish long-haired ewes were assigned for the experiment. Blood was collected every 28 days for 6 hours, every 20 minutes. The conducted research has shown that the secretion of the T3 hormone decreases with the milk yield. However, the secretion of the T4 hormone tends to increase with the progress of lactation.*

Key words: *sheep, thyroid hormones, lactation, milk*

INTRODUCTION

Lactation is the process of milk production and secretion by the mammary glands of female mammals (Lakhani et al., 2017). In sheep, it is the last stage of the reproductive cycle (Błasiak and Molik, 2014). It is concluded that thyroid hormones play an important role of physiological significance to generation of the seasonal reproductive cycle of ewes (Dahl, 1995; Karsch et al., 1995). Thyroid hormones (triiodothyronine and thyroxine) regulate metabolic processes that control the body's energy balance ((Bauman and Currie, 1980; Cheng et al., 2010) and body weight (Iwen et al., 2013). The hypothalamus - pituitary - thyroid (PPT) axis is responsible for the secretion of these hormones. Triiodothyronine (T3) and thyroxine (T4) in milk originate from the blood plasma and T3 is produced in the mammary gland by the enzymatic process of T4 to T3 conversion (Chopra et al., 1978; Ślebodziński and Brzezińska-Ślebodzińska, 1991). They also take part in the neuronal development of the fetus, the functioning of the cardiovascular system and the initiation and maintenance of lactation (Ślebodziński et al., 1999; Todini, 2007; Błasiak and Molik, 2014; Ortiga-Carvalho et al., 2016; Carvalho and Dupuy, 2017; Molik et al., 2019). The process of initiating and maintaining lactation in seasonal animals requires the presence of many factors. Therefore, the aim of the research was to determine changes in thyroid hormones in sheep during the lactation period.

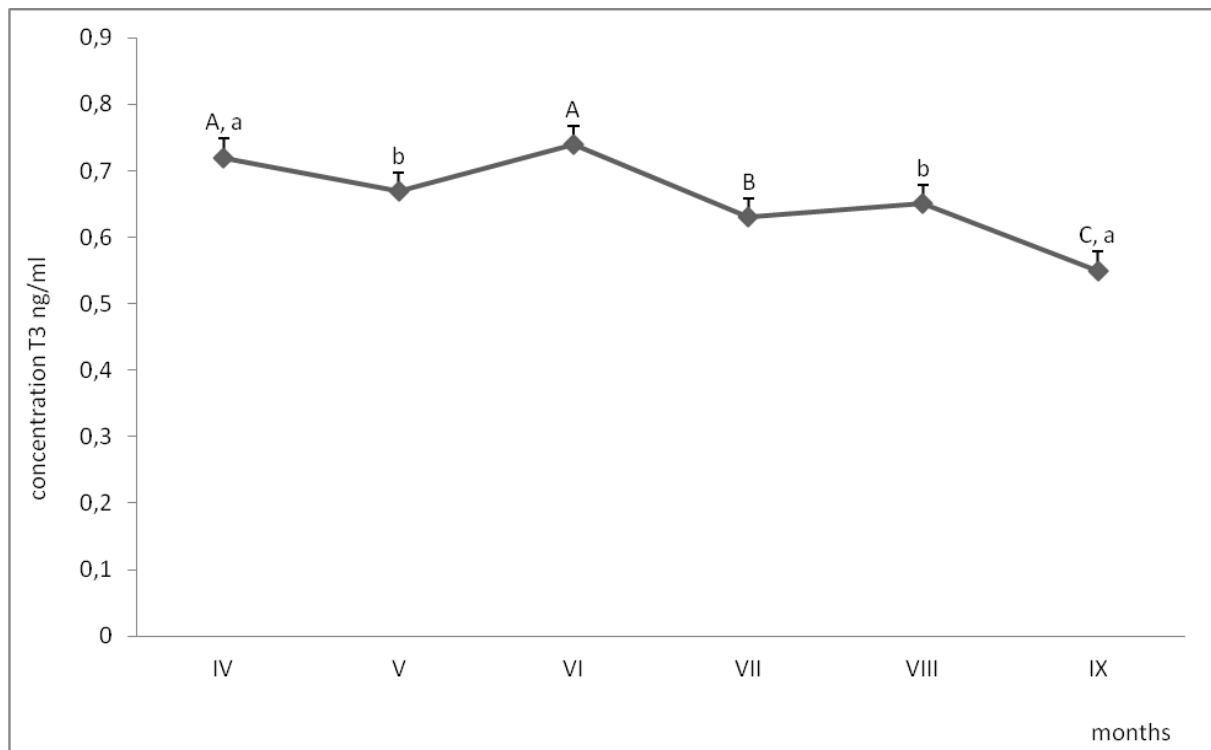
MATERIAL AND METHODS

The research was carried out with the approval of the Local Ethical Committee. 20 Polish long-haired sheep ewes were used for the experiment, which were similar in terms of body weight ($60 \text{ kg} \pm 5 \text{ kg}$) and age (4 - 5 lactation). During the experiment, ewes were housed in an alcove with the possibility of using runs. The estrus cycle of females was synchronized

using the Chrono-Gest method. For this purpose, polyurethane sponges impregnated with 40 mg of Cronolone (Intervet, the Netherlands) were introduced intravaginally to the animals for 14 days. On the day the polyurethane sponges were removed, the sheep were injected with 500 IU. PMSG (Serogonadotropin, Biowet, Drwalew). Estrus in sheep appeared after 48 - 72 hours after dosing PMSG. The heat time was additionally verified with a ram. Breeding took place from 10 to 15 September, while lambing and starting lactation took place in mid-February. Rearing of lambs with their mothers lasted 56 days. The mothers' nutrition was adjusted to their current physiological state; until the end of the 4th month of pregnancy, the sheep were fed according to the rules and standards of the National Research Institute of Animal Production (1993). In order to unify the nutrition, from the 5th month of pregnancy until drying off, sheep were given a granulated mix and meadow hay ad libitum.

For the determination of thyroid hormones, blood was collected from the mothers every 28 days using a catheter inserted into the jugular vein. Catheters were introduced 5 hours before blood sampling to allow the animals to get used to and to eliminate the stress factor. Blood was collected every 28 days for 6 hours, every 20 minutes. T3 and T4 hormone determinations were made using the radioimmunoassay (RIA) method (Cisbio kit) described by Kokot and Stupnicki (Kokot and Stupnicki, 1985). The Mann-Whitney-Wilcoxon test was used to determine the differences in lactation parameters. Changes in hormone concentration were analyzed using one-way analysis of variance (ANOVA) using the Scheffe test.

RESULTS AND DISCUSSION

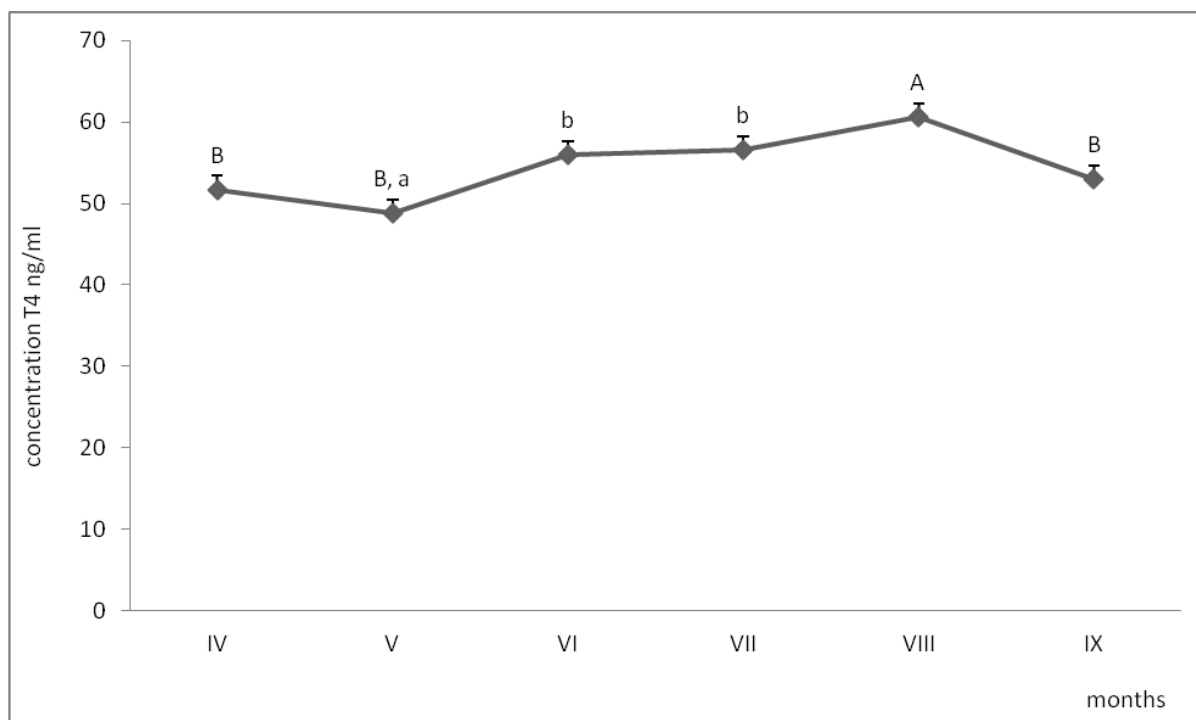


A, B, C - means marked with different letters differ significantly at $P \leq 0.01$

a, b - means marked with different letters differ significantly at $P \leq 0.05$

Fig. 1. Changes in triiodothyronine (T3) secretion in sheep during lactation.

Significantly ($P \leq 0.01$) the highest level of T3 hormone concentration was found in the first and third month of milking (April - 0.72 ± 0.02 ng / ml and June 0.74 ± 0.02 ng / ml). In the second month of milking (in May), the concentration of T3 decreased significantly ($P \leq 0.05$) (0.67 ± 0.02 ng / ml). Since June, the concentration of T3 continued to decrease, in July it was (0.63 ± 0.02 ng / ml). In the last month of milking, the lowest concentration of T3 (0.55 ± 0.01 l) was observed in relation to the remaining months (Fig. 1.).



A, B - means marked with different letters differ significantly at $P \leq 0.01$

a, b - means marked with different letters differ significantly at $P \leq 0.05$

Fig. 2. Changes in thyroxine (T4) secretion in sheep during lactation.

The conducted studies on changes in the T4 profile significantly ($P \leq 0.01$) showed the lowest concentration in the first two months of milking (April - 51.7 ± 9.8 ng / ml and May 48.8 ± 7.4 ng / ml, respectively). In the next two months milking T4 significantly ($P \leq 0.05$) increased, and was, in June (56.4 ± 8.2 ng / ml) and July (56.6 ± 8.3 ng / ml). Significantly ($P \leq 0.01$) the highest concentration of T4 was recorded in August (60.55 ± 10 , ng / ml). In the last month of milking, the concentration of T4 decreased significantly and amounted to (52.93 ± 8.7 ng / ml) (Fig. 2.).

The studies conducted so far have shown that triiodothyronine modulates milk production by inhibiting the transcription of the prolactin gene. It is caused by blockage of signal transducer and activator of transcription 5a (STAT5) which acts as an intermediary in the signaling pathways of prolactin in the mammary gland (Campo Arboccó Verde et al., 2015). This study showed an increase in T3 concentration in the third and fifth month of milking, while in the last month of lactation, during the short day, the lowest T3 concentration was observed. Similar results were obtained in in vitro studies on thyroid explants. It has been shown that the concentration of triiodothyronine was higher in the summer months, and decreased during the shorter day (Klocek-Górka et al., 2010). The thyroxine profile showed the opposite tendency to the triiodothyronine concentration. With the passage of lactation from April to August, the concentration of thyroxine increased. Previous studies have shown that in sheep with the progress of lactation, the concentration of thyroxine increases, however, in goats this relationship is inversely proportional (Ślebodziński and Twardon, 2004; Błasiak and Molik, 2014). The lowest level of T4 was observed in May, this result seems to be consistent with in vitro studies which showed a lower concentration of the hormone in the spring when the day

is longer, and a higher concentration during the short day (Klocek-Górka et al., 2010). High T4 concentration and low T3 level were also obtained during the dry period in dairy cows (Pezzi et al., 2003). It has been demonstrated, that during the transition from late pregnancy to early lactation constantly T4 increase with a parallel decrease in T3 in high yielding dairy cows indicating higher requirements of thyroid hormones during gestation for fetal development and lactation (Steinhoff et al., 2019).

CONCLUSION

In conclude, the studies have shown that the concentration of triiodothyronine decrease as lactation progresses. However, the concentration of thyroxine increases. Changes in the concentration of thyroid hormones (T3 and T4) affect the milk yield of sheep during lactation, especially in seasonal sheep. It is suggested that further research is needed to determine the importance of metabolic hormones in the initiation and maintenance of lactation.

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BIOACTIVITY OF THE *ORIGANUM VULGARE* L. ESSENTIAL OIL AGAINST *CALLIPHORA VOMITORIA* (DIPTERA: CALLIPHORIDAE), A THREAT TO THE FRESH AND CURED MEAT INDUSTRY

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Calliphoridae flies are vectors of pathogenic microorganisms and lay eggs on fresh and cured meat, representing, therefore, a threat to the food industry. In this paper, the essential oil extracted from *Origanum vulgare* L. was tested for its insecticidal activity against adults and eggs and its deterrence to the oviposition of *Calliphora vomitoria* (Linnaeus, 1758) (Diptera: Calliphoridae). The results showed that oregano essential oil is effective with a dose-dependent activity. The application by EO-charged mist in meat processing factories is, hence, to be encouraged.

Key words: Essential oil, oregano, botanical, blowflies

INTRODUCTION

Calliphoridae flies, known as blowflies or carrion flies, are vectors of potentially pathogenic microorganisms, such as *Cronobacter* spp., *Cryptosporidium parvum*, *Escherichia coli*, *Giardia duodenalis*, *Listeria monocytogenes*, *Mycobacterium avium* subsp. *avium*, *M. a. paratuberculosis*, *M. a. hominissuis*, *Salmonella* spp., *Shigella* spp., *Taenia* spp., etc. (Daniel et al., 1990; Olsen, 1998; Fischer et al., 2001 and 2004; Graczyk et al., 2001; Förster et al., 2007; Pava-Ripoll et al., 2012). Because of the adults' feeding behaviour, these filth flies come into contact with decomposing organic matter, excrements, secretions, and tissues from dead animals (Greenberg, 1973). The promiscuous landings on different substrates, food included as these blowflies are synanthropic, cause the contamination through direct contact with parts of their bodies (legs, hairs, thorax, head, mouthparts) or by faeces and vomit (Olsen, 1998; Förster et al., 2007).

The spread of these microorganisms is one of the causes of the insurgence of foodborne diseases, both in industrialized and developing countries. According to the World Health Organization (WHO, 2020), each year worldwide there are 600 million cases of illnesses related to the consumption of contaminated food and 420 000 deaths, 40% of which are represented by children under the age of five.

Additionally, blowflies are attracted to the smell of fresh and cooked meat, on which gravid females lay hundreds of eggs. The resulting maggots voraciously feed on the meat, causing the rapid rotting and consequent spoilage of the infested products (Bedini et al., 2018 and 2020).

Blowflies' control currently relies on synthetic insecticides, mainly organochlorines, organophosphates, and insect growth regulators (French et al., 1992; Baker et al., 2014), products anyhow not admitted on food and in meat factories. Moreover, their use has negative drawbacks on the environment and human and animal health (Deb, 2018), and it can lead to the development of resistance in the repeatedly treated species (Campbell, 1998). It is necessary to find safer and eco-friendly alternatives, especially for the control of blowflies in facilities where meat is processed, stored, and sold.

Essential oils (EOs) proved to be effective insecticides and repellents against several insect pests (Bertoli et al., 2012; Bedini et al., 2017, 2018, 2019, and 2020) and are characterized by low persistence in the environment and low-risk for humans and non-target animals (Regnault-Rogers et al., 2012). Moreover, food-grade EOs extracted from cooking spices are safe for human consumption, and their use in foodstuffs is allowed by the Regulation (EC) No 1334/2008 of 16 December 2008.

Origanum vulgare L. is employed worldwide as a fresh culinary herb or dried spice, to season dishes with its pungent and spicy flavour. Its EO it is widely employed as a flavouring agent and preservative in several culinary, pharmaceutical, and cosmetic preparations (Kokkini et al., 2003). *O. vulgare* is well known, since ancient times, in the traditional medicine of many European and Middle Eastern countries. In Italy and Serbia, for example, the infusion of aerial parts treats respiratory infections and gastrointestinal disorders, as it is a mild antiseptic, spasmolytic, expectorant, aperitive, and digestive (Tucakov, 1971; Leporatti and Ivancheva, 2003). The insecticidal properties of *O. vulgare* have been tested against different harmful insects, with promising results in the control of some Coleoptera and Lepidoptera species (Aslan et al., 2005; La Pergola et al., 2017; Nasr et al., 2017). In Albania, dried and ground oregano is traditionally used in the houses and facilities where meat or dairy products are processed, to repel Diptera flies (Xhuvel and Lipe, 1997). A comprehensive investigation of the activity of oregano on the blowflies is, therefore, necessary to validate this folkloristic application.

In the present work, for the first time, we tested the insecticidal and repellent effect of the oregano EO against the Diptera Calliphoridae *Calliphora vomitoria* (Linnaeus, 1758), one of the filth flies, a vector of pathogenic microorganisms and a threat to the fresh and cured meat industry.

MATERIAL AND METHODS

The *Origanum vulgare* L. inflorescences and leaves were collected in Almeria, Andalusia, Spain. The vegetal sample was air-dried in the shadow at room temperature, until constant weight.

Essential oil extraction and characterization: 100 g of the vegetal sample were extracted by hydrodistillation in a Clevenger apparatus, for three hours, according to the current European Pharmacopoeia (2018) procedure. The oil was separated from the aqueous layer and dried using anhydrous sodium sulphate, then stored in dark sealed vials at -18 °C until required.

For the GC/EI-MS chemical characterization, a Varian CP-3800 gas chromatography (Agilent Technologies Inc., Santa Clara, California) equipped with a DB-5 capillary column (30 m $\times$ 0.25 mm internal diameter, coating thickness 0.25 μ m) and a Varian Saturn 2000 ion-trap mass detector (Agilent Technologies Inc.) were used. According to Giuliani et al. (2017), the operating conditions were the following: oven temperature from 60 to 240 °C with an increase of 3 °C/min; injector temperature 220 °C; transfer-line temperature 240 °C; carrier gas helium (1 mL/min); volume of injection 1 μ L (5% EO v/v in n-hexane HPLC grade). The identification of the compounds was based on the comparison of their linear retention indexes (l.r.i.) with the ones of authentic samples, relative to the series of C9–C22 n-hydrocarbons. The mass spectra were also computer matched against spectral data from the commercial NIST (2014) libraries and a laboratory-made library, developed from pure substances and components of identified EOs and literature data (Adams, 2007).

Calliphora vomitoria L. adult blowflies were reared in 47.50 $\times$ 47.50 $\times$ 93.0 cm cages (BugDorm-4M4590DH, MegaView Science Co. Ltd., Taiwan) under laboratory conditions (23 °C, 60-70% RH, natural photoperiod), and fed with sucrose and yeast 4:1 and water *ad libitum*, according to Bedini et al. 2017. To stimulate the oviposition, warm beef mince was provided to adult females.

Contact toxicity tests: 2 μ L of 0.0 (control), 10.0, 20.0, and 30.0% ethanol (EtOH) solutions of the oregano EO (corresponding to 0.0, 0.20, 0.40, and 0.60 μ L EO/fly) were applied, using a Burkard micro-dispenser (Burkard Scientific Ltd., United Kingdom), on the thorax of 20 unsexed adults, 7-10 days old. Before the topical application, flies were anesthetized by cold treatment for 180 s at -18 °C. Then, flies were maintained in Plexiglas cages 20 cm of diameter $\times$ 30 cm with sucrose and water *ad libitum*. Mortality was checked after 24 hours, and values were corrected using Abbott's formula (Abbott, 1925). For each dose tested, three replicates were performed.

Oviposition deterrence tests: according to Bedini et al. (2017 and 2019), 150 unsexed adult flies (10-14 days old, sex ratio 1:1) were moved in a 47.50 $\times$ 47.50 $\times$ 93.0 cm cage (BugDorm-4M4590DH) and fed with sucrose and yeast 4:1 and water *ad libitum*. For each cage, 16 cylindrical microscopy embedding moulds (3.50 cm of diameter $\times$ 5.0 cm, surface area 9.61 cm²) were filled with 8 gr of minced pork meat and water (20% w/w) to maintain the humidity. Meat surface was flattened and sprayed with 150 μ L of 0.0 (control), 0.50, 1.0, and 2.0% EtOH solution of the oregano EO (corresponding to 0.0, 0.08, 0.16, and 0.32 μ L EO cm⁻² of meat). In every cage, in correspondence of the inner corners, were placed four moulds for each of the three concentrations tested and control. The entire experiment was replicated three times, under a bank of uniform lights (14 lux), at 23 °C and 75% RH. The number of eggs laid was counted after 24 hours.

Ovicidal activity tests: according to Bedini et al. (2019), 4.5 $\times$ 4.5 cm squares of filter paper (surface area 20.2 cm²) were treated with 100 μ L of 0.0 (control), 0.10, 0.20, 0.50, 0.75, and 1.0% EtOH solution of the oregano EO (corresponding to 0.0, 0.005, 0.010, 0.025, 0.037, and 0.050 μ L EO cm⁻² of filter paper). EtOH was evaporated under an airflow for 3 minutes, then the squares of filter paper were wetted with 380 μ L of water. 0-12 hours old eggs were distributed, in the number of 50, on the treated squares of filter paper and put in glass Petri dishes (10 cm of diameter), then incubated at 25 °C in the dark, in a climatic chamber. Eggs'

hatchability was checked daily for three days (24, 48, and 72 hours) with the help of a dissecting microscope, and values were corrected using Abbott's formula (Abbott, 1925). The squares of filter paper were wetted with 380 μ L of water after 24 and 48 hours. For each concentration tested, five replicates were performed.

Statistical analysis: since the variance of the toxicity bioassay data was not homogeneous, they were analysed by Kruskal-Wallis test with the EO concentration/dose as a factor, and the means were separated by Dunn-Bonferroni pairwise comparisons. The oviposition deterrence data were processed using one-way ANOVA, with the EO concentration as a factor. The ANOVA assumptions of variance homogeneity and normality were confirmed by Levene's test and the Shapiro-Wilk test, respectively. Statistics were performed by SPSS 22.0 software (IBM SPSS Statistics, Armonk, North Castle, New York).

RESULTS AND DISCUSSION

Origanum vulgare EO chemical composition: the GC-EIMS analysis identified 15 compounds in the oregano EO, corresponding to 100.0% of the total composition. The main components are carvacrol (81.5%), p-cymene (8.0%), and γ -terpinene (2.4%) (Table 1). Oxygenated monoterpenes are the most represented chemical class, followed by monoterpene hydrocarbons (Table 1).

Table 1. Chemical composition (compounds $\geq 0.5\%$) of the *Origanum vulgare* essential oil hydrodistilled from inflorescences and leaves.

Compounds	I.r.i. ^a	Relative abundance (%)
1-Octen-3-ol	980	1.8
3-Octanone	987	0.8
β -Myrcene	993	1.1
α -Terpinene	1020	0.8
p-Cymene	1027	8.0
γ -Terpinene	1062	2.4
Carvacrol	1298	81.50
β -Caryophyllene	1420	1.7
Caryophyllene oxide	1581	0.5
Monoterpene hydrocarbons		12.80
Oxygenated monoterpenes		82.40
Sesquiterpene hydrocarbons		1.70
Oxygenated sesquiterpenes		0.50
Non-terpene derivatives		2.60
Total identified (%)		100.0

^a linear retention indexes

Based on the chemical composition, the oregano EO here tested belongs to the carvacrol chemotype, according to the classification given by Kokkini (1994). Carvacrol and thymol have been reported as the main components of *O. vulgare* EO by several authors (Baser et al., 1993 and 1994; Vokou et al., 1993). The prevalence of one phenol over the other can depend on the geographical location of the collection area. For example, Turkish samples mainly contain carvacrol (Baser et al., 1993 and 1994) like the EO here analysed, while Greek ones have high contents in thymol (Vokoul et al., 1993).

Contact toxicity tests: the oregano EO showed a clear toxic dose-dependent action 24 hours after the topical application on the thorax of adult *C. vomitoria* flies. The increase in the percentage of mortality is statistically significant (Figure 1). The mortality at the dose of 0.60 μ L EO/fly is almost 100%.

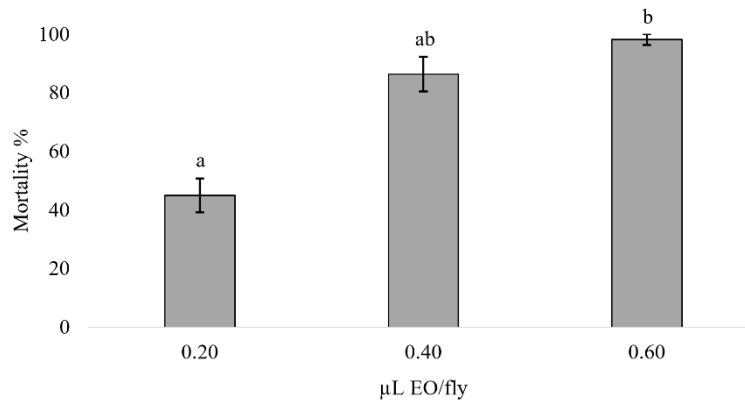


Figure 1. Mortality % given by the *Origanum vulgare* essential oil by contact on adults of *Calliphora vomitoria*. Bars represent standard error. Different letters indicate statistically significant differences between the doses.

Oviposition deterrence tests: all the three concentrations tested of the oregano EO strongly reduced the number of eggs laid by adult females of *C. vomitoria* on the pork meat samples, after 24 hours from the start of the assay. The reduction was higher than 70% in all the cases if compared to the control, without statistically significant differences (Figure 2).

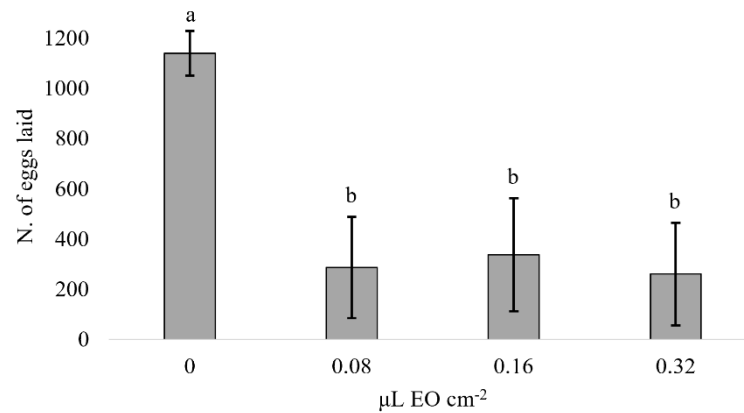


Figure 2. The number of eggs laid by *Calliphora vomitoria* females on pork meat samples treated with different concentrations of *Origanum vulgare* essential oil. Bars represent standard error. Different letters indicate statistically significant differences between the concentrations.

Ovicidal activity tests: the percentage of eggs hatched after 72 hours from the beginning of the trial decrease as the concentration of the oregano EO applied on the squares of filter paper supporting the eggs increases, in a statistically significant way. At the concentration of 0.037 µL EO cm⁻², about 50% of the eggs hatch (Figure 3).

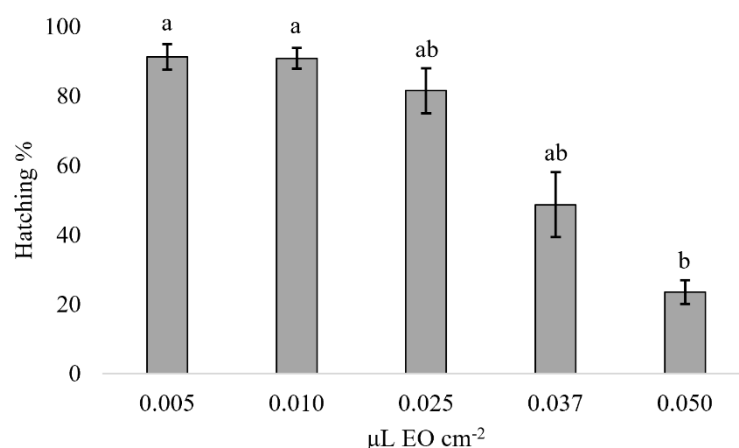


Figure 3. Hatching % of the eggs of *Calliphora vomitoria* treated with different concentrations of *Origanum vulgare* essential oil. Bars represent standard error. Different letters indicate statistically significant differences between the concentrations.

Despite the folkloristic use in Albania of oregano grounded leaves to repel synanthropic flies (Xhuvel and Lipe, 1997), the investigations about the effect of *O. vulgare* on Diptera flies are scarce. Only Xie et al. (2019) assessed the toxicity of an *O. vulgare* EO against second age larvae and pupae of the housefly *Musca domestica* L. (Diptera: Muscidae), by contact and fumigation. The EO used contained 58.13% of carvacrol, 17.85% of p-cymene, and 8.15% of thymol, hence, it is considered a carvacrol chemotype like the EO tested in this work. The LC₅₀ for the contact toxicity on larvae was 0.23 μL EO cm⁻² of filter paper, and the one for the fumigation toxicity was 9.52 μL EO L⁻¹ air. Pupal toxicity, evaluated based on the inhibition of the number of emerged adults, was 90.9% at 0.25 μL EO cm⁻² by contact and complete (100%) at 20 μL EO L⁻¹ air by fumigation.

Regarding the use of EOs extracted from other cooking herbs to control *C. vomitoria*, there are few studies reported in the literature. Bedini et al. (2020) tested the toxicity by contact given by garlic (*Allium sativum*), rosemary (*Rosmarinus officinalis*), and sage (*Salvia officinalis*) EOs on adult *C. vomitoria* flies, reporting an LD₅₀ of 0.44, 1.10, and 1.97 μL/fly for the three culinary spices, respectively. For the tarragon (*Artemisia dracuncululus*) EO, LD₅₀ by contact was 0.485 μL/fly, and, for sweet annie (*Artemisia annua*), it was 0.786 μL/fly (Bedini et al., 2017). The results here obtained for the oregano EO show that the percentage of mortality at 0.40 μL/fly is more than 80%, so this EO has broader contact toxicity on *C. vomitoria* adults.

At the highest concentration tested by Bedini et al. (2020), corresponding to 5 μL EO cm⁻² of meat, garlic, rosemary, and sage EOs were all able to avoid the oviposition of *C. vomitoria*. At the concentration of 1.25 μL EO cm⁻², the complete oviposition deterrence (100%) was achieved only by the *A. sativum* EO. The tarragon EO was able to completely (100%) prevent the oviposition starting from the concentration of 0.05 μL EO cm⁻², while sweet annie EO did not exceed the 79% of deterrence at the same concentration (Bedini et al., 2017). For the oregano EO here tested, at the concentration of 0.32 μL EO cm⁻² of meat, the reduction of the laid eggs was 77%.

About the ovicidal activity, Bedini et al. (2020) reported that *A. sativum* EO completely (100%) arrested the eggs' hatching at the concentration of 0.16 $\mu\text{L EO cm}^{-2}$ of filter paper, while the EOs extracted from *R. officinalis* and *S. officinalis* were significantly less effective. At the concentration of 0.050 $\mu\text{L EO cm}^{-2}$, the oregano EO reduced to about 23.5% the hatchability of the treated eggs, underlining its high ovicidal activity.

CONCLUSION

EOs have been proved to be effective in insect pests' control, blowflies included. In particular, the EOs extracted from cooking herbs, like oregano, are allowed by law to be used in the presence of food as they are already part of several recipes as preservatives or flavouring agents. Despite these advantages and the promising characteristics of the EOs, such as the high biodegradability and low toxicity on mammals, their application is currently not as common as desired. The results obtained suggest a possible use of *O. vulgare* EOs in factories and facilities where meat is processed, stored, or sold. Oregano EOs could represent a tool to avoid blowflies' oviposition and the loss of the infested meat products and to prevent the spreading of pathogenic microorganisms with the consequent insurgence of foodborne diseases. A possible application of the oregano EOs could be in the formulation of an EO-charged mist, to be sprayed by an automatic settable dispenser to prevent the entrance of the blowflies in the meat processing rooms. Further investigations are however needed to ascertain the best parameters for such administration.

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INFLUENCE OF EPIBRASSINOLIDE AND ITS SULFONATES ON MORPHOMETRIC PARAMETERS OF *HELIANTHUS ANNUUS* IN FIELD CONDITIONS

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Summary presents results of the effect of epibrassinolide and its sulfated conjugates on germination, stem length, green mass yield, seed yield of sunflower in the field conditions. The hormones impact dynamics in certain concentrations on the development, stability and productivity of sunflower plants, depending on different types of processing with brassinosteroids was revealed.

Key words: *epibrassinolide, sulfated brassinosteroids, Helianthus annuus L., field conditions.*

INTRODUCTION

In modern applied ecology, methods for increasing plant stability and productivity through the use of phytohormones are becoming increasingly important. Several brassinosteroids (BS) have shown good results in crop adaptation [1]. Thanks to instead due to their environmental friendliness, large range of biological effects and low consumption rates, studies of brassinosteroids [2] and their new synthesized derivatives are continuing. The goal is to determine the peak efficiency of their influence on functional parameters, especially on the growth and yield of agricultural crops under the influence of various environmental stress factors.

Like most hormones, brassinosteroids acquire new biological properties as a result of various biosynthetic and metabolic processes [3-6]. In this regard, a number of water-soluble 24-epibrassinolide (EB) sulfates, including 2 α , 3 α -disulfate and all possible monosulfates, were synthesized in the laboratory of steroid chemistry of the Institute of Bioorganic chemistry of the National Academy of Sciences of Belarus [7].

Due to the extensive use of contaminated soils in the food industry and in phytoremediation [8], we selected a promising agricultural crop, annual sunflower (*Helianthus Annius* L.), to study the effect of various concentrations of epibrassinolide and its sulfonates on plant growth and development in field conditions. According to the screening of data obtained earlier in laboratory and in field conditions (2019 season) [9], the most effective epibrassinolide sulfates and their concentrations were selected. In addition, the introduction of hormones was carried out in two ways: pre-sowing seed processing and foliar processing (spraying) of plants.

MATERIAL AND METHODS

The following hormones were selected for sunflower: epibrassinolide and its conjugates – 2,3-disodium sulfate-24-EB (960) at concentrations of 10^{-11} M and 10^{-9} M, 2 - sodium sulfate-24-EB (1010) at concentrations of 10^{-9} M, 23-sodium sulfate-24-EB (990) at concentrations of 10^{-10} M and 10^{-9} M, and 3-sodium sulfate-24-EB (993) at a concentration of 10^{-11} M [7]. The assessment was performed using the following morphometric parameters: germination [10], shoot length, green mass (biomass), and seed yield.

During the pre-sowing processing, the seeds were pre-soaked for 5 hours in solutions of epibrassinolide and its sulfonates. Then the seeds were planted in the experimental field of the Agrobiology Department of the ecology Center (Brest). During the foliar processing, the seeds were planted without processing, and hormones were introduced twice: at the stage of the second pair of real leaves formed and at the stage of budding. As a form of control, seeds soaked in distilled water (Kd) and without (K) were used. In total, 16 variants were obtained for sunflower (8 for each processing method).

RESULTS AND DISCUSSION

During the growing period of annual sunflower (May-August 2020), the following weather data were recorded: maximum and minimum air temperature, average air temperature and precipitation amounts. Data analysis showed that high temperatures and drought in July and August played a significant role in limiting growth (figure 1). During this period, a significant anti-stress effect on growth parameters was detected during pre-sowing processing of 17 % for 990 at a concentration of 10^{-9} M and 12 % for 960 at a concentration of 10^{-9} M, as well as during foliar processing of 36 % for EB at a concentration of 10^{-9} M and 14 % for 960 at a concentration of 10^{-9} M (figure 3).

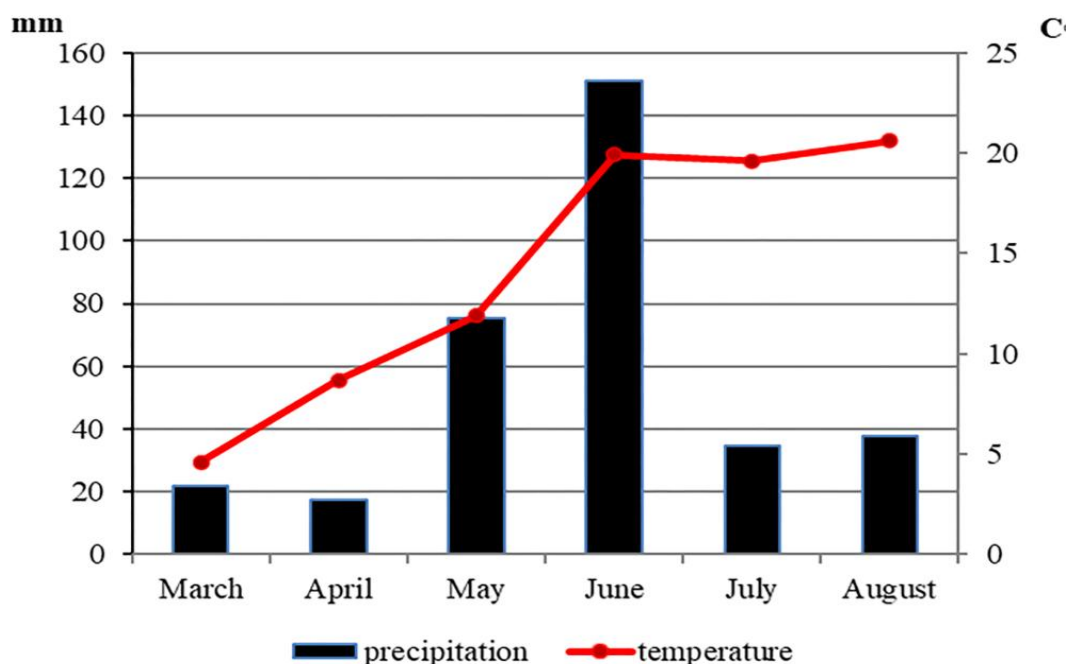


Fig. 1 Precipitation and temperature indicators in 2020 (monthly total)

Pre-sowing processing of seeds showed a positive effect of epibrassinolide and its conjugates on the germination of annual sunflower in the range from 14 to 34 % relative to the control, with the exception of $993 \cdot 10^{-11}$ M and $990 \cdot 10^{-10}$ M. During foliar processing at the time of germination measurement, the introduction of hormones did not occur, and therefore there are no changes in the germination parameter relative to the control (figure 2).

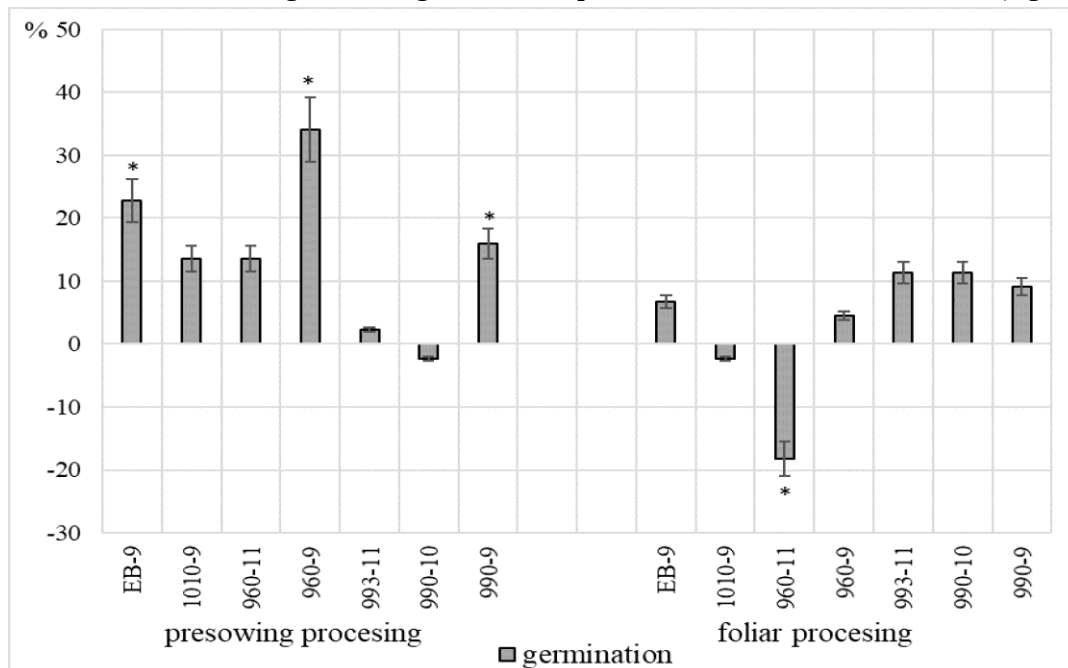


Fig. 2 Influence of epibrassinolide and its sulfonates on the germination of sunflower depending on the type of processing in the field conditions (relative to control)

At the beginning of the growing season (June), there was no significant difference in the shoot length parameter relative to the control, but after foliar processing of plants and repeated measurement (July), a positive effect of the sulfonated hormone 960 at a concentration of 10^{-9} M – 12 % and 990 at a concentration of 10^{-9} M – 18 % was noted. By the end of the growing season, foliar processing showed its effectiveness with an increase in shoot length by 14 % when exposed to 960 at a concentration of 10^{-9} M and distinctively high and reliable values are observed under the influence of EB 10^{-9} M – 36 % relative to the control (figure 3).

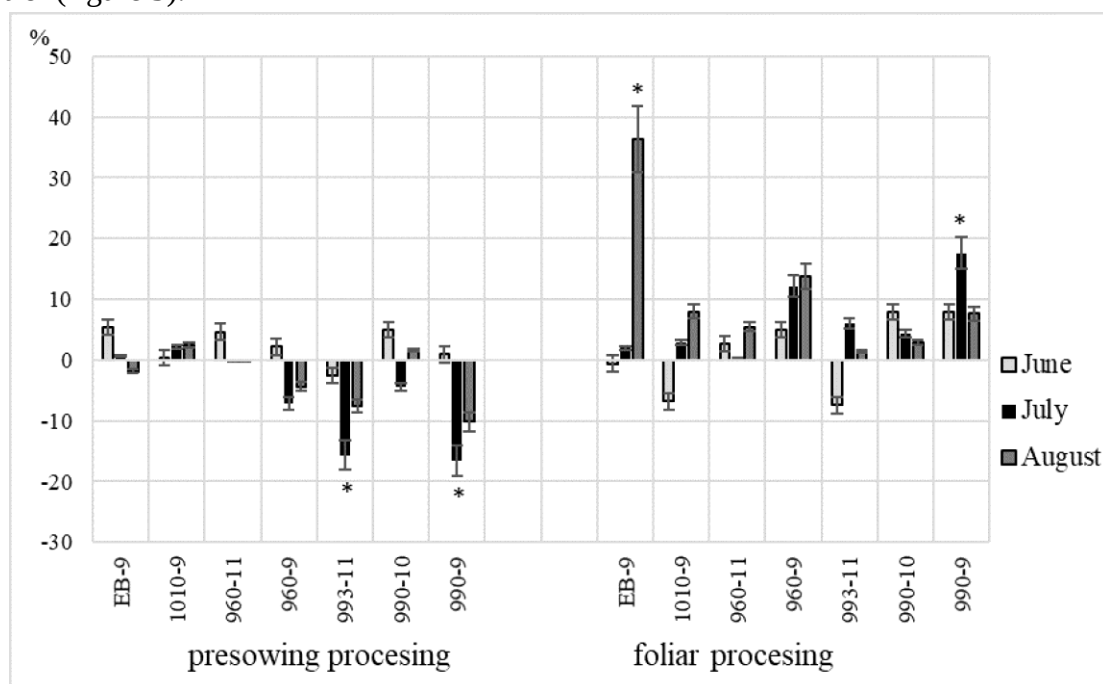


Fig. 3 Effect of epibrassinolide and its sulfonates on the length of sunflower shoots depending on the type of processing in the field conditions (relative to control)

The obtained data on the biomass parameter (figure 4) indicate the predominant efficiency of foliar processing for sunflower. A significant increase is observed under the influence of 960 at a concentration of 10^{-9} M – 54 % and 993 at a concentration of 10^{-11} M – 62 %. The crop yield (figure 4), namely the mass of seeds, indicates low biological activity relative to the control of epibrassinolide and its sulfonated hormones except for 993 at a concentration of 10^{-11} M – 46 % and 960 at a concentration of 10^{-9} M – 114 %. However, a full yield assessment for this crop will be provided after a biochemical analysis to determine the oil content of the obtained seeds.

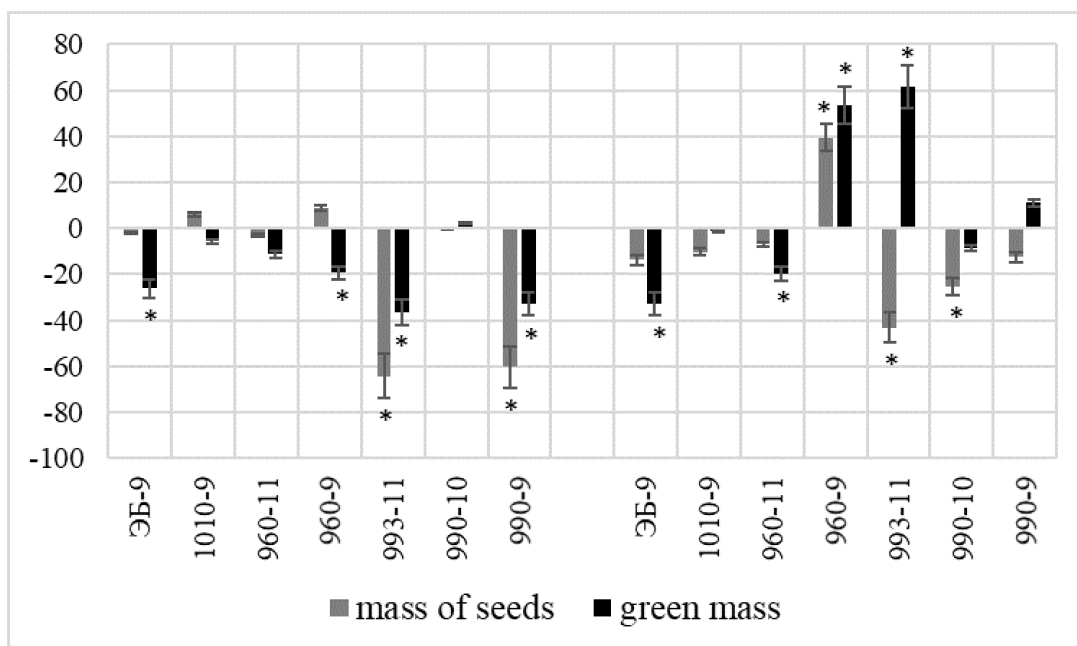


Fig. 4 Effect of epibrassinolide and its sulfonates on the length of sunflower shoots depending on the type of processing in the field conditions (relative to control)

CONCLUSION

Studies have shown that in the field conditions, for most of the studied parameters, the greatest stimulating effect on the morphometric parameters of annual sunflower was exerted by epibrassinolide at a concentration of 10^{-9} M and 2,3-disodium sulfate-24-epibrassinolide at a concentration of 10^{-9} M. Epibrassinolide monosulfates exhibit inhibitory properties relative to morphometric parameters in most cases. It is recommended that brassinosteroids are added by foliar processing of plants and immediately before exposing them to the adverse environmental factors. That is due to the high adaptive and stress-protective properties of this group of phytohormones.

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EXAMINATION OF THE EFFECTS OF CORN SMUT INFECTION ON PLANT PHYSIOLOGICAL PARAMETERS IN A SMALL SCALE FIELD EXPERIMENT

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SUMMARY

The corn smut (*Ustilago maydis* DC. Corda) is an ubiquitous pathogen of corn. It can infect corn at any phenology phases. The main symptoms are: chlorosis, necrosis, growth inhibition and presence of tumours. The aim of this study was to investigate the effects of corn smut infection on physiological parameters of the host plant, such as relative chlorophyll content, plant height, stem diameter, and leaf senescence. The parameters were measured seven and 14 days after the infection. The infection was taken at the plants' 4-5 phenology phase. The inoculum was created under laboratory conditions. The spore numbers were set up 2.500; 5.000 and 10.000 sporidium/millilitre in a Burkholder chamber.

In all treatments 14 days after the infection the relative chlorophyll content significantly decreased. Average plant height was reduced by the 10.000 sporidium number of corn smut at both sampling times. Stem diameter was thickened by the 10.000 sporidium number of corn smut only 14 days after the infection. As a result of the leaf senescence study, significant differences were observed among the treatments.

The 2.500 and 5.000 sporidium number of corn smut did not decrease the examined parameters significantly, except the relative chlorophyll content 14 days after the infection.

Key words: Corn smut, sporidium number, field experiment, relative-chlorophyll content, plant height, stem diameter

INTRODUCTION

Plant protection is one of the most important factors of agriculture (Pogăcean and Gavrilăscu, 2009). Diseases, pests, and weeds can cause large economic damage (Thomson et al., 2020). Today, Integrated Pest Management (IPM) is the most promising strategy of plant protection. It harmonizes the use of chemicals, biological control types and cultivation techniques to protect our plants, thus minimizes the effect of harmful substances on the environment and human health (Ha, 2014). Diseases are the major problem in plant protection.

Pathogen genus, host plant resistance, and environmental conditions also affect the infection (Lapin and Van den Ackerveken, 2013). Most often environmental conditions are favourable for the infection (Colhoun, 1973). High temperature and humid weather conditions are ideal for the diseases' proliferation (Velásquez et al., 2018). Host plant resistance depends on some

special molecules such as phosphatases and protein kinase (Yang et al., 1997). Resistance breeding is very important against diseases (Aydogu et al., 2015).

The corn smut (*Ustilago maydis* (DC) Corda) is one of the most important fungal diseases in the world (Robin et al., 2009). It can damage corn at all phenological phases (Banuett, 1995). The main symptoms of the infection are: chlorosis, necrosis, growing decrease, and tumour formation on leaves or on cobs (Skibbe et al., 2000). Direct protection against corn smut is impossible, preventive plant protection can be ideal such as, avoiding mechanical and insect damage. Economic damage is mainly caused by cob infection.

The aim of this research was to examine the effects of different sporidium numbers on the corn smut infection intensity, and the changes of some morphological parameters of corn under field conditions.

MATERIAL AND METHODS

The experiment was set up under field conditions. The experimental site was located in the garden of Institute of Plant Protection (University of Debrecen). The test plant was *Zea mays* cv. P9578. The seeds were sown at the end of April in small field trials, the number of repetitions was three. Before the sowing, the pre-emergence weeding was done by Successor T 300 (which agents are: 300 g L⁻¹ petoxamid and 187.5 g L⁻¹ terbutilazin) and the post weeding was done by Laudis (which agents are: 44 g L⁻¹ tembotrion and 22 g L⁻¹ izoxadifen-etil). After the sowing the irrigation was continuous with garden hose (approximately 5-6 litre a day). The pest-control was done with Karate Zeon 5CS (which agent is 50 g L⁻¹ lambda-cihalotrin). Fungicide was not used during the experiment.

The inoculum was created under laboratory conditions. The infected corn cobs were collected from fields, and then the teliospores were sprayed on the corn smut specific substrate. Then the pure culture was created. Next the pathogen was replicated in the liquid specific substrate and the dilution series (one -, ten -, one hundred -, one thousand -, ten thousand -, one hundred thousand- and one million fold) was made. (Figure 1).

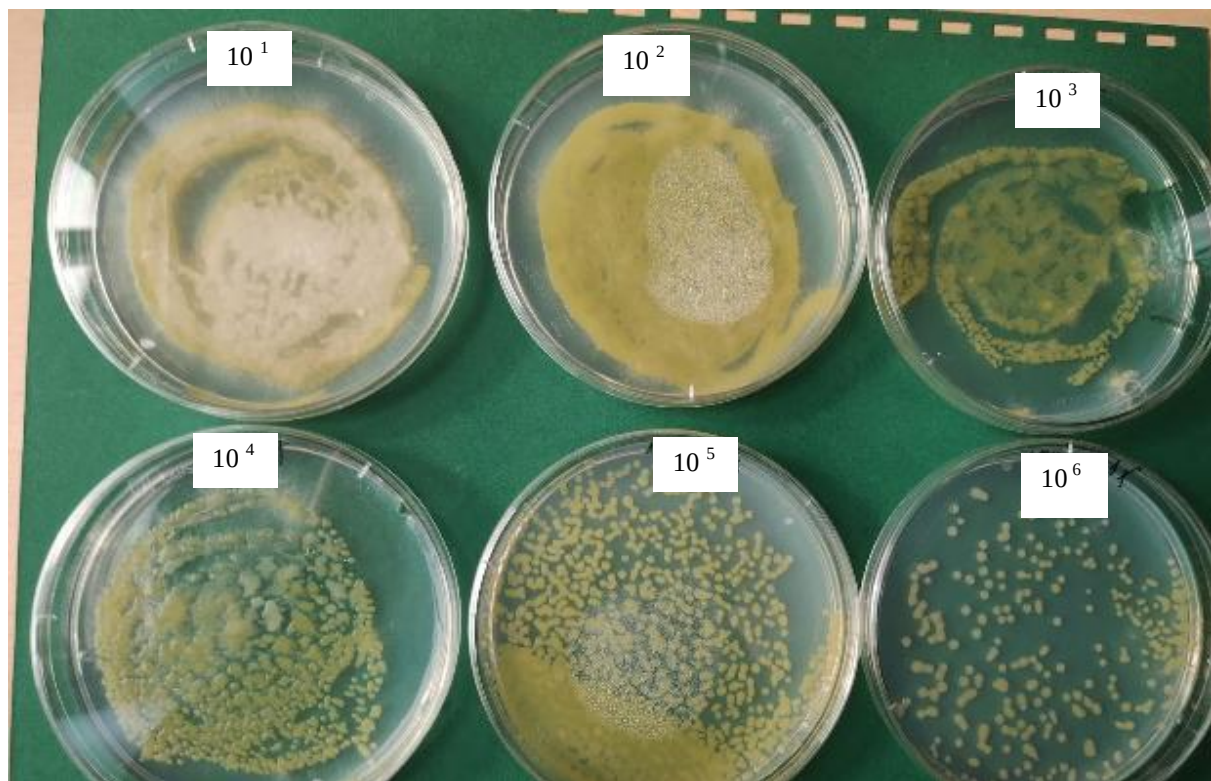


Figure 1. The colonies form after the dilution

Twenty-two strains were created and the first, seventh and twelfth strains were compatible with each other. For the infection, the monosporidial strains were propagated on a liquid medium, grown for 48 hours and then mixed into a ratio of 1:1 before the inoculation. The cell count was set up to 2,500; 5,000 and 10,000 sporidium number/ml in the Burker chamber.

The infection was taken in the 4-5 leaf stage of the plants. The inoculum was injected in the host plant's stem which was between the second and third nodes (in the case of control distilled water was injected instead of inoculum). There was ten plant/treatment.

The effects of infection were studied. The relative chlorophyll contents, plant height, stem diameter and leaf senescence was measured seven and 14 days after the infection. The relative chlorophyll content was measured on the sixth leaf by SPAD 502 PLUS (KONICA MINOLTA, Japan). The stem diameter was measured with slide calliper on the stem part between the second and third nodes. The plant height was scaled from the soil surface to the origin of the youngest leaf. The senescence of the leaf was determined by observing the yellowing of the cotyledon.

IBM SPSS Statistic 25 software was used for statistical analysis. The data normality was tested by Kolmogorov-Smirnov and Shapiro-Wilk tests. Results of the experiments were subjected to one-way ANOVA (for relative chlorophyll content, plant height, stem diameter

and leaf senescence) and the means were compared by Tukey HSD test at $p < 0.05$. Significant differences were marked by letters (a, b, c, d) in the conference paper.

RESULTS

The relative chlorophyll content was significantly decreased (with 16.81 %) when the plants were infected with 10.000 sporidium number of corn smut seven DAI. Significant difference was not observed among the other treatments compared to the control plants at the first sampling time. At the second sampling time the relative chlorophyll content was reduced by all treatments of corn smut infection. Relative chlorophyll content was reduced by 38.53% in the 2.500 sporidium number samples, 34.49% in the 5.000 sporidium number samples, and 57.35% in the 10.000 sporidium number samples when compared to the control plants. The result is illustrated in Figure 2.

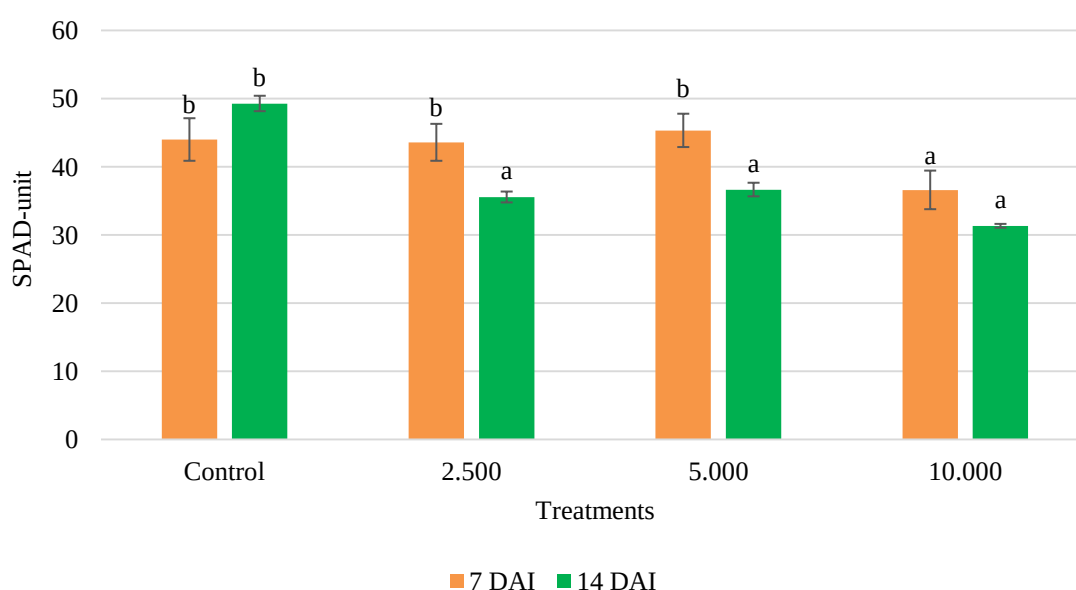


Figure 2: The effect of the corn smut infection on the relative chlorophyll content (SPAD-unit $\pm$ SD, $n=50$) in the sixth leaf of maize seven and 14 day after infection (DAI). Small letters (a, b, c) show significant differences between treatments based on Tukey HSD's test ($p<0.05$)

Analysing the effect of corn smut infection on plant height, only the 10.000 sporidium number of corn smut infection reduced the plant height significantly compared to the control plants at both sampling times. Seven DAI the plant height was reduced by 28.71 %, 14 DAI it was reduced 40.65% (Figure 3).

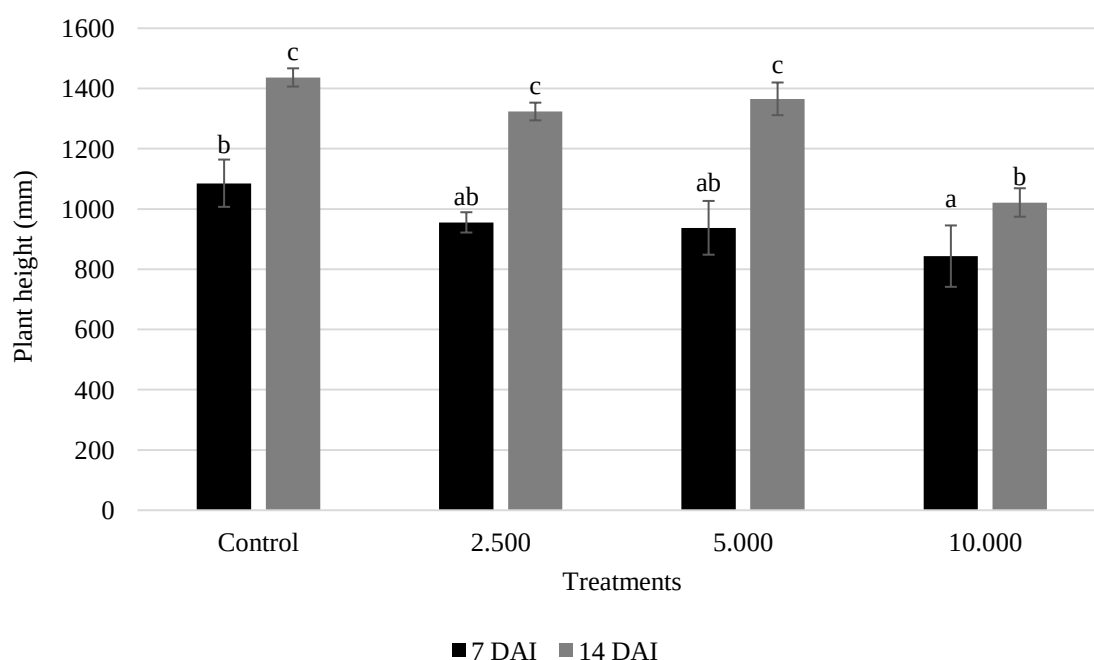


Figure 3: The effect of the corn smut infection on the plant height (mm $\pm$ SD, n=10) seven and 14 day after infection (DAI). Small letters (a, b, c) show significant differences between treatments based on Tukey HSD's test ($p < 0.05$)

Analysing the effects of corn smut infection on stem diameter, the 10.000 sporidium number had significantly thicker stem (by 23.74%) than the control plants 14 DAI. There were no significant differences among the treatments seven days after the infection (Figure 4).

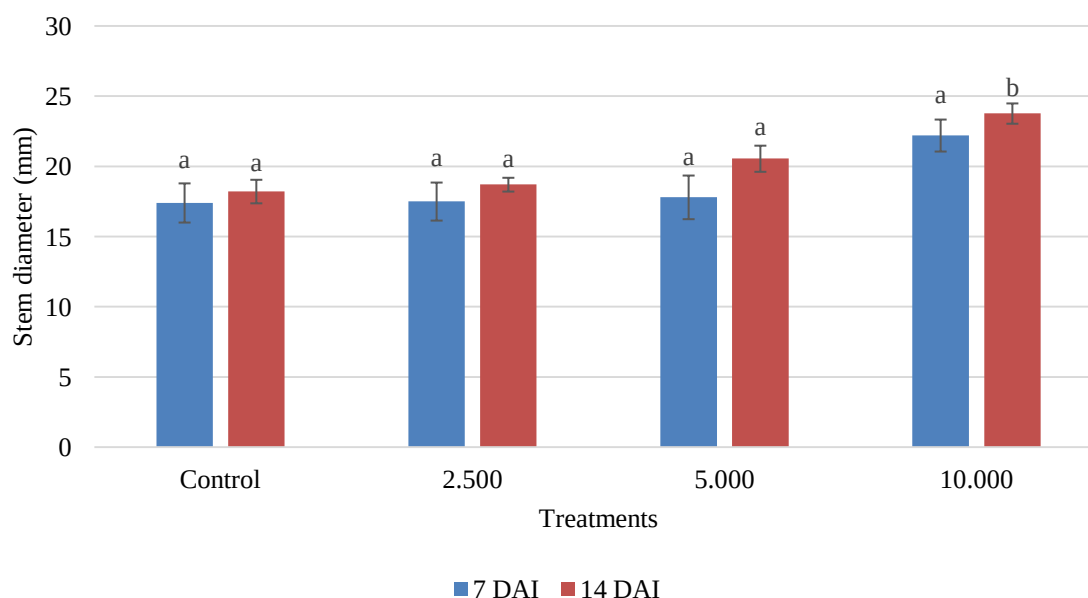


Figure 4: The effect of the corn smut infection on the stem diameter (mm $\pm$ SD, n=10) seven and 14 day after infection (DAI). Small letters (a, b) show significant differences between treatments based on Tukey HSD's test ($p < 0.05$)

As a result of the leaf senescence study, significant differences were observed among the treatments. Yellowing was observed on all seed-leaves of plants (results are not shown).

DISCUSSION

In this study the effects of corn smut infection on plant physiological parameters were studied. The experiment was set up under field conditions. The aim of this research was to investigate how the sporidium number affects the infection intensity. The 2.500 and 5.000 sporidium number of corn smut infection had no significant influence on the examined parameters (except the relative chlorophyll content at the first sampling time). According to the study of Frommer et al. (2019) the 2.000 sporidium number of corn smut infection decreased the relative chlorophyll contents of sweet corn hybrids two weeks after the infection. In this study all of the treatments reduced the relative chlorophyll content two weeks after the infection, but in the first sampling time only the 10.000 sporidium number of corn smut infection reduced the SPAD-unit. For the stated reasons, low concentration inoculums can also have negative effects, but the host plant can tolerate lower concentration inoculums.

Infection intensity is affected by the plant pathogen (pathogenicity ability, genus, species, and so on) and the susceptibility of host plant. Sweet corn is more susceptible for corn smut than

fodder corn (Frommer et al., 2015). Weather conditions also affected the infection intensity. In this experiment the weather was warm and humid which was beneficial for the plant pathogen. Moreover, insect and hail damage were observed.

The results showed that there was a correlation between the sporidium number and the infection intensity. If environmental factors are suboptimal for the pathogen, the infection intensity can be reduced. The spore number can reach one hundred thousand in field conditions. Prevention is very important, because there isn't any effective fungicide against corn smut.

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MAPPING OF NATIONAL PARKS IN THE REPUBLIC OF SERBIA

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Summary: *Program R is primarily used for statistical analysis, but it is possible to create interactive maps using certain packages. The program is getting upgraded on a daily basis, so today we have several packages in the R program that we can use for mapping. This paper presents the application of the R programming language in the mapping of five national parks in the Republic of Serbia. The “leaflet” is one of the most suitable packages for creating interactive maps, and it requires using of GPS data. The package “leaflet” works with open street maps and that allows us to create our own maps using certain adds within the package, such as polygons, dots, pop-up tags, lines, and other features within the package.*

Key words: *RStudio, the package “leaflet”, mapping, national parks, Serbia*

INTRODUCTION

Mapping is extremely important for the protection of the environment because it is crucial to mark a certain area that is placed under protection, and thus facilitates the management of a certain area. The two most suitable programs used in mapping are the GIS program and the R program. R program with its environment RStudio is extremely suitable for creating maps in the field of environmental protection. The virtual database in the R program is also suitable for mapping protected endemic species and for zoning the protected area. Mapping is a commonly applied procedure in landscape ecology, which deals with all segments of the environment. This paper presents the mapping of five national parks in the Republic of Serbia. One of the procedures to protect natural assets, in this case, a national park, is *in situ* protection, which means the protection of endangered plant species in their natural habitat, and taking into account that the protection of plant species is not possible without the protection of their natural habitats. *In situ* protection includes the conservation, maintenance, and recovery of species populations in their natural habitat, as well as the preservation of native ecosystems (Lakićević, 2018). A protection regime is a set of measures that determines the manner and degree of protection, arrangement, and improvement of a natural asset. Separation of protection regime is performed based on the degree of preservation of natural values, and there are three protection regimes, namely the protection regime of the I stage, the protection regime of the II stage, the protection regime of the III stage. The protection regime of the first stage represents strict protection, prohibits the use of natural resources and the construction of facilities, and limits works to scientific research and monitoring of natural processes. The protection regime of the II stage represents active protection, prohibits the

construction of facilities, works, and activities provided by the Law on Nature Protection. The protection regime of the III stage is proactive protection and management interventions can be performed in order to restore, improve, and revitalize the protected area. The protection strategy represents certain measures, goals, and priorities related to the protection of natural resources. There are several categories of protected areas, and these categories define management objectives and influence the choice of management type. In national parks, the protected area is mainly managed for the purpose of ecosystem protection, promotion, education, and recreation.

MATERIAL AND METHODS

The paper presents the mapping of five national parks in the Republic of Serbia. The National Park is an area which, according to its ecological and biogeographical characteristics, as well as natural ecosystems, represents a natural whole of exceptional importance with landscapes of special value in terms of the diversity of flora and fauna (Lakićević, 2018). The goal of mapping is easier management and improvement of the area, which is under a certain degree of protection. The program R version 4.0.2. and the “leaflet” package (Cheng et al., 2018) was used for mapping national parks. Before mapping, it is necessary to process GPS data, by creating a numerical vector in the program R for longitudes and latitudes of a certain national park, and then it is necessary to create a data frame, which is a table with aggregated data for each national park individually. A total of 382 GPS coordinates were needed for mapping the Fruška gora National Park, 224 GPS coordinates for the Tara National Park, 371 GPS coordinates for the Đerdap National Park, then 186 GPS coordinates for the Kopaonik National Park and for the Šar planina National Park we needed 500 GPS coordinates. By processing GPS coordinates and merging them into a data frame, as well as using the “leaflet” package, and applying specific codes we create a polygon on the map, that represents the exact boundaries of each national park. The color of the polygon and the position of the legend can be adjusted. The main goal of the paper is to map the borders of each national park in Serbia and briefly describe its basic characteristics related to their floristic elements.

RESULTS AND DISCUSSION

Table 1 contains basic data on national parks in the Republic of Serbia, such as their name, year of protection, and the area they occupy, expressed in hectares. The largest national park in the Republic of Serbia is the Đerdap National Park, which covers an area of 63,608 ha, and the smallest is the National Park Kopaonik.

Table 1. Data on national parks in the Republic of Serbia

National park	Year of protection	Area [ha]
Fruška gora	1960	25,393
Tara	1981	19,175
Đerdap	1974	63,608
Kopaonik	1981	11,809
Šar planina	1993	39,000

The massif of the Fruška gora National Park is built of different types of rocks. Over 1,400 plant species have been recorded on Fruška gora, and 12% of them are relict or endemic. As far as vegetation is concerned, the most represented are mixed forests *Quercus petraea* and *Carpinus betulus*, herbaceous vegetation consists of steppe communities of relict character. In the Fruška gora National Park, there are 5 paleontological, 14 botanical, and 12 ornithological reserves of high scientific value (Vujić, 2008). Borders of the National Park Fruška gora is shown in Fig. 1.

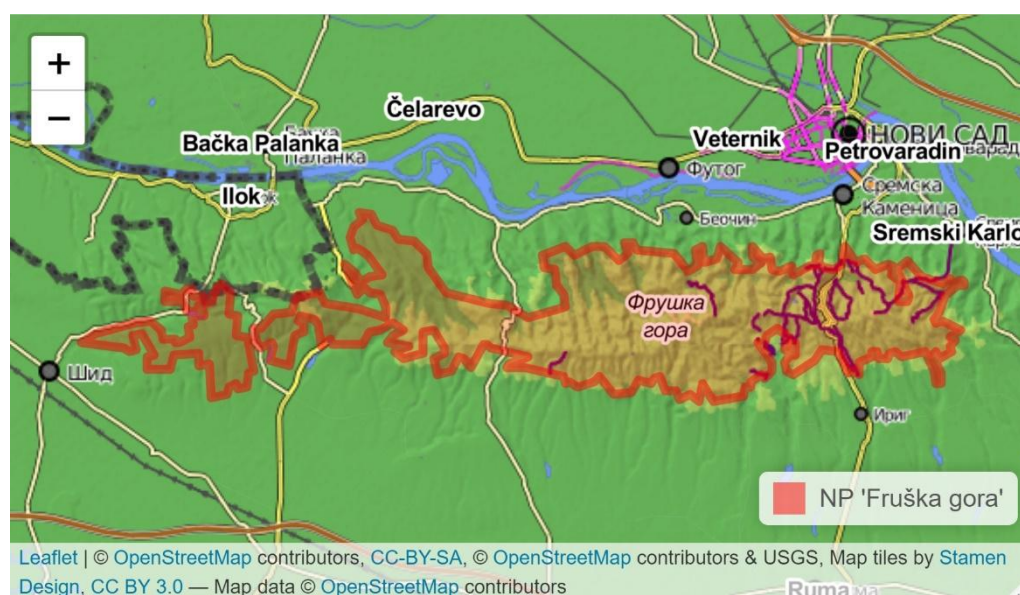


Fig. 1 Fruška gora National Park

Tara National Park is built of a Triassic massif formed of Paleozoic limestone and shale. Over 1,000 plant species have been recorded on Tara, of which 20 are endemic, especially valuable is the species *Picea omorika*. As far as vegetation is concerned, the most represented are mixed autochthonous communities *Picea abies*, *Abies alba*, and *Fagus silvatica* (Vujić, 2008). Borders of Tara National Park are shown in Fig. 2.



Fig. 2 Tara National Park

Đerdap National Park is built of crystalline shales, metamorphic and igneous rocks, and Mesozoic and Cenozoic deposits. The flora of the Đerdap National Park is characterized by diversity, richness, and relict character, and over 1,100 plant species have been recorded, many of which are remnants of tertiary flora, as well as 35 relict forest and shrub communities. Of the vegetation, the most common are the forests of the modern type (Amidžić, 2010). Borders of Đerdap National Park are shown in Fig. 3.

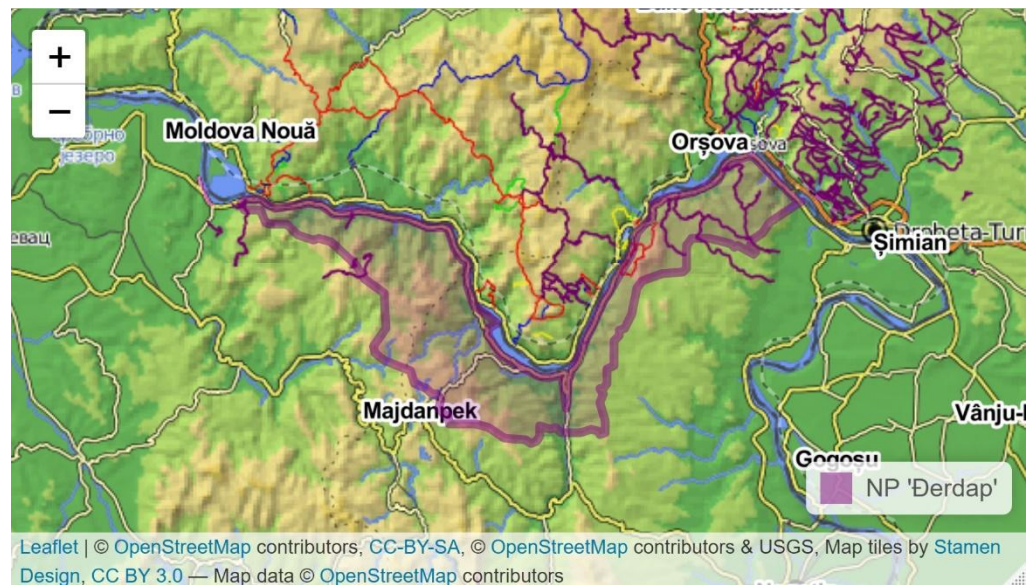


Fig. 3 Đerdap National Park

Kopaonik National Park is built of granite, peridotite, and serpentinite, and a small proportion of limestone fragments. There are over 1,500 plant species in the Kopaonik National Park, of which 91 are endemic. From the bottom of the mountain to the top, forest ecosystems change from genus *Salix* sp. and the genus *Populus* sp. , then thermophilic communities of *Pinus nigra*, genus *Quercus* sp. and genus *Fagus* sp., while above them is a belt of shrubby vegetation, built of the genus *Juniperus* sp., *Picea* sp., and *Vaccinium* sp. In the Kopaonik National Park, 13 localities are under the first protection regime (Vujić, 2008). Kopaonik National Park is shown in Fig. 4.

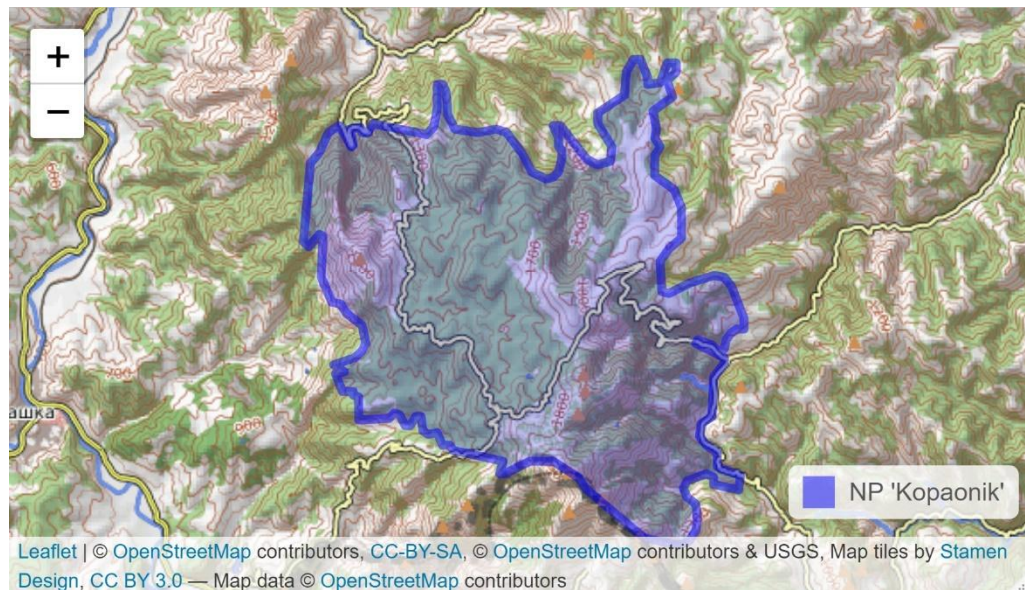


Fig. 4 Kopaonik National Park

Šar planina National Park is an area with about 2,000 plant species, while a special value is represented by numerous endemic and relict species of the tertiary and glacial age. 332 endemic species have been registered, of which 20 are local endemics. Šar planina National Park, together with Prokletije, is one of the six European centers of endemism (Amidžić, 2010). Using package the “leaflet” each national park is shown in a different color, for the best possible visual presentation, and each of these colors is shown in the legend, along with the name of the national park. Šar planina National Park is shown in Fig. 5.



Fig. 5 Šar planina National Park

CONCLUSION

The paper presents the application of the R programming language in the mapping of all five national parks in the Republic of Serbia, using the package “leaflet”. Package “leaflet” works with open street maps and GPS data that we use in this case to create polygons, that represent the border of a certain national park. A total of five national parks located in the Republic of Serbia have been mapped using GPS data. The “leaflet” package allows us to use latitude and longitude, which are previously merged into one data frame complete with the names of national park, to apply points to the map, which are then merged and form a polygon, or in this case the border of the national park. Each national park is shown in a different color, for the best possible visual presentation, and each of these colors is shown in the legend, along with the names of the national parks. Future research may address the tasks of mapping of the zones under the first, second, and third protection regime in each national park.

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IMPACT OF SOIL PROPERTIES ON SOIL MOISTURE MAPPING AND IRRIGATION REQUIREMENTS IN MONTENEGRO: THE CASE OF ANCIENT OLIVE GROVES IN THE COASTAL MEDITERRANEAN REGION

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Rainfed olive growing is an ingrained tradition in Montenegro. Due to the extensive agricultural practices, insufficient quantities of olive fruit and oil are obtained. Soil moisture mapping is crucial for sustainable land and water management. Therefore, available soil water capacity was computed for 5030 soil horizons, testing the performance of six pedo-transfer functions and two spatial interpolation techniques. The results showed good performance of both interpolation techniques, while the pedo-transfer functions of Rawls and Brakensiek (1985) and Saxton and Rawls (2007) proved to be the most suitable for the soils of Montenegro. The prediction model for irrigation management confirmed satisfactory results of soil moisture estimations and relative yield of ancient olive trees.

Key words: soil moisture, olive groves, water requirements

INTRODUCTION

Despite olive cultivation being an ingrained tradition in the coastal part of Montenegro, extensive agricultural practices, parcels' fragmentation, undeveloped market channels, inadequate promotion and limited scope of domestic production lead to unsatisfactory quantities of olive fruit and olive oil, insufficient to meet the national consumption requirements. National public and private bodies concerning olive growing and olive oil production recognize numerous opportunities, such as connection between tourism and olive growing; establishment of new intensive olive plantations on uncultivated lands; strengthening identity of Montenegrin olive industry at the international frames; development of olive products' assortment; energy production, designation of origin, refinery development and improvement of domestic "know-how" through the intensification of scientific research.

In Montenegro, due to the unequal distribution of precipitation, critical hydrological periods occurring in the warmest part of the year coincide with the periods with greatest demand for water and energy, thereby facing the "resource paradox". This environmental problem refers to the strong need for nation-wide predictive management and digital soil moisture mapping. The proper implementation of soil and water conservation practices in the

olive groves requires detailed knowledge governing hydrological processes. Soil water serves as a solvent and carrier of nutrients for successful plant growth and furthermore, acts as a nutrient itself, regulates soil temperature, soil forming processes and weathering; enables metabolic activities of microorganisms, participate in chemical and biological activities of soil and provides an essential component for plant growth and photosynthesis. Therefore, soil moisture information may be used for water storing management, early warning of droughts, irrigation scheduling and crop yield forecasting.

Hence, to investigate soil moisture over a large area is a very difficult, laborious, time and money consuming practice. In addition, the lack of information on soil data at different scales is an issue due to its extreme variability in both the spatial and temporal domain and at different observation scales. On the other hand, estimation models have the advantage of achieving simultaneously satisfactory sampling frequency and national coverage. There is a variety of such methods which are proposed according to the specific data types, available inputs and/or specific variables to be interpolated.

This research is primarily focused on assessing the feasibility of using spatially variable soil information for soil moisture mapping at the country and field scale, in order to provide more detailed inputs to simulation models so as to evaluate irrigation water requirements for olive groves in the southern part of Montenegro. The study provides a comparative analysis of soil moisture variation monitored at the experimental site with soil moisture estimated using pedo-transfer functions and interpolation techniques on small and large scale. This research shows an indisputable need for accurate and continuous real-time in situ soil moisture monitoring covering a wide range of agro-climatic zones and cropping patterns in Montenegro (*Ben-Gal et al., 2006; Arampatzis et al. 2018*).

MATERIAL AND METHODS

The research was conducted on both national (large) and experimental site (small) level. After producing final database of Montenegrin soils available in GIS, pedo-transfer functions were tested, as well as different spatial interpolation techniques in order to obtain soil moisture maps. Climatic data for the experimental site were collected on daily basis. After site preparation including pruning of the olive trees, soil sampling and laboratory analysis were carried out. The results obtained for the specific site were used for statistical analysis of soil spatial variables accuracy in order to test the feasibility of their use for soil moisture prediction purposes. Finally, an Excel-based tool was used for real-time irrigation management which enabled development of soil water balance in the root zone, estimation of the amount of water to be applied according to different irrigation practices together with potential yield reduction.

National soil database of Montenegro has been complemented from two main data sources. A great effort was made to precisely accomplish the task of data digitization, filtering and validation as well as soil profiles' coding and inserting into GIS database. The final produced database, filtered and validated, represents the basis for further research of Montenegrin soils. It contains 5030 soil layers (distributed within 2347 soil profiles) with

clay, sand, silt and organic matter percentage content presented. Given that Montenegro's territory is about 13812 km², one sample represents an area of 5.9 km². Soil texture data for Montenegrin soils, classified according *Atteberg (1905)* system were transferred into USDA/FAO soil particle distribution range (clay <2 µm, silt 2-50 µm and sand 50-2000 µm), according to the model proposed by *Minansy and McBratney (2007)*.

Taking into consideration required input variables, place of origin and evaluation performance, six different pedo-transfer functions (PTFs) and/or models were tested (*ROSETTA*, *Chakarborty et al. (2011)*, *Adhikary et al. (2008)*, *Saxton and Rawls (2006)*, *Rawls and Brakensiek (1985)* and *Tomsella and Hodnett (1998)*). PTFs requiring soil texture data as input variables were ran for the complete dataset (5030 layers), while the PTFs requiring both soil texture and organic matter were ran for the layers having information about organic matter content (305 layers miss organic matter percentage data info).

After testing six PTFs, values of Field Capacity (FC) and Permanent Wilting Point (PWP) were obtained. After that, Available Water Content (AWC) was calculated as the difference between these two parameters (Eq.1).

$$AWC = FC - PWP \quad \text{Eq. 1}$$

Where:

AWC is available water content (%);

FC is field capacity (%);

PWP is permanent wilting point (%).

In order to present AWC volume according to the depth of soil layer/horizon per unit area (1m²), the following equation (Eq. 2) was used:

$$AWC_h = AWC / 100 \cdot [(d_2 - d_1) / 100] \quad \text{Eq. 2}$$

Where:

AWC_h is volume of available water content per soil layer/horizon (m³);

d_2 is the depth of the upper soil horizon per unit area (cm³);

d_1 is the depth of the lower soil horizon per unit area (cm³).

However, according to the fact that coarse textured soils have lower AWC values due to large pores subject to free drainage, content of coarse fragments had to be taken into consideration. Reduction of AWC was carried out as shown by Eq. 3:

$$AWC_{rs1} = AWC_h \cdot (1 - CF / 100) \quad \text{Eq. 3}$$

Where:

AWC_{rs1} is reduced available water content (m³) – first method;

CF is percentage (%) of coarse fragments (>2mm).

After computing AWC_{rs} for each soil horizon (layer) within the profile, values were summed in order to obtain final available water content reduced for coarse fragments for each soil profile existent on the Montenegrin territory.

For the purposes of this research, two different interpolation techniques were implemented: Inverse Distance Weighting (IDW) and Ordinary Kriging (OK) (*Li and Heap, 2014*). Therefore, two interpolation methods were run for six different values of AWC_{rs} for 2347 soil profiles within Montenegrin territory. Apart from those twelve AWC maps on the national level, chosen interpolation techniques were implemented at soil subtype level as

well. Experimental site, located in the olive orchard *Džidžarin* in municipality of Bar, Montenegro was examined, and soil subtype (*Eutric Cambisol on phlich*) was assigned according to the national soil classification system (*Fuštić et al., 2000*). Finally, twenty-four maps were produced and compared with measured AWC value on-field, in order to determine the most feasible PTF and interpolation technique for moisture mapping of Montenegrin soils. Statistical indices for analyzing the performance of the selected interpolation techniques and therefore pedo-transfer functions were as following: Absolute Error (ΔA), Relative Error (RE), Mean Absolute Error (MAE) and Root Mean Square Error (RMSE).

The study area is located approximately 5 km from the Adriatic Sea, between 75 and 265 meters a.s.l. and it has the following coordinates - 42°09'N, 19°14'E. The total surface area of the olive orchard is 682,827 m². From January to September 2019, daily weather data was collected regularly for the following parameters: average air temperature (°C), minimum air temperature (°C), maximum air temperature (°C), rainfall amount (mm), sunshine duration (hours), average relative humidity (%), average wind speed (m/s), maximum wind speed (m/s) and wind direction. Opening of the soil profile in order to take disturbed and undisturbed samples and perform laboratory analysis was carried out at the representative parcel within the olive orchard, according to the methodology described by *Dugalić and Gajić (2005)*. 18 undisturbed soil samples were taken, 4 per first three horizons and 3 per last two horizons, from top to bottom. Furthermore, soil sampling for field moisture monitoring of greater soil depths has been conducted with a hand auger, at four depths: 0-20 cm; 20-40 cm; 40-60 cm and 60-80 cm. Soil moisture has been monitored at the experimental site where six old olive trees were taken into consideration – three rainfed and three irrigated trees.

Irrigation time and water amount was coordinated by the outputs of Excel-based tool for real-time irrigation management at field scale. This model was used in order to develop soil water balance in the root zone on a daily basis, estimate amount of water that needs to be applied and assume eventual olive yield reduction in case of moderate water stress conditions (*Todorović M., 2006*).

RESULTS AND DISCUSSION

Trends of daily weather data - The data was collected for the eight-month period, from the 1st of January to the 1st of September 2019. While the average daily temperature for this period was equal to 17.9 °C, minimum average was recorded in January (2.1°C), and maximum average in June (30.3°C). The total sum of precipitation during the experimental period equaled to 605.5 mm and it occurred in 65 events. Total sunshine duration for eight month period was equal to 1810.3 hours (≈ 76 days), justifying the fact that Bar is the sunniest town in Montenegro. Daily relative humidity ranged between 23% and 91%, with the average equal to 56.8%. The most frequent wind direction is north (N) or north-east (NE); however, north-west (NW) and north-north-west (NNW) occurred often as well. Average reference evapotranspiration was equal to 6.44 mm/day. On the other hand, crop evapotranspiration, calculated for the olive trees, had an average value of 3.74 mm/day.

Description of soil profile - According to the national soil classification system, the subtype of the examined soil is *Eutric cambisol on phlich*. This soil subtype occupies the area of about 35-40.000 hectares, mainly in the coastal, southern Mediterranean part of Montenegro. Terrains where *Eutric cambisol on phlich* is mainly present are usually mountainous, stretching from the sea shore to the surrounding carbonate hills, up to 800-900 meters above sea level. These conditions led to the development of shallow, erodible soils with alluvium-colluvium properties. Therefore, the majority of landscapes with *Eutric cambisol on phlich* are terraced with dry-stone walls. Distinguished soil horizons of the soil profile at the study site (Fig. 1) are: organic (O), mineral topsoil horizon (Ac), first subsoil illuvial horizon (Bca), second subsoil illuvial horizon (Bt) and bedrock (C). Soil processes that were recorded during sampling are initiation of gleization and lessivage (from Bt horizon), indicating leaching of clay particles to deeper horizons, where poor drainage conditions started to occur due to waterlogging and reduction properties.

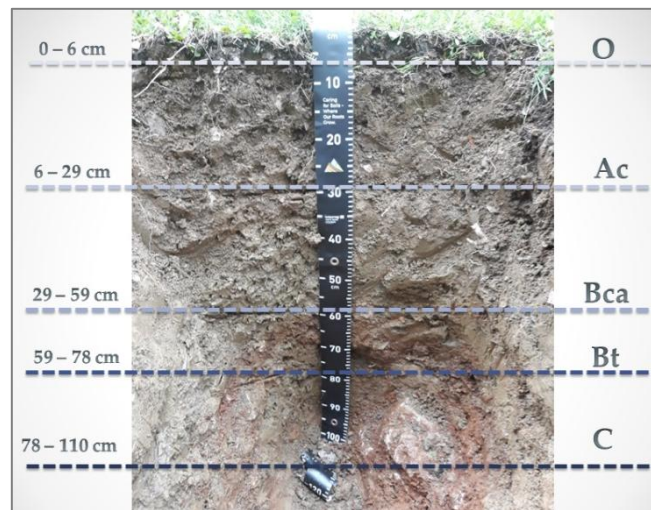


Fig. 1: Experimental soil profile with distinguished horizons

Soil physical properties - Mechanical composition of rainfed *Eutric cambisol on phlich* indicates textural differentiation and heterogeneity within the profile depth. Clay content increases by depth, and its highest content is observed in Bt horizon (silty clay horizon). Particle density varied from 2.3 g/cm³ from the most upper soil horizon (O) to 2.68 g/cm³ in illuviated soil horizon, while bulk density varied irregularly up to 1.72 g/cm³. Average soil porosity equaled 40.6%, and therefore, may be classified as low porous soil. Average air capacity values equaled to 20.32%, which is moderate to high air capacity. Average value of field capacity for the examined soil profile equaled to 34.46%. Motions of FC values are irregular and negative trend was observed until Bt horizon, when FC increased up to 38.8%. Observed soil moisture motion may be defined as declining with depth, while the peculiarities of Bt horizon (high clay content) lead to significant increment of soil water contents at different matric potentials. The average Available Water Content of the profile is 0.090152 m³. Within the whole soil profile, macro-aggregates are the pre-dominant structural aggregates, indicating homogenous soil structure.

Soil chemical properties - Agro-chemical analysis of the sampled soil showed that pH reaction (in 1M KCl) is neutral throughout the whole soil profile. Calcium-carbonate content increased with depth, ranging from carbonate to highly carbonate soil (highest in Bca horizon). The content of organic matter decreases with depth, ranging from very humic in thin surface layer, decreasing to poorly humic and terminating with very poorly humic class. The content of easily available phosphorous (P_2O_5) and potassium (K_2O) significantly decreased with depth and soil was being poorly provided with P_2O_5 and medium provided with K_2O . At depth greater than 30 cm, the soil was characterized with poor availability of both elements. This situation may be explained by seldom and shallow fertilizer application, due to the specific terrain configuration, parcels fragmentation, soil erosion and olive trees age as strong limitations.

Soil moisture maps of Montenegro and *Eutric Cambisol on phlich* - Working in MapInfo Professional 12.5 and Vertical Mapper 3.7, we managed to produce six maps of AWC in Montenegro with IDW technique and six additional with OK technique. According to the statistical performance indicators, the IDW outperforms OK in the first case scenario (level of Montenegro) with MAE of 0.0244 m^3 and RMSE 0.0872 m^3 . On the other hand, the OK outperforms the IDW in the second case scenario (level of *Eutric cambisol on phlich* soil subtype) with MAE of 0.114 m^3 and RMSE of 0.0569 m^3 . On the other hand, the OK outperforms the IDW in the second case scenario (level of *Eutric cambisol on phlich* soil subtype) with MAE of 0.114 m^3 and RMSE 0.0569 m^3 . Compared to the interpolation at the larger, national scale, it may be concluded that smaller scale estimations are more correct and relevant. It is very important to mention influence of different PTFs, since determination of input AWC values were obtained with six different functions that required either percentage of sand, clay and silt and/or organic matter content as input parameters. Therefore, it may be concluded that PTF of *Saxton and Rawls (2016)* is suitable for national scale interpolation, while for the soil subtypes, more specific PTFs, such as *Rawls and Brakensiek (1985)* should be used and/or, even better, developed for the experimental site. Best case scenarios are showed in Fig. 2 (large scale) and Fig. 3 (small scale).

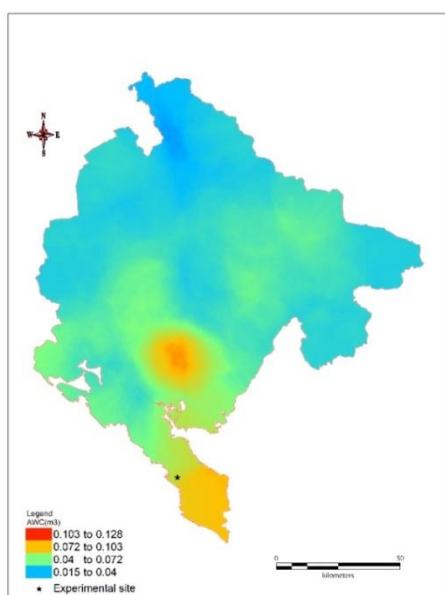


Fig. 2: Soil moisture map of Montenegro
OK_AWC_{rs5}

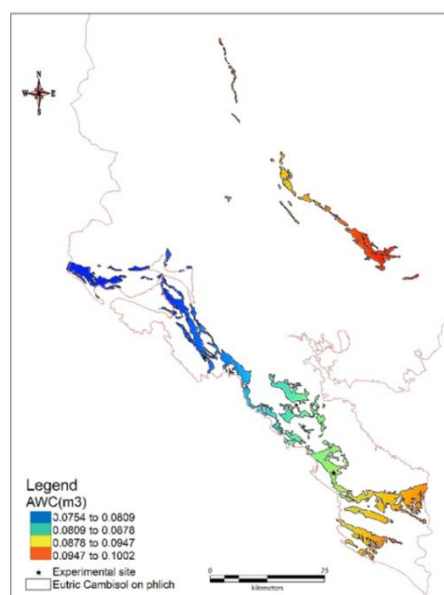


Fig. 3: Soil moisture map of *Eutric cambisol*
on phlich IDW_AWC_{rs4}

Olive irrigation management at the experimental site - The soil water balance was simulated on a daily basis for both irrigated (Fig. 4) and rainfed parcels and the soil water content was given by means of the root zone water depletion. Total net irrigation supplied was 194 mm. It was equal to gross amount of 21.5 m³ of water over an irrigated area of 40 m². 100 liters bucket was filled up before irrigation in order to measure the time of filling and to supply the proper amount of water. It must be noted that it was very difficult to introduce irrigation supply at the experimental site, due to the complex terrain configuration and almost non-existent infrastructure.

Yield forecasting of *Žutica* olive variety with Excel-IRR - The olive fruits at the three branches collected from the three irrigated trees were compared with the additional tree branches, collected from the olive trees at the rainfed parcel. Parameter of interest was the olive fruit mass (gr) and it was compared with the yield predicted by the model. According to Excel-IRR, for the irrigated plot, 10.15 t/ha of yield was predicted, while for the rainfed treatment yield dropped to 8.97 t/ha, facing the decrease of 1.18 t/ha or 11.6%. Olive yield obtained at the rainfed olive complex of *Džidžarin* equaled to 9.85 t/ha, while the estimations in the case of irrigation adoption, showed yield increase of 0.2 t/ha and it was equal to 10.05 t/ha. The comparison of model estimated relative yield data with the on-field small scale measurements and large-scale predictions, it might be concluded that the model showed excellent fitting for the irrigated case scenario and slight underestimation (8.94%) for the rainfed treatment.

Comparison of measured and estimated soil moisture for rainfed and irrigated treatments - The comparison of model estimated soil moisture in terms of depletion with the on-field regular soil moisture monitoring showed significantly accurate results, with slight underestimation. Average gravimetric soil water content difference between the rainfed and

irrigated parcels equaled to 10.15%, while the difference in depletion at the effective root zone estimated by Excel-IRR tool, if expressed in percentages, amounted to 8.11%. A slight underestimation of 2.04% may be observed, implying suitability of using this irrigation management tool at the field scale for the olive orchard soil moisture monitoring at the coastal Montenegro.

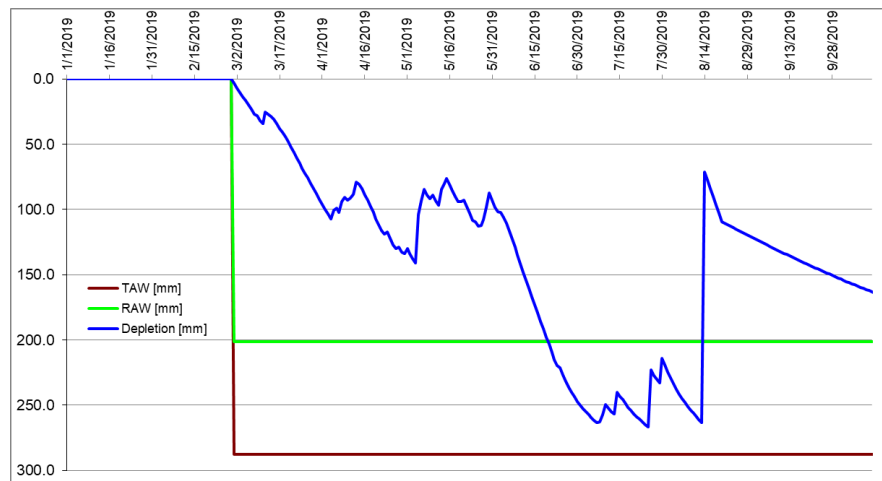


Fig. 4: Soil water balance at irrigated experimental olive parcel

CONCLUSION

This is the first research that adopted approach of soil moisture mapping in Montenegro, thus helping the regulation of cropping pattern and acting as useful tool for overcoming the resource paradox. Particular importance of this study is reflected through the contribution given to the completion of the national soil database. An added value to this study was given through the particle size conversion from the national classification system to the USDA/FAO model, expanding the spectrum of possibilities for further data processing and usage. High accuracy of the final results proved excellent performance of *Rawls and Saxton (2006)* and *Rawls and Brakensiek (1985)* pedo-transfer functions for the soils of Montenegro, as well as of deterministic and geo-statistical spatial interpolation techniques used. Excellent performance of the model in domain of yield predictions and soil moisture variations for both treatments implies suitability of its using for the olive orchard management at the coastal Montenegro. Numerous spatial interpolation techniques and pedo-transfer functions for soil moisture estimations are available and their usage potentials for the soils of Montenegro should be tested in the future. The results of Available Water Holding Capacities (AWC) at the territory of Montenegro and *Eutric Cambisol on phlich* should be further validated in order to test the accuracy of the selected methods. Furthermore, it is highly recommended to continue soil moisture mapping on the scale of the remaining 20 soil subtypes. Development of site-specific pedo-transfer functions would improve the accuracy of the AWC inputs used for spatial interpolation. The quantification of soil moisture in Montenegro provided in this research opened a whole new spectrum of interpretation and analysis of data which will lead to more intensive management and deeper insights. Precise prediction of soil moisture, as one

of the main factors in the agricultural production and hydrological cycle, acts as an important tool for the rational use and management of water resources. The knowledge of the spatial distribution of soil moisture is a rather important prerequisite for the modeling of many hydrological processes such as infiltration, soil erosion and runoff. Soil moisture maps might be also particularly useful for policy making in domain of agriculture. Finally, the concept of soil moisture estimation may provide a positive indirect impact on the overall agro-economy of Montenegro.

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STUDIES ON BIOACTIVE COMPOUNDS OF MUSHROOMS AND THEIR POTENTIAL ANTIVIRAL EFFECTS AGAINST COVID-19

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SUMMARY

The fungi's antiviral activity has been well-known for a long time. Among them, the Ascomycota deserves mentioning in particular, although antiviral compounds are also present in the Basidiomycota. Mushrooms with medicinal potential were historically used in Asia, where the ancient tradition of alternative medicine makes good use of them. The elements responsible for bioactive properties due to specific chemical compounds, with unique properties are mycelium of medicinal mushrooms, as well as extracts and chemical derivatives. The chemical composition of mushrooms extracts can vary in accordance with the mushroom species that served for their preparation. Their antiviral potential has been highlighted in the present paper, showing their high efficiency in the prevention and the treatment of viruses, such as HIV, SARS, herpes simplex, enterovirus, hepatitis C, the infectious hematopoietic necrosis virus (IHNV), cytomegalovirus and others. This work related on published scientific papers brings a contribution consisting in an exhaustive evaluation of the most efficient mushrooms with antiviral and anti-inflammatory potential.

Key words: SARS, antiviral, medicinal mushrooms, coronavirus, bioactive compounds

INTRODUCTION

In spite of modern medicine progress, viruses continue to take many lives around the world. The new Coronavirus SARS-CoV 2 has infected a great number of people, among its effects ranging from milder symptoms such as fever and muscular pain to very severe and even fatal complications of the condition of the respiratory system and the heart (Shahzad et al., 2020). In December 2019 in China this new Coronavirus SARS-CoV 2 produced many cases of pneumonia, that led to complications and even to death (Bai et al., 2020; Huang et al., 2020). The last data presented by Worldometer about Covid-19 testify about a number of 56 million cases of infected persons globally, of which 1.3 million deaths. In March 2020, when statistics started to be released, there were around 60 to 70 thousand newly infected

persons daily, while in November it reached a tenfold of 600,000 cases per day (Worldometer, 2020). At present there is research under way for a vaccine, seen as a solution in the face of this scourge. It is therefore the right time to also consider the development of efficient viral natural inhibitors, from different sources (plants, algae, mushrooms).

The mushrooms' nutritious value has always been appreciated, yet their therapeutic qualities came to be studied much later. However, the mushrooms' healing properties were known in China's traditional medicine for thousands of years, and they are still being used today (Kosanić et al., 2016). Mushrooms have the capacity to accumulate minerals and vitamins, and also to produce a great variety of secondary metabolites, such as organic acids, alkaloids, phenols, terpenoids with immunostimulatory, antioxidant, anticancer, antimicrobial and antiviral effects (Jayakumar et al., 2011). At the same time, mushrooms are a rich pharmacologic and dietary source of natural polysaccharides/ glucans, antioxidants and antibiotics (Akyuz and Kirbag 2009).

For a virus-destructive treatment to be deemed efficient and safe, it must pass the screening test, yet very often the trials fail, because they are toxic for humans and not safe enough to be administered (Stamets, 2006). Mushrooms, on the other hand, are a very good source of bioactive components from a pharmacological point of view and are at the same time safe for human consumption (Aras et al., 2018).

This paper brings together several scientific works based on studies aimed at limiting the virus' growth and spread in humans, with the help of antiviral components extracted from mushroom mycelia, especial from Basidiomycetes.

DESCRIPTION AND HISTORY OF FUNGI IN MEDICINE

Fungi as moulds, yeasts and macrofungi (mushrooms) are abundantly spread, in soils and in water or food as well. In nature, they have a beneficial contribution, as they are associated with decomposition of organic waste from plants and dead animals, sometimes even helping to clean waters from oil spills. The fungi thrive in all kinds of natural habitats.

The term mushroom is being widely used to mean the mycelium's fructiferous body that has a cap and a stem and can be seen and touched. Mushrooms are part of the kingdom Fungi. Taxonomically, the terms Macromycetes or Macrofungi are used. The best known Macromycetes belong to the branches Basidiomycota and Ascomycota. Currently, it is estimated that there exist a number of 140,000 known species, which represents 10% of the existing fungi species (Lindequist et al., 2005), yet few of them have been domesticated and can be cultivated.

The discovery of penicillin isolated from *Penicillium notatum*, belonging to the branch of Ascomycota, in 1928 by Alexander Fleming was an important breakthrough in the fight against bacteria. This discovery of the antibiotic has revolutionized medicine and all biology sciences, getting to save thousands of lives during the war. Building on this experience, research can be done to discover and produce new types of antibiotics in microbiology. A large percent of already known antibiotics in the world are being produced from Actinomycete (Jong and Donovick, 1989).

Fungi are producing secondary metabolites that in an ecological sense represent an essential component in nature and at the same time support the human organism, with their capacity to release their bioactive compounds into the bloodstream, from where they are taken to all the body's organs and exert beneficial chemical and pharmacological effects on the human organism. These secondary metabolites produce many substances with antibiotic effects, some of them being proven to even have antiviral action. Given the current Covid-19 pandemic, it is worth trying to exploit these mushrooms as potential sources of antiviral substances.

Mushroom metabolism consists of two phases: the primary metabolism (the balanced growth phase) in which mushrooms gather energy, proteins and macromolecules to grow, and secondary metabolism (the stocking and maintenance phase) when cell growth and functions stop because they have accumulated enough energy to develop, and when they start releasing specific compounds (Rose, 1987). In this static phase of stocking and maintenance secondary metabolites are synthesized, among which antibiotic substances as well. These antibiotics are not playing a very important role in mushroom development, since they are developing after the organism has passed its growth phase, hence their denomination as secondary metabolites.

This review offers a perspective on the importance of mushrooms as ingredients with antiviral potential, that can help human organisms in stimulating the immune system or even in healing.

CHEMICAL COMPOSITION AND ANTIVIRAL MOLECULES OF MUSHROOMS

Fungi contain a wide range of nutrients that are beneficial for human health, such as proteins, fats, carbohydrates, valuable minerals and vitamins (fat soluble vitamins and provitamins, water soluble vitamins). Concerning the chemical composition that is responsible for the taste and flavor, it includes pigments, phenolic compounds, sterols, indoles, purines, biogenic amines and polyamines. Other chemical compounds with a beneficial effect in stimulating health are antioxidants, beta-glucans, carbohydrates, proteins (lectins and hemolysins), lovastatin, eritadenine, gamma-aminobutyric acid, ergothioneines (Kalač, 2016). These bioactive compounds vary according to the analyzed species and the kind of substratum it grew on, either in nature or in cultivated areas, and also depending on the way it was stored and dried (Guillamon et al., 2010).

Most mushrooms contain polysaccharides that include chitin, glucans and heteroglycans. These polysaccharides are being described as biological products from the growth and development process of the mushroom's fructiferous body. Polysaccharides are playing an important role in regulating the immunity of human cells (Kalač, 2016). Immunomodulating polysaccharides such as β -glucans are not toxic for the human body and have no secondary effects while being used against bacteria and viruses (Masterson et al., 2020), showing also antitumor and immunostimulating properties (Ren et al., 2012).

Clinical studies confirm the fact that β -glucans can reduce a series of symptoms of the respiratory apparatus caused by various infections, as well as the fact that they can lower systolic and diastolic artery blood pressure (Dharsono et al., 2019).

It is well-known that the symptoms caused by the COVID-19 infection are severe, and there are studies that proved alleviation of symptoms and considerable improvement of the patient's state, following administration of β -glucans, particularly in most vulnerable cases within ICU's (Geller and Yan, 2020). It is presumed that oral administration of β -glucans could be an efficient and inexpensive way to support the immune system of COVID-19 infected patients, yet this needs more clinical studies to be performed for confirmation.

SCIENTIFIC EVIDENCE ON MUSHROOMS ANTIVIRAL PROPERTIES

A large part of Basidiomycetes contains a variety of bioactive compounds such as organic acids, fats, polysaccharides, polyphenols, steroids, triterpenes with antitumoral, immunomodulating, antibacterial, antiviral and antifungal effects, thus having various applications in medicine.

The discovery of a new chemical compound against a new virus requires a great amount of time, work and financial support. If the substance that is attacking the virus could be identified, the process of creating the drug would be much easier and the quantity of required energy much lower.

In this paper we focused on studies confirming the antiviral properties of mushrooms (Jong and Donovan, 1989). Previous research showed that bioactive substances from certain mushrooms extracts have antiviral activity (Lindequist et al., 2005; Glamoclija et al., 2015).

Ganoderma lucidum is well-known for its antitumoral, antiviral, anti-inflammatory qualities, being used in different forms, such as tea, food supplements or pharmaceutical ingredients. This mushroom is one of the most widely used in studies on antiviral qualities. The active substances, such as the α ganoderic acid, were extracted from the mushroom's fructiferous body using methanol and tested against the HIV 1 virus, showing an inhibitory effect on it (El-Mekkawy, 1998).

The antiviral action of a few species of wild mushrooms from Russia, such as *Daedaleopsis confragosa*, *Datronia mollis*, *Ganoderma valesiacum*, *Irpex lacteus*, *Ischnoderma benzoinum*, *Laricifomes officinalis*, *Lenzites betulina*, *Phellinus conchatus*, *Piptoporus betulinus*, *Trametes gibbosa*, and *Trametes versicolor* was tested on two type A viruses, the A/chicken/kurgan/05/2005 (H5N1) bird virus, and the A/Aichi/2/68 (H3N2) human virus. The study revealed that these mushrooms are producing antiviral substances that are blocking the synthesis of viral enzymes, at the same time boosting the human organism's immunity (Teplyakova et al., 2012). One other study in Russia, performed in 2020 with mostly the same team of researchers Ilyicheva et al., has been studying the antiviral effect of ethanol and water extracts of *Pleurotus pulmonarius* fructiferous body against the A/California/07/09(H1N1pdm) virus, the results showing that the ethanol extracts had a more powerful antiviral effect than the water extracts. This mushroom's fructiferous body is an important source of active substances, especially polysaccharides with an important antiviral activity against infections caused by this flu virus. Out of the water extract from *Pleurotus tuber-regium*, the antiviral activity of β -glucans on the herpes simplex virus type 1 (HSV-1), herpes simplex virus type 2 (HSV-2), respiratory syncytial virus (RSV) and

influenza A virus (Flu A) was tested by Zhang et al., in 2004. The tests showed that the powerful antiviral effect against HSV1 and HSV-2.

Water and methanol extracts from *Boletus edulis*, *Lentinus edodes* and *Pleurotus ostreatus* were tested against the herpes simplex type 1 (HSV-1) virus. These mushrooms' antiviral activity was correlated with the presence of β -glucans. Along with these aspects, the methanol extracts showed low antiviral activity compared to the water extracts. The largest quantity of cytotoxins with great inhibitory effect was found in the extract of the mushroom *Lentinus edodes* with an inhibition of 50%, while *Boletus edulis* and *Pleurotus ostreatus*, at similar concentrations, showed 25% and 15% inhibition (Santoyo et al., 2012).

Agaricus brasiliensis is one of the best known and most consumed mushrooms across the world, both gastronomically and pharmaceutically, in form of supplements due to its properties. This mushroom's antiviral activity was proven in a study made in 2007 in Brazil by Faccin et al., in which polysaccharides were isolated from water and ethanol extracts and tested against the type 1 poliovirus.

Grifola frondosa is one of the mushroom species that grows on tree stumps, having been domesticated in order to be cultivated. In 2007, Gu et al., and in 2016 Zhao et al., conducted studies regarding its antiviral capacity against the enterovirus 71 and HSV-1. Both studies proved the fact that the polysaccharides in the mushroom's composition have antiviral effects and can be used as therapeutic antiviral agents against the two investigated ones.

Inonotus obliquus is a parasite mushroom that can live up to 20 years, less known in Europe given its un-mushroom-like shape. Due to this, it is harder to find in nature, yet its therapeutic importance recommends it on account of its antioxidant, anti-inflammatory and immunostimulatory properties. (Ling et al., 2009, Vang et al., 2009, Zhong et al., 2011). Extracts of this mushroom are used for their hepatoprotective, antitumoral, anti-inflammatory, antioxidative, antibacterial qualities, being also preferred by athletes for the tonus boost they are giving (Lemieszek et al., 2011). In 2011 Shibnev et al., used water extracts of this mushroom against the virus of hepatitis C. The extract's antiviral effect indicated that it inhibits the production of infectious viruses in the kidney cells of a pig embryo.

The most recent study shows that methanolic extract from *Pleurotus sp.* and *Lentinus sp.* were used against the cytomegalovirus (HCMV). The inhibition was of 80%-90% for both mushroom types, which proves that both were capable to efficiently inhibit the propagation of *in vitro* human cytomegalovirus (Roy et al., 2020). One other study in which β -glucans of the *Pleurotus ostreatus* was used, significantly reduced the infection of the respiratory tract in children, due to a very similar infection to influenza (Stovall et al., 2008).

In 2018, Ren et al., made a study on an extract from the *Lentinus edodes* mushroom's mycelium. From this study resulted the fact that the lentinan polysaccharide, isolated from the fructiferous body has important antiviral activity and is directly involved in inhibition of the replication of the infection with the infectious hematopoietic necrosis (IHNV). Along with the viral inhibition, administration of lentinan led to the regulation of pro-inflammatory cytokines, IL-2, IL-11 and TNF- α , as well as IFN-1 and IFN- γ , that induce antiviral and immunomodulating effects (Shahzad et al., 2020). This LNT-1 regulates the human body's

inborne autoimmune response, which makes it play a very important role in the fight against the new COVID-19.

Other further similar study performed in 2007 by Ohta et al., proves that an acidic polysaccharide (APS) isolated from the extract of the *Cordyceps militaris* mushroom has antiviral effects against the influenza virus A from the body of infected mice. The nasal administration of this polysaccharide increased the survival rate in mice and lowered the viral titer in the broncho-alveolar liquid. Along with the viral inhibition, the level of TNF- α and IFN- γ , as well as IL-1 β , IL-6 and IL-10 cytokines also rose. The results proved that this polysaccharide has therapeutic effects in infections with the type A influenza virus, particularly by regulating the immune function. People infected with COVID-19 have high titers of inflammatory cytokines, which testifies of the fact that the lentinan polysaccharide from *Lentinus edodes* (Shahzad et al., 2020) and the acidic polysaccharide (APS) of *Cordyceps militaris* should be given more attention in the fight against SARS CoV-2.

Table 1. Mushrooms with antiviral properties against described targeted viruses, and their bioactive compounds

Mushroom species	Extract/ Bioactive compound	Target virus	References
<i>Agaricus brasiliensis</i>	β -glucans	Poliovirus type 1	Faccin et al. (2007)
<i>Boletus edulis</i>	chitin-binding lectins; β -glucans	HSV-1	Santoyo et al. (2012)
<i>Cordyceps militaris</i>	arabinogalactan (APS)	type A virus (mice)	Ohta et al. (2007)
<i>Daedaleopsis confragosa</i>	protein, polysaccharides, terpenoids	H5N1, H3N2	Teplyakova et al. (2012)
<i>Datronia mollis</i>	protein, polysaccharides, terpenoids	H5N1, H3N2	Teplyakova et al. (2012)
<i>Ganoderma lucidum</i>	α ganoderic acid	HIV-1	El-Mekkawy et al. (1998)
<i>Ganoderma valesiacum</i>	protein, polysaccharides, terpenoids	H5N1, H3N2	Teplyakova et al. (2012)
<i>Grifola frondosa</i>	polysaccharide (GFP1)	Enterovirus 71, HSV-1	Zhao et al. (2016); Gu et al. (2007)
<i>Inonotus obliquus</i>	polyphenylcarbonic acid	Hepatitis C Virus	Shibnev et al. (2011)
<i>Irpex lacteus</i>	protein, polysaccharides, terpenoids	H5N1, H3N2	Teplyakova et al. (2012)
<i>Ischnoderma benzoinum</i>	protein, polysaccharides, terpenoids	H5N1, H3N2	Teplyakova et al. (2012)
<i>Laricifomes officinalis</i>	protein, polysaccharides, terpenoids	H5N1, H3N2	Teplyakova et al. (2012)
<i>Lentinus edodes</i>	β -glucans; lentinan	HSV-1; IHNV	Santoyo et al.

			(2012); Ren et al. (2018)
<i>Lentinus sp.</i>	Glycoprotein	Cytomegalovirus	Roy et al. (2020)
<i>Lenzites betulina</i>	protein, polysaccharides, terpenoids	H5N1, H3N2	Teplyakova et al. (2012)
<i>Phellinus conchatus</i>	protein, polysaccharides, terpenoids	H5N1, H3N2	Teplyakova et al. (2012)
<i>Piptoporus betulinus</i>	protein, polysaccharides, terpenoids	H5N1, H3N2	Teplyakova et al. (2012)
<i>Pleurotus ostreatus</i>	β -glucans	HSV-1	Santoyo et al. (2012)
<i>Pleurotus pulmonarius</i>	polysaccharides	H1N1pdm	Ilyicheva et al. (2020)
<i>Pleurotus sp.</i>	Glycoprotein	Cytomegalovirus	Roy et al. (2020)
<i>Pleurotus tuber-regium</i>	β -glucans	HSV-1, HSV-2, RSV, Flu A	Zhang et al. (2004)
<i>Trametes gibbosa</i>	protein, polysaccharides, terpenoids	H5N1, H3N2	Teplyakova et al. (2012)
<i>Trametes versicolor</i>	protein, polysaccharides, terpenoids	H5N1, H3N2	Teplyakova et al. (2012)

Note: HSV-1 - Herpes simplex virus; HSV-2 - herpes simplex virus type 2; RSV - respiratory syncytial virus; Flu A - influenza A virus; H5N1 - Virus of birds A/chicken/kurgan/05/2005; H3N2 - Virus of humans A/Aichi/2/68; H1N1pdm - Virus A/California/07/09; IHN - Infectious hematopoietic necrosis virus.

From the very beginnings of history, the world's populations have been repeatedly confronted with fatal diseases caused by viruses. Being important sources of biologically active substances, mushrooms can contribute to the development of efficient and safe antiviral medicines. This is to say under the circumstance that not all existing fungi species are yet known (Hawksworth and Lücking 2017).

FINAL REMARKS

Extracts and chemical derivatives obtained from Macromycetes represent a new source of compounds capable to prevent and inhibit a number of viruses that affect the human body. Polysaccharide, terpenoids or proteins can inhibit the proliferation of these viruses to a great percent. This study can come to support new studies focused on the mushrooms' antiviral activity, both in prevention and cure. Moreover, polysaccharides extracted from mushrooms showed great intracellular inhibition of the virus. As patients infected with COVID-19 have high titers of inflammatory cytokines, polysaccharides extracted from mushrooms should be

considered for both future studies and for possible treatments of infections with SARS CoV-2 and other types of viruses. Certain medicinal mushrooms showing antiviral properties have a positive effect on the human organism, boosting immunity.

Therefore, further advanced research on taxonomic groups from various natural habitats could offer a valuable source in developing new potential drugs.

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CAMPYLOBACTER SPP. IN FRESH BROILER CHICKEN MEAT AT ESTONIAN RETAIL LEVEL

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Campylobacter spp. is one of the most important zoonotic foodborne pathogens. The highest risk of campylobacteriosis is related with fresh broiler chicken meat. Our study showed that fresh broiler chicken meat of Estonian origin is generally free of Campylobacter, compared to Lithuanian and Latvian products available at Estonian retail. The prevalence of Campylobacter spp. in fresh broiler chicken meat samples of Lithuanian origin was consistently high during the study period, which poses a risk of human Campylobacter infection through the year. Campylobacteriosis risk from domestic fresh broiler chicken meat products remain very low.

Key words: *Campylobacter* spp., prevalence, counts, fresh broiler chicken meat

INTRODUCTION

Campylobacter spp. is the most common cause of human acute gastroenteritis. Campylobacteriosis is the most often reported gastrointestinal disease in humans in EU with 246,571 confirmed cases in 2018 (EFSA, 2019). The most common causes of human campylobacteriosis are related to the consumption or handling of poultry meat, especially consumption of insufficiently heat treated broiler chicken meat. Therefore, the high prevalence and counts of *Campylobacter* spp. in fresh broiler chicken meat sold at retail level is a serious risk to consumer health. Commission regulation (EC) No 2073/2005 on microbiological criteria for foodstuffs lays down process hygiene criteria for carcasses of broilers with the limit of 1000 cfu/g of neck skin samples, however no food safety criteria have been established. High prevalence and high concentration of *Campylobacter* spp. remains the most important cause of human campylobacteriosis, therefore the aim of the present study was to investigate prevalence and counts of *Campylobacter* spp. in fresh broiler chicken meat of Estonian, Latvian and Lithuanian origin at Estonian retail level.

MATERIAL AND METHODS

The samples were collected from the Estonian biggest supermarket retail outlets in Estonia, Tartu. A total of 399 fresh broiler chicken meat samples were collected on a monthly basis between September 2018 and August 2019. *Campylobacter* spp. enumeration and detection from the meat samples was performed according to the International Organization

for Standardization method described in ISO 10272–1:2017. Detection of *Campylobacter* spp. was done by using enrichment method, according to ISO standard 10272–1:2017. All the analyses were performed in the laboratory of the Chair of Food Hygiene and Veterinary Public Health of the Estonian University of Life Sciences.

RESULTS AND DISCUSSION

Campylobacter spp. was isolated in 141 (35.3%) broiler chicken meat samples of Estonian, Latvian and Lithuanian origin, collected at Estonian retail (Table 1). In total, 2 (1.5%) of Estonian origin, 50 (37.6%) of Latvian origin and 89 (66.9%) of Lithuanian origin fresh broiler chicken meat samples were positive for *Campylobacter* spp. Results are indicating that during the study period *Campylobacter* was very seldom present in broiler chicken meat of Estonian origin. Contrary, it was found that imported broiler chicken meat was often contaminated, especially in the case of broiler chicken meat of Lithuanian origin.

Table 1. The prevalence of *Campylobacter* spp. in broiler chicken meat samples in 2018 and 2019

The origin country of samples	Total No. of samples	No. of positive samples / positive % (CI95% of positive %)
ESTONIA	133	2 / 1.5 (0.2-5.8)
LATVIA	133	50 / 37.6 (29.5-46.4)
LITHUANIA	133	89 / 66.9 (58.2-74.7)
TOTAL	399	141 / 35.3 (30.7-40.3)

In present study the observable seasonal peak in *Campylobacter* prevalence was confirmed only for fresh broiler chicken meat samples of Latvian origin: there was less contamination in winter and spring months (0%-21.4%) compared to the summer or autumn months (28.6%-90%) (Figure 1). *Campylobacter* in samples of Estonian origin was found only in July, when two samples were positive. It can be speculated that only few broiler chicken farms were *Campylobacter* positive in July in Estonia. In contrast, broiler chicken meat samples of Lithuanian origin showed high prevalence throughout the year (28.6%-100%), including in the winter months (80%-85.7%). This could be related with low effectiveness of biosecurity measures applied on Lithuanian farms, which allows the spread of *Campylobacter* at farm level, also in broiler chicken meat samples sold at retail level.

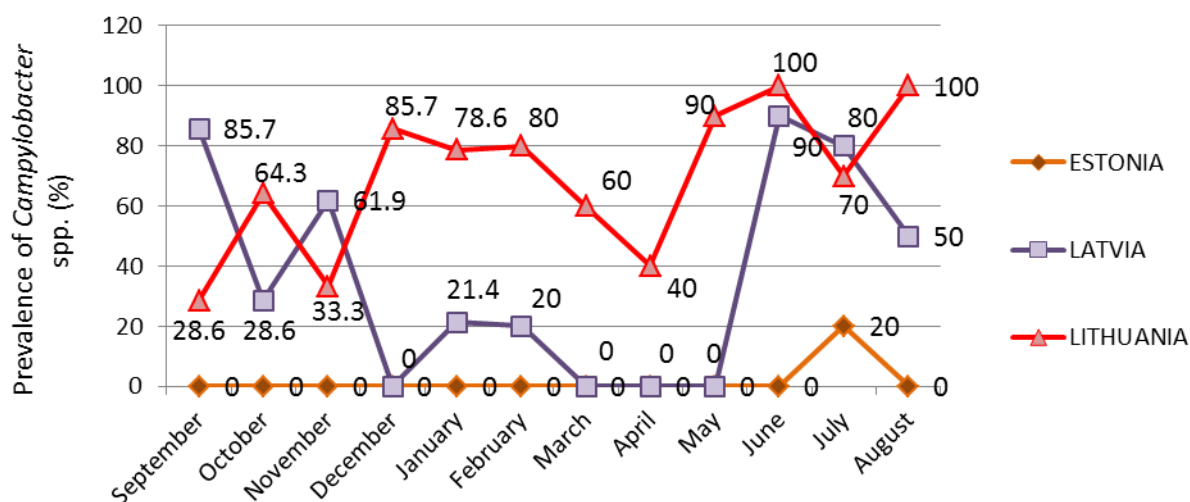


Fig. 1. Proportion of *Campylobacter* spp. positive fresh broiler chicken meat samples in different months

In addition to the prevalence of *Campylobacter*, it is important to assess the number of *Campylobacter* spp., because the high numbers of *Campylobacter* in fresh broiler chicken meat is one of the most important risk factors for human campylobacteriosis. According to the European Food Safety Authority, the risk to human health is highest with *Campylobacter* spp. >1000 cfu/g product (EFSA, 2013). Present study revealed that 20.3% ($n = 27$) of samples originating from Lithuania and 0.8% ($n = 1$) of Latvian broiler chicken meat samples contained *Campylobacter* spp. ≥ 1000 CFU/g (Figure 2). While in fresh broiler chicken meat of Latvian origin the presence of *Campylobacter* spp. in high concentration was found in one sample in February, then in Lithuanian products high *Campylobacter* concentrations were found from October 2018 to August 2019. Few ($n = 2$) samples of Estonian origin that were *Campylobacter* positive had *Campylobacter* numbers below the detection limit (<100 CFU/g).

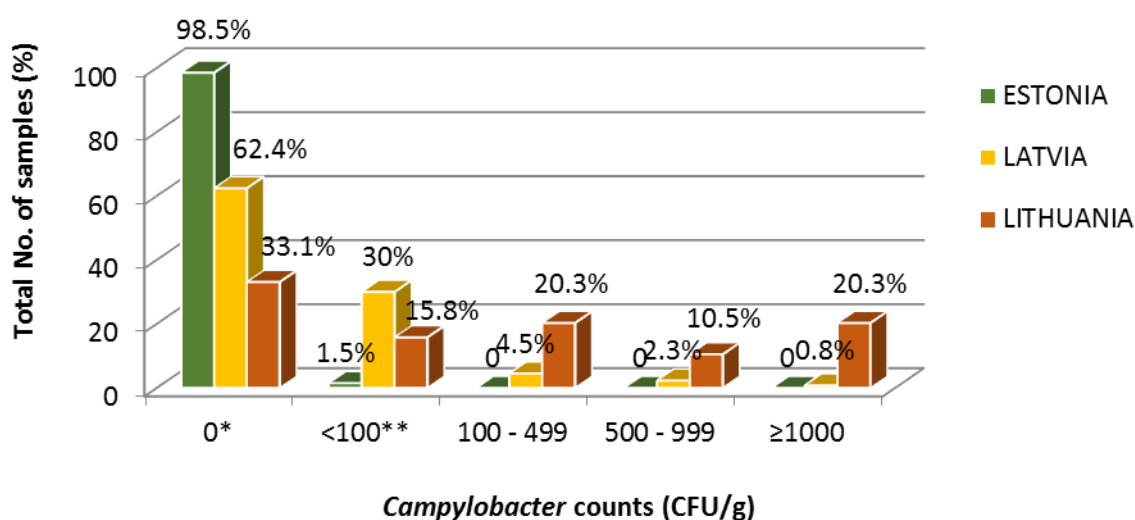


Fig. 2. *Campylobacter* concentration in fresh broiler chicken meat of Baltic origin at Estonian retail (0* means negative detection and negative enumeration; <100** means negative enumeration, but positive detection)

CONCLUSION

In conclusion, fresh broiler meat of Estonian origin is generally free of *Campylobacter*. However, fresh broiler meat of Latvian and Lithuanian origin available at Estonian retail is often contaminated. While the number of *Campylobacter* in products of Latvian origin remain mostly low, products of Lithuanian origin were found to be highly c by a high number of *Campylobacter* through the year. Although the seasonal peak of *Campylobacter* contamination is mainly in the warm summer months, there may be a risk of *Campylobacter* spp. infections through the year, depending in particular on whether to consume domestic or imported broiler chicken meat products. The effective implementation of hygiene and control measures throughout the broiler production chain is essential to keep the prevalence and numbers of *Campylobacter* in broiler chicken meat as low as reasonably achievable.

ACKNOWLEDGMENTS

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BIOEFFICACY OF CORIANDER ESSENTIAL OIL AGAINST *SITOPHYLUS ORYZAE* (COLEOPTERA: CURCULIONIDAE)

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Summary: Rice is the most important cereal crop in Turkey and rice weevil (*Sitophilus oryzae*) is a serious stored product pest. Syntethic chemicals are used to manage weevil infestation in rice. In an attempt time to find alternative control essential oil of coriander (*Coriandrum sativum* L.) was evaluated repellency effect and fumigant toxicity activity under laboratory conditions against *S. oryzae*. In the present study first experiment mortality of the exposed adults was assessed after 1, 3, 5 and 7 days. 5 adult weevil were used each treatment and three different concentrations of 3, 5, 10 μ l were tested. The study also determined the repellency effect of coriander essential oil against the *S. oryzae* as a second experimen. And of this study the highest mortality after 9 days ...%. The effective essential oil coriander of after 48 hours was showed highest toxicity (10 μ l) as 80%. The three different essential oil (3, 5, 10 μ l) of *C. sativum* recorded over the 50% repellent effiancy. In conclusion, essential oil of *C. sativum* was the most effective in rice weevil control and therefore can be used in formulating eco-friendly bio-pesticide against *S. oryzae*.

Key words: *Sitophilus oryzae*, coriander, biopesticide, Turkey

INTRODUCTION

Cereals are included in the human food chain. Some products from the cereal and grain groups are consumed and some are stored in storages. There are many pests in stored grains depending on storage conditions. These pests cause food damage such as reduced product quality, germination and nutritional value (Zawalska et. al., 2016). One of these pests, *Sitophilus oryzae* (L.), is one of the most serious pests on stored crops. Adults and their larvae are damaged, in the grain of the products such as rice and wheat (Yıldırım et. al., 2001).

Medicinal aromatic plants have been used for many years for nutrition and treatment of diseases. They obtained from different parts of plants are also used in agriculture, pharmacy, cosmetics, spice and paint (Hakverdi and Yiğit, 2017).

There are different control methods such as cultural precautions and physical struggle to control the pests in stored products (Chonwaski et al., 2014). The use of synthetic pesticides has raised concerns related to the environment and human health (Nerio et al., 2010), and promising alternatives control methods such as plant extract and essential oils (Araújo et al., 2016) Metabolites in plants can be both attracting and repellent for pests. In previous studies, plant metabolites such as insecticide, fungicide, acaricide or nematocide have been used.

(Gülsoy and Yeşilayer, 2017). Essential oils are produced as secondary metabolites in plants. There are several reports on the insecticidal and repellent activities of essential oils against stored product pests (Park et al., 2003) including their activity as a repellent (Boeke et al., 2004; Weaver and Subramanyam, 2000). It was recently shown that essential oils and their related constituents might have potential as alternatives to currently used fumigants (Tun et al., 2000; Park et al., 2003)

Coriandrum sativum is a medicinal and aromatic plant, widely grown in several parts of the world. The essential oil of coriander is obtained from seeds or coriander plants leaves using steam distillation. The oil is a colorless or pale yellow liquid with a characteristic linalool odor and mild sweet, warm, and aromatic flavor. *C. sativum* (coriander, ashuti) is a plant from the family of Apiaceae (Akgül, 1993; Baytop, 1999).

MATERIAL AND METHOD

This experiment was carry out in entomology laboratory at the Tokat Gaziosmanpaşa University, Agricultural Faculty, Plant Protection, Tokat-Turkey. The essential oil of coriander (*Coriandrum sativum* L.) were obtained from commercially from local market. *Sitophilus oryzae* adults were obtained from commercially available packed rice in Tokat, and reared for several generations on rice.

Insect culture

Rice weevil, *Sitophilus oryzae* (L.), was reproduced in 500 ml plastic containing wheat and rice as a source of food. The insects were maintained in the container at a room temperature of $25 \pm 1^{\circ}\text{C}$ and 75 % RH. They were laboratory reared with laid eggs on wheat, one week after laying eggs, the insect parents were removed. The eggs were kept in the same condition until adult emergence. Ten to fourteen-day olds of adults were used for bioassay tests.

Fumigant assay

Fumigant assays used 3 μL , 5 μL , 10 μL essential oil of *C. sativum* in 2mL ethanol; before quantities were established in preliminary trials. The oils were applied using an automatic pipettor on 3X3 cm filter paper that was fixed to the lower surface of the container lid. Individual tests were performed for each essential oil with a completely randomized design and each treatments had six replicates, and mortality was evaluated 1, 3, 5, 7 and 9 day after essential-oil exposure. control treatments used the equivalent volumes of ethanol. And then 5 unsexed adult *S. oryzae* were relased in to the central of box.

Repellency assay

Repellency assays were used Petri-dishes. Half the bottom of a Petri dish was covered with treated filter paper, while the other half was covered with a half filter paper disk treated with ethanol only. Five mixed-sex adults were released at the centre of each Petri dish, and the lid was sealed with stretch film. six replicates were run for each tested concentration, so that 30 adults/concentration were assayed. Observations were taken after 24 and 48 hrs from the beginning of the test: in each of them, the number of insects on the two half paper disks was recorded.

RESULTS AND DISCUSSIONS

Repellent effect

Many essential oils are known to possess ovicidal, fumigant repellent, and insecticidal effects against various stored-product insects (Silva et al., 2003; Yildirim et al., 2005).

The coriander essential oils showed repellent activity on *S. oryzae* adults (Figure 1). The increasing doses of essential oils caused a significant increase in the repellency effect when the rice weevil adults were exposed to these oils 48 hours.

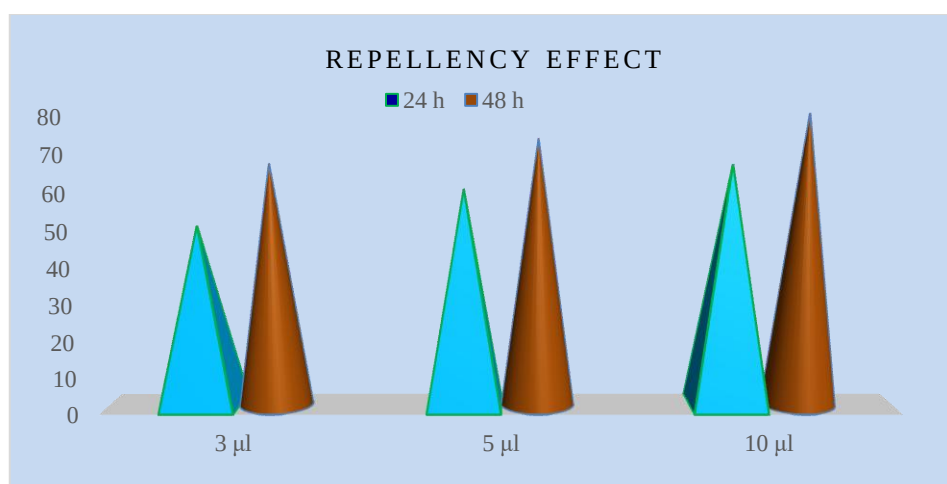


Figure 1. Repellency effect of essential oil of *Coriandrum sativum* against *Sitophilus oryzae*

Persantage repellent effect of rice weevil 50%, 60% and 66.6% respectively at the 3 µL, 5 µL, and 10 µL dose after 24 hour. The caraway and grapefruit oils have been reported repellnt effect against the weevils at a dose of 1 µL repelled only 77.8% and 61.1% of, respectively. However, the repellent activity was increased to more than 80% at a dose of 10 µL (Yoon et al., 2007).

Fumigant effect

The insecticidal activity of coriander essential oil, against *S. oryzae* was assessed by the fumigation toxicity, and the results are shown in Figure 2. Fumigant treatments have shown that insect mortality rate increased with the increase in essential oils concentration, which concurs with Bittner et al. (2008). Complete insect mortality was detected at the highest concentration within after 216 h of exposure for the criander essential oils against *S. oryzae*. It was observed the 10 µl of essential oils from pine and basil %97.5 mortality after 72 h (Pangnakorn and Chuenchooklin, 2018).

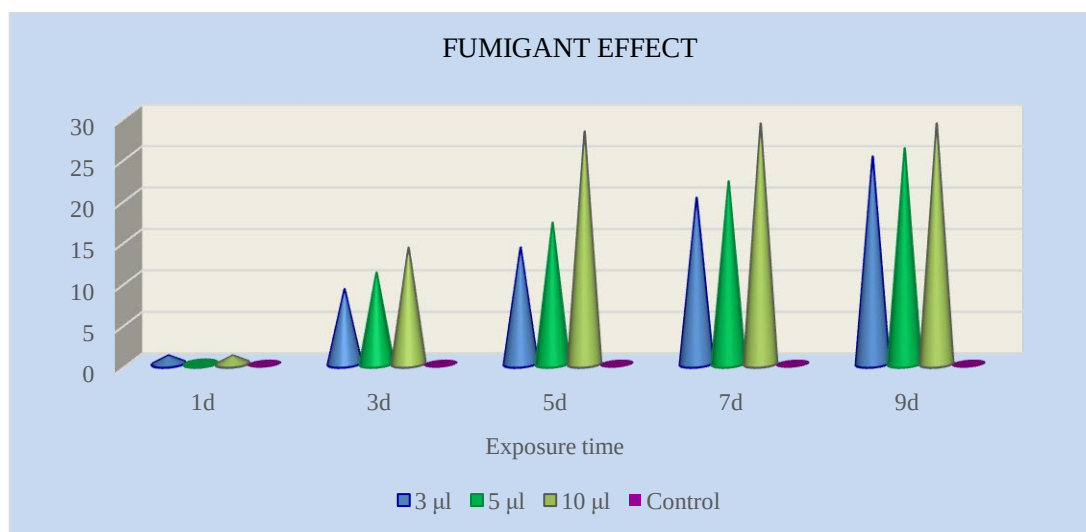


Figure 2. Under laboratory conditions of fumigant toxicities essential oil of *Coriandrum sativum* against *Sitophilus oryzae*

There are no previous citations concerning the insecticidal effects of coriander essential oil from Turkey on adults of *S. oryzae*. But the insecticidal activity of coriander against the stored product beetles, *Ephesia kuehniella* has been previously reported by AlpKent, et al., 2013. In this study was found mortality rate of the eggs was 98% at the dose of 125 µl/l of coriander oil.

In this study fumigant effect more toxic than repellent effect against *S. oryzae*. And essential oil coriander were shown repellent and insecticidal effect to *S. oryzae*. These repellent and fumigant toxicity results could be important in order to the efficacy of essential oil of *C. sativum* and could be potential for use in the insecticidal activities

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IS BIOCHAR APPLIED IN 2014 STILL ABLE TO IMPROVE SOIL PH FOUR YEARS AFTER ITS ADDITION TO THE SOIL?

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Summary: Soil acidity may severely reduce crop production. Biochar may increase soil pH and cation exchange capacity. The objective of the study was to evaluate, in a four-year field experiment, the influence of biochar on the pH of silt loam Haplic Luvisol. Biochar used in this experiment was produced from grains husks and paper fiber, at the temperature of 550 °C. Soil samples were collected from April to November in 2018, once a month. Our results showed that biochar addition to the soil statistical significantly increased soil pH in all treatments except, treatment B10N0, where biochar insignificantly decreased soil pH compared to control treatment B0N0. According to our results, we can say that biochar applied in 2014 is still able to improve soil pH after four years of its application.

Key words: biochar, pH of the soil, soil

INTRODUCTION

Soil acidity is a global problem in the world because it reduces soil fertility. Acidification is a slow natural process which considerably affects the sustainability of agriculture (Guo et al., 2010). Moreover, around 50 % of the world's arable land is considered acidic (Kochian et al., 2015). The most important physico-chemical process is soil sorption, which ensures the retention of nutrients, heavy metals and other different chemical substances in soil (Thompson et al., 2012). Application of biochar into the soil may prove to be a valuable instrument in the aspiration to increase the soil fertility and its use is deemed important to counter the risks of soil acidification (Dai et al., 2017). Biochar or charcoal produced from pyrolysis of biomass is gaining recognition around the world due to its unique properties when applied as a soil amendment. Also, biochar can play important role in controlling the mobility of pathogens in soil and water environment. Biochar has a large surface area and also is highly porous. Its impact can be important on the soil's cation exchange capacity (Ling et al., 2006). Biochar can improve plant growth by improving soil physical and biological properties (Steiner, 2009; Goenadi, 2010). Biochar applied in the soil is a residence or habitat for soil microbes (Saleh, 2017). The biochar alkalization effect can be recommended to increase the pH of the soil in acidic soils and provide solid organic matter (OM) in the long run. Feedstock and biochar production conditions are two factors that greatly influence biochar physicochemical characteristics (Setiawati et al., 2019), and subsequently also the properties of the soil. The determination of soil pH is one of the most widely used and standard pedological measurements. Biochar application into the soil resulted a significant

increase in soil pH and cation exchange capacity (Jatav et al., 2017; Niggusie et al., 2012). The change of pH in the soil upon biochar addition was related to the initial soil pH and the alkalinity of the biochar (Martinsen et al., 2015). In addition, information on application doses for different crops and soils is lacking (Lehman and Joseph, 2009). The main goal of this study was to investigate the effect of biochar four years after its application on the pH of the soil. Our main question was: To what extent does pure biochar or biochar in combination with N-fertilizer cause an increase in soil pH with increasing concentrations of biochar and N-fertilizer

MATERIAL AND METHODS

Experiment description

The field site used in this study is located in Dolná Malanta of Nitra region in Slovak republic (48°19' N; 18°09' W). The altitude of the locality is from 175 to 180 m above sea level. Before the experiment, the soil at study site was classified as the silt loam Haplic Luvisol with soil organic carbon content being 9.13 g.kg⁻¹ and had on average slightly acid pH (pH KCl = 5.71) and silt loam soil texture (Tobiašová, Šimnský, 2009). The content of loam was 15.2 %, clay 24.9 % and silt 59.9 %. The study was carried out during year 2018 (April – November) on the biochar experimental field established in March 2014, followed by biochar application (at 0, 10 and 20 t.ha⁻¹) and N-fertilizer application (at 0, 40 and 80 kg.N.ha⁻¹). The experimental site is located in a temperate zone with an average annual air temperature of 9.8 °C and an average annual precipitation of 539 mm. The average annual air temperature in 2018 was 11.4 °C with annual total precipitation of 367.4 mm. The experiment included 9 variants with three replicates. It was arranged in 27 experimental plots (4 m x 6 m) in a random division of blocks separated by 0.5 m wide protective strips. In 2018, for the other experimental objective, the original plots were divided into two parts (two subplots with dimensions 4 m x 3 m). Biochar was again re-applied in April 2018 at rates of 0, 10 and 20 t.ha⁻¹ (plots “B”). However, this study reports results measured during the growing season from April to November in 2018 at plots “A” where the biochar was applied only in 2014. The crop rotation in 2018 included spring barley (*Hordeum vulgare* L. variety *Malz*) which was harvested in late July). Nitrogen fertilizer was applied at rates of 0, 40 and 80 kg.N ha⁻¹ in 2018 (acronym “N0, N40 and N80”). spring barley). Treatments are shown in following Table 1.

Table 1. Treatments of the field experiment with application rates of biochar and N-fertilizer

Treatments	Biochar doses in 2014 (t.ha ⁻¹)	N-Fertilizer doses in 2018 (kg.N ha ⁻¹)
Not fertilized group (kg.N ha⁻¹)		
B0N0	0	0
B10N0	10	0
B20N0	20	0
Fertilized group 1 (kg.N ha⁻¹)		
B0N40	0	40
B10N40	10	40
B20N40	20	40
Fertilized group 2 (kg.N ha⁻¹)		
B0N80	0	80
B10N80	10	80
B20N80	20	80

Biochar properties

Biochar used in the experiment was obtained from paper fiber sludge and grain husks by pyrolysis at 550 °C - pyrolysis time: 30 min in Pyreg reactor (Pyreg GmbH, Dörth, Germany). The parameters of applied biochar are in Table 2.

Table 2. Physical and chemical parameters of applied biochar (Sonnenerde Company, Riedlingsdorf, Austria)

pH (KCl)	C (%)	N (%)	C:N	Bulk density (g.cm ⁻³)	Specific surface area (m ² .g ⁻¹)	Ash (%)
8.8	53.1	1.4	37.9	0.21	21.7	38.3

Soil sampling and soil analysis

Soil sampling was performed once a month from April to November 2018, from a depth of 0 - 10 cm. The pH (KCl) of the soil was determined potentiometrically in a suspension of soil and KCl solution using a pH meter (Hanna Instruments).

Statistical analysis

Statistical analysis was performed by Statgraphics software. A one-way analysis of variance (ANOVA) was used to evaluate the effects of different biochar application rates on the measured parameters. Differences between the treatment means were compared using least significant difference test (LSD) at the probability level $P < 0.05$.

RESULTS AND DISCUSSION

Soil reaction

Table 3 presents the effect of biochar treatments on soil pH over the growing season in 2018 (April – November).

Table 3. Statistical evaluation of soil pH (means $\pm$ standard error; n = 3)

Treatments	pH of the soil (-)
Not fertilized group (0 kg. N ha⁻¹)	
B0N0	5.67 $\pm$ 0.1 a
B10N0	5.61 $\pm$ 0.1 a
B20N0	5.93 $\pm$ 0.1 b
Fertilized group (40 kg. N ha⁻¹)	
B0N40	5.10 $\pm$ 0.1 a
B10N40	5.85 $\pm$ 0.4 b
B20N40	5.62 $\pm$ 0.2 b
Fertilized group (80 kg. N ha⁻¹)	
B0N80	4.75 $\pm$ 0.1 a
B10N80	5.32 $\pm$ 0.3 b
B20N80	5.37 $\pm$ 0.2 b

Generally, all biochar treatments with N-fertilization significantly ($P < 0.05$) increased soil pH compared to its individual control treatments (B0N0, B0N1 and B0N2) (tab. 3). However, the addition of 10 t.ha⁻¹ of biochar alone did not have significant influence on soil pH. It was found that treatment B10N0 decreased (insignificantly) soil pH in comparison with control treatment B0N0. We found that values of pH increased with increased biochar addition rate in the following order: B10N0 < B0N0 < B20N0 for treatments without N-fertilization. Also, the similar case was found in treatment with N-fertilization at a rate of 80 kg.N ha⁻¹ where the order was as follows: B0N80 < B10N80 < B20N80 (Fig. 2). However, in treatment with N-fertilization at a rate of 40 kg.N ha⁻¹, the highest influence of biochar was found at a rate of 10 t.ha⁻¹ (B10N40 treatment). The lowest values of pH (4.75) were measured in B0N80 treatment. After addition of N alone at a rate of 40 kg.N ha⁻¹ (B0N40), the pH value was 5.10, while pH increased from 5.10 to 5.85 after the addition of 10 t.ha⁻¹ biochar with N. The highest value of pH (5.93) was found in the case of treatment B20N0. Our results are particularly in line with authors Zong et al. (2018), who stated that higher rates of biochar addition to the soil are more beneficial than lower rates of biochar addition. Cybulak et al. (2019) found that the soil pH decreased with increasing biochar application rate what is in contrary to observations in our study. Our data confirmed that biochar addition to the soil can be beneficial for soil pH improvement (Obia et al., 2015; Horák et al., 2017; Šimanský et al., 2018).

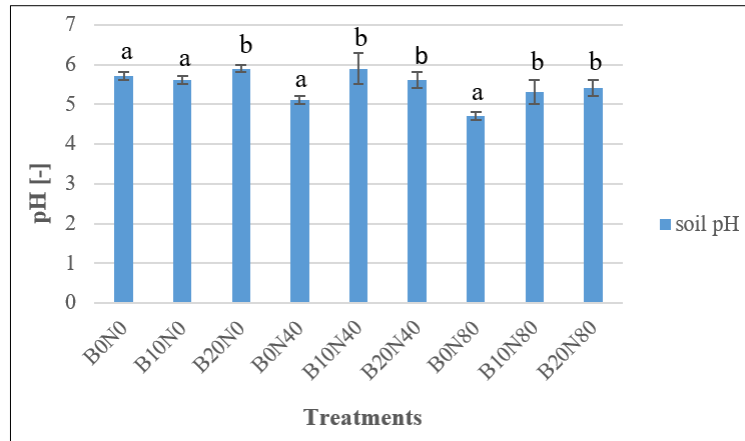


Figure 2. Statistical evaluation of soil pH amendment with or without biochar and N-fertilizer. Error bars indicate standard errors of the means (n=3) Different letters indicate significant differences between different treatments according to the LSD test ($P < 0.05$).

CONCLUSION

Generally, statistically significant changes ($P < 0.05$) of soil pH were primarily observed in all biochar treatments. Only treatment B10N0 did not result in statistically significant change of soil pH. The addition of higher biochar rate (20 t.ha^{-1}) had bigger effect than biochar at lower rate. The highest value of pH was observed in treatment B20N0 and the lowest in treatment B0N80. Based on the results of this study, we can conclude that addition of biochar into soil mainly at higher rate of biochar or in combination with N-fertilizer is still able to improve soil pH four years after its addition into the soil. In addition, we expect that shorter intervals between biochar additions could be more favorable. This study will assist producers to find optimal biochar application doses for their soils to improve soil pH and also agricultural sustainability.

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IMPORTANCE OF SUGAR BEET PRODUCTION OF VIROVITICA-PODRAVINA COUNTY IN CROATIA

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Summary: *The sugar industry in the Republic of Croatia is a very significant segment of the food industry and thus sugar beet as the main raw material has gained important position. This study analyzes sugar beet production in area around sugar factory Viro d.d. located in Virovitica, a town that is the county seat of Virovitica-Podravina County (Republic of Croatia). The factory processed sugar beet, sugar cane and produced various primary and secondary products important for the supply of the European agri-food market. Focus of this study was production during the period 2015-2018. In this study packaging and distribution capacities, data on total contracted and sown areas, description of agrotechnical measures taken during certain production years and climatic and agroecological conditions, the average amount of sucrose in sugar beet root, product and by-product quantities and also the total processed amount of sugar beet were included. Business year 2016, Viro d.d. stands out with record results in above parameters, while in 2017 it was quantitatively the most numerous, but as it was accompanied by poorer agro-climatic conditions before sowing, the consequences were visible at the end of the business year. By unification of three Croatian sugar factories (Osijek, Županja, Virovitica) into one joint company, the Croatian sugar industry d.d., results in the cessation of sugar beet production for Virovitica, so the weaker result from 2018 can be attributed to that situation. The research was made on the basis of primary, internal data provided by the factory Viro d.d. and secondary data available on the official website of the factory as well as relevant existing articles in proceedings based on the sugar industry in the Republic of Croatia.*

Key words: *sugar beet, Virovitica-Podravina County, sugar beet by-products, case study Viro d.d.*

INTRODUCTION

Numerous benefits and value of sugar beet are initially reflected in the primary product-white sugar, but from the industrial production point of view, many authors have seen the value of by-products as which are beet pulp, molasses, saturation sludge (Kanisek et al., 2008; Jurišić et al., 2011; Pospišil, 2013, Aboudi, 2017). Also, crown with leaves can have yield around 20 t/ha (Marinković et al., 2004.) and can be a nutritionally valuable food for the livestock.

This paper presents the results of the analysis made on these products and by-products of the sugar factory Viro d.d. placed in Vitovitica. As the youngest and most modern sugar factory in the Republic of Croatia, it is capable of processing alternative raw materials for sugar production of which the most important was once imported raw sugar from sugar cane, while

today it is sugar beet of domestic subcontractors and producers. Viro sugar factory d.d. has been producing white sugar, liquid sugar, molasses and dry briquetted beet pulp till 2018. Sugar is produced by processing sugar beet (so-called green campaign) and processing raw (cane) sugar (so-called yellow campaign). The technological process of sugar beet processing is at a high degree of automatization; moreover, the company has its own complex for the production of technological steam from which it produces electricity for its own needs. The aim of this study was analysis of the factory itself including production, packaging and distribution capacities, data on total contracted and sown areas, description of agrotechnical measures taken during certain production years and climatic and agroecological conditions of average sucrose root sugar production, product and by-product quantities and also the total processed amount of sugar beet.

MATERIALS AND METHODS

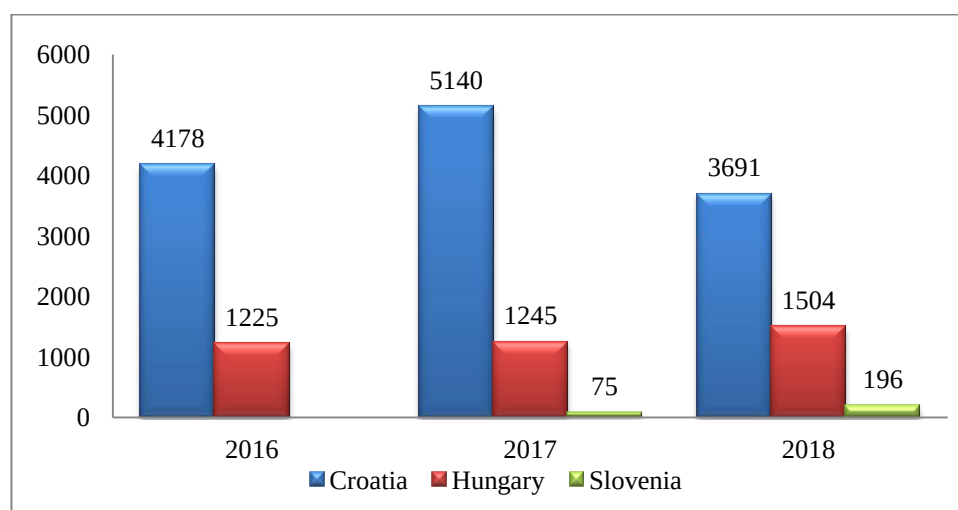
For the purposes of making this paper, the sugar factory Viro d.d. provided some of the internal data mentioned in the research, but also secondary data sources have been used such as the official website of Viro sugar factory and the relevant database of previously conducted scientific and professional research in domain of the production and analysis of sugar beet in the Republic of Croatia. Qualitative and quantitative analysis will cover mostly production years from 2016 until 2018, but for some indicators a longer time span is available and mentioned. By the method of analysis and synthesis based on the processed data of Viro sugar factory d.d. results were obtained that are applicable as a one part for the future assessment of the situation in the total production potential of sugar beet in the Republic of Croatia.

RESULTS AND DISCUSSION

Frequent monitoring of the quality of input and output raw materials guarantees products of uniform quality, which in the case of the factory Viro d.d. quality control is necessary considering the large amount of production. The factory warehouse has a capacity of about 12,000 tons of sugar, which depends on the amount of packaged sugar which quality is previously controlled in the laboratory. Viro d.d. also has a silo with a capacity of about 35,000 tons and can hold bulk sugar, obtained from sugar beet or sugar cane. It is important to mention that factory cooperates with numerous partners in the European Union in all segments of its business, so 80% of sugar production from Viro sugar beet is placed on the demanding European market, mostly in Germany, England, Slovakia, Italy, Hungary and Slovenia. It also exports most of its production of dried briquetted beet pulp, mainly to the Italian market, and they are also present on the market of Slovenia and Hungary, which is one of the main processing suppliers of sugar beet, which can be seen in *Fig.1*. Furthermore, there are three production years shown (2016, 2017, 2018) both descriptively and numerically

according to the relevant internal data of the factory, but also in the content of the website attached to the literature.

In 2016 was favorable agro-climatic conditions and also beet germination was uniform. In that year, 5.414 ha were contracted, and 5.357 ha were sown. The contracted price of beet was 270 kn/t (cca 36 EUR/t). Sowing began on March 21st 2016 and lasted until April 5th 2016. Plant assemblies ranged from 100.000 – 115.000 plants per hectare, and according to Kristek (2012) the correct distribution and optimal number of plants per unit area does not only affect on root yield but also on technological properties and is therefore an extremely important factor in sugar beet production. During the germination period, weed protection and prevention procedures were carried out and the interventions were successfully implemented. As the *Cercospora beticola* has created problems in recent years, in 2016 meteorological stations were set up and electron microscopes were procured, which enabled the monitoring of the course, intensity and development of disease. The sugar beet extraction campaign began on September 7th 2016 while processing began on September 10th. Viro sugar factory processing in 2016 was the best in analyzed period. In 2016 sucrose content in the root was 15.96%, root yield 74.35 t/ha, biological sugar yield 11.39 t/ha and the total processed amount of sugar beet root was 404.023 tonnes. It can be concluded for 2016 that the investment in these preventive technological tools paid off in the analyzed business year.



Graph 1: Contracted areas (ha) of VIRO sugar factory d.d.

Source: Adjusted according to internal data of Viro sugar factory d.d.

In 2017, it was planned to increase the sown area to over 8.000 hectares, but the contracted area was smaller (6.460 ha). A novelty in 2017 is the contracting of agricultural land in Slovenia, although so far it has been the practice of doing business in Croatia and Hungary. Sowing of sugar beet started in mid-March and the agro-climatic conditions were not as favorable as the year before, which resulted to more difficult germination because the soil could not be better prepared previously (dry period). This had a direct effect on the number of assemblies, which in that year were in the range of 90.000-120.000 plants per hectare. They also recorded more unfavorable agro-ecological conditions during the

vegetation, mostly the influence of high temperatures until the end of August when heavy rain improved the situation. Sowing was done earlier that year and in a short time, and it took root well in time, and thus was ready for the dry period which means it endured it more resiliently. During the sugar beet extraction campaign, the factory operated at full capacity, cutting and producing sugar on a daily basis of over 1000 tons. Despite unfavorable weather conditions, during sowing and vegetation period, a record high content of sucrose in the root of 16.34%, an average yield of 69.83 t/ha and impurities of 12.42% were achieved.

In 2018, the lowest area was contracted with sugar beet producers and sugar beet was sown only on 4.667 hectares were sown. The reason for this is the previously low price of sugar, and in 2017 the agreed price was 230 kn/t (cca. 30 EUR/t). Sowing begins in the first week of April, which is a little later than the previous one, and ends in the last week of April in more difficult dry conditions. Poor soil preparations for sowing and inadequate dry conditions result with a smaller number of plots – from 80.000 to 100.000 per hectare. Protection against pests and weeds was carried out successfully, while the fight against the disease was a little bit weaker, namely the development of rot and the attack of *Cercospora beticola* were the result of high temperatures and drought. The extraction campaign starts at the end of September and, as in previous years, work was done at full capacity (> 1.000 t per day). Given the above difficulties during the breeding process and the results are weaker. The average root yield was 58.41 t/ha, average digestion 14.75% and impurities 13.22%.

For the purpose of comparing the described years, main years in this analysis are from 2016 until 2018, but data from 2015 were added to Fig. 2. on which can be seen a positive growth of processed quantities of sugar beet. Furthermore, on Fig. 4 can be seen 2014 and 2015 which show the sucrose content in the sugar beet root.

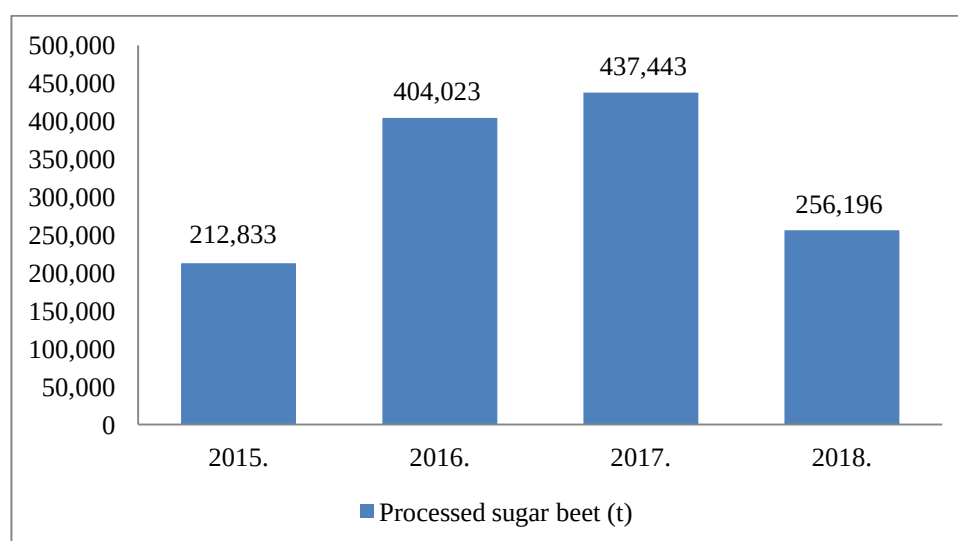


Fig. 2: Total amount of processed beets in tons (2015-2018)

Source: adjusted according to internal data of Viro sugar factory d.d.

It can be concluded that the amount of processed beets, and thus its products and by-products, is certainly variable from year to year. The instability of market prices is mostly pointed out as the cause of the oscillations, and according to media sources, in 2015 there was a "sugar crisis" where market prices dropped dramatically to approximately 220 kn/t (cca. 30 EUR/t).

The situation is already visibly improving in 2016, which can be concluded from all the above data (amount of sown areas, satisfactory sucrose content in the root ...). A significant "jump" is evident after 2015, when the amount of processed beets doubled, which at the end of 2016 almost tripled the results for finished products and by-products, adding to the increase in yield or sugar content in sugar beet root (Fig. 3), certainly a good quantitative and qualitative result was expected.

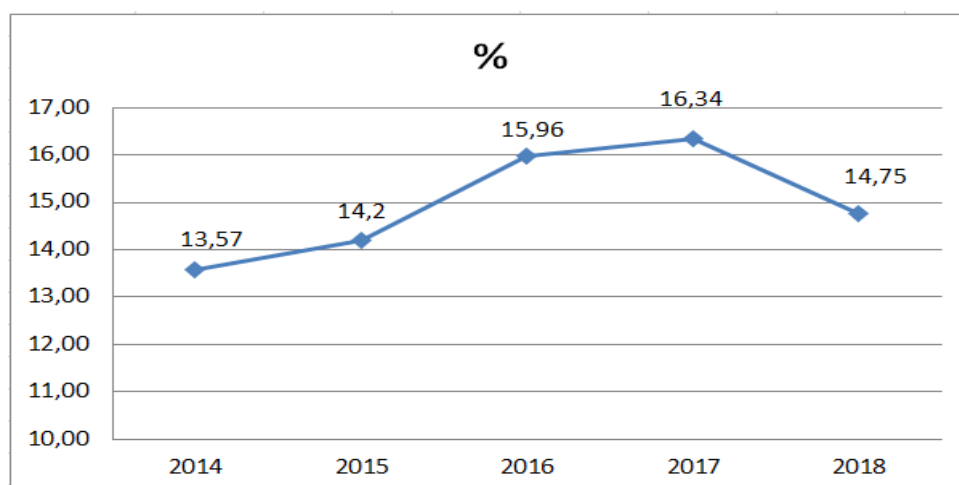


Fig. 3. Sucrose content (%) in sugar beet root (2014-2018)

Source: Adjusted according to internal data of Viro sugar factory d.d.

The record year in terms of sucrose content in sugar beet root was 2017, when most white sugar was produced, but the following year its number decreased by half (Fig. 4). 2017 also stands out in terms of the amount of molasses and dry noodle (by-products) produced, while 2018 indicates a significant reduction in the amount of sugar beet, and thus the number of products and by-products.

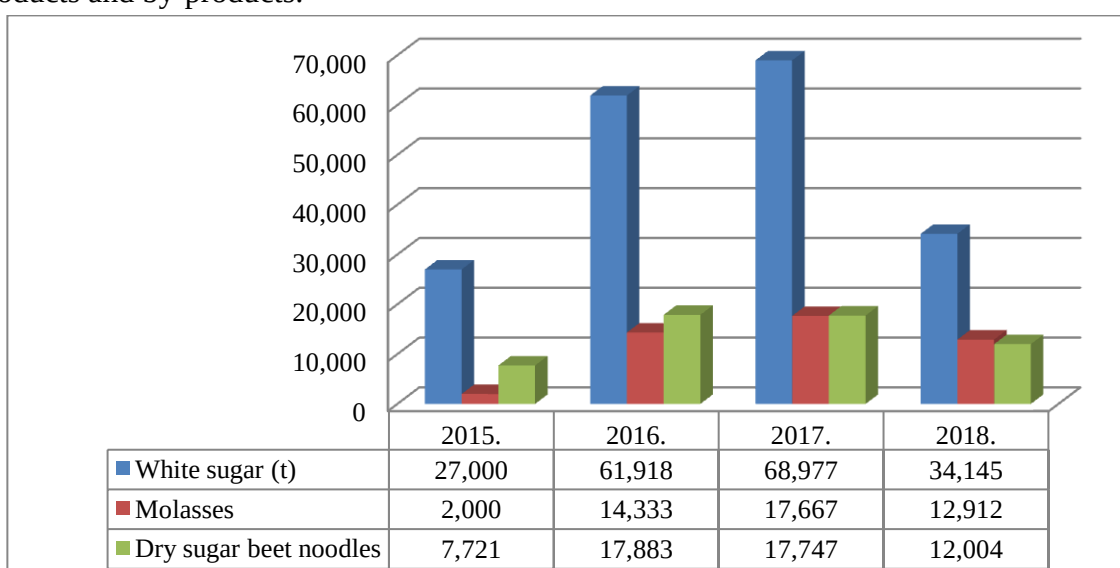


Fig. 4. Quantity of products in tonnes (t) and by-products of Viro d.d. (2015-2018)

Source: Adjusted according to internal data of Viro sugar factory d.d.

From the figures it is evident that the production of white sugar dominates, especially in the production year 2016 and 2017 and the production ratio of by-products during all observed years is relatively constant, except in 2015, which recorded the lowest amount of molasses produced. Marković and Zekić (2011) stated that in Serbia prices of inputs grow faster than price of sugar beet root, which leads to deteriorated parity of this agricultural product. This situation is similar in Croatia for sugar beet producers.

The publicly announced annual report of Viro d.d. for 2019 states that as a result of the unification of the three sugar factories (Županja, Osijek and Virovitica) into a joint business system, the Croatian Sugar Industry d.d., Viro sugar factory permanently stopped producing sugar beet and the main focus for the future was placed on sugar cane, but within the newly formed organization.

CONCLUSION

Based on the analyzed data in this paper, it can be concluded that 2016 is certainly a historic year for the Viro sugar factory, which then achieved a record production of white sugar 11 t/ha, also an average yield of 74.35 tonnes sugar beet per hectare of sown area and all accompanied by a very good percentage of sucrose in the root of beet in the amount of 15.96%. The next two years (2017 and 2018) are not negligible in terms of results - they certainly did well, but in 2018 there was a significant shortage of contracted and sown areas, which from one perspective indicated that it is also the last year for sugar beet. Viro sugar factory d.d. by the decision of the Management in 2019 has announced that it will no longer process sugar beet, but they will processing of sugar cane (which has been part of the business so far) remains under the jurisdiction of the newly formed organization Croatian Sugar Industry d.d., but still not so significant like it was sugar beet as the primary raw material.

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APPEARANCE OF *ASPERGILLUS FLAVUS* AND *ASPERGILLUS NIGER* TOXIN-PRODUCING FUNGI ON MAIZE AROUND DEBRECEN IN 2019 AND 2020

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In 2019 and 2020 appearance of two toxin-producing fungi were detected on maize hybrids near Debrecen in Eastern Hungary. First pure culture of the fungi were grown, then A. flavus and A. niger were identified under microscope. Sequencing of the two fungi were determined by molecular biology methods. A. niger showed 98% identity with the sequence KF753944 and A. flavus showed 98% identity with the sequence MH237666.

Key words: *Aspergillus flavus*, toxin-producing, Hungary, *Zea mays*, molecular biology

INTRODUCTION

Maize (*Zea mays* L.) is one of the most cereal crops in Hungary. Maize has a lot of fungal diseases such as *Ustilago maydis*, *Fusarium* spp., *Stenocarpella maydis*, *Macrophomina phaseolina*, *Nigrospora oryzae*, *Aspergillus* spp., *Bipolaris zeicola* and *Exserohilum turcicum*.

In Hungary, recently several diseases, pests and weeds appear what prefer subtropical or tropical climate such as *Macrophomina phaseolina*, *Aspergillus* species, *Scaphoideus titanus* and *Phytolacca americana*. *Aspergillus flavus* is a toxin-producing fungus what mainly causes toxin contamination in tropical and subtropical areas. Aflatoxins produced by *Aspergillus* species are among the most significant mycotoxins, which are harmful to animals and humans, and cause various disease symptoms (VARGA et al., 2009). Aflatoxins are carcinogenic secondary metabolites which produced by *Aspergillus* species, especially *A. flavus*, *A. parasiticus* and *A. nomius* (MURPHY et al., 2006). Among the known aflatoxin forms (B₁, B₂, G₁, G₂, M₁, M₂) (DIAZ, 2005), aflatoxin B₁ has the strongest carcinogenic effect (MURPHY et al., 2006). Aflatoxin B₁ is a potent genotoxic carcinogen in laboratory animals such as salmonid fishes (SCHOENHARD et al., 1981) turkeys and chickens (WILLIAMS et al., 2011) and very harmful of liver in humans (WILD and TURNER, 2002). Aflatoxin B₁ is able to provoke hepatocellular carcinoma (HCC) which is one of the most lethal cancers worldwide (CABRERA and NELSON, 2010). Although aflatoxin B₁ can be detoxified in the liver, in high level quantity can induce acute hepatic necrosis (DAI et al., 2017). Africa and China have the greatest incidence of HCC compared with the rest of the world due to

aflatoxin exposure. In Brazil, HCC is now the 7th cause of cancer death (GONÇALVES et al., 1997).

Apart from *A. flavus*, other species of *Aspergillus* genus such as *A. fumigatus*, *A. niger*, *A. nidulans* and *A. glaucus* can produce mycotoxins (HANSEN and HAGEDORN, 1969). *A. niger* can produce essential enzymes, antibiotics and organic acid in food industry and pharmaceutical industry. Citric acid is produced by fermentation of *A. niger* and *A. wentii*. In addition to citric acid, *A. niger* is a rich source of enzymes such as pectinase, protease and amyloglucosidase (FROST and MOSS, 1987). Ochratoxin, the nephrotoxic and carcinogenic mycotoxin was first reported from the black *Aspergillus* species, *A. foetidus* (UENO et al., 1991). First ochratoxin A production by *Aspergillus niger* was reported by ABARCA et al. (1994) and it was later confirmed by TÉREN et al. (1996) and TANIWAKI et al. (1999). Ochratoxin A is primarily a nephrotoxin that has been shown to have nephrotoxic effects in mammalian species. Due to its nephrotoxic properties, Ochratoxin A causes great damage to the pig and poultry industry (KUIPER GOODMAN and SCOTT, 1989). Ochratoxin A contamination accumulated in food can be broken down by strain CBS120.48 of *A. niger* (RIGÓ, 2003).

MATERIAL AND METHODS

The experiment were carried out near Debrecen in Eastern Hungary at the field experimental station of the Institutes for Agricultural Research and Educational Farm of University of Debrecen in 2019 and 2020.

In the first step, the presence of *Aspergillus* species was examined visually in 4 different maize hybrids (Kamária, P9903, DKC4351 and Kampraris). The sowing data were as follows: the maize hybrids were sown to similar spacing (75 cm) with different plant density. The plant densities were 65-75-85 and 95 thousand plant/ha. Half of the area were irrigated close to tasseling.

Before harvest, 3-3 corn samples were taken randomly from the four hybrids. After crumbling, the samples were stored in paper bags in 4° C.

Laboratory tests were performed on the collected samples. First, the pathogen was cultured. 50-50 grains from each bags were taken and were put on PDA medium. Then Petri dishes containing isolates were incubated at 25° C. On the day 7 of incubation *A. flavus* and *A. niger* fungi appeared on the medium.

In the next step, mycelium fragments from the fungus developed on the grains were put into PDA medium to produce pure cultures. Then *A. flavus* and *A. niger* were identified by microscopic examination based on colonial morphology marks on the developed pure cultures.

After conventional identification of the pathogen, molecular biological analysis was performed on the samples. Total genomic DNA was extracted from the mycelium using the *NucleoSpin Plant II* (Macherey-Nagel) method according to the protocol, followed the manufacturer's instructions. Mycelium was added to a *BashingBead Lysis Tubes* (Zymo Research). Beads are used with *Analog Disruptor Genie* (Scientific Industries). DNA concentrations were measured by *NanoDrop* (Thermo Scientific) and used in PCR

amplification of the internal transcribed spacer region (ITS) with primers ITS1/ITS4 (White et al. 1990). Amplification were performed on *Primus 96 Thermal Cyclers* (MWG Biotech) using the following parameters: predenaturation at 94° C for 5 min, followed by 35 cycles of denaturation at 94° C for 2 min, annealing at 53° C for 2 min, and elongation at 72° C for 2 min, followed by a final extension at 72° C for 30 min (Zarrin and Erfaninejad, 2016). The PCR products were purified by using *NucleoSpin Gel and PCR Clean-up* (Macherey-Nagel). Purified amplification products were sequenced by Microsynth Austria GmbH.

RESULTS AND DISCUSSION

Presence of the pathogen was determined in field studies in all 4 maize hybrids tested (Kamária, P9903, DKC4351 and Kampraris). Samples were collected from these for laboratory testing. In laboratory, after 7 days incubation *A. flavus* and *A. niger* appeared on corn seeds of every types of the examined hybrids (Figure 1, Figure 2).

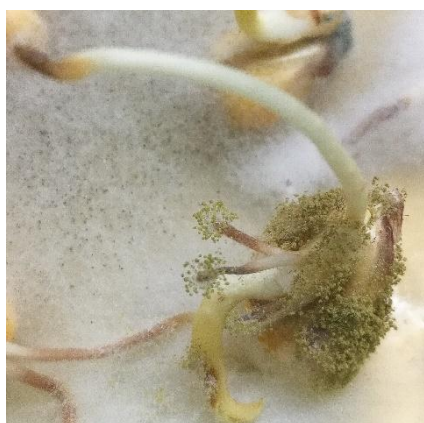


Figure 1: Aspergillus flavus on maize after 7 days incubation

Photo: by Kitti Csüllög



Figure 2: Aspergillus niger on maize after 7 days incubation

Photo: by Kitti Csüllög

A total of 50 grains from 3 corn-cob were examined for each hybrid tested (4 hybrids), from each plots of different stocking densities (4 different stocking densities/hybrid). A total of 50 grains from 3 corn-cobs were examined 16 times under non-irrigated conditions and the same number under irrigated conditions. Infected seeds were counted of each 50 grains tested. The results are shown in Table 1.

Table 1: Occurrence of *A. flavus* and *A. niger* on the tested maize grains (50 grain) in 2019.

65 thousand plant/ ha					
Non-irrigated	<i>A. flavus</i>	<i>A. niger</i>	Irrigated	<i>A. flavus</i>	<i>A. niger</i>
<i>Kamaria</i>	17	4	<i>Kamaria</i>	4	4
<i>P9903</i>	2	-	<i>P9903</i>	9	-
<i>DKC4351</i>	8	-	<i>DKC4351</i>	8	-
<i>Kamparis</i>	21	2	<i>Kamparis</i>	11	2
75 thousand plant/ ha					
Non-irrigated	<i>A. flavus</i>	<i>A. niger</i>	Irrigated	<i>A. flavus</i>	<i>A. niger</i>
<i>Kamaria</i>	12	4	<i>Kamaria</i>	6	-
<i>P9903</i>	2	-	<i>P9903</i>	4	-
<i>DKC4351</i>	16	-	<i>DKC4351</i>	8	-
<i>Kamparis</i>	36	2	<i>Kamparis</i>	24	6
85 thousand plant/ ha					
Non-irrigated	<i>A. flavus</i>	<i>A. niger</i>	Irrigated	<i>A. flavus</i>	<i>A. niger</i>
<i>Kamaria</i>	4	2	<i>Kamaria</i>	4	-
<i>P9903</i>	1	-	<i>P9903</i>	14	-
<i>DKC4351</i>	17	2	<i>DKC4351</i>	14	-
<i>Kamparis</i>	11	-	<i>Kamparis</i>	26	-
95 thousand plant/ ha					
Non-irrigated	<i>A. flavus</i>	<i>A. niger</i>	Irrigated	<i>A. flavus</i>	<i>A. niger</i>
<i>Kamaria</i>	6	-	<i>Kamaria</i>	7	-
<i>P9903</i>	3	2	<i>P9903</i>	7	2
<i>DKC4351</i>	9	-	<i>DKC4351</i>	15	-
<i>Kamparis</i>	21	4	<i>Kamparis</i>	30	3

The most infected maize hybrid was Kamparis with 75 thousand plant/ha density. From the 50 grains tested, 36 were infected with *A. flavus*. 30 grains were infected with *A. flavus* with 95 thousand plants/ha density under irrigated conditions of Kamparis hybrid. In the case of spots with higher plant densities (85 and 95 thousand plant/ha) the number of infected grains were higher under irrigated conditions. DKC4351 hybrid was the most infected at 85 thousand plant/ha density. Infection with *A. flavus* was confirmed in 17 of 50 tested grains. Under irrigated conditions more grains were infected with *A. flavus* in the case of denser plant vegetation. The least infected hybrid was the P9903 under non-irrigated conditions. Infection

was higher under irrigated conditions. In the case of Kamaria, the number of infected grains was the highest at 65 thousand plants per hectare under non-irrigated conditions. Under irrigated conditions of only from 4 to 7 seeds were infected. *A. niger* appeared only in a few case. Both *Aspergillus* species were found by microscopic examination (Figure 3, Figure 4). In 2020, there was no significant damage to the hybrids tested.



Figure 3: Spores of *Aspergillus flavus*
Photo by: Kitti Csüllög



Figure 4: Conidiophore of *Aspergillus niger*
Photo by: Kitti Csüllög

According to molecular biological studies, NCBI BLAST analysis of the 549-bp sequence (GenBank accession no. MW314793) showed 98% identity with the sequence MH237666 of *A. flavus*. NCBI BLAST analysis of the 539-bp sequence (GenBank accession no. MW314791) showed 98% identity with the sequence KF753944 of *A. niger*.

CONCLUSION

Aspergillus species are extremely important pathogens due to their toxin production. During drought years, infection of these fungi can lead to severe toxin poisoning in the field. After field infection, corn crop stored in warehouses may become infected again, which can lead to severe toxin poisoning. The fact that the disease occurred in both years during the study shows that the pathogen has adapted to the Hungarian climate. In the future, in the increasingly frequent hot summers, they should be expected to perform almost every year. According to BATTILANI et al. (2016), if the average temperature rises by 5 ° C, the whole of Europe (except the Scandinavian parts) will be heavily contaminated with aflatoxin. Climate change has been reported as a driver of food and feed safety issues worldwide and its impact in the presence of mycotoxins in food and feed may be a matter of serious concern (BATTILANI et al., 2016).

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THE DEVELOPMENT OF DIGESTIVE SYSTEM IN NEWLY HATCHED CHICKENS AND THE ROLE OF GHRELIN IN THIS PROCESS

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Summary: *After hatching significant changes occur in the anatomy, physiology, size and, roles of the digestive system. Growth in poultry has been extensively explored, many studies focused on the development of the gastrointestinal tract (GIT) during the hatching growth period. The GIT has a considerable role in activating growth while the hatching period during which the GIT segments develop in size faster than the rest of the body. Many studies are based on gastrointestinal hormones, one of which is ghrelin. Ghrelin in avian species inhibits the appetite. The major source of ghrelin synthesis and release is the stomach. Ghrelin affects the growth, development, and motility of the digestive system. Due to differential anatomy and histology of the digestive tract, the question arises about the role of ghrelin in each part of the gastrointestinal tract of chickens. In spite of extended research, there is still a lack of information about the degree of ghrelin secretion from the crop, stomach, and intestine in naive chickens i.e. without interaction with other peptides/hormones/enzymes. The aim of the study was to investigate the changes of ghrelin concentration in the crop, stomach, and intestine of newly hatched chickens. The in vitro experiment was carried out on newly hatched chickens (n=24), 3 hours after hatching. Sections of tissues were stained with hematoxylin and eosin (H&E), then evaluated with a light microscope. Tissues were homogenized and the supernatant was stored at -80°C until ghrelin and protein levels were measured by RIA and Lowry method, respectively. The highest concentration of ghrelin was in the stomach followed by crop and intestine. In contrast, the highest concentration of proteins was in the intestine, then in the crop, and in the stomach. The microscopic evaluation clearly showed diversified cells in each part of the digestive system. The tissues variability was correlated with different ghrelin and proteins concentrations in tested GI fragments. The obtained results suggest that ghrelin might be a stimulator of growth and development as well as a modulator of GI tissues functions.*

Key words: *chickens, ghrelin, digestive tract*

INTRODUCTION

Newly hatched chicks change the way of feeding that based on energy given by the endogenous nutrients of the yolk to the exploitation of exogenous carbohydrate-rich feed through the small intestine. After hatching significant changes occur in the anatomy, physiology, size, and roles of the digestive system (Cook and Bird, 1973; Bayer et al., 1975; Uni et al., 1996). The intensive growth of the GI tract starts about 24 hours after the ingestion of exogenous nourishment. The growth of birds/poultry has been extensively explored, many

studies focused on the development of the gastrointestinal tract (GIT) during the hatching growth period (Laird, 1965; Lilja, 1983; Nir et al. 1993). Gastrointestinal tract segments grow more intensively than the whole body during the hatching period (Katanbaf et al., 1988). The GIT plays a much more important role in the development of meat-type broiler (BT) than in egg-type Leghorn (ET) chickens. It has also been reported that after hatching BT chicks, duodenal epithelial cells are almost mature at hatching time and could be able to digest and consume nutrients more efficiently than ET chicks (Sklan, 2001). In the immediate post-hatch period dynamic changes occur in the small intestines of poultry (Dror et al, 1977). In birds, the substrates for pre-hatching growth are from the yolk, showing priority apportioned to intestinal growth in the newly hatched chickens (Nitsan et al., 1991). The duodenum growth starts earlier than either the jejunum or ileum. During hatching, the enterocytes of the small intestine are immature and round, after hatching they increase in length and develop a pronounced polarity and defined brush border. The alimentary tract consists of a buccal cavity, pharynx, esophagus, proventriculus, ventriculus, small intestine (duodenum, jejunum, ileum), large intestine (caecum, rectum), and cloaca. The esophagus is a thin-walled distensible tube and has a single outpouching to form the crop (Fig.3.). The stomach (Fig.4.) is built of two distinct parts: the muscular portion and the glandular portion. A meaningful characteristic feature of the esophagus is the presence of a mucous gland in the crop. The two parts of the stomach are connected by an intermediate zone. Microscopically, the wall of the stomach consists of four distinct functional tunicae namely, mucosa, submucosa, muscosa, outermost serosa. The small intestine (Fig.5.) extends from the pyloric end of the stomach to the junction of the small intestine, caeca, and colon. The mucosa of the intestine is thrown into villi, the submucosa is thin, hardly distinguished in some regions. The muscosa consists of two smooth muscle layers: a thick circular muscle layer and an outer longitudinal layer, they surround the intestine and allow mixing and propulsion of the digesta through the intestinal tract (Zaher et al., 2012).

In recent years, the presence of various endocrine cells with assigned hormones has been demonstrated in the digestive tract of birds (Gribble and Reimann, 2019). This was done by using immunocytochemical methods. At the same time, several hormones were isolated from the extracts and their physiological properties were determined. Ghrelin (Dyaczynski et al., 2018) is a feeding-promoting peptide in mammals, opposite to avian species where inhibits the appetite. The major source of ghrelin synthesis and release is the stomach. Ghrelin (Pierzchała-Koziec, 2012) in avian species affects the growth, development, and motility of the digestive system. Due to differential anatomy and histology of the digestive tract, the question arises about the role of ghrelin in each part of the gastrointestinal tract of chickens. In spite of extended research, there is still a lack of information about the degree of ghrelin secretion from the crop, stomach, and intestine in naive chickens i.e. without interaction with other peptides/hormones/enzymes. The aim of the study was to investigate the changes of ghrelin concentration parallelly with protein changes in crop, stomach, and intestine of newly hatched chickens.

MATERIAL AND METHODS

The *in vitro* experiment was carried out on newly hatched chickens ($n = 24$), 3 hours after hatching. Fragments of crop, stomach, and intestine were immediately taken for histological preparations and estimation of ghrelin and proteins concentration. Sections of tissues were stained with hematoxylin and eosin (H&E), then evaluated with a light microscope. Tissues were homogenized and the supernatant was stored at -80°C until ghrelin and protein levels were measured by RIA and Lowry method, respectively.

RESULTS AND DISCUSSION

The highest concentration of ghrelin (Fig.1.) was in the stomach (626 ± 75 pg/mg protein) followed by crop (510 ± 61 pg/mg protein) and intestine (345 ± 41 pg/mg protein). In contrast the highest concentration of proteins (Fig.2.) was in the intestine (206 ± 25 $\mu\text{g}/\text{mg}$ tissue), then in crop (102 ± 12 $\mu\text{g}/\text{mg}$ tissue), and in the stomach (22 ± 3 $\mu\text{g}/\text{mg}$ tissue). The table shows the ratio of ghrelin to proteins concentrations (Tab.1.).

In agreement with Leznicka (1971) the secretion of the crop mucosa glands could serve to moisten seeds for a certain degree which helps the bird to avoid the distasteful feeling of swallowing dry seeds. In line with the previous study (Okamoto and Yamada, 1981; Prasad and Kakade, 1990; Imai et al., 1991; Liman et al., 2010) the histological studies of the present investigation revealed that proventricular folded mucosal surface is not smooth, it is covered by several projections or papillae. The stomach is a smooth muscular organ, located between the esophagus and the intestine and it is made up of two distinct parts: glandular and muscular and its gastric ventriculus is created by four semi-autonomous muscles, two dark-colored and thick (the cranioventralis and the caudodorsalis), and two with thickness and clear colored (the caudoventralis and the craniodorsalis), their function is crush the victuals ingested. This observation is analogous to that of Bailey et al., 1997. The small intestine is differentiated into the duodenum, jejunum, and ileum. The results presented by Taylor (2000) and Vukicevic et al., (2004) are in agreement with those obtained in the present study.

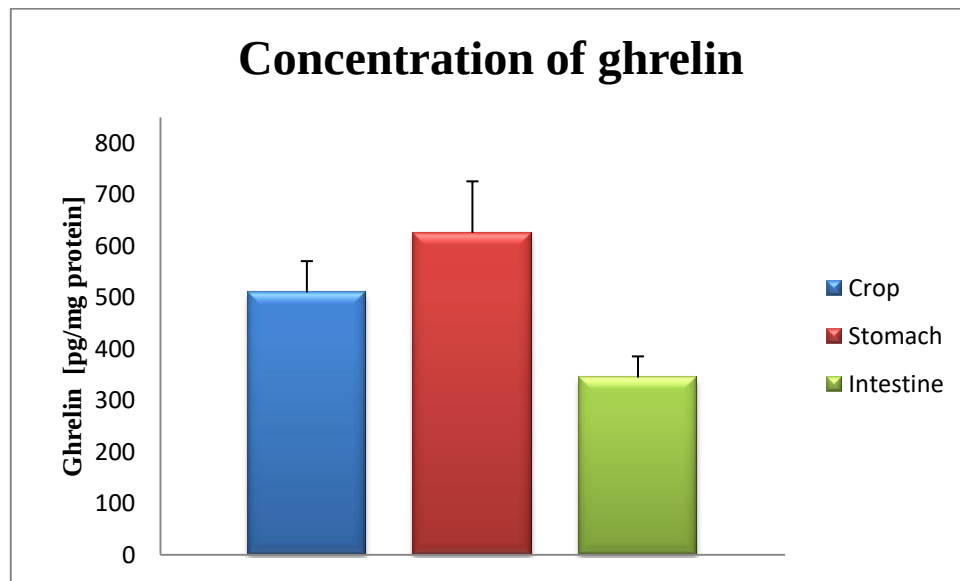


Fig. 1. The concentration of ghrelin in the crop, stomach, and intestine.

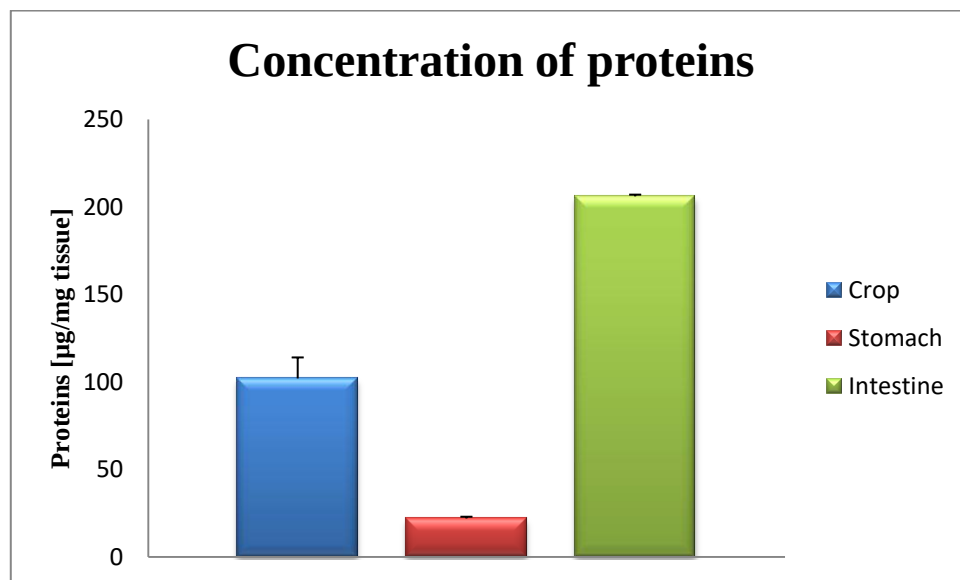


Fig. 2. The concentration of proteins in the crop, stomach, and intestine.

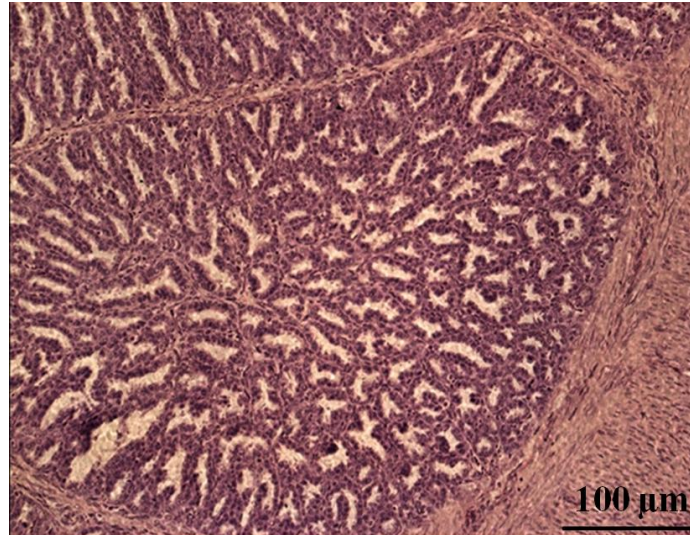


Fig. 3. Histology of the crop X20. H&E stain.

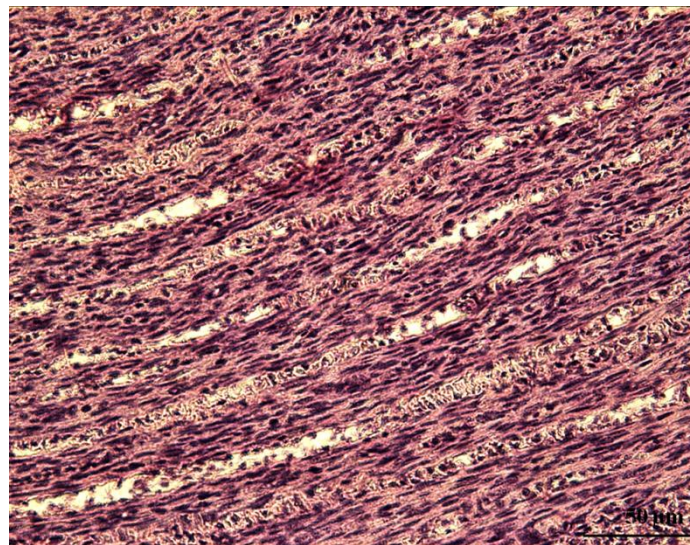


Fig. 4. Histology of the stomach X40. H&E stain.

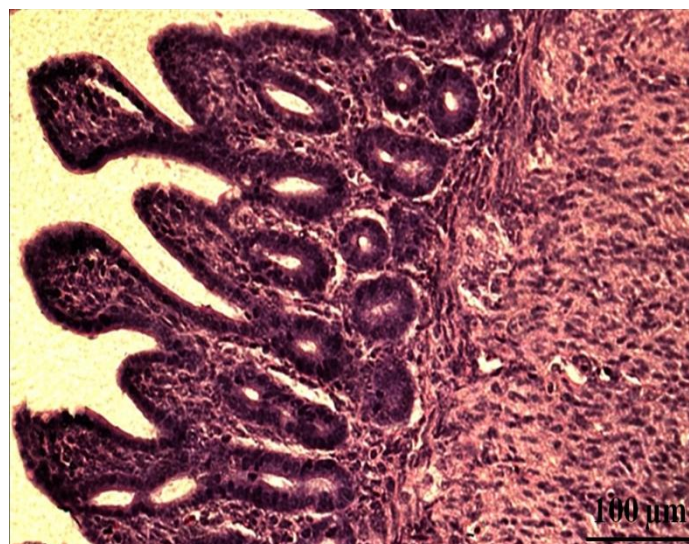


Fig. 5. Histology of the intestine X20. H&E stain.

Type of tissue	Ratio ghrelin/proteins $\times 10^{-6}$
Crop	5:1
Stomach	28,5:1
Intestine	1,68:1

Tab. 1. Ratio ghrelin/proteins concentrations in the tested tissues.

CONCLUSION

The microscopic evaluation clearly showed diversified cells in each part of the digestive system. The tissue variability was correlated with different ghrelin and proteins concentrations in tested GI fragments. The obtained results suggest that ghrelin might be a stimulator of growth and development as well as a modulator of GI tissues functions.

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EFFECT OF ACETYLSALICYLIC ACID ON THE CONTENT OF 20-HYDROXYECDYSONE AND POLYPODINE B IN THE PLANT *LEUZEA CARTHAMOIDES* DC.

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Summary: *Acetylsalicylic acid (ASA) was applied on the plant Leuzea carthamoides DC. in three different concentrations with water as a control. 20-hydroxyecdysone and Polypodine B were the main subjects studied. The results were evaluated by HPLC analysis and statistical methods. It has been shown that low and medium concentrations of ASA have a significant effect on increasing the content of 20- hydroxyecdysone and Polypodine B in the root and leaf mass.*

Key words: *acetylsalicylic acid, ASA, 20-hydroxyecdysone, Polypodine B*

INTRODUCTION

Leuzea carthamoides DC. is a perennial medicinal herb of the *Asteraceae* family. It is known from Chinese, Russian Tibetan and Mongolian medicine, where its root has been used for its healing effects for hundreds of years. *Leuzea carthamoides* DC. extracts have been shown to help treat nervous, mental and cardiovascular diseases (Kokoska, Janovska, 2009). It treats diseases of the lungs and kidneys, the stimulating effects are known (Abramchuk et al., 2014; Selivanova 1979; Łotocka, Geszprych 2004., Постников 1995., Федеральный закон 2009., Монгуш, Монгуш 2012., Сосков 1959., Кубан, Дорогина, Жмудь 2018).

In recent years, *leuzea carthamoides* DC. has been the subject of studies, mainly influenced by steroids called phytoecdysteroids, which are attributed to its healing ability. One of the best known phytoecdysteroids from *leuzea carthamoides* DC. is 20-hydroxyecdysone, for which studies have shown the ability to help treat various types of cancer. (Kormosh et al. 2006., Devillers, 2013., Sláma, Santiago- Blay 2017., Skala et al. 2018., Sláma 2019).

Acetylsalicylic acid (ASA) affects physiological processes, photosynthesis, nitrogen metabolism and water management. It affects plant function at specific doses when certain plant mechanisms are induced or inhibited. This can be achieved by using a low or high concentration of elicitors. Kužel et al. (2003., 2007., 2008., 2009.), Hruby et al. (2002), Tlustoš et al. (2005) and Cigler et al. (2010) reiterated that it is not easy to use elicitors if they cause them to increase the content of bioactive substances. ASA is one of the elicitors that are protected, including research results, by the Czech patent CZ-296300 (Cígler et al., 2006) and the European patent EP-1750507 (Cígler et al. 2015).

The use of acetylsalicylic acid (ASA) in three concentrations therefore serves to verify, increase or decrease the monitored substances - 20-hydroxyecdysone, Polypodine B.

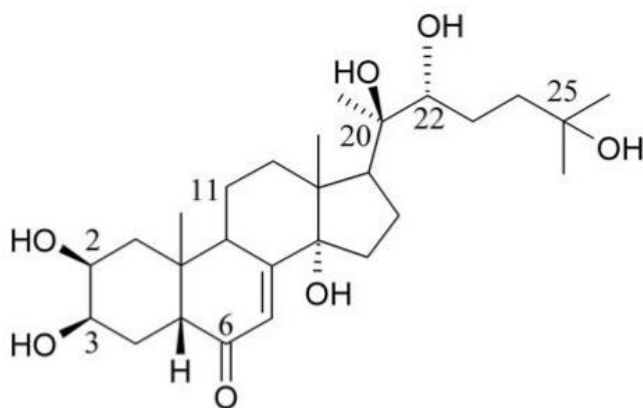


Fig 1. Chemical structure of 20-Hydroxyecdysone (Browning, Christopher, et al. 2007)

MATERIAL AND METHODS

Leuzea Carthamoides DC. was grown on a field where it was treated with ASA acid and water. The plants were divided into four groups, each group containing 20 plants. The growing units were marked as "A1, A2, A3 and W". Group A1 was treated with 500 ml of a solution with a low concentration (10-5 mol.l-1) of acetylsalicylic acid (ASA), group A2 was treated with 500 ml of a solution with a medium concentration (10-4 mol.l-1) of acetylsalicylic acid (ASA), group A3 was treated with 500 ml of a solution with a high concentration (10-3 mol.l-1) of acetylsalicylic acid (ASA) and group W was treated with only 500 ml of water. The W group was for control only. Every group was treated every three weeks, at the same time.

In the third week after the last application of ASA and water, representative samples were taken. 50 grams of aerial and root parts of plants were removed from group A1. 50 grams of aerial and root parts of plants were removed from group A2. 50 grams of aerial and root parts of plants were removed from group A3. 50 grams of aerial and root parts of plants were removed from group W. The samples were dried for 7 days at a temperature of 20 degrees and a humidity of 60 %.

SOLVENTS AND CHEMICALS

Methanol (LC-MS grade, Honeywell, USA), ammonium fluoride (Sigma-Aldrich, USA), 20- hydroxyecdysone, polypodine B, cortisol-d4 (Sigma-Aldrich, USA).

INSTRUMENTATIONS AND MEASUREMENTS A UHPLC

Infinity 1290 kit (Agilent Technology, USA) coupled to a Q-TOF 6550 mass detector (Agilent Technologies, USA) was developed for UHPLC-MS / MS analyzes. Chromatographic separation takes place on a Zorbax analytical column Eclipse Plus C18 RRHD, 2.1 mm x 100 mm; 1.8 (m (Agilent Technologies, USA) with mobile phases of 1 mM ammonium fluoride (mobile phase A) and methanol (mobile phase B). Gradient elution with the following course

was used: 0 min - 80:20 (A: B); 1 min - 50:50; 5 min - 30:70; 6 min - 0: 100; 7 min - 0: 100; 7.2 min 20:80; 8.5 min 20:80. The flow rate of the mobile phase was 0.3 ml / min, the column temperature was 40 ° C and sample injection 2 µl.

Electrospray ionization in positive was used for mass detection module. The parameters of the ion source were as follows: drying gas temperature (nitrogen) - 150 ° C; drying gas flow - 15 l / min; misting gas temperature (nitrogen) - 375 ° C; misting gas flow - 12 l / min; nebulizer pressure - 25 psi; voltage on capillary - 3500 V. In the so-called auto MS / MS mode, m / z in the range of 50–1500 were monitored 80 and MS / MS spectra were collected at a collision energy of 20 V.

PREPARATION OF SOLUTIONS

Agilent MassHunter software version B.05.01 (Agilent Technologies, USA) was used to obtain data and evaluate the results. Calibration curves for quantitative analysis of 20-hydroxyecdysone and polypodine B were composed of eight points (concentrations 200, 100, 50, 10, 5, 1, 0.5, 0.1 g / ml).

The methodology developed in the Laboratory of Forensic Analysis of Biologically Active Substances, ICT Prague was used for the analysis. Analyzes were performed by laboratory staff under the leadership of Ing. Kuchař, Ph.D.

SAMPLE PREPARATION

The dried samples were ground finely using an electric grinder (IKA A 11; IKA Werke GMBH & Co.KG, Germany). Half a gram of plant material was extracted 10 ml of 70% methanol containing deuterated cortisol at a concentration of 1 µg / ml as internal standard. The samples were shaken at room temperature for two hours. The extracts were then centrifuged (10 minutes, 10,000 rpm, Hettich Universal 320R Centrifuge, Hettich, Germany), then 60 µl of the extract was diluted with 940 µl 20% methanol. For the determination of 20-hydroxyecdysone in root samples, this solution was further diluted 20-fold. The solutions were used for UHPLC-MS / MS analysis. The samples were prepared in three repetitions.

RESULTS AND DISSCUSIONS

A statistically significant increase in the observed substances occurred after the use of ASA low (a) and ASA medium (b) on the content of 20-Hydroxyecdysone in the aboveground part of the plants.

ASA low (a) - 10⁻⁵ mol.l⁻¹

ASA medium (b) - 10⁻⁴ mol.l⁻¹

Statistics of a significant increase in 20-Hydroxyecdysone in the aboveground part of plants occurred when using ASA low (a) and ASA medium (b).

In contrast, application of ASA high (c) in high concentration caused a decrease in 20- Hydroxyecdysone in the aboveground part compared to the control group Water (d).

ASA high (c) - 10⁻³ mol.l⁻¹

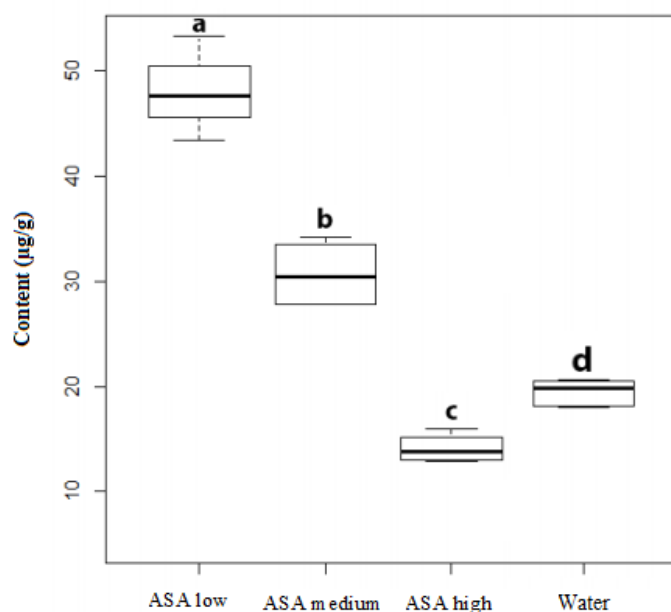


Table 1. Content of 20-Hydroxyecdysone in aerial parts after treatment

Statistically significant increase of the content of 20-Hydroxyecdysone in the root of *Leuzea carthamoides* DC. became ASA low (a) at a concentration of 10^{-5} mol.l⁻¹.

ASA medium (b) and ASA high (c) showed a statistically significant reduction in the content of 20-Hydroxyecdysone in the root.

ASA medium (b) - 10^{-4} mol.l⁻¹

ASA high (c) - 10^{-3} mol.l⁻¹

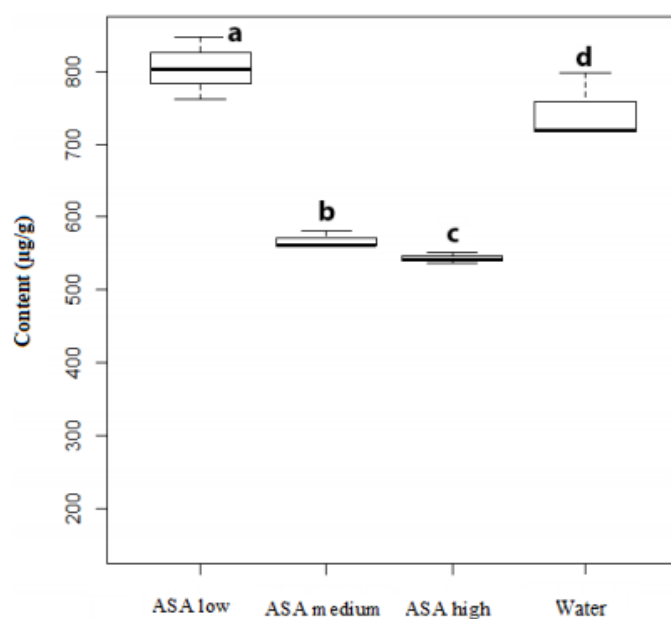


Table 2. Content of 20-Hydroxyecdysone in root part after treatment

A statistically significant increase in Polypodin B in the aboveground part of the plant was achieved by application of ASA low (a) and ASA medium (b).

ASA low (a) - 10^{-5} mol.l⁻¹

ASA medium (b) - 10^{-4} mol.l⁻¹

A statistically significant reduction in Polypodin B compared to the water control (d) was achieved in the supernatant portion of the ASA high (c).

ASA high (c) - 10^{-3} mol.l⁻¹

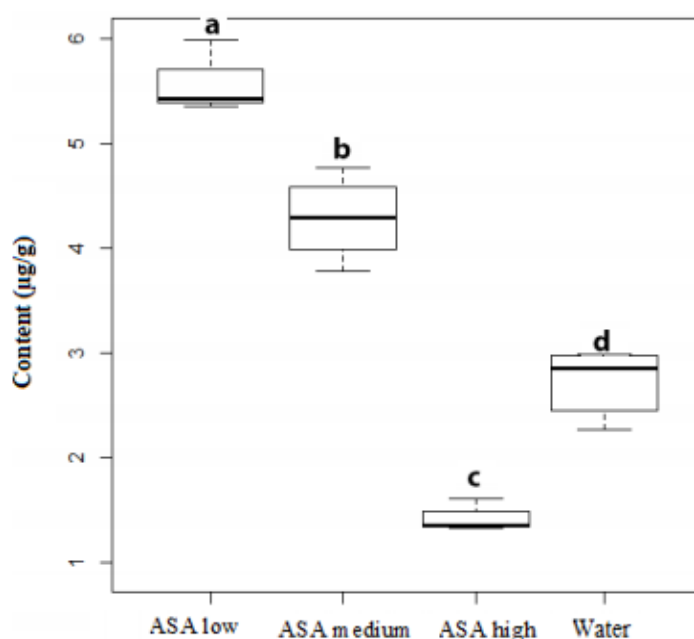


Table 3. Content of Polypodine B in aerial parts after treatment

Polypodin B at the root statistically increased its content the most after ASA application low (a) compared to water control (d).

ASA low (a) - 10^{-5} mol.l⁻¹

Conversely, a reduction in Polypodin B content compared to the water control (d) was achieved by the application of ASA medium (b) and ASA high (c).

ASA medium (b) - 10^{-4} mol.l⁻¹

ASA high (c) - 10^{-3} mol.l⁻¹

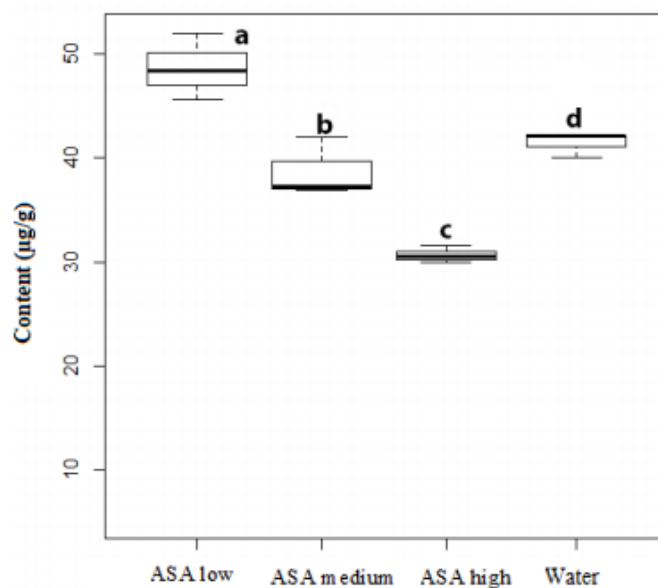


Table 4. Content of Polypodine B in root part after treatment

CONCLUSION

Content of 20-Hydroxyecdysone and Polypodine B in the plant *Leuzea carthamoides* DC. is statistically affected by the interaction between the plant and the treatment. The treatment has a different effect on the content of 20-Hydroxyecdysone and Polypodine B in the aerial parts and roots of the plant. Tested by two-way analysis of variance method (two-way ANOVA).

ASA in low concentration (a) demonstrably increased the content of 20-Hydroxyecdysone and Polypodine B in the aboveground mass and in the roots.

ASA low (a) - 10^{-5} mol.l⁻¹

Mean ASA (b) demonstrably increased the content of both observed substances - 20-Hydroxyecdysone and Polypodine B, in the above-ground mass.

ASA medium (b) - 10^{-4} mol.l⁻¹

In contrast, the high concentration of acetylsalicylic acid (c) caused a decrease in both the monitored substances 20-Hydroxyecdysone and Polypodine B, both in the aboveground and root mass compared to the control group - water (d).

ASA high (c) - 10^{-3} mol.l⁻¹

For a complete understanding of the effect of acetylsalicylic acid on the content of biologically active substances in the plant *Leuzea carthamoides* DC. further attempts will be needed.

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USE OF NATURAL SWEETNERS IN THE DEVELOPMENT OF HYPOCALORIC PRODUCTS

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Summary: *Overweight and obesity are global health problems that affect more and more people every year, often being associated with an increased risk of chronic diseases such as cardiovascular disease, type 2 diabetes or metabolic syndrome. One of the incriminating factors for the occurrence of these diseases is excessive sugar consumption. Therefore, both the food industry and the specialized forums are looking for feasible solutions in an effort to reduce sugar consumption. For example, the Dietary Guidelines for Americans 2015-2020 recommended limiting sugar to 10% of total energy. Another alternative is to use natural sweeteners, which in addition to the role of sugar substitutes, have other functional properties (sources of fiber, prebiotics, increase the melting resistance of the product, lower caloric value). The aim of this work was to develop a product similar to chocolate, but with a low caloric value, ideal for diets, which can also be used by people with diabetes, because it doesn't increase level of glucose in blood. Inulin and isomalt were used as sweeteners for the development of the product, the products obtained being sweet, but with a low caloric content. Following the sensory analysis performed, the product was highly appreciated by consumers, who expressed their interest in buying healthier products with a lower caloric value.*

Keywords: *Isomalt, Inulin, Hypocaloric, Natural, Fiber.*

INTRODUCTION

In order to limit the excessive consumption of sugar, a healthier alternative are the natural sweeteners, which can have multiple roles, they are sources of fiber and prebiotics, having a low caloric value.

Inulin can improve certain markers of heart health, it can reduce triglyceride and cholesterol levels, keeping them between normal values. The normal triglyceride level is less than 150 mg /dl and the total cholesterol level for adults should not exceed 200 mg /dl, while HDL cholesterol should have a maximum value between 40-60 mg /dl for adults, and LDL cholesterol should be kept below 100 mg /dl. The results indicated that the consumption of inulin alone improved the lipid profile and lowered the LDL cholesterol. (Sandra A. Reis et al., 2014).

The purpose of this work is to develop a chocolate-like, but healthier, product with a reduced number of calories, ideally in diets, which can be used by diabetics, as it does not increase blood glucose level. In this product the sweeteners used were inulin and isomalt, the

obtained products being sweet, with a low caloric content and maintaining the blood glucose at normal values.

The objectives of this paper are the following:

- Development of a healthy alternative to classic sweets
- Creating a low-calorie “homemade chocolate” product that is easily accepted by consumers
- Development of an innovative product with a low caloric content
- Making a product for diabetics
- Acceptability of finished products in terms of consumer perception

MATERIAL AND METHODS

To develop a hypocaloric product, similar to chocolate powdered milk, cocoa, inulin, isomalt and water were the main ingredients. The product obtained being was sweet, with a low caloric content, having also functional properties because of inulin content.

During the optimization process the product went through several variants, more precisely 8. The first recipes of the product was made with sugar, butter and just with isomalt. After we try 2 recipes like this, we excluded butter and included inulin, after that we excluded sugar and start working on the proportions of each ingredient. If the proportion of the ingredients was wrong the product could not solidify. Lately, we made 8 recipes of this product until we found the perfect one, with low caloric value, sweet taste, creamy texture but also which solidified.

In order to perform an evaluation of the product from the consumers' perspective, a sensory analysis is performed, using affective tests, more precisely the hedonic test. This test, which is also one of acceptability, is defined as an affective method (hedonic = pleasure), which allows the evaluation of the acceptability or pleasure of the product on the hedonic scale (Lawless, 2010).

For sensory analysis, naive evaluators (consumers) will be used to perform this test. The samples/ products will be 3 in number (the product version with inulin and isomalt, the isomalt version and the classic version, with sugar), they will be coded with 3 digits so as not to transmit any evaluative information. The samples are prepared in this way and be identical in terms of physical appearance: shape, size, appearance/ presentation, weight, temperature, texture (Laura Stan, 2018). Prior to tasting, the evaluators are instructed on how to complete the form and how to evaluate the sample (chew gently in the mouth for a sufficient time to fully perceive the food flavor before and after swallowing. Not on the evaluation sheet. A break between samples is recommended until the remaining taste disappears (Laura Stan, 2018).

Inulin is a polysaccharide soluble in water, which belongs to the group of indigestible carbohydrates called fructans, which means that it is made up of fructose molecules taken to prevent digestion in the small intestine. Inulin has status GRAS, which means Generally Recognized As Safe, being part of over 3.000 species of plants, and having a nutritional value around 1,5 kcal/g. Roots of chicory are considered the main sources of inulin. This

polysaccharide can be also found in: bananas, oats, onions, garlic, asparagus and leeks (Bornet et al., 2008). This polysaccharide is considered a dietary fibre and among the dietary fibres widely used in a variety of foods, inulin represents a noteworthy ingredient (D'Antuono I. et al., 2018).

Inulin is composed of multiple parts of fructose with a terminal glucose molecule. The β - (2-1) D-fructosyl fructose bonds are present between fructose units of inulin and β -configuration of anomeric carbon, which makes it indigestible in small human intestine (Apolinario A. C. et al., 2014). Compared to digestible carbohydrates, inulin gives only 25–35% energy and is widely used as sugar replacer in the processed foods (Shoaib M. et al., 2016).

Instead to be digested, inulin arrived in intestine, where it acts just like a prebiotic. In 2008, in the 6th Meeting of the International Scientific Association of Probiotics and Prebiotics (ISAPP) “dietary prebiotics” were defined as a selectively fermented ingredient that results in specific changes in the composition and/or activity of the gastrointestinal microbiota, thus conferring benefit(s) upon host health” (Gibson G.R. et al., 2010). Prebiotics helps feeding and increase the number of beneficial bacteria in our digestive tract. Inulin can also be fermented in the human large intestine by the intestinal microflora, due to the stimulation of growing bifidobacteria (Apolinario A. C. et al., 2014). Inulin has a series of health benefits, it improves digestion health, helps to control level of blood sugar, contributes to calcium absorption, helps to control weight, improves health of heart.

Due to its capacity of water absorption inulin, as well as other prebiotics, has the ability to prevent the problems associated with constipation and reducing the risk of many diseases of the intestinal tract such as: ulcerative colitis, Crohn's disease, inflammatory bowel diseases (IBD) and colon cancer. The decrease of this risk is possible due to the development in the intestine of bifidobacteria and lactobacilli (Shoaib M. et al., 2016; Giuseppe Tripodo and Delia Mandracchia, 2019). Inulin is resistant to gastric juice from the small intestine, it helps to stabilize the microflora of the large intestine by stimulating the growth of bifidobacteria, which is why it is used more and more frequently in the daily diet. Approximately 90% of the inulin that passes into the colon enters the digestion of the bacteria present in it (Shoaib M. et al., 2016). One of the most evident effects after inulin fermentation by bifidobacterium, is the formation of short chain carboxylic acids (SCFAs) such as acetate, butyrate and propionate, lactate and gases. It has been reported that butyric acid could result in maintaining the health of colonic mucosa while propionic acid seems implied in carbohydrate and lipid metabolism also influencing the motility of the colon (M. Nyman, 2002).

Inulin has a series of beneficial effects to human organism, this dietetic fibre is considered a key ingredient in preparation of easily digested food, which can be consumed as a dessert or as a sweet snack, with no effect on blood sugar (Shoaib M. et al., 2016). Different studies point on the positive effect of inulin on diabetic patients, in particular for type 2 diabetes. Furthermore, inulin has a glycaemic index lower than soluble carbohydrates as well as lower caloric value which helps in obesity treatment (C.M. van der Beek et al., 2018). Interestingly, inulin and oligofructose have been shown to modulate hormones which may play a role in satiety (F. Liu et al., 2017). Inulin acts through specific changes in hormones

that control appetite and helps reduce the number of calories consumed daily by reducing the feeling of hunger, and the results can be impressive. For example, studies have shown that individuals are able to lose between 1 and 6 kilograms within 12-18 weeks by adding inulin powder to their diet (Shoaib M. et al., 2016). Not only did these individuals have the potential to reduce body weight, but when they used inulin powder, a reduction in ectopic fat deposits was also observed, fat that is stored in certain areas of the body, such as the liver or muscle tissue (Grant Koch, 2019).

Once inulin arrives in the body, it increases the absorption of calcium, an important mineral in the formation of teeth and bones, relaxation and constriction of blood vessels, the functioning of nerves and muscles, hormonal balance (C Coudray and colab., 1997). Modified the pH in the large intestine (which become more acidic), inulin increases the surface area that can absorb nutrients and produces more proteins who makes bond with calcium. Differently from others dietary fibres, inulin does not negatively influence the bioavailability of ions of nutritional importance like magnesium, calcium or iron. On the contrary, it has been demonstrated an increase in their adsorption in coadministration with inulin, an increase measured in about 60% (A. Franck, 2006).

These findings lead to the *in vivo* assessment (on either animals or humans) of the effect of co-administration of inulin and calcium on bone remineralization in patients suffering from osteoporosis (U. Krupa-Kozak and colab., 2016). Decreasing risk of osteoporosis and bone fractures is a critical issue worldwide, with >28 million people in the US having osteoporosis or low bone mass, and 1 in 8 EU citizens over 50 years of age fracturing their spine each year (Cashman KD et al., 2002). Increasing the bioavailability and absorption of calcium with the intake of prebiotics is a critical target for healthy bone structure in adolescent and elderly populations (Whisner CM. et al., 2013). The distal intestine is one of the primary sites of calcium absorption, and absorption is stimulated by the chemical changes and increases in acid fermentation of the prebiotic dietary fibers by various bacteria.

Lately, inulin is increasingly used in the development of innovative products due to its properties: fiber source, prebiotic, fat and sugar substitute (Shoaib M. et al., 2016).

RESULTS AND DISCUSSION

Because we wanted to obtain a hypocaloric product, we try a lot of recipes until we found the best from a nutritional and technological point of view. The final recipe has several pluses, both in terms of organoleptic and physical properties. The taste is sweet, pleasant smell of cocoa, the texture is creamy, fine, without agglomerated particles. Visually, the product is in the form of a homogeneous, chocolate-brown mass, similar to homemade chocolate.



Fig. 1. Hypocaloric product

We made the caloric analysis first to see the caloric value that we obtain for our product, because from the beginning the aim was to obtain a variant with a low caloric content but without influencing the taste and aroma of the finished product. Thus, for the final version the values are the following:

Table 1. Ingredients for product and their caloric value

Ingredients	Quantity	Caloric Value
Isomalt	50g	100 kcal
Inulin	50g	100 kcal
Powdered milk	40g	193 kcal
Cocoa	15g	42 kcal
Water	70ml	0 kcal
Total	225g	435 kcal

$$\text{Total caloric value/ 100g product} = \frac{100 * 435}{225} = 193 \text{ kcal /100g product}$$

We obtained a product that has 193 kcal/ 100g, less than half of the caloric value of a conventional chocolate, which has approximately 470-530 kcal/ 100g. In addition to the large difference in calories, the fiber content also differs, a conventional chocolate has about 1.8% fiber, our product has 22.7% fiber. At the same time, the fat content is significantly lower in the low-calorie product than in another conventional product.

Another calculation was the one referring to the nutritional value. From a nutritional point of view, the product is rich in fiber, on the strength of inulin content. The following tables present the nutritional information of the product ingredients.

Table 2. To get 100g of final product we needed:

Ingredients	Quantity
Inulin	23g
Isomalt	22g
Powdered milk	17g
Cocoa	7g
Water	31ml

Table 3. Chemical composition and nutritional value of inulin

Nutritional Value	100g Inulin	23g Inulin
Energy / energy value	200 kcal	46 kcal
Fats	0g	0g
Carbohydrates, of which:	8g	1.84g
- sugar	8g	1.84g
Fiber	89g	20.47g
Protein	0g	0g
Salt	0g	0g

Table 4. Chemical composition and nutritional value of isomalt

Nutritional Value	100g Isomalt	22g Isomalt
Energy / energy value	200 kcal	44 kcal
Fats	0g	0g
Carbohydrates, of which:	97g	21.34g
- sugar	0g	0g
Fiber	0g	0g
Protein	0g	0g
Salt	0g	0g

Table 5. Chemical composition and nutritional value of cocoa

Nutritional Value	100g Cocoa	7g Cocoa
Energy / energy value	285 kcal	20 kcal
Fats, of which:	10.5g	0.74g
- saturated fatty acids	6.4g	0.45g
Carbohydrates, of which:	8.5g	0.6g
- sugar	0.4g	0.028g
Fiber	31.2g	2.2g
Protein	23.5g	1.65g
Salt	1.1g	0.077g

Table 6. Chemical composition and nutritional value for powdered milk

Nutritional Value	100g Powdered Milk	7g Powdered Milk
Energy / energy value	486 kcal	83 kcal
Fats, of which:	26g	4,4g
- saturated fatty acids	16,6g	2,8g
Carbohydrates, of which:	37g	6,3g
- sugar	37g	6,3g
Fiber	0g	0g
Protein	25g	4,3g
Salt	0,74g	0,12g

Thus, following the calculations made, the nutritional value of the final product is presented in the table below

Table 7. The nutritional value of the hypocaloric product developed

Nutritional Value	100g
Energy / energy value	193 kcal
Fats, of which:	4,4 g
- saturated fatty acids	3,25g
Carbohydrates, of which:	30g
- sugar	8,16g
Fiber	22,7g
Protein	6g
Salt	0,2g

The hedonic test aims to evaluate the emotions and feelings of consumers by determining the acceptability of the product, by indicating an option available on the hedonic scale (Laura Stan, 2018). The evaluator's form must contain identification data of the person (age, sex, level of education), purpose of the test and hedonic scale. This scale is a bipolar one, the extremities representing the positive and negative appreciation of the product, and in the centre of it is a neutral value. This scale can be in 7 or 9 points (Laura Stan, 2018). Following the sensory analysis performed on 14 people, that are usually consuming chocolate, with the help of a bipolar scale in 9 points, the grades presented in figure 2 were obtained. During the analysis, 3 variants of the product were made: variant 1 was the one with isomalt, variant 2 contained both isomalt and inulin, and in variant 3 the natural sweeteners were replaced with sugar.

	Sample 1	Sample 2	Sample 3
Pers. 1	9	8	9
Pers. 2	8	9	8
Pers. 3	7	8	6
Pers. 4	8	9	7
Pers. 5	9	8	5
Pers. 6	7	9	6
Pers. 7	8	9	7
Pers. 8	8	7	5
Pers. 9	8	8	9
Pers. 10	6	8	7
Pers. 11	8	9	7
Pers. 12	7	8	6
Pers. 13	7	7	8
Pers. 14	7	7	6
Arithmetic mean	7,64	8,14	6,86
Standard deviation	0,84	0,77	1,29

Fig. 2. The grades, arithmetic mean and standard deviation from sensory analysis

After collecting the notes, the arithmetic mean was calculated, using the formula = AVERAGE, and for the calculation of the standard deviation the formula = STDEV was used. Thus, with the help of the obtained results, the column type graph from figure 3. was

made, where it can be seen that the participants in the analysis preferred the second variant (inulin + isomalt), obtaining a score of 8.14 out of a maximum of 9 points. As we can see in the graph, variant 1 (isomalt) was also to the liking of the participants, but it obtained a slightly lower score, namely 7.64, while variant 3 (sugar), obtaining the lowest score, the one of 6.86 out of 9 points.

Following this sensory analysis, the participants noticed a significant difference in the taste and texture of the 3 samples. Arguing that the texture of variant 2 is softer and more elastic than variant 1, which is stronger and more crumbly, and variant 3 had a sandier texture than variant 1 and 2, which were finer.

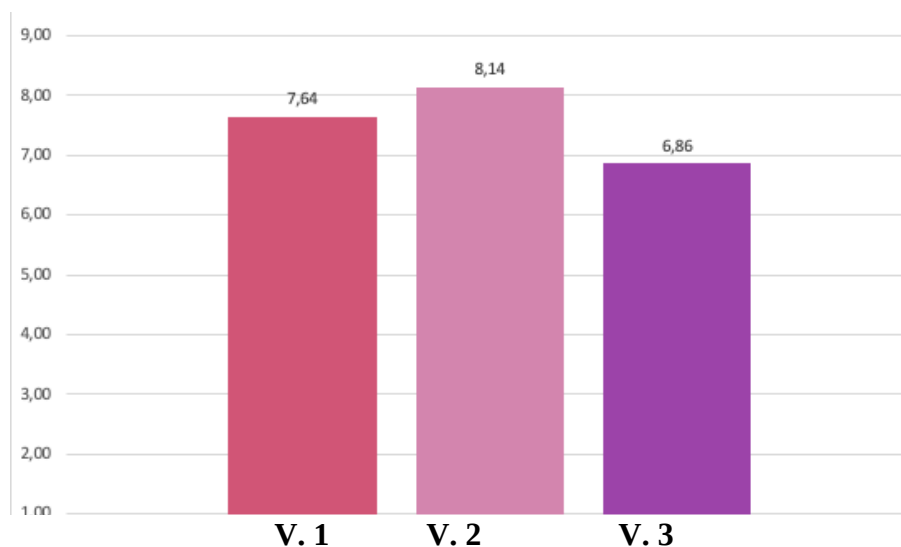


Fig. 3. The hedonic score obtained for the sensory analysis performed on the 3 product variants

Following the centralization of the data obtained during the analysis, it can be concluded that the participants preferred samples containing natural sweeteners instead of the sample containing sugar.

CONCLUSIONS

Following the study performed on natural sweeteners, it can be concluded based on the results obtained that the caloric difference is not perceived from a sensory point of view by consumers when they consume the product manufactured with natural sweeteners. Following the nutritional analysis, it was found that the product has a high fiber content and low in fat, hence its low caloric value.

Finally, it is worth mentioning few of the benefits of the final product, namely: product obtained only from natural ingredients; high fiber content that helps digestion; does not increase blood glucose levels, so it can be consumed even by diabetics; low glycemic

index; low caloric value, which makes it ideal in diets; due to the isomalt content it stimulates the peristaltic movements of the stomach, thus accelerating digestion.

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ENVIRONMENTALLY FRIENDLY ALTERNATIVES FOR WHEAT WEEVIL CONTROL

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SUMMARY

Summary: *This work investigated the efficacy of some ecologically acceptable insecticides against the wheat weevil (*Sitophilus granarius* (Linnaeus, 1758)), an important storage pest. Azadirachtin, spinosad and diatomaceous earth were tested, while cypermethrin served as a standard insecticide and water as control. The experiment was performed in laboratory conditions, testing each insecticide applied to wheat grain in three different doses. A total of 13 variants were set up in four repetitions. In each repetition, 10 adult wheat weevils were added. The mortality of weevils was determined every 24 hours during five days. Azadirachtin, diatomaceous earth, and spinosad showed high efficacy (84.28%-100.00%) 96 hours after treatment, which did not differ significantly from efficacy of cypermethrin. This work showed that these ecologically friendly insecticides have great potential in wheat weevil control on stored wheat.*

Key words: *azadirachtin, diatomaceous earth, efficacy, *Sitophilus granarius*, spinosad, wheat*

INTRODUCTION

Storage of agricultural products is a very important measure that represents the final intervention in the process of agricultural production. In some cases, it is more important than the technology used in production, because poor storage leads to a loss of quality and quantity of goods, which reduces its market value. Global losses caused by pests amount to 42.1%, and in Europe they amount to 28% (Oerke et al., 1994). In the Republic of Croatia, pests reduce agricultural production by 29.2% of potential yield, worth about USD 500 million per year (Maceljčki i Igrc, 1991). In addition to reducing yield, pests also reduce the quality of agricultural products and limit their turnover and distribution in the world. The emergence of resistance and the reduction in the number of registered active substances have led to the need to find new, environmentally friendly solutions in the control of storage pests. Ecologically acceptable measures are measures that, when properly applied, do not have a toxic effect on humans and beneficial organisms, do not pollute the environment, minimally disturb the established balance of organisms, and rarely have a negative effect on the diversity of species in nature. These measures can achieve better success, despite the fact that they may be less effective than other measures (Maceljčki, 2002).

The aim of this study was to investigate the efficacy of some environmentally friendly insecticides derived from plants (azadirachtin), microorganisms (spinosad) and naturally occurring substances like minerals (diatomaceous earth) against the wheat weevil (*Sitophilus granarius* (Linnaeus, 1758)), an important storage pest.

MATERIAL AND METHODS

Sampling of wheat and test insects

Common wheat (*Triticum aestivum* L.) samples were taken from dryers and silos in Veliki Bukovec, located in northwestern Croatia (46°17'33.0"N 16°42'49.8"E). The wheat was used as a growing medium for wheat weevil (*S. granarius*). Wheat weevils used as test insects in this experiment were collected in stored wheat of a private storage space located near Ludbreg (46°15'05.5"N 16°37'54.3"E). The sampling of both wheat and insects took place in February 2019. Weevils were reared on the sampled wheat under laboratory conditions (temperature of 24 ± 2 °C, relative air humidity of 50-60%) for about a month until the experiment was set up. The research was then conducted at the Department of Agricultural Zoology of the Faculty of Agriculture in Zagreb in March 2019 under the same laboratory conditions.

Insecticides used in the experiment

The wheat used as a growing medium for wheat weevil was treated with three environmentally friendly insecticides: azadirachtin (NeemAzal[®]-T/S, Trifolio-M GmbH), spinosad (Laser[™], Dow AgroSciences) and diatomaceous earth (Silico-Sec, Biofa AG). In order to be able to compare the results, cypermethrin (Cythrin Max, Arysta LifeScience) as a standard insecticide for storage pest control was used, as well as untreated control (pure water). Each insecticide was tested in three doses. A total of 13 variants were set in the experiment, with each variant set in four repetitions. Insecticides and doses used in the experiment are shown in Table 1.

Table 1. Insecticidal treatments used in the laboratory experiment with wheat weevil (Zagreb, 2019)

Doses	Azadiracht in (ml/200 g)	Spinosad (mg/200 g)	Diatomaceo us earth (g/200 g)	Cypermeth rin (mg/200 g)	Untreated control
Double the recommended dose	6.25	1.74	4.00	0.80	/
Recommended dose	3.125	0.875	2.00	0.40	/
Half the recommended dose	1.56	0.437	1.00	0.20	/

Experiment implementation

Wheat was cleaned of impurities and chaff, and physical values were measured using a grain moisture and density meter (Draminski GMDM, DRAMINSKI S. A.) in a volume of 250 ml. Wheat moisture of 11.9%, weight of 167.5 g, and hectolitre weight of 79.1 kg/hl were

determined, i.e. the physical values of the grain were within the standard values. Then 200 g of wheat was weighed for each variant and placed in plastic storage bags (volume 3 l). Then, the grain in each storage bag was treated with insecticide or water (control), by adding the amount shown in Table 1 for each variant. After application of the insecticides, each plastic bag was tightly closed in such a way that there was air left. The contents of the bag were then stirred with a vigorous shaking for 30 seconds to distribute the insecticide well over the grain. In the same way, water was applied to the control variant. After mixing, the grain was placed in an open bag to air dry for 24 hours. Each sample of 200 g of treated grain was then divided into four repetitions and placed in 250 ml plastic vials (50 g of treated grain per vial). After the insecticide had dried, 10 adult wheat weevils were added to each repetition (a total of 40 weevils per variant). The vials were then closed with a mill cloth to allow the insects sufficient air access. The variants were kept under controlled conditions at a temperature of 24 ± 2 °C and a humidity of 50-60% during the whole experiment.

Experiment reading

The experiment was conducted over five days. The efficacy of insecticides was assessed through the mortality of wheat weevil that were inspected and recorded for each variant. The first reading was performed 48 hours after the treatment and every 24 hours for a further three days. Every day during the reading, the contents of the vial were poured on a test surface and the number of dead individuals was determined. The dead insects were removed, and the remaining material (live insects and grain) was returned to the vial. For each reading day for each repetition, the efficacy was calculated according to the Schneider-Orelli formula (1) (1947 cit. Püntener, 1981).

$$(1) \quad \text{Efficacy (\%)} = \frac{(\text{Mortality (\%)} \text{ on treatment} - \text{mortality (\%)} \text{ on control})}{(100 - \text{mortality (\%)} \text{ on control}) \times 100}$$

Data analysis

Efficacy data were processed by analysis of variance (ANOVA) using the statistical program ARM 9[®] (GDM Solutions, 2020). Mean data were ranked using Duncan's new multiple range test to determine differences in efficacy between the variants. In the case of uneven data distribution, the data in the ARM program are transformed using an automatic square root transformation of $x+0.5$.

RESULTS AND DISCUSSION

The efficacies of different doses of azadirachtin (NeemAzal), spinosad (Laser), diatomaceous earth (Silico-Sec), and cypermethrin (Cythrin Max) as a conventional insecticide on adult wheat weevils are shown in Table 2.

Table 2. Efficacy of environmentally friendly insecticides on wheat weevil determined in the laboratory experiment (Zagreb, 2019)

Insecticide	Dose	Efficacy (%) after			
		48 h	72 h	96 h	120 h
Azadirachtin	1.56 ml/200 g	3.40 c*	4.77 cd	87.45 a	100.00 a
Azadirachtin	3.13 ml/200 g	1.30 c	2.26 d	100.00 a	100.00 a
Azadirachtin	6.25 ml/200 g	100.00 a	100.00 a	100.00 a	100.00 a
Spinosad	0.44 mg/200 g	100.00 a	100.00 a	100.00 a	100.00 a
Spinosad	0.88 mg/200 g	100.00 a	100.00 a	100.00 a	100.00 a
Spinosad	1.74 mg/200 g	100.00 a	100.00 a	100.00 a	100.00 a
Diatomaceous earth	1.00 g/200 g	3.40 c	13.27 c	21.37 b	27.69 b
Diatomaceous earth	2.00 g/200 g	1.30 c	30.40 b	84.28 a	92.00 a
Diatomaceous earth	4.00 g/200 g	26.15 b	84.28 a	100.00 a	100.00 a
Cypermethrin	0.20 mg/200 g	100.00 a	100.00 a	100.00 a	100.00 a
Cypermethrin	0.40 mg/200 g	100.00 a	100.00 a	100.00 a	100.00 a
Cypermethrin	0.80 mg/200 g	100.00 a	100.00 a	100.00 a	100.00 a

*Differences within the same group (hours) marked with the same lower case letter statistically do not differ (Duncan's new multiple range test, P = 5%).

Of all the environmentally friendly active substances tested, spinosad proved to be very effective and caused complete mortality of wheat weevils after only 48 hours. The same results were achieved with the conventional chemical insecticide cypermethrin, which was expected. Lower and recommended doses of azadirachtin and diatomaceous earth had poor initial efficacy, but by the end of the experiment, both insecticides achieved high efficacy (92-100%) when used in recommended doses. The results also showed that all active substances, except for diatomaceous earth, caused high mortality of weevils even at half the recommended dose five days after treatment.

As the number of registered active substances decreases and resistance to standard insecticides occurs, it is necessary to find new, more environmentally friendly solutions in the control of storage pests. Some of these solutions we can search in testing natural compounds found in plants (botanical insecticides), some organisms such as bacteria (spinosyns and other derivatives) or some other naturally occurring substances like minerals for the control of harmful organisms. Azadirachtin is a botanical insecticide known and used since ancient times, but still not at full potential. It acts on insects as a growth regulator, rejects and reduces insect nutrition and reduces female fertility. It is effective on about 200 species of insects, mites, and nematodes (Gahukar, 1995). In our study, azadirachtin administered at the recommended dose was shown to be 100% effective 96 hours after the treatment. A similar study was conducted by Bohinc and Trdan (2017), where after 14 days efficacy of 65.74% at 30 °C and 100% at 35 °C was recorded. The highest efficiency (100%) was achieved by using the combination of azadirachtin and wood ash after 14 days at 30 °C. High efficacy of azadirachtin was also achieved by Kavallieratos et al. (2007) in the control of *Sitophilus oryzae* and *Tribolium confusum* species in wheat and maize.

Another natural insecticide tested in this work was spinosad, derived from soil bacteria. Spinosad affects the nervous system of insects that ingest the insecticide or come in contact with it (NPIC, 2013). This causes uncontrolled flexion of the muscles, leading to paralysis and eventually to their death, usually within one to two days. It is used to control a large number of pests: thrips, lice, mites, mosquitoes, ants, fruit flies, and others (NPIC, 2013). In our study, spinosad achieved maximum efficacy (100%) on wheat weevil after only 48 hours in all three doses administered, showing an acute toxicity for the weevil. Good efficacy of spinosad was also achieved by Velez et al. (2018) on corn rootworm. Deltamethrin was also used in that study, which proved to be somewhat more effective than spinosad. In our research spinosad was as effective as the conventional insecticide cypermethrin. Good results (efficacy of 95%) of spinosad in the control of another important storage pest, the rice weevil, were achieved by Andrić et al. (2011).

Besides insecticides derived from plants or living organisms, we also tested diatomaceous earth as naturally occurring compound used in pest control. Diatomaceous earth is a natural powder used as an insecticide. Small particles, i.e. diatoms, stick to the body of the insect and physical forces, mainly sorption and abrasion, damage the waxy layer on the body which protects the insect from moisture loss from the body. Through damaged places, insects lose moisture from the body and after some time, due to dehydration, die (Ebeling, 1971). Also, diatomaceous earth is known to have repellent properties for insects, which also protects stored products (White et al., 1966). In our study, diatomaceous earth administered at the recommended dose proved to be highly effective (92.00%) against the wheat weevil five days after the treatment. High mortality (83% to 100%) was achieved by applying a combination of spinosad and diatomaceous earth in the control of rice weevil on wheat (Chintzoglou et al., 2008).

CONCLUSION

The environmentally friendly insecticides azadirachtin, spinosad, and diatomaceous earth show great potential for controlling the wheat weevil on stored wheat. However, this research was conducted in a laboratory in a controlled atmosphere and should be repeated and conducted in real conditions. Realistic conditions refer to storage areas, where different temperature and humidity conditions prevail that could affect the grain weevil and the efficacy of insecticides. If the tested insecticides prove to be effective in storage areas also, their use as environmentally friendly insecticides would greatly contribute to the integrated protection of cereals from this important storage pest.

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THE EFFECT OF CITRIC ACID AND SALT ON THE RHEOLOGICAL PROPERTIES OF LIQUID EGG YOLK

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Egg yolk undergoes undesirable processes as a result of freezing. These include gelation, which is an irreversible process that also greatly reduces usability. In this experiment, table salt, a cryoprotectant in industrial practice, was used (0, 1, 2, 5 and 10%) to study the effect of freezing on the rheological behaviour of liquid egg yolk and citric acid was used to adjust the pH (pH 5.5 and pH 6.5). The tests were performed with a rotary rheometer. We found that for both pHs, the sample containing 5% salt had the lowest shear stress and apparent viscosity values, so it had the greatest inhibition of gelation.

Key words: liquid whole egg, rheological properties, gelation, pH adjustment, salt concentration

INTRODUCTION

Eggs are very versatile in the food industry due to their excellent nutrient content and various functional properties. Processed egg products, such as separated, homogenized and pasteurized liquid egg products, are generally used for industrial purposes. Liquid egg yolk (LEY) is a popular ingredient in the food industry due to its excellent gelling, emulsifying, colouring and coagulating properties, high nutritional value and unique organoleptic properties. It is often used as an ingredient in confectionery products such as lemon curd, ice cream, cream pudding and mayonnaise (Cook and Briggs, 1986; Uysal et al., 2019).

Liquid egg yolk serves as an excellent medium for microorganisms, so its shelf life is limited. Even without the addition of preservatives, it is only a few weeks (Dawson and Martinez-Dawson, 1998). Pasteurization is used to increase the shelf life of these products, but a relatively mild combination of pasteurization time and temperature can be used. Heat treatment above 60°C leads to transitions that cause changes in the microstructure, appearance, calorimetric and rheological properties of egg yolk (Cordobés et al., 2003). The rheological behaviour of liquid egg products (liquid egg white, liquid whole egg, LEY) at different temperatures was studied by other authors (Atılgan and Unluturk, 2008). The Herschel-Bulkley model was found to fit well with the shear rate – shear stress data.

Freezing is a traditional physical preservation process that can increase the shelf life of liquid egg yolk for up to 1 year. However, egg yolk undergoes an irreversible fluid loss, when it is frozen at -6°C or a lower temperature. This leads also to a gelation process, which affects the functionality of egg yolk (Primacella et al., 2020). Freezing of liquid egg products is a time-temperature-related process (Lechevalier et al., 2017). By increased freezing and

thawing rates, smaller ice crystals are formed, that results in lower drip loss and better texture and proteins are less dehydrated (Powrie, 1968). Due to the gelation, the usability of frozen and thawed egg yolk changes.

In industrial practice, the addition of 10% salt or sugar is currently preferred to reduce the degree of gelation (Primacella et al., 2019). The specific impact of additives on protein interactions can vary greatly and is dependent on the chemical nature, additive concentration, protein type, and pH (Primacella et al., 2018).

Citric acid is a naturally occurring antioxidant that prevents the green discolouration of egg products and lowers pH (recognized as safe (GRAS)) (Schuman and Sheldon, 2003). Further studies demonstrated that the functional and gel characteristics of liquid egg products can be altered by pH alteration (Chang and Chen, 2000; Sousa et al., 2007).

In this experiment, we investigated how the combination of pH and salt affects the rheological properties of LEY. Liquid egg products were frozen and their rheological properties were re-examined. Our goal was to determine the pH and salt concentration at which we achieve flow properties similar to fresh LEY.

MATERIAL AND METHODS

Materials

Pasteurized LEY produced under industrial conditions (Capriovus Ltd., Szigetcsép, Hungary) was used in this study, which was made by the separation of fresh hen eggs, homogenization and pasteurization (600 s holding time at 65°C with a flow rate of 600 kg/h). LEY was cooled to 3°C and it was filled into PET bottles. Industrial citric acid was used to adjust the pH, while household salt was used to adjust the NaCl concentration.

Sample preparation and experimental design

In this study, 5 different NaCl concentrations (w/w%) were used (0; 1; 2; 5 and 10%). Different pHs were adjusted, these were 5.5, 6 and 6.5. The rheological properties of the samples were examined fresh on the day after sample preparation and after freezing and thawing. Freezing was performed at -18°C in a laboratory freezer for 30 days. Samples tested after freezing were removed from the freezer 24 hours before testing and placed in a laboratory refrigerator at 4°C. Prior to measurement, they were tempered to 20 °C in a water bath.

Determination of the flow curve

Examination of the rheological behaviour of LEY samples was performed by MCR 92 rheometer (Anton Paar, Les Ulis, France) in rotational mode equipped with a concentric cylinder (cup diameter 28.920 mm, bob diameter 26.651 mm, bob length 40.003 mm, active length 120.2 mm, positioning length 72.5 mm). Anton Paar RheoCompass software was used to control the equipment. The temperature of rheological measurements was 20°C. Shear stress was measured by increasing shear rate from 1 to 1000 1/s.

Calculation of the rheological parameters

Herschel-Bulkley model (1.) was fitted to flow curves (shear rate-shear stress diagrams) by using the least square fit method of Excel Solver, where τ_0 , K and n are the changeable values.

$$\tau = \tau_0 + K \left(\frac{d\gamma}{dt} \right)^n$$

where τ is shear stress (Pa), τ_0 the yield stress (Pa), γ the shear rate (1/s), K is the consistency coefficient (Pa·sⁿ) and n the flow behaviour index (dimensionless).

Statistical analysis

Statistical analysis was performed by using IBM SPSS Statistics 24 software. Kolmogorov-Smirnov test was used to test the normality of the error terms. Levene's test was used to test the homogeneity of variances. Multivariate test (Wilk's Lambda) for the rheological parameters showed a significant effect of the cross effect for pH, salt concentration and freezing, so univariate test was performed. Because of the significant cross effect for pH, salt concentration and freezing in univariate tests, we examined the effect of pH, salt concentration and freezing one by one. Tukey test was used to decide which group differ among the samples with the same pH before and after freezing. Paired-sample T-test was used to decide if the values with the same pH differ.

RESULTS AND DISCUSSION

Figure 1. shows the flow curves of LEY samples at pH 5.5 in the fresh state and frozen-thawed after 30 days. It can be seen that LEY samples have pseudoplastic flow behaviour, which means that the shear stress value increases less and less with increasing shear rate, thus obtaining a convex curve with decreasing slope as a flow curve. This means that as the shear rate increases, the sample develops increasing resistance to the probe. This may be due to a decrease in the degree of interaction between the molecules. It can be observed that the flow curves of LEY samples were greatly influenced by both freezing and thawing and different salt concentration. The flow curve of the salt-free sample showed the lowest shear stress values, so it had the lowest apparent viscosity value during the measurement. However, it can be observed that after freezing, this sample had the highest shear stress values. From this it can be concluded that for samples with pH 5.5, the addition of salt before freezing increased the shear stress and the apparent viscosity in all cases, however, freezing yielded samples with lower shear stress and apparent viscosity in all cases. The effect of salt is therefore clearly visible. To some extent, it reduced gelation at all concentrations during freezing.

For the samples adjusted to pH 5.5, it can be seen that their rheological behaviour did not differ much before freezing. It is interesting to note that the sample containing 5% table salt had the highest shear stress and, consequently, apparent viscosity values. However, after freezing, large differences in their rheological behaviour can be seen. It is obvious that after freezing, the shear stress value for the first measuring point changed for each sample. This means that a yield stress value has appeared that must be reached during forcing to flow in order for the sample to start moving at all. This stress value can be given as the intersection of the y-axis when approximating with different models, such as the Herschel-Bulkley model. As mentioned above, the salt-free sample had the highest values. The sample with the highest concentration in our experiment (10%) had a similar flow curve. This means that this concentration is already too high to be used effectively to inhibit gelation. The lowest shear stress and consequently the apparent viscosity values were obtained for the sample containing 5% table salt.

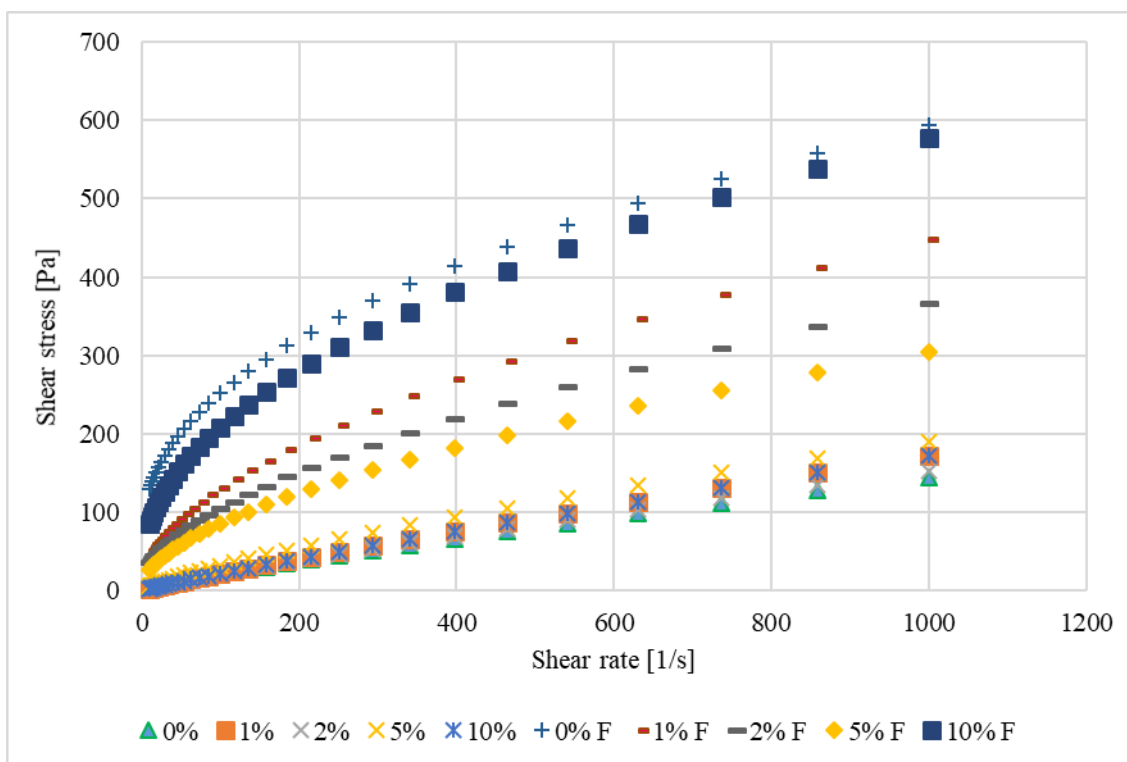


Fig 1. Flow curves of the liquid egg yolk samples (pH 5.5) with different table salt concentrations before and after freezing (F)

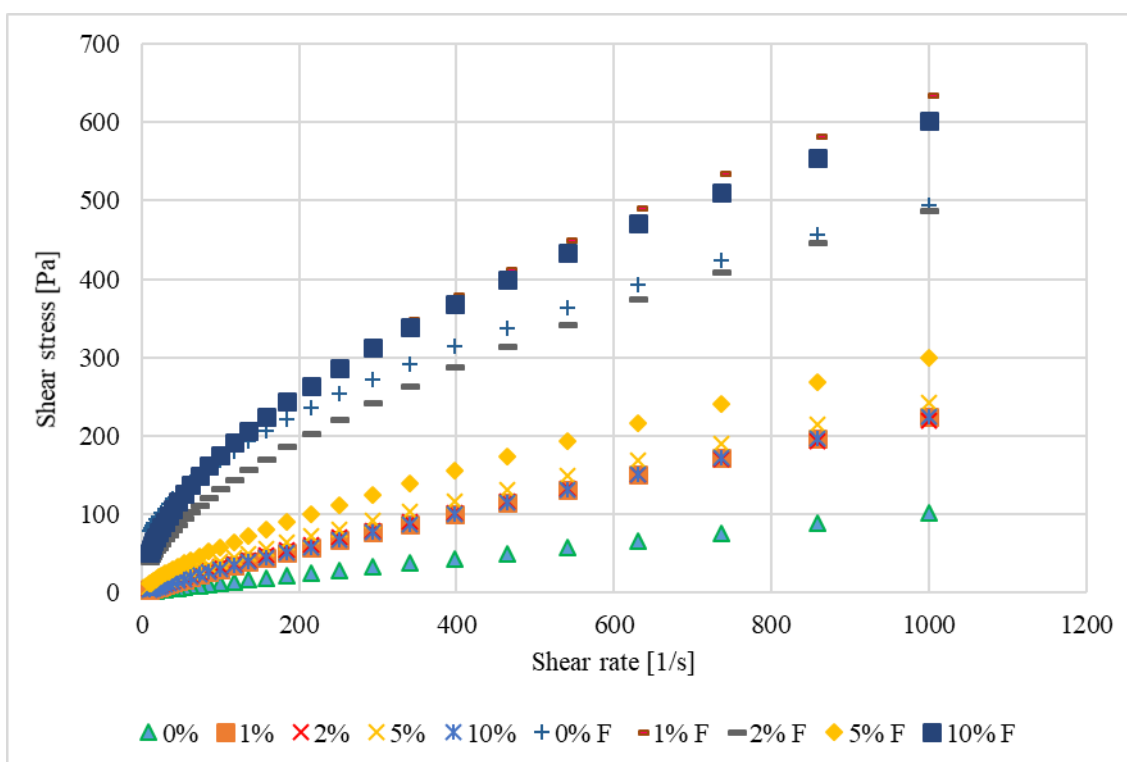


Fig 2. Flow curves of the liquid egg yolk samples (pH 6.5) with different table salt concentrations before and after freezing (F)

The other adjusted pH was 6.5, and the associated flow curves are shown in Figure 2. In this case, it can be observed that the flow curve of the salt-free sample had lower values than that of the pH 5.5 sample. From this, it can be concluded that lowering the pH increases the shear stress and the apparent viscosity, resulting in a denser product. The addition of salt in this case results in a higher increase in shear stress and apparent viscosity. Even after freezing, the flow curve of each sample developed differently, as the sample without salt had lower values than the pH of 5.5. In contrast, samples containing 1% and 10% table salt were found to be denser upon freezing and thawing. The effect of the 2% salt was also negligible, however, the flow curve of the sample containing 5% salt was very similar to that observed at pH 5.5. In this case, this concentration proved to be the most effective against the gelation caused by freezing, as well.

In addition to the analysis of the flow curve, we also examined the rheological behavior of the samples using the Herschel-Bulkley model. With this model we can get knowledge about the yield stress, flow behavior index and consistency coefficient of the liquied egg yolk samples. Figure 3 shows the evolution of the consistecy coefficient during the experiment. A large difference between fresh and frozen-thawed samples can be seen in all cases as a result of freezing. As can be seen, for each sample group (fresh or frozen thawed samples with the same pH), there was a significant difference between samples with different salt concentrations. For the fresh sample at pH 6.5 alone, there was no significant difference between the samples containing 0% and 1% salt. There was a significant difference between fresh and frozen-thawed samples with the same salt concentration at different pH in each case. It can also be seen that, based on the consistency coefficient, samples containing 5% salt are most similar to fresh samples without salt after freezing and thawing. Of the two, pH 6.5 proved to be better.

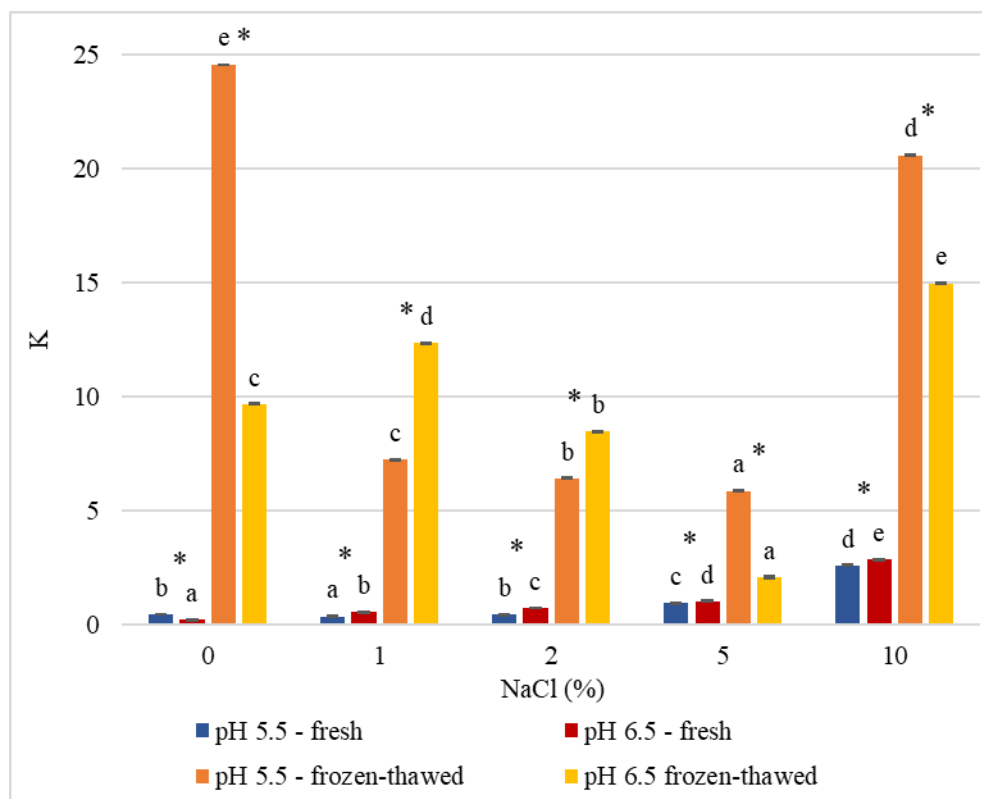


Fig 3. Consistency coefficients (K) of the pH adjusted (pH 5.5 and pH 6.5) egg yolk samples with different NaCl concentrations before and after freezing; Different letters show significant differences among samples with the same colour ($p < 0.05$); * shows significant difference between the fresh samples with different pH and frozen-thawed samples with different pH

If we observe the yield stress values shown in Table 1, it can be seen that this value did not appear in any of the fresh, non-frozen samples. However, after freezing, this value appeared in almost all samples. For samples with a pH of 5.5, it can be seen that the sample with 5% salt had the lowest yield stress value. Of the pH samples of 6.5, the yield stress value was also 0 for the sample containing 5% and 10% salt.

The parameter describing the nature of the flow is the flow behavior index. If its value is 1, we can speak of Newtonian fluid, if the value is below 1, the flow of the sample is pseudoplastic, and if it is above 1, it is dilatant. The examined samples were pseudoplastic in all cases, this was already established during the analysis of the flow curve. Overall, the flow behavior index of the samples before freezing is closer to 1 than after. Prior to freezing, increasing the salt concentration resulted in a flow behaviour that was increasingly different from Newtonian fluid. The effect of pH can also be observed, pH 6.5 resulted in higher n values. In the case of frozen samples, the sample containing 5% salt with a pH of 6.5 stands out, which had the highest n value.

Table 1. Rheological parameters (yield stress, τ_0 and flow behaviour index, n) of the pH adjusted (pH 5.5 and pH 6.5) egg yolk samples with different NaCl concentrations before and after freezing

Sample	pH	NaCl concentration [%C]	τ_0 [Pa]		n	
			Mean	S.D.	Mean	S.D.
fresh liquid egg yolk	5.5	0	0.00	0.00	0.844	0.000
		1	0.00	0.00	0.900	0.002
		2	0.00	0.00	0.858	0.001
		5	0.00	0.00	0.771	0.001
		10	0.00	0.00	0.698	0.001
	6.5	0	0.00	0.00	0.917	0.001
		1	0.00	0.00	0.876	0.002
		2	0.00	0.00	0.829	0.001
		5	0.00	0.00	0.793	0.001
		10	0.00	0.00	0.712	0.001
frozen-thawed liquid egg yolk	5.5	0	61.39	0.77	0.446	0.002
		1	21.72	0.41	0.578	0.001
		2	12.06	0.17	0.582	0.001
		5	4.04	0.07	0.570	0.001
		10	24.04	2.65	0.485	0.001
	6.5	0	42.43	0.80	0.552	0.001
		1	9.08	0.25	0.573	0.002
		2	6.55	0.11	0.573	0.001
		5	0.00	0.00	0.712	0.002
		10	0.00	0.00	0.535	0.002

CONCLUSION

In our experiment, we investigated the effect of pH adjustment with citric acid and table salt concentration on the gelation process undergoing when freezing of LEY. We found that the use of 5% table salt proved to be the most effective in reducing gelation. Among the different pH levels with the salt concentration of 5%, the higher pH (6.5) gave the most similar flow properties of the salt-free fresh sample. In this experiment, only an approximate salt concentration and pH were determined, however, further studies are required by using other pH values and salt concentrations (pH 6 and pH 7; 4.5 and 6% salt). In addition to rheological properties, it is necessary to study changes in colour, protein denaturation and in properties that affect usability, such as emulsion stability.

ACKNOWLEDGEMENT

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THE YIELD OF MAIZE HYBRIDS IN DIFFERENT CROP YEARS

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Summary: *In our experiment we examined four maize hybrids in two different years. In 2019 we had irrigated and non-irrigated plots, but in 2020, due to the rainy summer, we didn't apply irrigation. The hybrids were sown on four different plant density. The hybrids' yield increase under good water supply was an average 0.3 t ha^{-1} in 2019. The yield under non-irrigated conditions was more favourable in 2020, an average of $8.9\text{-}11.3 \text{ t ha}^{-1}$.*

Keywords: *maize yield, crop years, plant density, hybrids*

INTRODUCTION

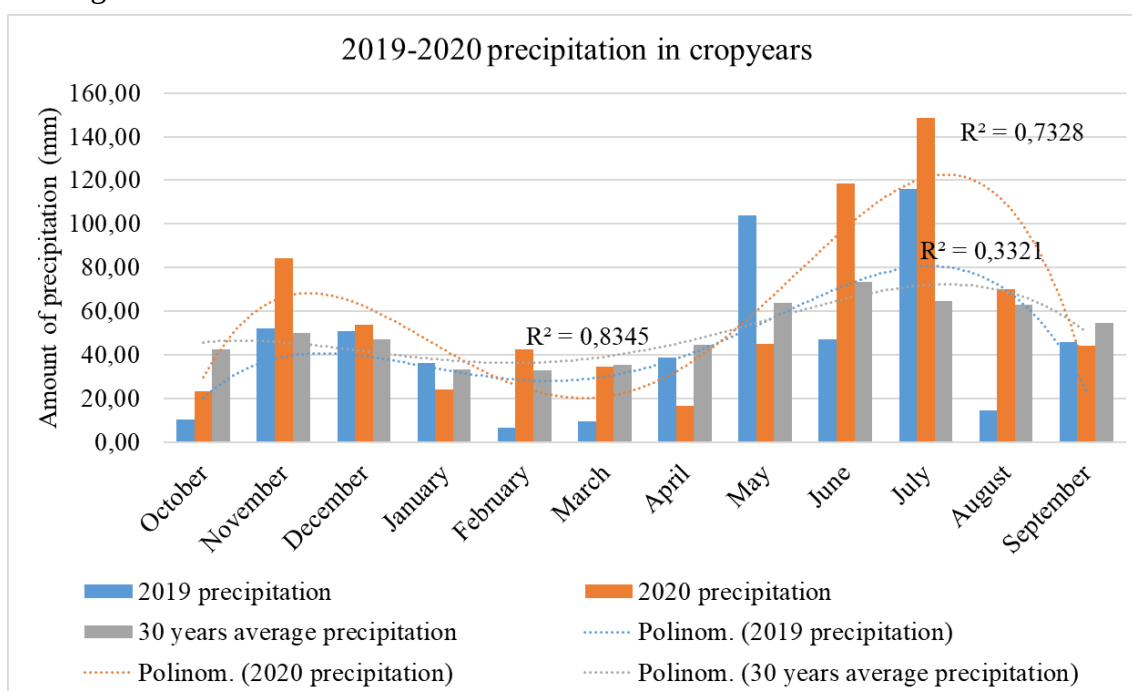
Pepó and Csajbók (2014) based on their examinations concluded, that the various agrotechnical elements had the following impact on maize yield: irrigation: 14%, plant density: 7%. Ruzsányi and Csajbók (2001) and Csajbók and Kutasy (2001) found that a good soil and modern technology significantly reduces the effect of the crop year. According to Csajbók et al (2005, 2014) and Virág et al. (2020) results, the correlation among nutrient and water utilization of maize and the temperature is more expressed in dry years. Examining the effect of irrigation on productivity and water use efficiency of maize, researchers found that under favourable conditions the maize use the water less efficiently (Tibin et al, 2021; Yufeng et al, 2021; Li et al, 2019). Fu et al. (2020) and Huailin et al (2020) experienced larger dry matter content under irrigation.

MATERIAL AND METHODS

In autumn before ploughing $30 \text{ kg ha}^{-1} \text{ N}$, $72 \text{ kg ha}^{-1} \text{ P}_2\text{O}_5$ $72 \text{ kg ha}^{-1} \text{ K}_2\text{O}$, and in spring, $135 \text{ kg ha}^{-1} \text{ N}$ were applied and incorporated before sowing in both years. Date of sowing was on 18.04.2019 and 16.04.2020 using a pneumatic seed drill. Each hybrid was sown in 4 replicates, 3-3 rows and 4 plant density ($65\text{-}75\text{-}85\text{-}95$ thousand plants ha^{-1}). Harvest was carried out on October 17. 2019 and October 8. 2020, with a Sampo parcel grain harvester. Results were analysed by Pearson correlation analysis.

RESULTS AND DISCUSSION

2019 (Fig 1) year's weather, which was partly favourable for maize and partly unfavourable in other respects, was also reflected in the development of maize and yields achieved. As the early spring was dry and warm, after that April was already well supplied with rainfall. There was abundant rainfall in April and June, but from July to November less rain fell than the many-year average. Weather conditions in 2020 varied from the unfavourable spring drought to the summer with abundant water supply. January precipitation was below, while February-March was close to the 30 years average. April-May was extremely dry for the spring-sown crops, as little or negligible rainfall fell during the critical germination period. In April, the precipitation was only one-third, and in May about 70 percent of the average. The June and July were extremely rainy. In August and September the water supply was around the average, but in October the ample rainfall hindered the harvesting.

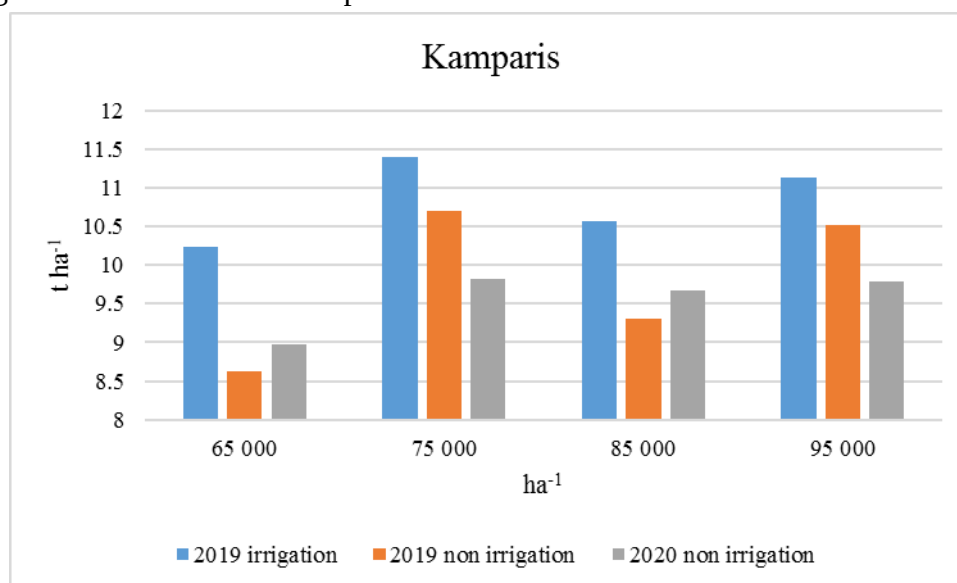


1. Fig. 2019-2020 precipitation in cropyears

Due to the dry soil condition, the sowing was delayed, nevertheless, the hybrids emerged ideally because the rain arrived soon after sowing, which facilitated initial development. Young plants evolved rapidly and dynamically on each of the hybrids. In terms of heat-demanding breed, the month of May was unfavourable to temperatures below the annual average, but higher rainfall helped the development. However, due to the drier period in early July, 25-25 mm of irrigation water was applied to half of the experimental area on July 1 and 15.

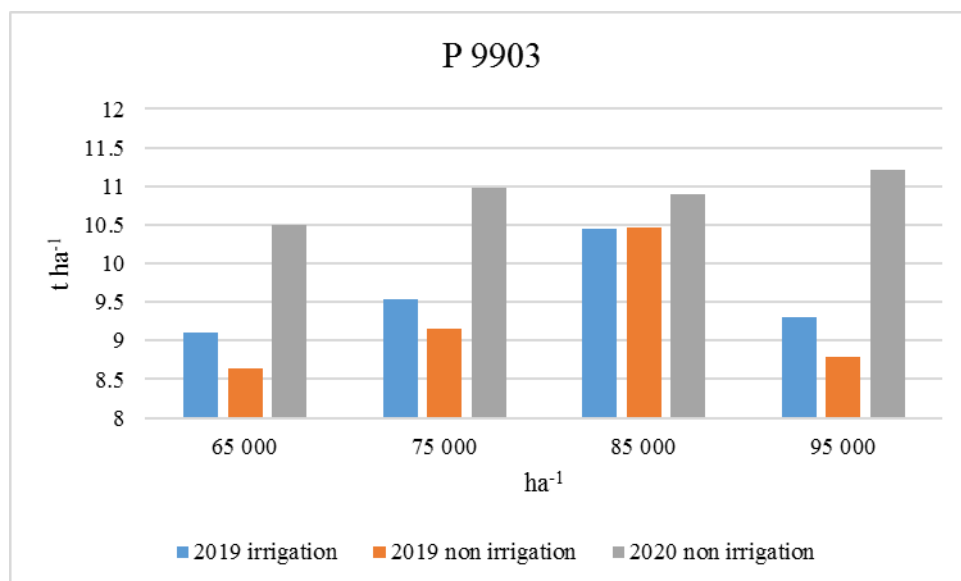
Despite the year full of adversity, the maize yield was excellent. Following the spring-drawn germination and slow development in the spring, due to the rains in June, the growth of the studied crop vegetation has become rapid, thus, optimal development stage has been

achieved. The weather was favourable at tasseling and pollen shedding, thus the number of developing kernels on each ear was optimal.



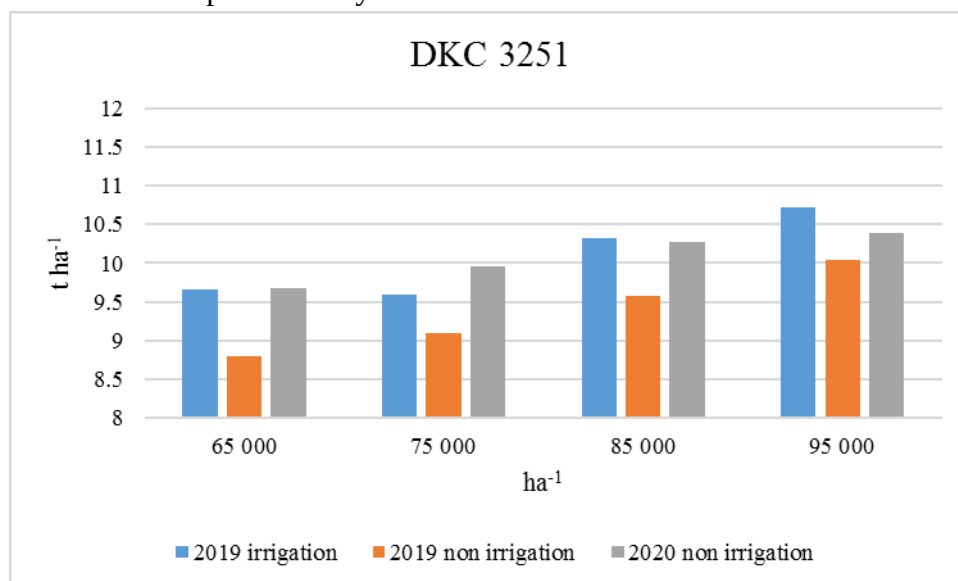
2. Fig.: Kamparis hybrid yields in different crop years (Debrecen 2019-2020)

Due to the rainy summer, irrigation near tasseling and kernel development could not be performed, as the soil was filled with water. Biological maturity of the ear was delayed, as the weather conditions determine the rate of dry-down. Harvesting depends on the moisture loss in order to reach an average grain moisture of 17.5 and 18.5% optimal for harvesting. We achieved an average of 8.9-11.3 t ha^{-1} yield. Compared to the previous year, all this resulted in a minimum yield surplus of 0.3 t ha^{-1} . Most of the examined hybrids achieved yields even exceeding the irrigated yield of the previous year (2019). However, Kamparis (Fig 2.) produced -0.7 -1.31 t ha^{-1} less on average compared to the previous year. Kamparis reached its maximum yield at 75,000 plant ha^{-1} plant density, but I experienced yield depression at 85,000 plant ha^{-1} plant density although at 95,000 plant ha^{-1} plant density its yield increased again in all treatments and cropyears.



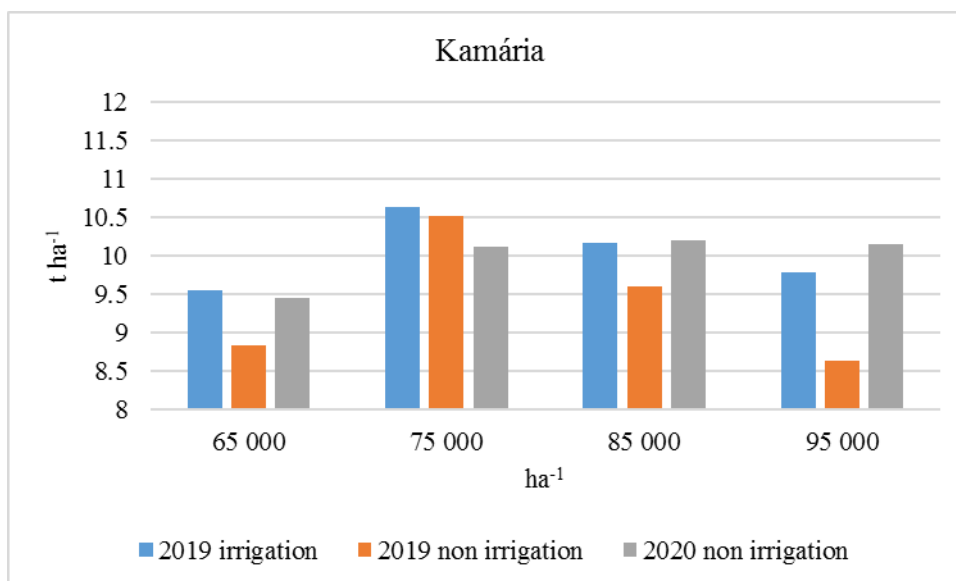
3. Fig.: P 9903 hybrid yields in different crop years (Debrecen 2019-2020)

The highest yield increase was achieved by P9903 (Fig 3.) with a yield surplus of 0.7-2.34 t ha⁻¹, and at the plant of 95 thousand with 11.3 t ha⁻¹, it also reached the highest yield among the hybrids by higher plant densities. DKC 4351 (Fig 4.) is a very stable hybrid as it only increased its yield by + 0.045-0.84 t ha⁻¹, but was able to stay stable above 9.5 t ha⁻¹. In addition to its stability, it was only able to gradually increase its yield in 2019 as well as in 2020 as a result of plant density increase from 9.6 to 10.47 t ha⁻¹.



4. Fig.: DKC 3251 hybrid yields in different crop years (Debrecen 2019-2020)

More favourable conditions in the precipitation contributed to higher yields in most cases in Kamária hybrid (Fig. 5.) compared to the previous year. It also tolerated better higher plant density, as in 2019 it reached its maximum yield at the average number of 85,000 seedlings per hectare in 2020 compared to 75,000 plant density ha⁻¹, however, above it yield depression was observed.



5. Fig.: Kamária hybrid yields in different crop years (Debrecen 2019-2020)

CONCLUSION

To increase the yield stability, it is important to keep the plant density instructions recommended by the breeding companies. For Kamária hybrid, taking into account the yield stability, the number of 75,000 plant ha⁻¹ plant density is optimal even in the droughty and rainy-rich summer period. On the other hand, the P9903 hybrid could achieve its greatest yield at 85,000 plant ha⁻¹ plant density. In case of DKC4351, linear correlation was between increasing of plant density and yield, however, in 2020 there is no significant difference between the yields of 85,000 and 95,000 plant ha⁻¹ plant density. The optimal number of Kamparis plants is around 75,000 plant ha⁻¹ plant density, as it could produce the best results on this plant density. Above it significant yield reduction was experienced, followed by a slight but not significant yield increase on the 95,000 plant ha⁻¹ plant density.

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EFFECT OF DIFFERENT TEMPERATURE AND TIME COMBINATIONS ON QUALITY ATTRIBUTES OF SOUS VIDE COOKED CHICKEN BREAST

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Summary: The combination of temperature and time in sous vide cooking could provide lower cooking loss, good water-holding capacity and tender cooked chicken breast meat. In this study, chicken breast treated with single step sous-vide (cooked at 45 °C and 60 °C) and double step sous-vide (first cooked at 45 °C and then at 60 °C) methods for 2h and 3h were studied. Our results showed lower cooking loss for the innovative double step SV treatments than single step SV treatment in both 2h and 3h treatments. Expressible moisture (%) was inversely related with cooking loss, except for one of the double step SV treatments (45 °C for 80 min and 60 °C for 40 min). All of the double step SV treatments had lower peak force than the single step sous vide treatments. The innovative double step sous vide treatment appears to be an attractive cooking method to produce tender chicken breast with lower cooking losses compared to the common single step SV treatment.

Key words: *Sous vide, double-step, chicken, texture*

INTRODUCTION

Low-temperature heating methods are becoming more and more widely used modern cooking techniques. They provide juicy meat because it improves the water holding capacity of the muscle. The benefits of these methods are low weight losses, improved juiciness and tenderness of LT cooked meat (Christien et.al., 2011b). Sous vide cooked meat products require little heating before consumption and the degree of freshness is high. This technology is already applied not only on a small scale but in the catering industry as well. The optimal time-temperature parameters are to a certain extent dependent on the raw material. According to the study of Becker et.al., (2016), sous vide (2h at 60 °C) method gave the lowest fiber transversal and longitudinal shrinkage values and cooking losses compared to other LT treatments. A greater effect of quality improvement of meat could be achieved employing the innovative sous-vide treatment (double-step treatment), such as improving water holding capacity, reducing cooking loss, and no changes on the color of meat (Ismail et.al., 2019). In case of chicken breast, 2h and 3h treatments at 60 °C have been observed to show better sensory quality attributes than 1h treatment (Park et.al., 2020). The aim of our study was to observe the effect of different temperature and time combinations on the quality attributes of single and innovative(double-step) sous vide cooked chicken breast.

MATERIAL AND METHODS

Fresh boneless, skinless chicken breast meat was obtained from a local commercial meat processing plant. Each raw chicken breast meat (95 ± 5 g) was vacuum packaged and cooked at different temperature and time combinations in water baths using single-step sous vide at 45°C and 60°C for 2 h and 3 h or double-step sous-vide at 45°C for the first temperature and 60°C for the second temperature for different time combinations. After the cooking process the chicken breast bags were quickly cooled in icy cold water for 1h. Samples were weighted before and after cooking to measure the cooking loss. Expressible moisture and texture features were also measured. All measurements were conducted in replicates.

RESULTS AND DISCUSSION

Our results showed lower cooking loss for the innovative two-step SV treatments than single SV treatment in both 2h and 3h treatments. Results showed also that cooking loss was lower as the time of the applied low temperature (45°C) was increased in the two-step SV treatments (Table 1). Similar results were also observed in 3h double-step SV treatments. The lowest cooking loss was observed in case of the 2h innovative double-step SV treatment when low temperature 45°C was applied for 80 min and the second temperature 60°C was applied for 40 min. In 3 h innovative double step-treatments, the lowest cooking loss was observed at 45°C for 2 hours + 60°C for 1 hour SV treatment. These results are similar with the observations of the Ismail et.al., (2019) on the cooking loss of sous vide cooked semitendinosus muscle from cattle and goat.

Table 1. Mean values of cooking loss (%) and expressible moisture (%) of 2h and 3h single and double-step sous vide cooked chicken breast

<i>Treatments</i>	<i>Cooking loss (%)</i>	<i>Expressible moisture (%)</i>
<i>60°C 2h</i>	14.91 $\pm$ 1.15	17.37 $\pm$ 1.03
<i>45°C 2h</i>	5.01 $\pm$ 0.21	24.81 $\pm$ 2.13
<i>45°C for 40 min and 60°C for 80 min</i>	10.53 $\pm$ 0.14	20.93 $\pm$ 2.13
<i>45°C for 60 min and 60°C for 60 min</i>	9.09 $\pm$ 0.14	24.27 $\pm$ 2.79
<i>45°C for 80 min and 60°C for 40 min</i>	8.43 $\pm$ 0.75	17.11 $\pm$ 0.94
<i>60°C 3h</i>	17.42 $\pm$ 0.99	12.91 $\pm$ 0.13
<i>45°C 3h</i>	5.25 $\pm$ 1.14	20.12 $\pm$ 0.45
<i>45°C for 1 h and 60°C for 2 h</i>	13.75 $\pm$ 1.48	14.51 $\pm$ 2.41
<i>45°C for 1.5 h and 60°C for 1.5 h</i>	13.15 $\pm$ 1.06	15.69 $\pm$ 0.77
<i>45°C for 2 h and 60°C for 1 h</i>	10.37 $\pm$ 0.43	17.35 $\pm$ 1.08

Expressible moisture (%) was inversely related with cooking loss, except for one of the double step SV treatments (45 °C for 80 min and 60 °C for 40 min). Higher expressible moisture can be observed in 2h double-step SV treatments that in single-step SV treatment (60 °C for 2h). The highest expressible moisture was observed in the double step SV treatment 45 °C for 60 min and 60 °C for 60 min. Higher expressible moisture can be also observed in 3h double-step SV treatments that in single-step SV treatments (60 °C for 3h). The highest expressible moisture was observed when chicken breast was cooked at 45 °C for 2 h and then at 60 °C for 1 h.

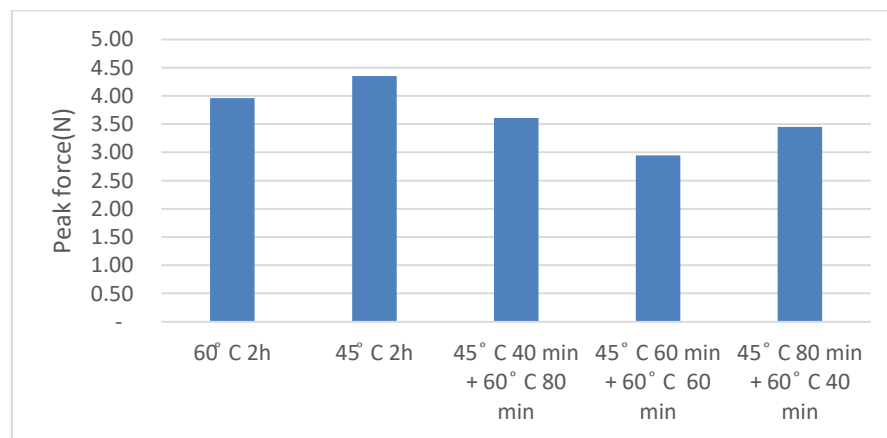


Figure 1. Mean values for peak force (N) of 2h single and double-step sous vide cooked chicken breast

All of the double step SV treatments had lower peak force than the single step sous vide treatments. These results the possibility of using the innovative sous vide cooking treatments for improving the tenderness of chicken breast meat. The lowest peak force (highest tenderness) was observed when chicken breast was cooked at 45 °C for 1h and that at 60 °C for 1h double-step SV treatment which was lower than the single step SV treatment (60 °C for 2h).

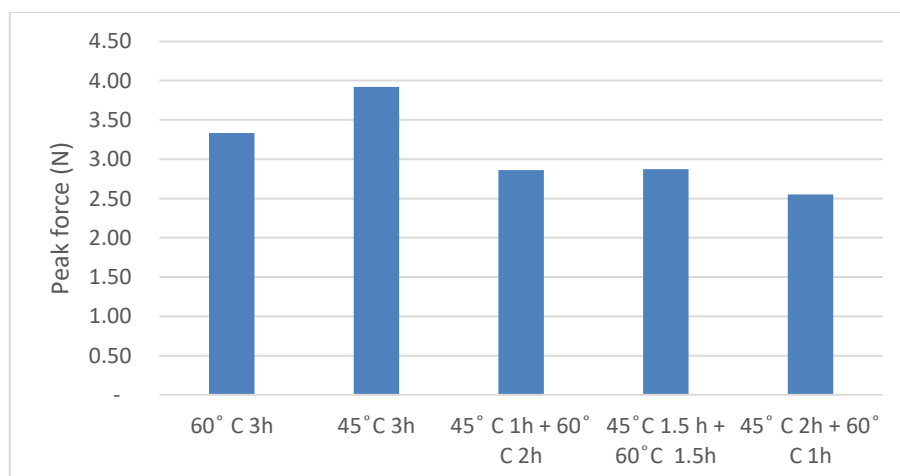


Figure 2. Mean values for peak force (N) of 3 h single and double-step sous vide cooked chicken breast

All of the 3h double step SV treatments had higher tenderness (lower peak force) than the single step sous vide treatment. The lowest shear force was observed at 45 °C for 2h and 60 °C for 1h double-step SV treatment which was lower than the single step SV treatment. No large differences were observed among 3h double-step-treatments. These results confirm the results from the study of Park et.al., (2020) that 2h and 3h treatments at 60 °C can show better sensory quality attributes of chicken breast.

CONCLUSION

The double-step sous vide treatments can provide a good alternative as it reduces the cooking losses while the meat preserve the water-holding capacity and improve the tenderness. Besides this our results showed also that cooking loss was lower and tenderness was higher as the time of the applied low temperature (45 °C) was increased in the two-step SV treatments. However, such observations were not seen in case of common single SV treatments. This innovative sous vide treatment appears to be an attractive cooking method, however further studies need to be done to find better temperature and time combinations to obtain a higher quality of cooked chicken breast.

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THE DIVERSITY OF YEASTS ASSOCIATED WITH GRAPES OF THE SMALL CARPATHIANS WINEGROWING REGION, SLOVAKIA

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Summary: The aim of this study was to isolate and identify microscopic fungi in different grape samples. We collected 13 grapes varieties samples (9 white and 4 red) from local Slovak winemakers in the end of the September 2020. Used 13 grape samples in this study: Alibernet, Irsai Oliver, Dornfelder, Blue Frankish, Feteasca regala, Green Veltliner, Pálava, Müller Thurgau, Rheinriesling, Cabernet Sauvignon, Pinot Blanc, Sauvignon Blanc, and Welschriesling. Yeasts in grape berries samples were detected on Malt extract agar by spread plate method. The number of yeasts ranged from 2.15 ± 0.09 in Pálava to $2.95 \pm 0.06 \log \text{cfu.g}^{-1}$ in Dornfelder. A total of 333 isolates of yeasts were obtained in this study. The most abundant yeasts belonged to species *Hanseniaspora uvarum* and *Metschnikowia pulcherrima* (100% frequency).

Key words: grape berries; mass spectrometry; Small Carpathians region

INTRODUCTION

The viticulture has a long tradition in Slovakia. There are six winegrowing regions in Slovakia, the Small Carpathians is one of them. Many factors affect the quality of the grape and also the wine. The microbiome of the grape can contribute to quality of the wine and is dependent on region, variety, and climatic factors of different winegrowing zones (Bokulich et al., 2013). Composition of the microorganism also plays an important role in wine production, as they metabolize the sugar from grapes and produce secondary metabolites that influence the aromatic quality (Fleet, 2003). The surface of grapes naturally contains a broad spectrum of the yeast species (Kántor et al., 2015). Previous research shown presence of genus *Candida*, *Rhodotorula*, *Cryptococcus* on different varieties of grapes from Small Carpathian region (Kačániová et al., 2019). Two types of grape, Blue Frankish and Green Veltliner from the same region, shown presence of genus *Hanseniaspora*, *Candida*, *Metschnikowia*, and *Saccharomyces* (Ženišová et al. 2014). Among the grapes from Small Carpathian region, the most abundant yeast is *Rhodotorula mucilaginosa* (Kačániová et al., 2018). It is necessary to detect the presence of non-pathogenic and pathogenic yeasts from

grape vines, which are able to affect the quality of wine production. The aim of the study was to isolate and identify the yeast microbiome from specific grape varieties grown in Small Carpathian region.

MATERIAL AND METHODS

Thirteen grape samples from vineyard in Vrbové located in the Small Carpathian wine region were used in this experiment. Ripe grape bunches were collected into sterile polyethylene bags and transported to the laboratory for microbiological analyses. The grape samples of following varieties were investigated: Alibernet, Irsai Oliver, Dornfelder, Blue Frankish, Feteasca regala, Green Veltliner, Pálava, Müller Thurgau, Rheinriesling, Cabernet Sauvignon, Pinot Blanc, Sauvignon Blanc, and Welschriesling. Each grape was one sample.

For detection of the yeasts in grape samples the Malt extract agar base (MEA) (BioMark™, India) was used and has been enriched with glucose (CentralChem®, Slovakia) (50 g/L), yeast extract (Conda, Spain) (3 g/L) and acid base indicator bromocresol green (Sigma-Aldrich®, USA) (0.020 g/L) (pH 3.8-5.4 yellow to blue). Yeasts were cultivated on Petri dishes at 25°C for 5 days in aerobic conditions.

We used MALDI-TOF Mass Spectrometer (Bruker Daltonics, Germany) for identification of yeasts isolated from federweisser samples. After incubation of yeasts at 25°C for 5 days, isolated colonies were picked and suspended in 300 µL of sterile distilled water and mixed thoroughly. 900 µL of absolute ethanol was added. The mixture was centrifuged at $13\,000 \times g$ for 2 min. After the supernatant was discarded, the pellet was centrifuged again. Residual ethanol was completely removed by pipetting and the pellet was allowed to dry at room temperature. Subsequently 10 µL of formic acid (70%) was added and mixed with the pellet with a sterile toothpick. Next, 10 µL of acetonitrile (100%) was added and mixed thoroughly. The solution was centrifuged at maximum speed for 2 minutes again, and 1 µL of the supernatant was placed on a polished MALDI target plate (Bruker Daltonics, Germany). Immediately after drying 1 µL of the matrix solution was added to each spot and allowed to air dry. The saturated solution of α -cyano-4-hydroxycinnamic acid (HCCA) (Bruker Daltonics, Germany) was used as a matrix. HCCA was dissolved in 50% acetonitrile with 0.025% trifluoroacetic acid (TFA) prepared with ultra-pure water. The samples on microplate were covered with 1 µL of matrix solution and were left to dry. Samples were then processed by the MALDI-TOF MS (Microflex LT/SH, Bruker Daltonics, Germany) with flex Control software. The results were obtained by Real-time Classification software (RTC) with use of “Taxonomy” (Bruker Daltonics, Germany) database.

RESULTS AND DISCUSSION

Yeasts are found throughout nature. However, they do not occur randomly, but are found in specific habitats where different species form communities (Lachance and Starmer, 1998). Within the winemaking environment (habitat), the vineyard (grape surfaces) and cellar (equipment surfaces and must) can be considered specialized niches where the wine related yeasts can form communities (Polsinelli et al., 1996). The yeast species found in different niches associated with grape growth (vineyards) and wine production (wineries, grape must, fermentation and wine) can be arbitrarily divided into two groups, i.e. the *Saccharomyces* group and the non-*Saccharomyces* group. The *Saccharomyces* group with its primary representative, *Saccharomyces cerevisiae*, is present on grape skins in low numbers (Rankine, 1972; Török et al., 1996, König et al., 2009), but they are found in greater numbers on winery equipment and in fermenting must (Fugelsang et al., 2007, Kántor et al., 2016). Non-*Saccharomyces* yeast are part of the natural microbiota present on grapes, and harvesting and winemaking equipment, and are present at least during the early stages of fermentation (Fleet and Heard, 1993; Renouf et al., 2005, 2007). While generally incapable of completing alcoholic fermentation, their application in co-inoculation or sequential inoculation with *S. cerevisiae* is increasingly popular (Ciani et al., 2006; Ciani and Maccarelli, 1998; Comitini et al., 2011; Jolly et al., 2006; Soden et al., 2000), particularly for their effects on wine composition, flavour and aroma (Benito et al., 2011; Ciani et al., 2006; Comitini et al., 2011; Cordero Otero et al., 2003; Di Maio et al., 2012; Domizio et al., 2011; Garcia et al., 2002; Jolly et al., 2006, 2014; Magyar and Toth, 2011; Morata et al., 2012; Soden et al., 2000; Toro and Vazquez, 2002).

In our study, the yeast counts isolated from different wine grape berries ranged from $2.15 \pm 0.09 \log \text{cfu.g}^{-1}$ in Pálava to $2.95 \pm 0.06 \log \text{cfu.g}^{-1}$ in Dornfelder (Table 1).

Table 1 Number of yeasts in grape varieties ($\log \text{cfu.g}^{-1}$)

Grape variety	average	SD %
Alibernet	2.36	0.08
Blue Frankish	2.54	0.07
Cabernet		
Sauvignon	2.25	0.05
Dornfelder	2.95	0.06
Feteasca regala	2.65	0.05
Green Veltliner	2.36	0.05
Irsai Oliver	2.16	0.07
Müller Thurgau	2.25	0.06
Pálava	2.15	0.05
Pinot Blanc	2.91	0.03
Sauvignon Blanc	2.43	0.06
Rheinriesling	2.52	0.07
Welschriesling	2.46	0.06

In our samples, different yeast species were found (Figure 1). The most abundant yeasts were: *Hanseniaspora uvarum* and *Metschnikowia pulcherrima* and were in all 13 (100 %) wine grape berries samples (Table 2). Altogether nine yeast species were isolated: *Candida magnoliae*, *Hanseniaspora uvarum*, *Issatchenkia orientalis*, *Issatchenkia terricola*, *Kazachstania exigua*, *Kluyveromyces marxianus*, *Metschnikowia pulcherrima*, *Rhodotorula glutinis* and *Yarrowia lipolytica*. The microorganisms can contaminate from different environmental sources. The origin of microorganisms could be the vineyard, can be residents of the winery flora, or can be transmitted with insects such as fruit flies and, bees (Fleet et al., 2002). Over twenty yeast species have been identified in wines (Renouf et al., 2007). *Hanseniaspora uvarum* (anamorph: *Kloeckera apiculata*), *Metschnikowia pulcherrima* (anamorph: *Candida pulcherrima*), and *Candida stellata* are thought to be the principal yeasts of grapes. In some studies, *Hanseniaspora* was reported to be the dominant species (Beltran et al., 2002; Combina et al., 2005; Hierro et al., 2006), while *Candida* assumed to be widespread as well (Clemente-Jimenez et al., 2004). The majority of *Candida stellata* isolates from wine is actually *Candida zemplinina* (Csoma and Sipiczki, 2008).

Table 2. Percentage of yeast in grape samples

Yeast species	Occurrence on	
	samples	Percentage
<i>Candida magnoliae</i>	8/13	61.54
<i>Hanseniaspora uvarum</i>	13/13	100.00
<i>Issatchenkia orientalis</i>	12/13	92.31
<i>Issatchenkia terricola</i>	10/13	76.92
<i>Kazachstania exigua</i>	7/13	53.85
<i>Kluyveromyces marxianus</i>	8/13	61.54
<i>Metschnikowia pulcherrima</i>	13/13	100.00
<i>Rhodotorula glutinis</i>	12/13	92.31
<i>Yarrowia lipolytica</i>	10/13	76.92

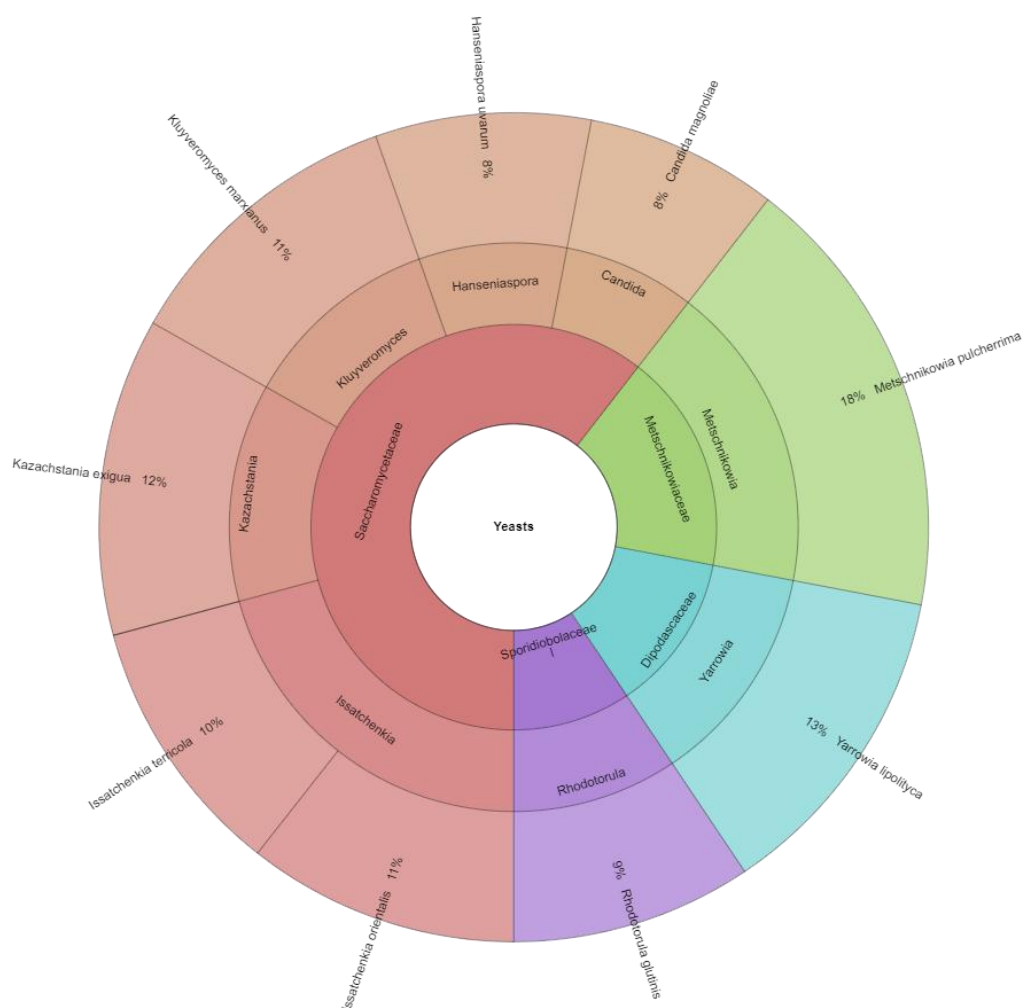


Figure 1. Diversity of yeast isolated and identified from grape berries samples

CONCLUSION

In this study we isolated and identified yeast species from 13 grape berries samples. We identified the yeast isolates by MALDI-TOF mass spectrometry Biotyper (Bruker Daltonics, Germany). The most dominant species were *Hanseniaspora uvarum* and *Metschnikowia pulcherrima* which were isolated from all 13 grape samples. It can be considered a very good result. Overall, we identified 9 different yeast strains that belong to 8 genera and 4 families.

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USE OF GENETIC ALGORITHMS IN INTEGRATING MULTIPLE IRRIGATION SPRINKLERS

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Summary: *Irrigation has a long history that has been practiced since the time of Mesopotamia. Use of sprinklers for irrigation was mentioned at the end of the 19th century and over time it increasingly applied in the fields of agriculture, industry, park irrigation and garden use. If the irrigation of a certain area is done by a combination of several irrigation systems, choosing the best combination is a best option. In this scientific research paper, we presented a solution, it is already known as genetic algorithms (GAs), a known method of heuristic optimization. For example: The goal is to provide irrigation for 20,000 m² of park with the lowest equipment costs using a combination of 4 types of sprinklers of different availability (number of pieces), different prices and different watering areas. In mathematical terms, the problem has two main functions (minimize costs and maximize area of irrigation, but with a limit, the area does not exceed 20,500m²) and four variables: x1 - number of sprinklers S100 in the range 0 to 23, x2 - number of sprinklers S50 in the range 0 to 53, x3 - number of sprinklers S075 in the range 0 to 125 and x4 - number of sprinklers S020 in the range 0 to 471. Mathematical problem modeling and GAs implementation was performed via the GANetXL application. Optimal solutions to the problem and goals set the amount of 107,110 dinars and 20,499 m².*

Keywords: *genetic algorithm; GANetXL; single objective; multiple objective; sprinklers; irrigation*

INTRODUCTION

Irrigation refers to the artificial application of water to assist in the growing of crops, trees and pastures. This can be done by letting water flow over the land (surface irrigation), by spraying water under pressure over the land concerned (sprinkler irrigation), or by bringing it directly to the plant (localized irrigation).

In 2012 over 324 million ha worldwide are equipped for irrigation, of which about 85 percent or 275 million ha are actually irrigated (FAO, 2014). Irrigated agriculture represents 20 percent of the total cultivated land, but contributes 40 percent of the total food produced worldwide.

In Europe the part of area equipped for irrigation that is actually irrigated represents 65 percent, which is low compared to the rest of the world. This is due to a moderate climate, in a large part, which allows agriculture to benefit from the available precipitation and thus not always needing to be irrigated. Localized and sprinkler irrigation account for about 14 percent of the total area equipped for irrigation worldwide. Localized irrigation has grown

rapidly since the invention of cheap plastic pipes in the 1970s: from almost 0.5 million ha in 1981 to almost 9 million ha in 2010 worldwide. Sprinkler irrigation equips more than 35 million ha in 2010. Although considered less efficient than localized irrigation, both its cheaper prices and potential mobility explain its wider expansion.

Irrigation in Serbia began at the end of the 19th century, the first irrigation systems covered about 43 ha, and construction on larger plots began after World War II. Irrigation in Serbia does not have much expansion as determined by statistical analysis conducted in 2019. That irrigation area is only about 46,863 ha of the total number of arable land which is 4,867,000 ha. Maximum number of coverage of irrigation system is shown as arable land and gardens with 94.9%, followed by orchards with 4.5%, and agricultural land with 0.6%.

The key to our task is based on “pop up” irrigation systems, which aims to present an artificially intelligent system in the field of agriculture, which brings the most optimal solution by selecting the best combinations of set parameters.

In scientific work „GANETXL: A DSS GENERATOR FOR MULTIOBJECTIVE OPTIMISATION OF SPREADSHEET-BASED MODELS” written by professor D. Savić, J. Bicik, M.S. Morley, explained that Nowadays computer-based modeling is used routinely as part of decision-support processes in many areas of business and engineering. Decision Support Systems (DSS), and model-driven DSS in particular, that could be defined simply as “computer-based models together with their interactive interfaces”, aim to support the user in addressing and solving particular unstructured problems in a timely manner. Most notably, DSS that integrate Evolutionary Multiobjective Optimizations (EMO) with simulation technologies have gained in popularity in the water resources literature over the last few decades.

Most recent work on GAs for water engineering has been on aspects of water distribution network optimization, including new design, rehabilitation, modeling and operation. Although design of a complete new network is in practice rare, extension and reinforcement of existing systems occur frequently, particularly for rapidly growing conurbations in developing countries, where shortages of water and inadequate pressures are commonplace. In long established systems, poor water quality (often associated with old iron pipes) and excessive leakage necessitate substantial investment in pipe replacement, often involving some quite complex decision making to obtain the best overall strategy for improvement.

The key advantage of EMO is that, unlike classical mathematical programming methods where objective functions may be optimized separately, these population-based methodologies have the ability to search effectively for many non-dominated (trade-off) solutions at the same time (in a single run), which makes them a very attractive tool for solving multiobjective optimization problems.

MATERIAL AND METHODS

An example that was processed in our research was done in Microsoft Excel 2010, using a tool that is used in addition to the mentioned program. The tool, which uses the Genetic Algorithm method to select the best solution and which we used in our example, is called GanetXL.

GANetXL- Savić, D. A., Bicik, J., and Morley, M. S. 2011 DSS generator for multifunctional optimization of spreadsheet-based models. *Environmental Modeling and Software*, 26 (5), 551-561. GANetXL is an add-in for Microsoft Excel for solving optimization problems based on a genetic algorithm. Using genetic algorithms, it serves to solve complex optimization and search problems. GANetXL is not a commercial program; its idea by the manufacturer is to serve for academic research. The features of this tool in its purpose are: optimization techniques for one or more goals, support for integration with simulation packages, then suspension, resumption and optimization and interactive browsing, browser with multiple goals, user-defined limits and penalty products, automatic population saving, serial duration, medium population backups, graphical user interface, result visualization, and progress.

Genetic algorithm

The genetic algorithm was proposed by Dr. John Henry Holland laid the foundations of this method in the early 1970s in his book *Adaptation in Natural and Artificial Systems* [HII75]. The genetic algorithm is a heuristic search method that mimics natural problem-solving processes. The principle of evolution "Darwin's theory" is applied in the search for the most optimal solution. The algorithm can solve simple and more complex problems. For natural processes, as well as for genetic ones, it can be established that: there is a population of individuals, some individuals are better adapted to the environment, better individuals have a higher probability of survival and reproduction, traits of individuals are recorded by genetic code genetic algorithm are the current solutions to the problem being solved. The total number of individuals makes up one generation called the population. When starting to solve the problem, the number of individuals in the population must be determined. In practice, from 5 to 500 individuals are usually used. The population represents the solutions from which the best ones are chosen. The genetic algorithm consists of the following processes: selection, crossing, and mutation. These processes help us to adapt as much as possible in a given environment, more precisely, these processes help us to reach the most optimal solution. The problem solving algorithm is presented in Figure 1.

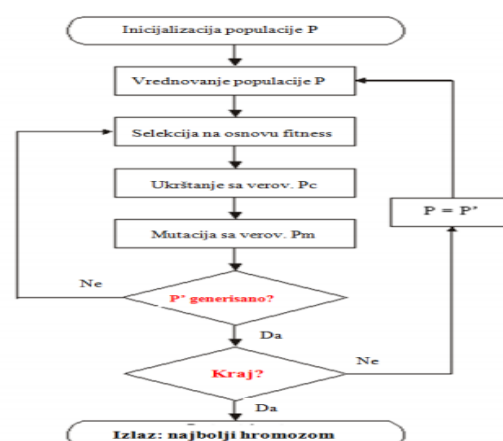


Figure 1: Basic version of the genetic algorithm Source: Z. Srđević, B. Srđević "Water Management System Analysis, p. 128.

When applying selection, bad individuals die out and better ones stay, and cross in the next step. Therefore, the worse individuals should be given a chance to have a small chance of survival, and the better individuals a greater chance. The genetic algorithm for parent selection uses: roulette wheel parent selection, stochastic selection, tournament selection. The tournament selection can be defined at random, several individuals can be selected and a tournament is organized, the winner of which is the individual with the best function of convenience. The winning unit will participate in obtaining the next generation. Which is explained in Figure 2. Tournament selection has the greatest use and cannot be applied in a simple genetic algorithm.

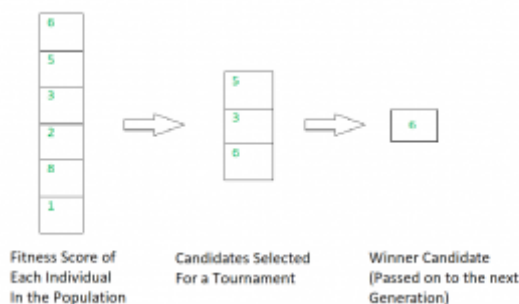


Figure 2. Illustration - tournament selection. (Source - <https://stamd.medium.com/šta-su-genetski-algoritmi-8af292758264>).

Crossover is basically the task of a genetic algorithm that is used to create new individuals, where there is an exchange of genetic material between two individuals. By crossing two individuals called "parents", one or two new individuals called "children" are created. The most important characteristic of crossbreeding is that children inherit the characteristics of their parents. This means that if parents have good genes, there is a high probability that children will inherit good genes, if not even better. There are three types of crossing: one-position crossing, two-position crossing, and uniform crossing. One-position crossing is the simplest way of crossing individuals. An integer is chosen at random from the interval $(0, l - 1)$, where l is the length of the parent. The gene i is uniformly selected. The first offspring inherits the genes of the first parent to the i -th and the genes of the second parent from the i -th. The second offspring inherits the genes of the second parent to the i -th and the first parent from the i -th.

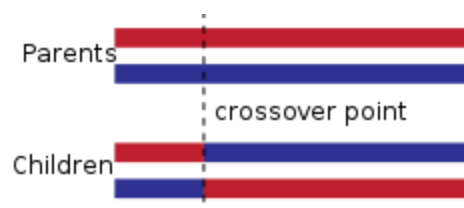


Figure 3. Illustration - single-position crossover. (Source - <https://stamd.medium.com/šta-su-genetski-algoritmi-8af292758264>).

Mutation is the most important task of a genetic algorithm whose action leads to a random change of selected genes. Mutation enables the return of useful genetic material that is lost in the selection and crossbreeding. This is an excellent tool for avoiding extremes. The mutation parameter is denoted by P_m . The exact number of genes to be changed has not been determined, explained in Figure 4. In genetic algorithms, mutations are applied with low probability.

Before Mutation						
A5	1	1	1	0	0	0
After Mutation						
A5	1	1	0	1	1	0

Figure 4. Illustration of mutation of randomly altered genes (Source - <https://towardsdatascience.com/introduction-to-genetic-algorithms-including-example-code>).

In order to define the stopping condition, it is necessary to first generate the initial population and processes that are performed on it, which consist of selection, crossing and mutation. When the stop condition is met, the solution to the problem does not have to be the optimal solution. Some of the stopping conditions are: maximum number of generations reached, similarity of individuals in the population, best individual repeated a certain number of times, predetermined optimal solution reached, proven optimality of the best individual (if possible), limited execution time of genetic algorithm or user interruption.

Configuration our example in GanetXL

Our task falls into difficult combinatorial problems because there are $23 * 53 * 125 * 471$ possible combinations and standard search methods cannot find the result, or at least not in an acceptable time. In the example discussed in this paper, a choice is made between several types of sprinklers, each of which represents one individual with different characteristics. Each individual is considered for the next generation, which is joined by certain quality measures - fitness, which is determined by the goal function. Based on these characteristics, GanetXL selects the most suitable number of sprayers in order to reach the optimal solution. We took information about the used sprayers from the sales catalog of the company "Savacoop" d.o.o. The manufacturer of irrigation sprayers is an Italian company called RAIN, and the sprayers themselves are called RAIN S020, RAIN S050, RAIN S075, RAIN S100. Each of these sprayers has an angle adjustment range of 40° to 360° . RAIN S020 spray is a sprayer that has a jet range of 2.7m to 5.2m, and in our example the maximum value or 5.2m was used. RAIN S050 is a type of gear sprayer that rotates and covers an area of 5.2 to 10.1m, and the adopted value is also a maximum of 10.1m. RAIN S075 is the first Rain s.p.a gear sprayer designed for basic home use and basic commercial use. It rotates counterclockwise and covers an area of 6.7m to 15.5m, while in the example we did the calculation with a jet range of 15.5m. The last in a series of used sprayers is the RAIN S100

sprayer, which has a jet range from 13.71m to 23.5m and the adopted value is also the largest - 23.5m. We received the prices of the mentioned sprayers directly from the employees of the company “Savacoop” d.o.o. Sales prices for S020, S050, S075 and S100 are 290 dinars, 1246 dinars, 1594 dinars or 7500 dinars in the order in which the sprayers are listed. The target function, in our example, is defined as the choice of the cheapest combination of sprinklers in order to irrigate as large an area as possible. In the task we processed, there are 4 individuals, more precisely 4 sprinklers with a population size of 50. For the type of selection we chose, tournament selection cannot be applied in a simple genetic algorithm, and the type of problem we solve is multiple objective. We decided for a single-position crossing because it has the greatest use in practice, and for the mutation in our case, a mutation rate of 0.4 was used.



Figure 5. Sprayer S075, Source - <https://savacoop.rs/pop-up-rasprskivac-rotor-s075-3-4>

RESULTS AND DISCUSION

Table 1. Shows the task setting as well as its solution in Microsoft Excel 2010. In fields B4: B7, the sprayer designations S100, S75, S50, and S20 (taken from the catalog) are entered. Field C3 represented as “Quantity” is the number of sprayers used. Column C is of great importance because it represents the variable we use in creating our task and mathematically we will present it below as x1-number S100 (field C4), x2 - number S75 (field C5), x3 - number S50 (field C6), and x4 - number S20 (field C7), shown in Table 1.

Table 1 Task with solutions presented in a table in Microsoft Excel.

Sprinklers	Quantitet	Cost	Sub- Total	Irrigated area of one sprinkler (m2)	Total Irrigated area (m2)
		7,500.00			
S100	0	дин.	0.00 дин.	867.00	0.00
		1,594.00	55,790.00		
S075	35	дин.	дин.	377.20	13202.00
		1,246.00	12,460.00		
S050	10	дин.	дин.	160.20	1602.00
			38,860.00		
S020	134	290.00 дин.	дин.	42.50	5695.00
		Total spent	107,110.00 дин.	Total Irrigated area	20499.00
			Optimize	107,110.00 дин.	20499.00
				Infeasibility	0

Fields D4: D7 are prices. One of the two objectives of the task is to minimize costs and this was done in column E. The calculation is presented as:

Min: $x_1 * 7500 + x_2 * 1594 + x_3 * 1246 + x_4 * 290$ and the solution of this column as seen in the picture is

Solution: $0 * 7500 + 35 * 1594 + 10 * 1246 + 134 * 290 = 107.110\text{din}$, in field E9.

Since each sprayer can water the surface of the whole circle, i.e. 360° , but in practice, the sprayers do not have that need due to the positions they are in the field, the average watering area of one sprayer is calculated as half the surface of the circle. In Microsoft Excel, it is represented as “Sprinkler surface of one sprinkler”, in column F and is obtained by the formula $P = \pi / 2 r^2$, where r represents the jet range. Spray jet range is: S100-23.5m, S75-15.5m, S50-10.1m, and S20-5.2m. The watering area covered by the sprayers used is:

PS020 = $5.2\text{m} * 3.14 / 2 = 42.5\text{m}^2$; PS050 = $10.1\text{m} * 3.14 / 2 = 160.2\text{m}^2$;

PS075 = $15.5\text{m} * 3.14 / 2 = 377.2\text{m}^2$; PS100 = $23.5 * 3.14 / 2 = 867\text{m}^2$

The second goal of the task is to maximize the flooding area calculated in column G.

The mathematical expression for this goal is:

Max: $x_1 * 867 + x_2 * 377 + x_3 * 160 + x_4 * 42$,

Solution: $0 * 867 + 35 * 377 + 10 * 160 + 134 * 42 = 20.499\text{m}^2$, but provided that it does not exceed $20,500\text{m}^2$, which is set as Infeasibility: = IF (G8> 20500, (G8-20500) * 100.0), located in field G13.

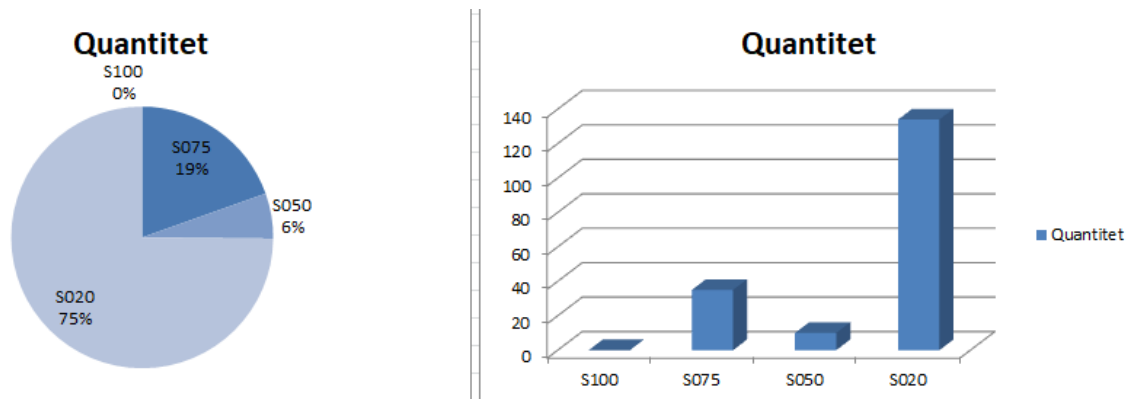


Figure 6. Represents the number of selected sprinklers on the histogram and structural circle.

The reason why the maximum watering area is 20,500 m², 500 m² higher than required in the task is that the watering area of one sprayer overlaps with the watering area of the neighboring sprayer, so we arbitrarily took a percentage of 2.5% of space to manipulate this value. The optimal solutions to the task and the set goals are given in fields F11 and G11 and they amount to 107,110 dinars and 20,499 m², respectively.

We came to this solution by trying several different problem configurations in GanetXL. The method we presented gave the most optimal solutions, as much as 15% better than the others offered, which means that by using this way of solving the task, we would save the amount of 19,644 dinars.

CONCLUSION

Working in GANetXL, we learned about the direct application of genetic algorithms and thus we had the opportunity to fully understand its sustainability in the field of agriculture and water management. Programs such as GANetXL are of great importance in making crucial decisions, by choosing the appropriate combinations that provide us with the most adequate solutions. Scientific papers on the implementation of the genetic algorithm through various applications and programs can lead to the improvement of the characteristics of the algorithm and thus lead us to new knowledge, which can greatly contribute to the scientific community.

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MEMBRANE EMULSIFICATION OF OLIVE OIL: SELECTION OF THE WALL MATERIAL

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Summary: *Besides being a key ingredient of the Mediterranean diet and cuisine, olive oil is related with many health benefits, including the prevention of many modern life-style diseases, like some kinds of cancer and cardiovascular diseases. Nevertheless, it is chemically unstable and susceptible to degradation. Microencapsulation has been adopted as an effective and important tool to protect the olive oil against oxidation during storage. Emulsification technology is a key step in the microencapsulation of oils. It is generally applied for the encapsulation of bioactive compounds in aqueous solutions, which can either be used directly in the liquid state or can be dried to form powders after emulsification. In this way, the oils' off-flavor can be masked and safety, quality, and the nutritional value of the product can also be improved. This study aimed at evaluating the potential of maltodextrin combination with different wall materials in emulsification technology process, in order to achieve better stability and better droplet size distribution. In first step, "membrane emulsification" technology was adopted to encapsulate olive oil using different combinations of gum Arabic (GA), maltodextrin (MD) with whey protein isolate (WPI) and carboxymethylcellulose (CMC). In second step, characterization of emulsions from different aspects, morphology, particle sizes, and stability had been discussed.*

Key words: *Membrane emulsification, olive oil, bioactive ingredient, wall materials*

INTRODUCTION

Health benefits associated with olive oil consumption have been extensively demonstrated, especially those related to cardiovascular diseases prevention and some kinds of cancer (Sousa et al., 2015). The nutritional benefits of virgin olive oil (VOO) are primarily related to the fatty acid composition, mainly due to the high content of oleic acid and also to the balanced ratio of saturated and polyunsaturated fatty acids that are known to show protective effects against cardiovascular diseases by preventing oxidation of low density lipoproteins. Besides, the content of phenolic compounds is an important factor to be considered in evaluating VOO quality. In fact, phenols are powerful antioxidants and a significant contributor to its stability against oxidants, and hence to extending its shelf-life relative to other vegetable oils (Veillet et al., 2010).

Nevertheless, because of its high instauration degree, olive oil is subject to oxidation during processing, distribution and handling. Oxidation leads to the formation of unpleasant tastes and odors and, consequently, to the reduction of oil's shelf life, besides promoting the generation of free radical, which may have negative physiological effects on the organism

(Aparicio and Harwood 2013). Thus, one of the solutions to preserve these bioactives is “the microencapsulation technology”: Microencapsulation is a method in which tiny particles or droplets are surrounded by a coating wall, or are embedded in a homogeneous or heterogeneous matrix, to form small capsules. The key step to release capsules is the “emulsification technology”. It is generally applied for the encapsulation of bioactives in aqueous solutions. It involves using a low pressure to force the dispersed phase to permeate through the pores of a microporous membrane into the continuous phase (Charcosset, 2009; Piacentini et al., 2016; Ramakrishnan et al., 2010).

Prior to this step, it is very important to make the choice of a suitable coating material, basically a film-forming biopolymer, in order to achieve the best microencapsulation efficiency (Jafari et al., 2008). Concerning olive oil microencapsulation, very little information is available in the literature. Calvo et al., 2010, used different wall components such as proteins (sodium caseinate and gelatin), hydrocolloids (gum arabic), and hydrolyzed starches (starch, lactose, and maltodextrin). The highest microencapsulation yield was observed when gelatine, maltodextrin and gum Arabic were used as wall materials. Sun-Waterhouse et al., 2011, also investigated the influence of microencapsulation and addition of a phenolic compound (caffeic acid) on the storage stability of olive oils encapsulated in alginate shell at a level of 50 mg oil/g alginate solution. More recently it has been encapsulated using the combination of maltodextrin, carboxymethylcellulose and lecithin as encapsulating agents which gave the highest microencapsulation yields (Calvo et al., 2012). Also, Devi et al., 2012, showed that olive oil could be encapsulated efficiently using gelatin A and sodium alginate complex, using as crosslinker glutaraldehyde. Koç et al., 2015, targeted the optimization of process variables to obtain extra virgin olive oil powder to have maximum microencapsulation efficiency and minimum particle size. Thus, wall material composition using a mixture of maltodextrin and whey protein isolate, homogenization rate, and feeding rate for spray dryer were optimized to find that high microencapsulation efficiency of extra virgin olive oil is achieved when the wall materials composition is 92 % maltodextrin, 7 % whey protein isolate, 1 % Tween 20, and 17,500 rpm of rotor-stator homogenization rate and 22 % of pump rate, respectively.

This study aimed at evaluating the potential of maltodextrin combination with different wall materials in emulsification technology process, in order to achieve better stability and better droplet size distribution. In first step, “membrane emulsification” technology was adopted to encapsulate olive oil using different combinations of gum Arabic (GA), maltodextrin (MD) with whey protein isolate (WPI) and carboxymethylcellulose (CMC). Three formulations were prepared: First, MD-GA contained a binary mixture of 83% maltodextrin and 17 % gum arabic. Second, MD/WPI-GA was a ternary mixture of 59% maltodextrin, 12 % gum arabic and 29% whey protein isolate. Third, MD/CMC-GA contained 53% maltodextrin, 38% gum arabic and 9% carboxymethylcellulose. In second step, characterization of emulsions from different aspects, morphology, particle sizes, and stability had been discussed.

MATERIAL AND METHODS

1. Extra virgin olive oil

Extra virgin olive oil sample was purchased from a local supermarket in Budapest – Hungary from the brand Bertolli.

2. Microencapsulation wall material

Maltodextrin was procured from Applichem panreac itw companies, Gum arabic was purchased from Bi-Bor Kft, whey protein isolate was procured from Buda Family Kft, carboxymethylcellulose (E466) was procured from Gréta- tortadekoracio- hungary and Tween 80 emulsifier was purchased from sigma Aldrich, France.

3. Process of membrane emulsification

The first step to achieve microencapsulation is the formation of fine and stable emulsion of the core material (olive oil) in the wall solution. In the present investigation, precursor microcapsules were prepared by tubular ceramic porous membrane, pore size 1.4 μm (PALL Austria Filter GmbH). The membrane material was α alumina and the active membrane surface area was 50 cm^2 . The emulsion was produced by Cross-flow operation method.

Table 1. Description of formulations

	MD- GA	MD/WPI-GA	MD/CMC- GA
Maltodextrin (g)	100	100	30
Gum Arabic (g)	20	20	21.48
Whey protein isolate (g)		50	
Carboxymethylcellulose (g)			5
Olive oil (g)	60	60	60
Tween 80 (g)	20	20	23.25
Deionized water (g)	500	500	650
Solid % w/v ^a	28.57	33.33	17.69
O/W Ratio (g/g) ^b	0.4	0.5	0.21
Wall material/oil ratio (g/g)	2	2.83	0.94

a solid content in emulsion including extra virgin olive oil.

b Ratio between dispersed (O) and continuous phases (W).

Three formulations were prepared to encapsulate olive oil with different wall materials. Microcapsules were formulated according to table 1. First, MD-GA contained a binary mixture of 83% maltodextrin and 17 % gum arabic. Second, MD/WPI-GA was a ternary mixture of 59% maltodextrin, 12 % gum arabic and 29% whey protein isolate. Third, MD/CMC-GA contained 53% maltodextrin, 38% gum arabic and 9% carboxymethylcellulose. Each of the three mixtures was considered as the continuous phase. In all cases, the dispersed phase (extra virgin olive oil) was pressed through the membrane pores under pressure which was fixed into 2.5 bars. The recirculation flow rate of the continuous phase was 190 $\text{dm}^3 \text{h}^{-1}$. The direction of flow of dispersed phase was tangential

with the membrane surface. Under the pressure, the dispersed phase formed as a drop at the pores opening in the membrane surface and subsequently those were detached by the cross-flowing continuous phase.

4. Microcapsule characterization

4.1. Size and morphology

The size, shape and morphology of microcapsules were analyzed by optical microscope (DELTA OPTICAL, USA), oil immersion objectives and associated microscope image reader.

4.2. Average droplet size and span value

Following the emulsion preparation, the average droplet size and span value were measured by a FRITSCH Laser Particle Sizer ANALYSETTE 22 NanoTec. Each sample was analyzed three times and the average data reported. The span value was considered as an indication of the dispersity of the droplet size. The lower the span value, the more monodisperse the emulsion.

4.3. Zeta potential

Zeta potential was measured by Malvern Zetasizer apparatus. The zeta potential is a key indicator of the stability of colloidal dispersions. It refers to the ability of aggregation of the dispersed particles. As a rule of thumb, suspensions with zeta-potential above 30 mV and below -30 mV is physically stable (Malvern instruments). For this analysis, bent capillary tube cell was used.

RESULTS AND DISCUSSION

1. Formulation and preparation of olive oil emulsion

Table 1 presents a description of the feeding formulations. Wall materials for the emulsification of oil must have high stability, high water solubility, drying properties, a tendency to form a fine and dense network during drying, emulsifying properties and should not permit lipid separation from the emulsion during dehydration (Calvo et al., 2012; Gharsallaoui et al., 2007). The selected wall material should also protect the core from possible degradation during storage. In our case, the challenge for the selected system is to protect olive oil microcapsules from oxidation upon storage. Finally, the costs and availability need to be considered. Almost no wall material can meet all the properties listed; therefore, they are used in combination with each other (Hogan et al., 2001; Jafari et al., 2008).

Gum arabic is preferred instead of other types of gums because of its excellent emulsifying capacity. It has an effective slow dispersion in water at room temperature (Gallardo et al., 2013). The use of gum arabic as an encapsulating matrix is common due to its characteristics of viscosity, solubility and emulsification. It was traditionally used for lipid encapsulation and can produce stable emulsions with most oils (Gharsallaoui et al. 2007). That is why gum arabic was used in all of the cases together with Tween 80 which was used as emulsifier for easier emulsification and drop stability. Maltodextrin, whey protein isolate and Carboxymethyl cellulose were used to provide protection from oxidation at emulsion droplet interfaces (Gallardo et al., 2013). Also, maltodextrin with the gum arabic was used in

all cases because it has been demonstrated that this combination of wall materials presents a good compromise between cost and effectiveness for the microencapsulation of oils in general (Rubilar et al., 2012).

High solid contents result in increasing the viscosity which reduces the migration of oil to the surface and provides a rapid skin formation (Jafari et al., 2008). In this work the solid content of O/W emulsions was around 30% w/v except for MD/CMC-GA, which was formulated with a solid percentage of around 17% w/v due to the high viscosity of CMC. In conclusion, it has been reported that for further step (the spray drying step) it is important to find a compromise between high air temperature and high solid concentration (Gharsallaoui et al., 2007).

2. Size and morphology

The description of the feeding formulations was presented in table 1. The bio-polymers have been selected according to different, complementary criteria. Wall material for microencapsulation of oil must have high stability, high water solubility, drying properties, a tendency to form a fine and dense network during drying, emulsifying properties and should not permit lipid separation from the emulsion during dehydration (Calvo et al., 2012; Gharsallaoui et al., 2007).

Figure 1 shows size and morphology of the three formulations. The average diameters of the microcapsules were 7 μm in the fig 2A, 11 μm in fig 2B and 9 μm in the fig 2C. Well defined spherical microcapsules were observed. CMC containing microcapsules presents more homogenous size distribution and higher mean size was observed in the emulsion containing WPI. Nevertheless, this later presented some agglomerates due to the flocculation of the proteins in the solution.

In all cases average size of the synthesized microcapsules is in line with the conclusion reached by Joscelyne and Trägårdh, 2000 which reported that for membrane emulsification, the size of synthesized emulsion particles might be 2-10 times greater than the pore size of the membrane. In the present work, the membrane pore size is 1.4 μm .

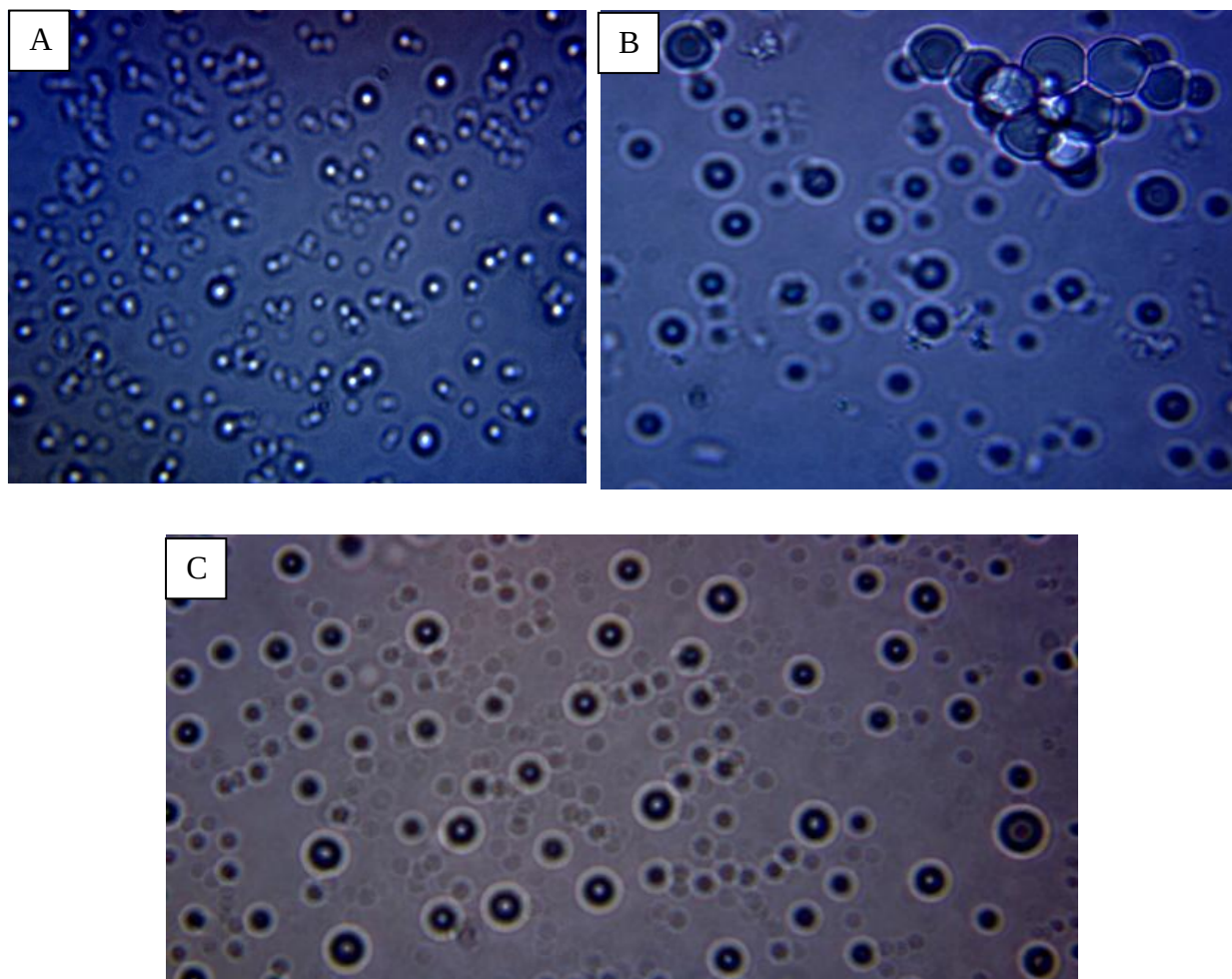


Fig. 1 Microscopic image of the samples after membrane emulsification. A) MD-GA, B) MD/WPI-GA, C) MD/CMC-GA.

3. Emulsification results

The dispersed average phase flux is an essential parameter for the membrane emulsification process. Increasing transmembrane pressure increases the flux of dispersed phase through the membrane, according to Darcy's law. At high fluxes, the average droplet size and the size distribution tend to increase because of increased droplet coalescence at the membrane surface (Charcosset 2009). In this investigation, transmembrane pressure was fixed to 2.5 bars and the dispersed phase average fluxes are $332 \text{ dm}^3 \text{ m}^{-2} \text{ h}^{-1}$, $60.6 \text{ dm}^3 \text{ m}^{-2} \text{ h}^{-1}$ and $40 \text{ dm}^3 \text{ m}^{-2} \text{ h}^{-1}$ for MD-GA, MD/WPI-GA and MD/CMC-GA respectively. In the literature, fluxes ranging from $50\text{--}250 \text{ dm}^3 \text{ m}^{-2} \text{ h}^{-1}$ were obtained with ceramic membranes of $0.2 \text{ }\mu\text{m}$ α -alumina membrane (Charcosset 2009). The drop of the fluxes when other wall materials are added into maltodextrin and gum arabic may cause a decrease in droplet size.

The results of the droplet mean diameter are summarized in figure 2. The average mean diameters D_{32} were $12.34\text{ }\mu\text{m}$, $5.22\text{ }\mu\text{m}$ and $9.16 \text{ }\mu\text{m}$ in the three different emulsification cases. The average mean diameter of microcapsules prepared with only MD-GA was $9.16\text{ }\mu\text{m}$. The addition of other wall materials into maltodextrin and gum arabic resulted in droplet size reducing in the emulsion containing CMC. The emulsion prepared from MD/CMC-GA

presented the smallest droplet size. This could be related to the high visible viscosity presented by this mixture which implies in a greater resistance to droplets movements, avoiding coalescence and resulting in smaller diameters.

In the emulsion containing WPI, the droplet size was greater than emulsion containing only MD-GA and this could be due to the whey proteins coalescence which is caused by the fact that the wall material is not enough to fully cover the droplets.

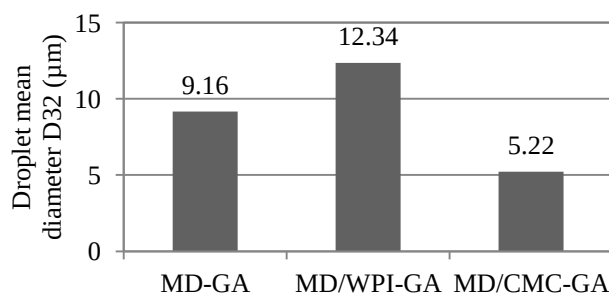


Fig. 2 Droplet mean diameter of emulsions with different wall materials.

The span value was considered as an indication of the dispersity of the droplet size. The lower the span value, the more monodisperse the emulsion. In this case MD/CMC-GA is more monodisperse compared to MD/WPI-GA and MD-GA respectively. This fact is also showed by the microscopic imaging.

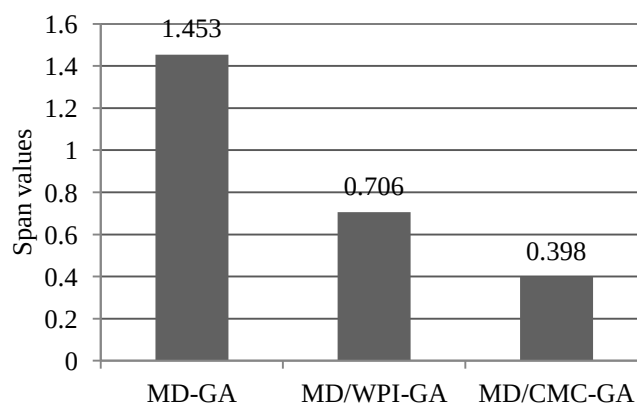


Fig. 3 Span values of emulsions with different wall materials.

4. Emulsions stability

The emulsion stability is expressed as the zeta potential of the emulsions. The results of zeta-potential for the different formulations are summarized in table 2. The zeta-potential of any material depends on its concentration in the solution. The average values of zeta-potential in three of the emulsions show the instability of the added wall materials in the continuous phase solution. The MD/CMC-GA formulation is closer to be stable compared to the three other formulations and this could be first due to the low concentration of maltodextrin in this emulsion. It has been proved previously by Albert et al., 2016 that low concentration of maltodextrin has a positive impact on the stability of maltodextrin in the solution. Consequently, it is better to use lower concentrations of maltodextrins to achieve better stability (toward -30 mV). Second, the higher stability of CMC containing emulsion could be

related to the higher viscosity of the continuous phase. It was previously proved by Tonon et al., 2012, that the increase in viscosity reduces the rate of sedimentation of the particles which results in better emulsion stabilization and thus avoiding droplet coalescence.

Table 2. Zeta potential results

Sample	Zeta-potential (mV)
MD-GA	-1.97
MD/WPI- GA	-0.895
MD/CMC- GA	-4.39

WPI containing emulsion is the most instable. The low stability of such emulsion can be attributed to the unfolding of protein molecules at the droplet surface, which enhances protein-protein interaction resulting in the flocculation phenomena during emulsification. The unfolding of protein molecules of the oil-water interface may result in changes in secondary and tertiary structure, and consequently exposure of their residues which would be linked (S-S- linkages or disulfide linkages) within the native globular structure, leading to the formation of intermolecular interaction at the oil-water interface and flocculation (Carneiro et al., 2013; Tonon et al., 2012).

CONCLUSION

In this work it was possible to evaluate the performance of different wall materials combinations in the olive oil emulsification process. The membrane emulsification of olive oil resulted in microcapsules with spherical shapes in all cases of formulation except the emulsion containing whey protein isolate which presented some agglomerates. This result may be due to the flocculation of WPI during the emulsification and consequently reducing the emulsion stability which was demonstrated by the zeta potential measurement. . The addition of CMC to the maltodextrin and gum arabic mixture resulted in a better stability of the emulsion and into a narrow droplet size distribution.

Limitations of the membrane emulsification process may be associated to the low fluxes associated to monodispersed emulsions, and to fouling phenomena. These disadvantages may be solved by recent processes, as rotating or vibrating membrane devices. It also appears that special membranes have to be developed to fulfill properties required for industrial applications: high fluxes, availability of large membrane area. Membrane emulsification should then appear as a very interesting technique for the food processing industry as well as for the pharmaceutical industry.

However further studies are still necessary in order to evaluate encapsulation efficiency after spray drying and the stability to oxidation of the encapsulated olive oil.

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THE INFLUENCE OF ACETYLSALICYLIC ACID ON THE CONTENT OF DIOSGENIN IN TRIBULUS TERRESTRIS PLANT

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SUMMARY: *The acetylsalicylic acid (ASA) was applied on the Tribulus terrestris plant in three concentrations with water as a control. Diosgenin was the main monitored substance. The HPLC analysis of plant samples was done and statistically evaluated. It was proven that the medium concentration of ASA increased the amount of diosgenin in fruit and decreased it in all concentrations in aerial parts of the plant.*

Key words: *acetylsalicylic acid, ASA, diosgenin*

INTRODUCTION

Tribulus terrestris as an annual remedial plant. Although this plant is in most countries in the world considered a noxious weed, it is popular for its ability to treat illnesses like diabetes, infertility, obesity or allergies. Bioactive compounds in the plant provide the ability to treat illness.

One of those substances is diosgenin, a steroidal saponin of spirostanol type. Many scientists confirm its theoretical use in medicine. This saponin has anti-inflammatory and anti-oxidant properties (Mannivan et al. 2013). It can be useful even against blood or brain diseases, allergies, diabetes and obesity (Roghani-Dekhorti et al. 2015). Probably, diosgenin can be used against cardiovascular diseases like thrombosis or atherosclerosis (Choi et al. 2011., Esfandriel et al. 2011., Liu et al. 2012., Yang et al. 2013., Kalailingam et al. 2014., Gong et al., 2011).

Quite convincing are also anti-cancer properties that depend on the type of the cancer. For example, anti-cancer effects were proven against prostate cancer PC-3 and DU-145 cells (Chen et al., 2011), colorectal cancer HCT-116 and HT-29 cells (Lepge et al. 2011). A lot of therapeutically usable and workable compounds, sexual hormones included, are made polysynthetically from natural precursors, mainly from diosgenin (Dong et al. 2015., Al Jasem et al. 2014).

Acetylsalicylic acid (ASA) affects physiological processes, for example photosynthesis, nitrogen metabolism and water management. It affects plant functionality under specific dosages, when some plant mechanisms are induced or inhibited. This can be done by using either low or high concentrations of elicitors (Kováčik et al. 2009). Kužel et al. (2003., 2007., 2008., 2009.), Hrubý et al. (2002), Tlustoš et al. (2005) and Cígler et al. (2010) researched, that it is not easy to use elicitors if desired to cause increasing the content of bioactive substances. ASA belongs among elicitors, which are protected, including research results, by

Czech patent CZ-296300 (Cígler et al., 2006), and by European patent EP-1750507 (Cígler et al. 2015).

Acetylsalicylic acid (ASA) was used to find out, if after the application the content of diosgenin will increase or decrease.

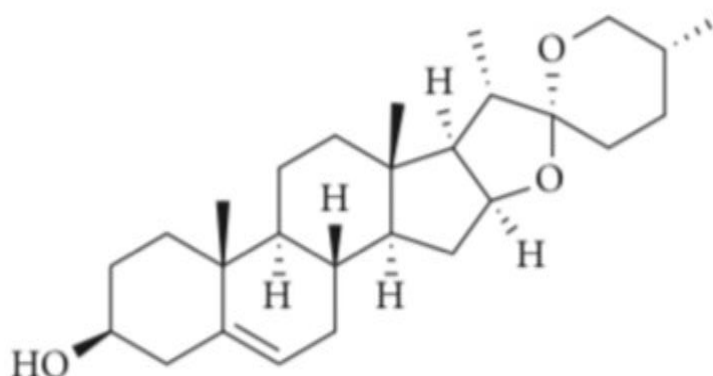


Fig 1. Chemical structure of diosgenin (Mafalda et al. 2016)

MATERIALS AND METHODS

Tribulus terrestris was grown on the field and were foliary treated with ASA and watter. Plants were divided in four groups, when in each group were 20 plants. Groups were marked as “A5, A4, A3 and W”. A5 group was treated with 500 ml of a low concentration solution (10^{-5} mol.l⁻¹) of acetylsalicylic acid (ASA), A4 group was treated with 500 ml of a medium concentration solution (10^{-4} mol.l⁻¹) of acetylsalicylic acid (ASA), A3 group was treated with 500ml of a high concentration solution (10^{-3} mol.l⁻¹) of acetylsalicylic acid (ASA) and W group was treated only with 500ml of watter. W group served for control only. Every group was treated every three weeks, at the same time.

Three weeks after the last application of ASA and watter, representative samples were taken. From A5 group has been removed 50 grams of aereial part of the plants and seeds – 5 grams of aerial parts and 5 grams of seeds from one plant. From A4 group has been removed 50 grams of aereial part of the plants and seeds – 5 grams of aerial parts and 5 grams of seeds from one plant. From A3 group has been removed 50 grams of aereial part of the plants and seeds – 5 grams of aerial parts and 5 grams of seeds from one plant. From W group has been removed 50 grams of aereial part of the plants and seeds – 5 grams of aerial parts and 5 grams of seeds from one plant. Samples were dried for 7 days at a temperature of 20 degrees and a humidity of 60%

SOLVENTS AND CHEMICALS

Diosgenin standard substance (5g) was purchased from Labicom (CZ). The acetonitrile (HPLC grade) was provided from Merck (Germany). Purified water was obtained by TKA water purification system (Germany).

INSTRUMENTATIONS AND MEASUREMENTS

UV spectra were scanned using instrument Lambda 12 spectrometer produced by Perkin Elmer (USA). Chromatography was performed on HPLC system 1200 series equipped with: binary pump SL, micro vacuum degasser, standard autosampler ALS SL, column compartment TCC SL, diode array detector DAD SL from Agilent Technologies (Germany). The separation was performed on HPLC column Purospher® RP-18e (150 x 4.6 mm i. d.; particle size 5 µm) with guard column LiChrosorb® (4 mm x 4 mm i.d.; particle size 7 µm) from Merck, (Germany) (Shishovska et al. 2015).

PREPARATION OF SOLUTIONS

All solutions were prepared using a mixture of equal volumes of acetonitrile and water as a solvent mixture. The stock standard solution was prepared by dissolving the standard substance diosgenin in volumetric flask. The solution was mixed in ultrasonic bath for 10 min. For UV spectrum scan, 181.6 µg/ml diosgenin solution prepared in solvent mixture was used.

Eight working solutions in a concentration range of: 10.90-545.88 µg diosgenin/ml were prepared for linearity testing. A working standard solution with diosgenin 54.49 µg/ml was used for precision testing. Extracts for accuracy test were prepared using 240 mg raw plant material and 10 ml solvent. Extraction was carried out in ultrasonic bath for 20 min. Extracts were filtered through a quantitative filter paper (black ribbon). The prepared solutions were spiked with diosgenin stock standard solution (by adding 0.25 ml; 0.5 ml and 1.0 ml standard solution to 0.5 ml extracts).

Four working solutions were prepared in a concentration range near approximate limit of detection (10.90-54.49 µg/ml diosgenin) for estimation of LOD and LOQ. All solutions were filtered through nylon 0.45 µm syringe Economy filter (Agilent Technologies) and then 10 µl of each solution were injected into HPLC system. This process were completed twelve times. (Shishovska et al. 2015)

SAMPLE PREPARATION

Aerial parts of the plant (flowers, leaves and stems) and seeds, dried and finely powdered, were used for sample preparation. Sample solutions were prepared using 240 mg raw plant material (240mg aerial parts and 240mg seeds) and 10 ml solvent. Extraction was carried on in ultrasonic bath for 20 min, and then cooled at room temperature. (Shishovska et al. 2015).

RESULTS AND DISSCUSIONS

Statistically inconclusive content of diosgenin in aerial part of the plant between treatments:

ASA high (bc) with ASA low (c)

Statistically conclusive content of diosgenin in aerial part of the plant between treatments:

ASA low (c) with ASA medium (ab) and watter (e)
 ASA medium (ab) with ASA low (c) and watter (e)
 ASA high (bc) with watter (e)
 Water (e) with ASA low (c), ASA medium (ab), ASA high (bc)

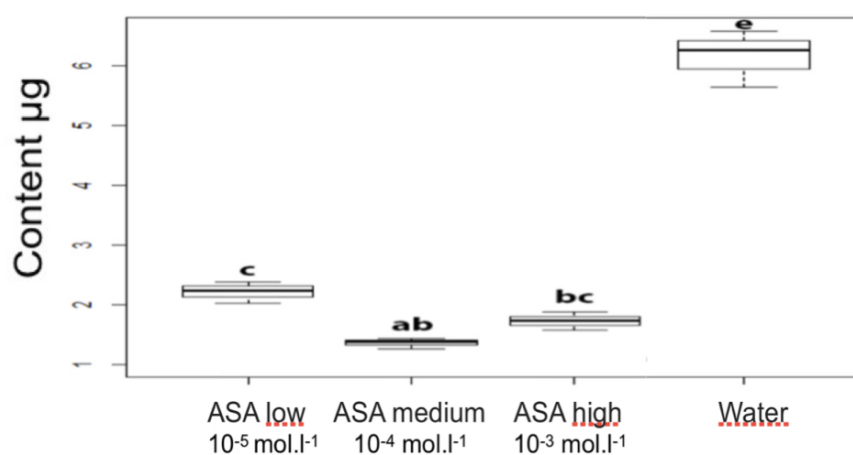


Table 1. Content of diosgenin in aerial parts after treatment

Statistically inconclusive content of diosgenin in seed between treatments:

ASA low (a) with watter (a)

ASA medim (b) with ASA high (b)

Statistically conclusive content of diosgenin in seed between treatments:

ASA low (a) with ASA medium (b) and ASA high (b)

ASA medium (b) with ASA low (a) and watter (a)

ASA high (b) with ASA low (a) and watter (a)

Water (a) with ASA medium (b) and ASA high (b)

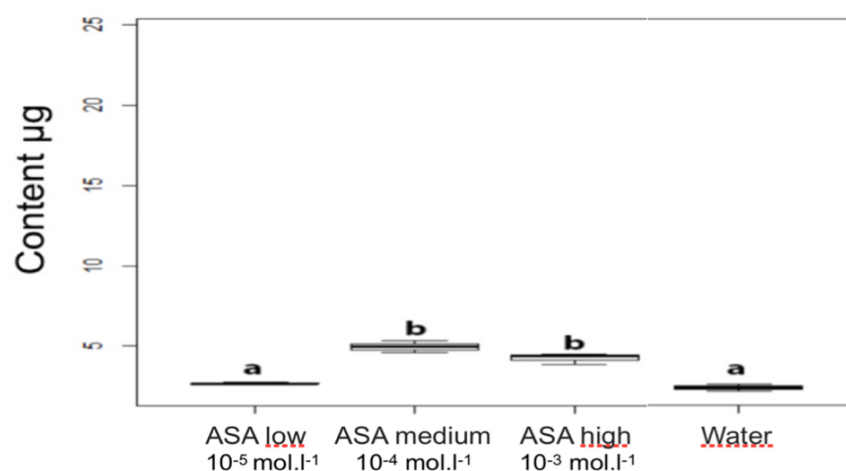


Table 2. Content of diosgenin in seeds after treatment

CONCLUSION

Content of diosgenin in *Tribulus terrestris* plant is statistically affected by interaction between plant and treatment. Treatments has a different effect on the diosgenin content in aerial parts and seeds in *Tribulus terrestris* plant. Tested using the two-way analysis of variance method (two-way ANOVA), [F (5,24) = 593.1, $P < 2 \times 10^{-16}$].

ASA compared with other concentrations and with watter statistically increased the content of diosgenin in seeds when treated with medium concentration $10^{-3} \text{mol.l}^{-1}$. On the other hand, ASA statistically decreased the content of diosgenin in all concentrations compared to watter.

More reaserch is needed to know how ASA affects the content of bioactive substances in *Tribulus terrestris* plant.

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THE IMPACT OF THE SEASON ON THE QUALITATIVE CHARACTERISTICS OF BEATEN CHEESE

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In our country, the interest of consumers for traditionally obtained cheeses is increasing and milk processing capacities are making efforts to industrialize their production process. The group of the most consumed traditional cheeses also includes beaten cheese.

Accordingly, the subject of analysis of our research was to determine the impact of seasonal changes in raw cow's milk on the qualitative characteristics of industrially obtained beaten cheese. Samples were taken over a period of one year. A total of 24 batches of beaten cheese were obtained in an industrial way of production, before every batch the quality of raw milk was analyzed. Milk samples were analyzed for physico-chemical composition (proteins, fats, lactose, solids-non-fat, minerals, added water, density, freezing point, and titration acidity) and hygienic condition (somatic cell count (SCC)). Meanwhile, samples of beaten cheese were analyzed for their physico-chemical composition (proteins, fats, salt, moisture, dry matter, and titration acidity).

The results obtained from this research indicate the fact that seasonal changes have an impact on the quality and hygienic condition of raw cow's milk. Whereas the quality (percentage of proteins and fats) and the hygienic condition of raw milk largely determine the quality and physico-chemical properties of the beaten cheese.

Key words: raw cow's milk, seasonal changes, beaten cheese

INTRODUCTION

Traditional cheeses are consumed daily in Macedonia, like white brine cheese and beaten cheese (Talevski G., 2012). Beaten cheese is an autochthonous dairy product that used to be made from sheep's milk and originates from the area of Mariovo (Dubrova Mateva N. et al., 2008).

With the development of science and technology, it has become known that the final quality of cheese largely depends on the quality of the primary raw material, i.e. raw milk (Osama K. et al., 2015). In general, the quality of raw milk depends on the diet, breed, and lactation period of the dairy animal, as well as on environmental conditions (season) (Guétouache M. et al., 2014). Additionally, the quality of beaten cheese depends on the production process, as well as the implementation of hygienic practices, standards, and

systems that along with the quality guarantee the safety of the final product (Levkov V. et al., 2014).

MATERIALS AND METHODS

The subject of analysis of this research were samples of industrially produced beaten cheese from raw cow's milk. Milk and cheese samples were taken for a period of one year, twice a month, from the milk processing capacity in the Pelagonia region. A total of 24 bulk tank milk samples were analyzed. From this milk was produced beaten cheese and 24 batches were obtained.

Sampling for raw milk analysis was performed in the morning, before the start of the working day. Representative samples were taken from bulk tank milk. Samples for analysis of physico-chemical parameters were taken in plastic cups (100ml). Samples for the analysis of the hygienic condition of milk were taken in sterile cups (40ml) with the preservative Broad Spectrum Microtabs II (one tablet for 40 ml of milk). After sampling, the samples were well closed and transported at 4 °C to the laboratory. Analyzes were performed on the same day. The samples were analyzed at room temperature (20°C to 22°C). The compositions of milk (proteins, fats, lactose, fat-free dry matter, minerals, density, added water, and freezing point) were determined using the Lactoscan MCC an ultrasonic milk analyzer. Soxhlet Henkel method was used for the determination of titration acidity. The somatic cell count was analyzed by LACTOSCAN SCC based on fluorescent microscope technique of counting cells.

The samples of beaten cheese were randomly chosen from several vacuum packages (50g). The samples were placed in plastic cups and transported to the laboratory. The determination of the percentage of proteins was done by the method of Kjeldahl, while the percentage of fats was determined by the method of Gerber. The titration acidity was determined by the method of Soxhlet Henkel. The percentage of NaCl was determined by the Mohr method. Mettler Toledo MJ33 was used to determine the percentage of moisture and dry matter.

The obtained results were grouped into four groups: Winter (January - March), spring (April - June), summer (July - September), and autumn (October - December).

Statistical significance among the examined categories was analyzed at a significance level of 5% ($p < 0.05$) using Student's t-test.

RESULTS AND DISCUSSION

The results obtained from the conducted analyzes in raw milk are presented in Table 1. The percentage of basic nutritional components in milk, for the winter season was 3.31% proteins and 4.5% fats, while the lowest values were obtained in the summer season, when the percentage of proteins and fats was 2.8% and 3.24%, respectively. Significant differences were observed in the percentage of proteins and milk fat during the summer season compared

with spring, autumn, and winter ($p < 0.05$). The percentage of proteins and fats is inversely proportional to ambient temperature (Ocak E. and Inci S., 2008), whereby, any increase in temperature leads to a reduction of these parameters (Nateghi L. et al., 2012). Additionally, the increase in the percentage of proteins in winter may be due to the use of a concentrated diet of dairy cows (Osama K. et al., 2015) or according to research by Rubporn K. et al., (2007) feeding of dairy cows with foods fortified with soy protein. Regarding the percentage of fats, similar results were obtained by Yang L. et al., (2013) who noticed that the percentage of milk fats is the highest in winter. According to Osama K. et al., (2015), at the beginning and at the end of the lactation phase of the dairy animal, there is an increase in the percentage of fats. Additionally, according to Vojtech K. and Sarka N., (2015), each increase in milk fats is followed by a decrease in solids-non-fat (SNF) and titration acidity. We obtained similar results, with the exception of the spring season (Table 1).

Table 2 - Physicochemical and hygienic analysis of raw cow's milk

Parameters	Spring	Summer	Autumn	Winter
Proteins (%)	3,16±0,22 ^b	2,80±0,30 ^a	3,22±0,27 ^b	3,31±0,18 ^b
Fats (%)	4,25±0,25 ^b	3,24±0,33 ^a	4,13±0,11 ^b	4,35±0,23 ^b
Lactose (%)	5,43±0,49 ^a	4,65±0,49 ^b	5,48±0,30 ^a	4,24±0,62 ^b
SNF (%)	8,06±0,40 ^a	6,77±0,30 ^b	7,81±0,48 ^a	7,94±0,49 ^a
Minerals (%)	1,01±0,16 ^{ab}	1,19±0,22 ^a	1,04±0,02 ^{ab}	0,93±0,90 ^b
Freezing point (°C)	-0,724±0,01	-0,730±0,01	-0,727±0,01	-0,723±0,01
Added water	0,0	0,0	0,0	0,0
Density (g/cm ³)	1,04±0,01	1,04±0,01	1,04±0,01	1,04±0,01
Titration acidity (°SH)	7,5±1,44 ^a	6,9±0,22 ^{ab}	6,8±0,26 ^{ab}	6,2±0,13 ^b
Total number of somatic cell (SCC/mL)	372.250 ± 46.964 ^a	623.166 ± 69.769 ^b	334.166 ± 139.041 ^a	812.000 ± 60.488 ^c

*The differences in values with different superscripts in the same column are statistically significant at the level of: a:b; a:c; b:c $p < 0.05$

The highest values for the percentage of lactose were observed in autumn (5.48%) and spring (5.43%), while the lowest in summer (4.65%) and winter (4.24%) and they are statistically significant at the level of $p < 0.05$. The percentage of lactose is generally conditioned by the lactation of the dairy animal (Rubporn K. et al., 2007) and is not

significantly affected by seasonal variations (Ocak E. and Inci S., 2008). According to Yang L. et al., (2013) the percentage of lactose is highest in winter, while according to Nateghi L. et al., (2012) in summer. In addition, the percentage of lactose is conditioned by the total number of bacteria and somatic cells (Yarabbi H. et al., 2014), so when the percentage of lactose is lower it means that the total number of bacteria is higher and inversely. In our results this correlation was observed in all studied seasons (Table 1).

A significant increase in somatic cells was recorded in winter (812,000 SCC/ml) as well as in summer (623,166 SCC/ml), while in spring and autumn the somatic cell count was below 400,000 SCC/ml. Significant differences were observed between the examined seasons ($p < 0.05$), with the exception only between the spring and autumn seasons ($p > 0.05$). Similar results were obtained by Nateghi L. et al., (2012) and Yang L. et al., (2013). According to Nateghi L. et al., (2012), the increase in the somatic cell count is most pronounced in winter, while according to Yang L. et al., (2013) in summer. In general, the main cause of an increase in the somatic cell count is an injury to the mammary gland and according to Yarabbi H. et al., (2014) an increase in the somatic cell count in summer is a result of favorable environmental conditions for the growth of bacteria (Yarabbi H. et al., 2014). Other factors that affect the increase in these two seasons are the temperature change, i.e., temperature stress as well as the change in diet (Nateghi L. et al., 2012; Yang L. et al., 2013).

Physico-chemical parameters and milk quality are influenced by season as well as temperature variations (Ocak E. and Inci S., 2008; Osama K. et al., 2015), diet (Yang L. et al., 2013; Rubporn K. et al., 2007), and individual characteristics of the dairy animal. Guetouache M. et al., (2014) states that the milk obtained in the winter has the most favorable chemical composition, and this reflects on the quality of dairy products, which fully coincides with the results obtained (Table 3).

In general, the nutritional characteristics of beaten cheese depend on the nutritional quality of the raw milk (Ramzan M. et al., 2010), i.e. proteins and fats are largely transferred to the cheese (Ramzan M. et al., 2010). As a result of the most favorable chemical composition of milk in the winter period, the cheese obtained in that period is characterized by the highest percentage of proteins (29.50%) and fats (35.90%). Similar results were obtained in the autumn period. While the percentage of proteins in the summer is 23.40% and in the spring 25.83%. The percentage of milk fats in cheese produced in spring is 33.50%, while in summer 32.24% (table 2). Also, no significant variations were recorded between the seasonal qualities of beaten cheese in our research. This is as a result of standardization of the parameters during industrial processing of beaten cheese (Levkov V. et al., 2014).

Table 3 – Physico-chemical analysis of the industrially obtained beaten cheese

Parameters	Spring	Summer	Autumn	Winter
Proteins (%)	25,83±1,74	23,40±1,34	28,40±1,25	29,50±1,22
Fats (%)	33,50±0,21	32,24±1,35	35,00±1,34	35,90±1,45
Salt (%)	6,6±0,51 ^{ab}	6,2±0,16 ^a	6,4±0,35 ^{ab}	6,8±0,25 ^b
Moisture (%)	41,52±0,24 ^a	44,62±0,88 ^b	44,64±1,30 ^b	36,32±2,67 ^c
Dry matter (%)	58,48±0,24	55,18±0,88	55,36±1,30	63,68±2,67
Titration acidity (°SH)	58±1,41 ^{ab}	56±4 ^a	58±2,58 ^{ab}	60±1,79 ^b

* The differences in values with different superscripts in the same column are statistically significant at the level of: a:b; a:c; b:c p <0.05

The percentage of salt ranged from 6.2% (summer) to 6.8% (winter) and these values were significant at the level of p <0.05. Slightly lower results were found by Talevski G., (2012). The percentage of salt in beaten cheese ranges from 5 to 10% (Dubrova Mateva N. et al., 2008). The high percentage of salt in this type of cheese is a result of the combined salting method (dry salting and salting in brine) (Levkov V. et al., 2014; Klopcevska J. et al., 2016).

The percentage of moisture ranged from 36.32% to 44.82%. In addition, the percentage of moisture in this cheese shows a significant difference (p<0.05), except between summer and autumn (p>0.05). Similar results were obtained by Klopcevska J. et al., (2016). Compared to Dimitrovska G. et al., (2018) the cheese obtained in this study had a higher percentage of moisture, which indicates the fact that the ripening process was faster (Dubrova Mateva N. et al., 2008).

The recorded values for the titration acidity range from 56°SH to 60°SH (Table 2). These values differs significantly at the level of p<0.05. Similar results were obtained by Talevski G., (2012). Additionally, changes in titration acidity are accompanied by changes in active acidity that most often depend on added starter cultures (Ramzan M. et al., 2010).

CONCLUSION

In general, the season and the hygienic quality of raw milk determine the nutritional value of beaten cheese.

The most favorable physico-chemical composition of raw milk was recorded in the winter period and it is 3.31% proteins, 4.35% fats, 4.24% lactose, 7.94% SNF, and 0.93% minerals, while the most unfavorable in the summer period (2.80% proteins, 3.24% fats, 4.65% lactose, 6.77% SNF and 1.19% minerals). In spring and autumn, the physico-chemical composition of the milk has almost equal quality. In addition, seasonal changes in the environment lead to a

decrease in its hygienic condition, as a result of the increase in the somatic cell count. The highest somatic cell count was recorded in winter (812,000 SCC/ml) and summer (623,166 SCC/ml). The hygienic condition of milk was significantly reduced in these two seasons. In spring and autumn, the somatic cell count was below 400,000 SCC/ml.

The best nutritional quality of raw cow's milk in winter resulted in the most favorable nutritional quality of industrially obtained beaten cheese which is 29.5% proteins and 35.9% milk fats. Additionally, the percentage of salt in the beaten cheese obtained in this period is 6.8%.

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DAMAGE AND PARASITISM RATE OF SEED WEEVILS (*CEUTORHYNCHUS OBSTRUCTUS*) IN WINTER OILSEED RAPE

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Summary: Oilseed rape is an economically important crop, and a valuable food source for insects. Cabbage seed weevil is considered as one of major oilseed rape pest and in its favourable conditions can considerably decrease yield. In this study, in total 11 winter oilseed rape fields were divided into two groups, five of the oilseed rape fields were located closer than 500 meters to the previous year's oilseed rape field and six were further than 500 meters from the previous year's oilseed rape field. In total, 11 000 pod were collected from the fields. The damage rate of cabbage seed weevil was higher in the fields located closer to the previous year's oilseed rape fields, whereas parasitism rate was not affected by this factor. Cabbage seed weevil can be effectively controlled by using integrated pest management approaches and naturally occurring biological control agents. In the control of cabbage seed weevil, hymenopteran parasitoids are considered as one of the most important agents, but also following crop rotation plans and spatio-temporal isolation between oilseed rape fields, is a natural way to decrease damage rate.

Key words: *Ceutorhynchus assimilis*, rapeseed, Brassica, pest management, natural enemies, parasitoid wasps

INTRODUCTION

Oilseed rape (*Brassica napus* L.) is one of the most important oil crops in the world (Hatzig et al., 2018). The increase in the cultivation area has also increased the abundance and distribution of oilseed rape pests. Cabbage seed weevil (*Ceutorhynchus obstrictus* (Marsham) syn. *C. assimilis* (Paykull)) is an important oilseed rape pest that mainly harms winter oilseed rape in Europe. Pest management of oilseed rape crop is mainly relied on the synthetic insecticides (Heimbach and Müller, 2013; Zimmer et al., 2014), which in addition to pests, also kills the non-target organisms including for instance natural enemies of pests (Jonsson et al., 2008; Kovács, 2018). Hymenopteran parasitoids are one of the most important group of pest's natural enemies. There are three key parasitoid species who can effectively control the abundance cabbage seed weevil: *Trichomalus perfectus* (Walker), *Stenomalina gracilis* (Walker) and *Mesopolobus morys* (Walker) (Haye et al., 2015; Ulber et al., 2010). The aim of this work was to find out the damage rate of winter rape pods and whether it was affected by the distance from the previous year's winter oilseed rape fields. In addition, to find out the parasitism rate of the cabbage seed weevils; whether the distance of the last years

fields affected it and whether the species composition of parasitoids was affected on the distance of the previous year's winter oilseed rape fields.

MATERIAL AND METHODS

The study was carried out in conventional winter oilseed rape fields in Tartu County, Estonia in 2018. In total, 11 fields were included in the study, five of them were located closer than 500 meters from the previous year's winter oilseed rape field and six more than 500 meters away from the previous year's winter oilseed rape field. The fields were on average 20 hectares in size, most of the fields had a previous crop of winter wheat or clover, but there was also horseradish and potato fields. Winter oilseed rape was sowed from 11 to 28 August 2017. Insecticides were used in four fields out of twelve.

Pods were collected in the pod ripening growth stage of plants (BBCH 76–84 (Meier, 2001), at five sampling points (12 meters from the field edge), within each point from ten plants, in total 20 pods per plant (ten from main raceme and ten from side branches). Plants and pods were chosen randomly within each site. In total, 1000 pods per field and altogether 11 000 pods were collected. Collected pods were marked and placed in a small plastic bag and brought to the lab. To identify the damage caused by cabbage seed weevil (*C. obstrictus*) larvae and the larval parasitism rate, the emergence traps were used (described in detail in Veromann et al., 2010). Pod samples were placed to traps – 500 ml carton bandbox, one side was covered with a thin fabric for air circulation and prevention of mould growth. In total 550 emergence traps were used, each of them containing all pods from one plant. Emergence traps were kept at room temperature for a month. After that, traps were opened and pods were examined very carefully one by one to find adult parasitoid wasps, weevils' larvae and the exit holes of weevils and/or parasitoids. All found parasitoid wasps were kept in 70% ethanol and at –20 °C till they were identified to species.

RESULTS AND DISCUSSION

Generally, the abundance of seed weevils were low in all fields and the damage rate of cabbage seed weevil did not exceed the economic threshold level that according to Free & Williams (1978) is 33% (Figure 1). However, our study revealed that the cabbage seed weevil damage rate was affected by the distance from last year's winter rape field (Wald $\chi^2(1)=26.41$; $p < 0.0001$; Figure 1), but the parasitism rate was not influenced on the distance of last year's winter oilseed rape fields (Wald $\chi^2(1)=0.92$; $p = 0.34$). According to the literature, the cabbage seed weevils overwinter in the field edges (Williams, 2010), and this explains why the damage rate was significantly greater in the fields that were closer than 500 meters to the previous years oilseed rape fields (Figure 1). In spring, the weevils emerged from their overwintering sites and started to seek suitable food and oviposition plants and found nearby located oilseed rape fields that offered unlimited food resource and oviposition sites. As an

average, the spatial isolation from previous year's oilseed rape fields decreased damage rate over 5% (Figure 1).

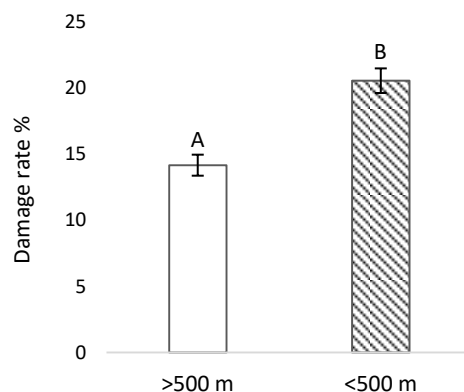


Figure 1. Mean ($\pm$ SE) rate of damaged pods by cabbage seed weevils per plant (%) in winter oilseed rape fields located closer than 500 m (<500 m) or more than 500 m (> 500 m) away from the previous year's winter oilseed rape field, in 2018 in Tartu County. Different letters indicate statistically significant difference between treatments ($p < 0.05$).

Despite the fact that the abundance of seed weevils' larvae was low, parasitoid wasps could successfully find their host and control the cabbage seed weevil abundance. In general, the parasitism rate was very high and reached up to 63%. The mean parasitism rate in fields that were closer than 500 meters from the previous year's winter oilseed rape field was 59.98% (± 2.3) and 63.21% (± 2) in fields located more than 500 meters away from the previous year's winter oilseed rape field. Overall, the parasitism rate was twice as high as the level required to the effective biological control (at least 32% (Hawkins and Cornell, 1994; Kovács et al., 2013)). In Estonia, there has been reported extremely high parasitism rates previously as well – even as high as 90% (Veromann et al., 2010). Therefore, we can conclude that there is no need to control cabbage seed weevils with insecticides because naturally occurring parasitoids can efficiently control the size of seed weevil's population. In addition, we found a weak but statistically significant negative correlation between the parasitism rate and insecticide treatments ($r = -0.13$; $p < 0.05$), which is also confirmed by several other studies (Gill et al., 2012; Heimbach and Müller, 2013; Moreby et al., 2001; Zimmer et al., 2014).

Trichomalus perfectus was the predominant parasitoid (651 specimens) but also *S. gracilis* (413 specimens) and *M. morys* (118 specimens) were found. While *T. perfectus* and *S. gracilis* were relatively equally present in fields closer to or farther than 500 meters from the previous year's winter oilseed rape field, *M. morys* was approximately 1.5 times more abundant in the fields that were closer than 500 meters from the previous year's winter oilseed rape field. As *T. perfectus* is well studied, the biology on *S. gracilis* and *M. morys* is not very well known and we can only assume, based on our study that probably *M. morys* is not so good flier as other two parasitoid species. Similarly to their host, parasitoid wasps also overwinter in the field edges (Ferguson et al., 2000; Haye et al., 2018; Murchie et al., 1999; Ulber et al., 2010), and by having the next year's oilseed rape field closer to the previous

year's one, increases the number of parasitoid wasps in the neighbouring field. Parasitoid wasps do not need to find their host somewhere else, which results in higher parasitism rate in the fields that were closer to the previous year's oilseed rape.

CONCLUSION

The damage rate of cabbage seed weevil was affected by the distance from last year's winter oilseed rape field, fields that were closer than 500 meters from last year's winter oilseed rape field, the damage rate was significantly higher than in the fields that were further than 500 meters from previous year's winter oilseed rape field. The distance from last year's winter oilseed rape field did not affect the parasitism rate of cabbage seed weevils. The species composition of the parasitoid wasps was also similar in both fields that were closer to the previous year's winter oilseed rape field and in the fields that were further away from the previous year's field. The most dominant species was *T. perfectus*, but *S. gracilis* and *M. morys* were also found. We found less *Mesopolobus morys* in the fields that were more than 500 meters from last year's winter oilseed rape field. Our study shows that by implementing integrated pest management approaches and offering biological control agent's necessary resources, they are capable of controlling cabbage seed weevil themselves and there is no need for insecticide.

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ST. JOHN'S WORT ESSENTIAL OIL: AN EFFICIENT ANTIFUNGAL AGENT AGAINST CARROT ROT CAUSED BY *PENICILLIUM* SPP.

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Summary: In the present study, the antioxidant and antifungal activities of St. John's wort essential oil containing the addition of tocopherol (*Hypericum perforatum*; HPEO) against three *Penicillium* (*P.*) species: *P. crustosum*, *P. expansum*, and *P. citrinum* were evaluated. The results showed that HPEO exhibited strong antioxidant activity (69.95 ± 0.42 %) and significant inhibitory effect (in the concentration $\geq 250 \mu\text{L.L}^{-1}$) on the growth of all inoculated fungi strains. Our results could provide useful data for developing methods to improve the shelf-life of fresh carrot using HPEO.

Key words: *Hypericum perforatum*, essential oil, antifungal activity, antioxidant activity, carrot

INTRODUCTION

St John's Wort (*Hypericum perforatum*; HP) is a perennial herbaceous plant of the *Hypericaceae* family, having many beneficial effects for improving the human health (Di Carlo et al., 2001). Hence, this plant is often used for its antidepressant capabilities in addition to its antioxidant, antiviral, anti-inflammatory, and antimicrobial properties (Barnes et al., 2001). Several phytochemical compounds with strong antioxidant activity such as hyperoside, chlorogenic acid (Gioti et al., 2009), rutin (Nöldner and Schötz, 2002), I3,II8-biapigenin (Colovic and Caccia, 2008), and catechins (Ploss et al., 2001) were identified in HP.

Essential oils (EOs), also defined as essences, volatile oils, etheric oils or aetheroleums, are a complex mixture of volatile constituents biosynthesised by plants (Hüsnü et al., 2007) which are commonly used as flavoring agents in food products, drinks, perfumeries, pharmaceuticals, and cosmetics (Chávez-González et al., 2016). These liquid, volatile, pure and colored mixtures of several aromatic compounds (characterised by a strong odour) can be

obtained from the different parts of plants, mainly from herbs and spices (Ravindran and Jaiswal, 2016; Wu et al., 2017). EOs isolated and identified from the vast number of plant species amount to over 3000, and of these, several hundreds have been produced commercially (Sankarikutty and Narayanan, 2003). The EO alone or in combination possesses significant medicinal properties (Chávez-González et al., 2016) and strong antimicrobial activities as they effectively destroy several bacterial, fungal, and viral pathogens (Swamy et al., 2016).

The most important limiting factors to long-term storage of carrots leading to considerable crop losses are postharvest diseases. Among the most prevalent storage diseases of carrots grown in temperate regions are grey mould rot and sclerotinia rot caused by *Botrytis cinerea* and *Sclerotinia sclerotiorum*, respectively (Kora et al., 2005). Also, *Penicillium* (*P.*) strains are the most isolated filamentous fungi from fresh carrots (Kłapeć et al., 2018).

Synthetic chemicals with antimicrobial properties (benzimidazoles, aromatic hydrocarbons, sterol biosynthesis inhibitors) are often used in the control of plant diseases in agriculture. However, there are many problems with the effective uses of these chemicals in the areas where the fungi have developed resistance (Brent and Hollomon, 1998). In order to overcome this problem, higher concentrations of these chemicals are still more and more applied leading to increase in their high-level toxic residues in the products. Thus, there has been a growing interest in the research of alternative uses of the plant EOs, which are less damaging to the human health and environment, for pest and disease control in agriculture (Costa et al., 2000).

The EOs of *Hypericum* species have been extensively examined by different chromatographic methods (Schwob et al., 2002; Bertoli et al., 2003; Cakir et al., 2004). However, there are no reports on the analyses of inhibitory effects of EO of *Hypericum* species against the growth of *Penicillium* spp. inoculated on the carrot. Thus, the aim of the present study was to determine the antioxidant capacity of *Hypericum perforatum* essential oil and to evaluate its antifungal activity against three pathogenic fungi of *P.* species (*P. crustosum*, *P. expansum*, and *P. citrinum*) vapor-phase method.

MATERIAL AND METHODS

Essential oil

St. John's Wort essential oil (*Hypericum perforatum*) with the addition of tocopherol was purchased from Gratis Company (Ltd, Istanbul, Turkey).

Radical Scavenging Activity—DPPH Method

The radical scavenging activity of the HPEO was measured using 2,2-diphenyl-1-picrylhydrazyl (DPPH) method according to Kačániová et al. (2020a). The 0.1 mL of EO sample was mixed with 3.9 mL of DPPH solution (0.025 g DPPH in 100 mL ethanol). Absorbance of the reaction mixture was determined with The Agilent Cary 60 UV-Vis

spectrophotometer (Agilent Technologies, United States) at 515 nm. The final scavenging activity was calculated as percentage (AA%) and determined according to the following formula: $AA\% = [(A_0 - A_1)/A_0 \times 100]$; where A_0 is the absorbance of the control reaction (DPPH radical); A_1 is the absorbance of the tested sample. The measurement was performed in triplicate.

Fungal strains

The fungi *P. crustosum*, *P. expansum*, and *P. citrinum* were isolated from grape samples and identified with the MALDI-TOF MS Biotyper (Bruker, United States) and 16S rRNA sequencing.

Vapor-phase of antifungal assay with carrot

Carrot samples were cut into slices of 5 mm. Warm PDA (potato dextrose agar; Oxoid, Basingstoke, UK) was poured into 60 mm Petri dishes (PD) and the lid also. A fungal spore of each strain was used for carrot inoculation. The EO was diluted in ethyl acetate to a final concentration of 125, 250, and 500 $\mu\text{L.L}^{-1}$ and transferred on a 55 mm round sterile filter paper using a micropipette. The filter paper was dried for 1 min for the evaporation of ethyl acetate and subsequently laid into the PD on the walls to ensure a 2 mm distance between the paper and the agar. Finally, the PD was hermetically closed with its lid containing the solidified agar. The PD were hermetically closed and kept at 25 °C for 14 days in the laboratory incubator. The colonies with visible mycelial growth and visible sporulation were evaluated (Kačániová et al., 2020a). The antifungal activity was expressed as percentage of mycelial growth inhibition (MGI) which was calculated using the formula: $MGI = [(C - T) / C] \times 100$, where C = fungal growth in the control and T = fungal growth in the treatment group, as was previously described by other authors (Talibi et al., 2012; Aman and Rai, 2015).

Statistical analysis

All assays were carried out in triplicate. Statistical evaluation of the obtained data was performed using the GraphPad Prism 8.0.1 (GraphPad Software Incorporated, San Diego, California, USA). Oneway analysis of variance (ANOVA) followed by Tukey test was used for statistical analysis.

RESULTS AND DISCUSSION

Antioxidant activity of HPEO

To assess the possible utilization of the EO as an antifungal agent, its antioxidant activity was firstly evaluated using the DPPH method. DPPH is a useful reagent for examining free radical scavenging activities of various bioactive compounds. Indeed, the reduction of DPPH

absorption is indicative of the capacity of the EO to scavenge free radicals, independently of any enzymatic activity (Lebeau et al., 2000). In our study, HPEO showed excellent percent inhibition of DPPH activity (69.95 ± 0.42 %). We assume that the strong antioxidant activity observed may be directly correlated to the phytonutrient composition of the EO and can be attributed to a hydrogen-donating ability of its components (Shimada et al., 1992). In accordance with our research, the evident antioxidant capacity of HPEO was also reported by Quassinti et al. (2013) which analyzed the EOs of three different *Hypericum* species. Their results revealed that HPEO is a good scavenger of free radicals, which can be attributed to the relatively high level of α -pinene.

***In situ* antifungal vapour contact assay**

The determination of MIG percentage of tested concentrations (125, 250 and 500 $\mu\text{L.L}^{-1}$) of HPEO was carried out in hermetically closed sterile PD for 14 days (at 25 ± 1 °C in the dark) in which one of the three *Penicillium* strains (*P. crustosum*, *P. expansum*, *P. citrinum*) used was inoculated individually on the carrot slice as a substrate for their growth and biomass production (Fig. 1; Fig. 2). The data is summarized in Table 1. Our results showed that HPEO strongly inhibited the growth of all three tested *Penicillium* strains at relatively low concentrations (above 250 $\mu\text{L.L}^{-1}$). In addition, we have found statistically significant ($P < 0.0001$) differences in the antifungal action of HPEO against *P. crustosum* and *P. citrinum* between the concentrations 250 $\mu\text{L.L}^{-1}$ and 500 $\mu\text{L.L}^{-1}$. This fact indicates a more prominent inhibitory impact of HPEO on *P. crustosum* and *P. citrinum* mycelial growth on carrot with an increasing level of the EO.

Table 1 Percentage of mycelial growth inhibition (MGI) of *Penicillium* strains expressing the antifungal potential of HPEO against the artificially inoculated fungi on carrot

MGI (%)	Fungal strains		
	<i>P. crustosum</i>	<i>P. expansum</i>	<i>P. citrinum</i>
125 ($\mu\text{L.L}^{-1}$)	9.11 ± 0.68^a	-5.26 ± 0.35^a	0.11 ± 0.09^a
250 ($\mu\text{L.L}^{-1}$)	56.75 ± 0.52^b	97.05 ± 0.74^b	83.95 ± 0.44^b
500 ($\mu\text{L.L}^{-1}$)	95.68 ± 0.83^c	98.01 ± 0.76^b	97.26 ± 0.97^c

Note: Values are expressed as mean of three determinations $\pm$ standard derivation. Means with different letter superscript within the same column are significantly different ($P < 0.05$). The negative values indicate a profungal activity against *Penicillium* strains.

Numerous studies deal with the antifungal and antibacterial effects (Canillac and Mourey, 2001; Elgayyar et al., 2001; Kačániová et al., 2020a; Kačániová et al., 2020b) of plant EOs (spruce, anise, angelica, basil, carrot, celery, cardamom, coriander, dill weed, fennel, oregano, parsley, rosemary, orange). Their inhibitory activities against fungi growth may be attributed to the diverse chemical composition of the EOs (Pattnaik et al., 1997), therefore, in our study, we decided to analyze HPEO with the addition of tocopherol. The results by Mašković et al. (2011) demonstrate that HP is a growth inhibitor of *P. canascens* which is in accordance with our study. Suppressed mycelial growth of *P. digitatum* by application of HP plant powder

was also reported by Balkan (2018). The antifungal activities of EOs from different *Hypericum* species are linked to their main components, such as α - and β -pinene, as well as β -caryophyllene (Crockett, 2010). We propose that these compounds contained in our HPEO used could also contribute to observed significant mycelial growth inhibition of all three *P.* species (*P. expansum*, *P. citrinum*, *P. crustosum*) growing on carrot slices.

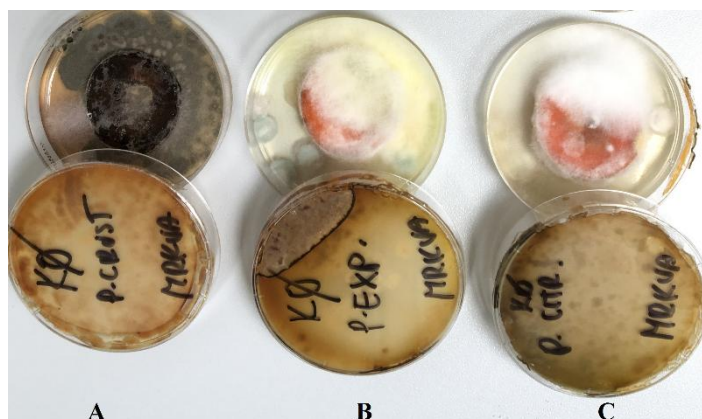


Fig. 1 *In situ* antifungal analysis of carrot inoculated with *Penicillium* spp. (A – *P. crustosum*; B - *P. expansum*; C - *P. citrinum*) without treatment with HPEO (control samples)



Fig. 2 *In situ* antifungal analyses of carrot inoculated with *Penicillium* spp. (1 - *P. crustosum*; 2 - *P. expansum*, and 3 - *P. citrinum*) after their treatment with different HPEO concentrations (A - 125 $\mu\text{L.L}^{-1}$; B - 250 $\mu\text{L.L}^{-1}$, and C - 500 $\mu\text{L.L}^{-1}$)

CONCLUSION

In conclusion, the results of the current study prove the antifungal activities of HPEO in vapor phase against *P. expansum*, *P. crustosum* and *P. citrinum* artificially inoculated on fresh carrot. Demonstrable inhibitory effects of HPEO in vapor phase against the growth of all three *Penicillium* spp. were shown in concentration $\geq 250 \mu\text{L.L}^{-1}$. The results provide information useful for developing alternative preservation methods to enhance the freshness and shelf-life of fresh products (including vegetables, e.g., carrot) using natural antifungal agents. However, additional researches are necessary to confirm the safety of recommended concentrations of HPEO with antifungal activity for human consumption.

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TEXTURE ANALYSIS OF FRANKFURTERS MADE OF INSECT MUSH WITH ADDED CHITOSAN

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Summary: During the research frankfurters made of insect mush with added chitosan were created and were refrigerated for twenty-four days. The changes of the texture and the effects on the shelf-life were examined. At the end conventional meat like texture was successfully created, therefore they have a better chance to consumer acceptance. The possible effect of chitosan on the texture has also appeared. On the contrary the chitosan antimicrobial effect was not present as the storage time was too short.

Key words: insect, frankfurter, texture, chitosan, storage

INTRODUCTION

Because of the rapid growth of the Earth's population the need for food products especially meat is increasing. However the industry won't be able to provide the sufficient amount of protein due to the escalating global environmental problems (e.g. pollution, green house gas emission, deforestation, erosion) (Lundqvist et al., 2008). As a result, we have to lean more on alternative protein sources such as insects. The production of food made of this material is environmentally friendly (Zanolli, 2014) as it has a lower ecological footprint (less land usage, less water usage, less green house gas emission). In addition, it is waste reducing because they can be fed on organic byproducts (Dossey et al., 2016), and they are a healthy protein source as they are high in vitamins, minerals, essential fatty acids and essential amino acids (Bukkens, 1997). Moreover, their exoskeleton is made out of chitin which has several advantages including antibacterial and antifungal characteristics. For example the chitin of mealworm was effective against certain pathogens such as *Listeria*, *Bacillus cereus*, *E. coli* (Shin et al., 2018). The main sources of chitin are crustaceans and they also utilise chitosan which is the digestible, broken down version of chitin (Wang et al., 2004). Due to the beneficial qualities, there are many possible applications of chitin/chitosan in the food industry. For example, in Ruiz-Navajas (2010) research, he coated a ham with chitosan film (with added thyme oil) which resulted in lower lipidoxidation and longer shelf-life. This effect was observed with quail eggs and beef as well (Hahn et al., 2019, Chen et al., 2014). On the other hand, chitosan also has a positive influence on the texture. Researchers found that the chitosan in fish sausages improves the texture, water holding capacity and shelf-life (Chattopadhyay et al., 2019). These results can be excellent starter points of further studies such as mine. In my research I developed a frankfurter made of mealworm mush with added chitosan and then I refrigerated them for 24 days. Through this storage I examined the

possible changes of texture and the effects on the shelf-life. My product development would be quite innovative as until this day there are no similar products on the market.

MATERIAL AND METHODS

Materials

The ingredients consisted of mealworm (*Tenebrio molitor*) from the Bugs World company in Budapest, Hungary; garlic powder, paprika, ground white pepper from Szilasfood Ltd., table salt mixed with nitrite (99.5% NaCl+0.5% NaNO₂), soluprat (tetrasodium-pyrophosphate), Na-ascorbate from Solvent Commercial House Ltd. and chitosan (from Nutriuniversum). I also compared my product to already existed ones which are vacuum packaged turkey frankfurter (55% turkey meat; sample F1), pork frankfurter (83% pork; sample F2) and cooked meatbread (42 % pork and 23% chicken meat; sample F3).

Sample preparation

At first, mealworms were cleaned then they were ground with a Robot Coupe R 201 Ultra E type cutter (3000 1/min) while gradually adding the soluprat. Then three fourth of the mush was passed through a Robot Coupe C80A type sieve (with 0.4 mm diametrical mesh) to get rid of the chitin particles as it has a negativ effect on customer acceptance. This way the two type (with and without of cithin pieces) of insect mush was produced. After this the other ingredients were also added and the following samples was developed: P= frankfurter made of 100 % pork meat (control sample), MC= frankfurter with mealworm mush with chitin pieces, M= frankfurter with mealworm mush without chitin pieces, M0.4= frankfurter with mealworm mush without chitin pieces and 0.4 % chitosan, M0.57= frankfurter with mealworm mush without chitin pieces and 0.57% chitosan. The samples were steamed at 185 °C for 15 minutes in a sous vide oven. After cooling they were vacuum packaged. Samples were stored at 5°C for 24 days. The tests were performed on 0th, 10rd, 17th, and 24th day.

Texture analysis

Stable Micro System TA.XTplus type device was used to monitor the changes in texture during storage after cooking. Warner-Bratzler V head was used (fix counterweight: 50 g, velocity: 2 mm/s). Data were to determine the maximum penetration force (N) using the avarage of nine parallel measurements. For the Texture Profile Analysis (TPA) a cylinder with 35 mm diameter was used (velocity: 2 mm/s, distance: 5 mm) The data was conducted using the avarage of six parallel measurement. The hardness, chewiness and cohesion were observed through the storage time (Friedrich, 2008). The results were compared with product from the market.

Microbiological analysis

The aerobic mesophilic microbe count (CFU/g) was analysed according to the MSZ 3640/4-86 Hungarian standard using TGE agar, with 48 hours incubation period at 30 °C.

Statistical analysis

Data were analysed by ANOVA ($\alpha=0,05$) with IBM SPSS Statistics 25 program the figures were created by Microsoft Excel. The texture analysis was conducted with the help of Texture Exponent 32 program.

RESULTS AND DISCUSSION

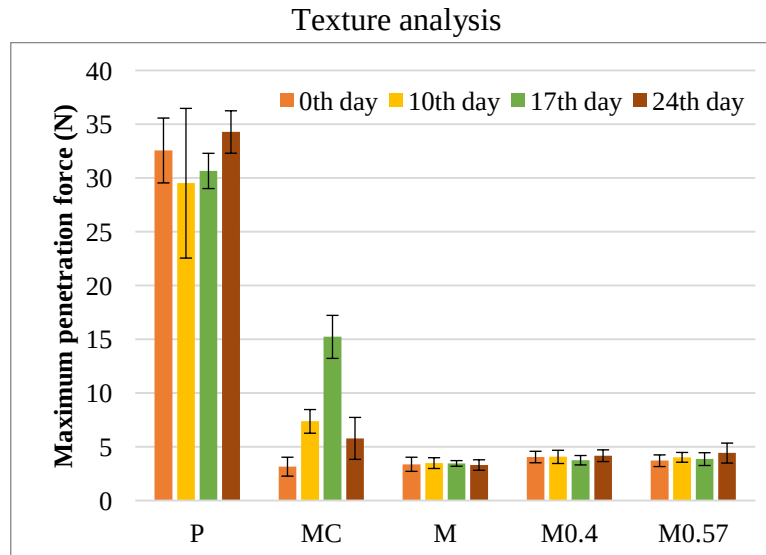


Fig. 1: Maximum penetration force during storage

The maximum penetration force (Fig. 1) shows the force which is needed for biting through the samples. The P sample (control) was eight times harder than the frankfurters with insect because of the strong protein structure the meat created and the sinewy parts. This result is similar to the one of Scholliers and colleagues (2020). On the other hand, the values of MC samples were not consistent as the cithin pieces broke the protein structure therefore they were quite inhomogenous. Furthermore, the values of M, M0.4 and M0.57 just slightly increased during storage. After the ANOVA there was no significant difference ($p > \alpha$) between the mealworm frankfurter with chitosan and without chitosan.

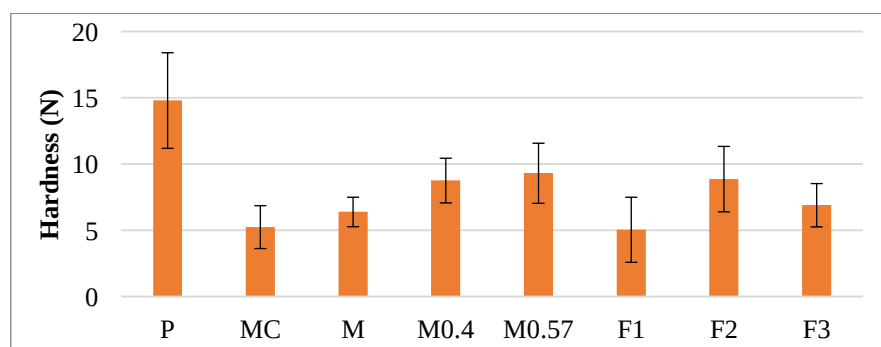


Fig. 2: Hardness values of the samples

These results were supported by the Texture Profile Analysis (TPA). The hardness (Fig. 2) was the highest at the P sample while the MC was the softest due to the inhomogeneous texture. The samples with chitosan (M0.4; M0.57) were slightly higher than M. This can refer to the chitosan ability to improve the texture but to be certain we have to conclude more

experiment in the future. Comparing the products from market to the samples, the F2 (pork frankfurter) sample resembles the most the M0.4 and M0.57. After the statistical analysis there was not a significant difference ($p>\alpha$) between these products.

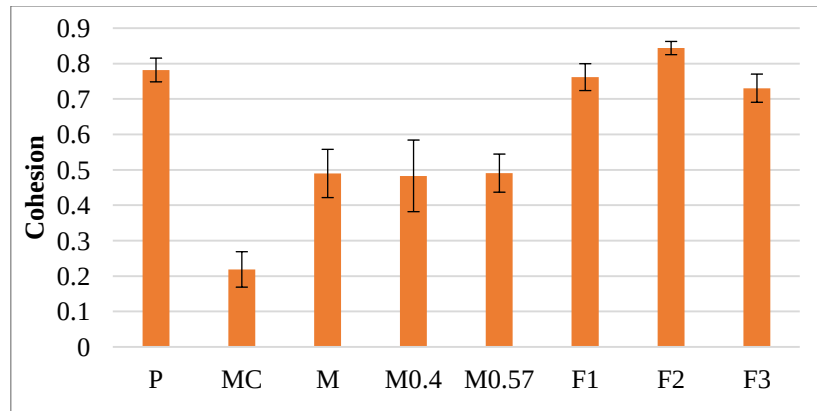


Fig. 3: Cohesion values of the samples

Examining the cohesion results (Fig. 3), it appears that the MC was the least cohesive due to the chitin pieces. The F1, F2, F3 samples had a higher cohesion compared to the M, M0.4; M0.57 samples. According to this the mealworm would create a less cohesive structure than conventional meat but there was not much of a difference. Moreover, the chitosan did not influence the cohesion significantly ($p>\alpha$) as the M, M0.4; M0.57 frankfurter was nearly identical in this regard.

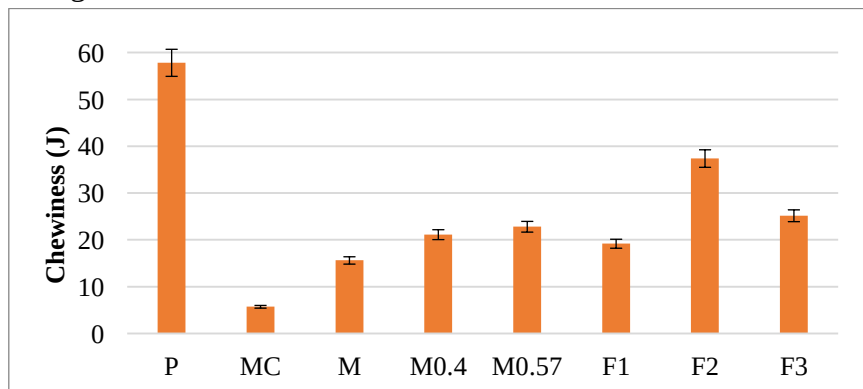


Fig. 4: Chewiness values of the samples

In regard of the chewiness (Fig. 4) the P sample has the highest result due to the strong protein structure. The MC was the least tough as the chitin pieces broke the texture. The samples with chitosan (M0.4; M0.57) were slightly higher than M. This can refer to the chitosan ability to improve the texture but to be certain we have to conclude more experiment in the future. Furthermore, the F1 and F3 were the most similar to M0.4 and M0.57 therefore the texture created by mealworm mush was similar to turkey (F1) and chicken (F3) meat in the regard of chewiness.

Aerobic mesophilic microbe count

Examining the aerobic mesophilic microbe count (Fig. 5), it appears that the starting microbe count was higher in case of the mealworm frankfurters but the heat treatment was successful in every case. During storage the samples did not spoil as the microbe count was around 200 CFU/g at each sample. As a result, the chitosan antimicrobial effect cannot be

declared clearly though my previous studies (Molnár, 2018) showed that the product containing more chitosan had a longer shelf-life. As a result, longer storage is needed to further investigate the antimicrobial effects of chitosan in insect frankfurters.

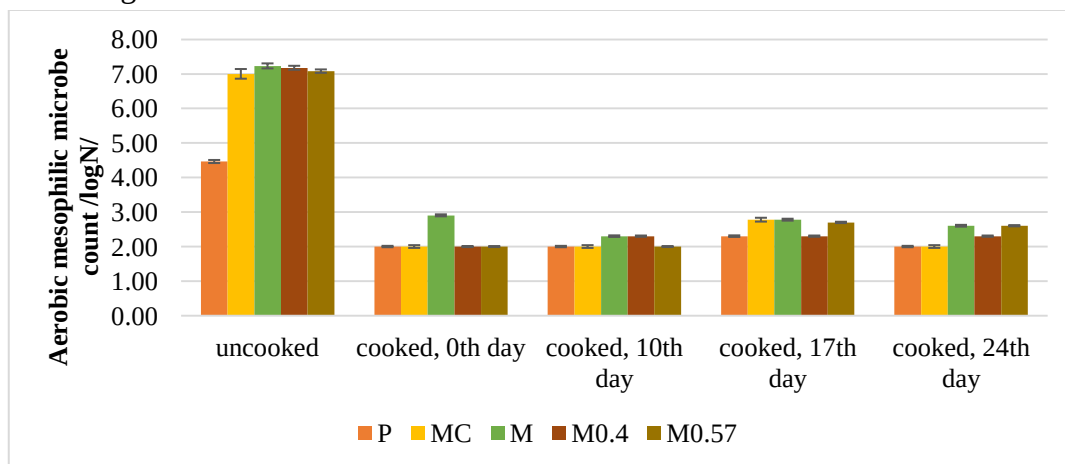


Fig. 5: Aerobic mesophilic microbe count during storage

CONCLUSION

In conclusion, a proper, conventional meat like texture was successfully created in case of the mealworm frankfurters without chitin pieces (M, M0.4, M0.57), therefore they have a better chance to consumer acceptance. Moreover, the possible effect of chitosan on the texture also appears, as the M0.4 and M0.57 samples had a stronger structure. On the other hand, the chitosan antimicrobial effect was not present in the experiment as the storage time was too short and the samples did not spoil at all. Because of this a longer storage would be beneficial in the future. It is also important to mention that this type of frankfurter (made of 100% insect mush) was created only once as we know of (University of Queensland, 2019) and scientific data has not been presented yet, therefore it is extremely difficult to compare these results to others. I hope this research contributes to the potential of insect ingredients in the food industry.

ACKNOWLEDGMENTS

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PREPARATION OF FOOD-GRADE OIL-IN-WATER EMULSIONS USING MEMBRANE EMULSIFICATION TECHNOLOGY - A REVIEW

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Summary: *Emulsions are delivery systems involved in the production of several food grade products such as salad dressings and margarines. To develop oil-in-water emulsions several emulsification processes have been adopted depending on the characteristics of the desired final product and the economical facilities. Conventional techniques for emulsification are commonly used although in the last years great attention has been given to membrane emulsification due to its simplicity, low energy consumption and its practical control of the final emulsion characteristics. In this review, comprehensive information about different emulsion types and technological aspects of membrane emulsification have been discussed. Furthermore, process parameters influencing the emulsification operation have been presented.*

Keywords: *Emulsion, Emulsification, Membrane emulsification, O/W emulsions.*

INTRODUCTION

Emulsions are widely present and used in different industries such as pharmaceutical industry, cosmetics industry, and food industry. It is considered a core system of food formulations that could encounter the new challenges food industries and scientists have been facing to develop suitable delivery systems capable of protecting and controlling the release of food active compounds. Emulsions are composed of at least two immiscible liquids where one of them is dispersed in the other in the form of small droplets and can be classified depending on the droplet size and the composition (Santana et al., 2013). Some examples of food grade emulsions include milk, margarine, mayonnaise, cream, ice cream and butter. Furthermore food emulsions characteristics depend directly on emulsion phases composition, the emulsification procedures and process parameters (Serdaroğlu et al., 2015).

Conventional methods for emulsions production using homogenizers, ultrasonics or stirring do not offer a decent control of the droplet size and stability of the emulsion which may lead to coalescence and to low quality products. The application of membrane emulsification is noteworthy for producing food grade emulsions as it is a simple process with low energy consumption which offer a good control of emulsion characteristics. Optimization and selection of the adequate membrane properties, the composition of dispersed and continuous phases and operating conditions is important for the control of droplet size and droplet size distribution in order to obtain a uniform and stable emulsion (Charcosset, 2009).

EMULSIONS

Emulsions are potential delivery systems commonly used in food industries and are acknowledged to have a considerable importance in food products fabrication (Roohinejad et al., 2018). An emulsion is a system of two or more immiscible liquids. It consists of two phases, a dispersed phase and a continuous phase where the dispersed phase is present as suspension in the continuous phase. Depending on the size, emulsions can be classified into nanoemulsions for droplet size between 1 nm and 100 nm, and miniemulsions for droplet size between 100 nm and 1000 nm which are both thermodynamically unstable and microemulsion for droplet size between 0.5 μm and 100 μm which is thermodynamically stable (Santana et al., 2013). They are used in the fabrication of a wide range of food products such as salad dressings, margarines, cream liqueurs and are as well evolved in the first step of encapsulation of bioactive compounds (Charcosset, 2009; Van Der Graaf et al., 2005).

Depending on the nature of the dispersed and continuous phases, emulsions can be classified into different types (Fig.1) including oil-in-water emulsions (O/W), water-in-oil (W/O) emulsions, double emulsions of water-in-oil-in-water (W/O/W) and oil-in-water-in-oil (O/W/O) (Van Der Graaf et al., 2005).

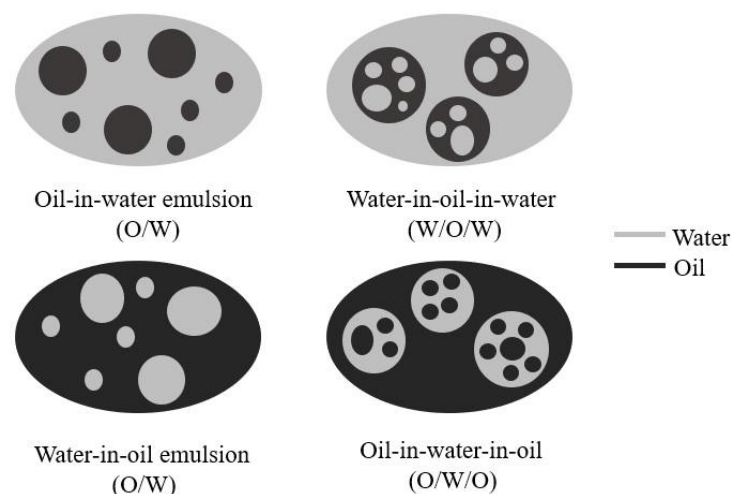


Fig. 1 Different emulsion types

Emulsions face different types of destabilization such as flocculation, coalescence, phase inversion, creaming and sedimentation. Resort to emulsifier has always been considered in order to enhance the stability of the emulsion by the reducing the interfacial tension. This is possible thanks to their amphiphilic nature consisted of both hydrophobic and hydrophilic parts (Rosen & Kunjappu, 2012).

EMULSIFICATION PROCESSES

Emulsification processes can be divided into low energy processes such as the spontaneous emulsification, membrane emulsification, and the phase inversion temperature methods, and high energy processes such as high-pressure homogenizers and rotor–stator devices (Rayner & Dejmek, 2015; Santana et al., 2013).

MEMBRANE EMULSIFICATION

Membrane emulsification has grabbed a lot of attention during the last years as a relatively new technique capable of producing emulsions with better control of droplet characteristics and low energy consumption (Fig.2). Premix membrane emulsification; during which a coarse pre-mix is pressed the membrane resulting in finer droplets and cross-flow membrane emulsification where a dispersed phase is pressed into a continuous phase flowing through microporous membrane pores with the help of an applied pressure (Fig.3), can be adopted for the production of oil in water emulsions (C. Charcosset et al., 2004; Van Der Graaf et al., 2005).

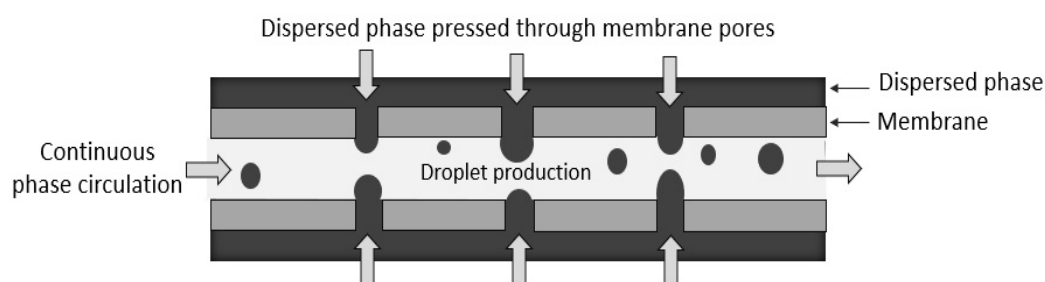


Fig. 2 Membrane emulsification process for O/W emulsion preparation

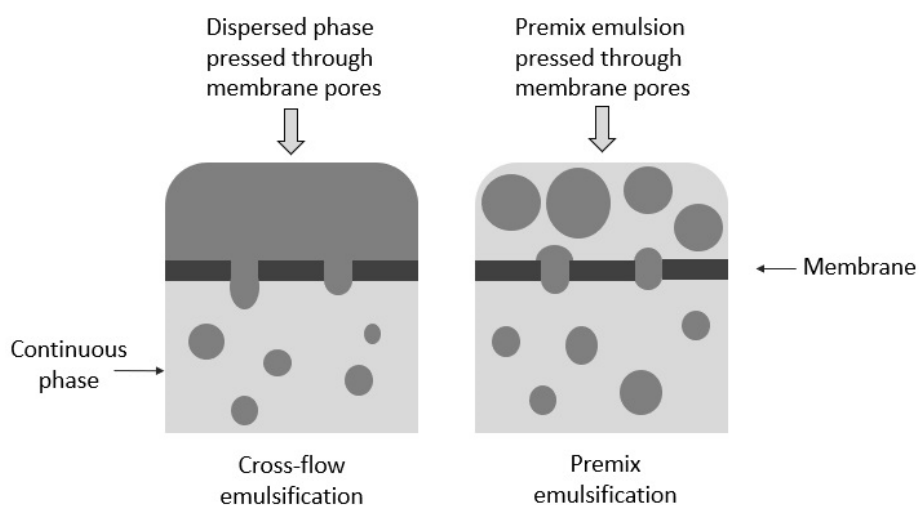


Fig. 3 Difference between crossflow and premix emulsification processes

Several membrane types have been used for the preparation of oil in water emulsions such as, Shirasu-porous-glass (SPG) membranes which are widely used for their narrow pores size distribution and tubular shape and are characterized by a high porosity and mean pore size ranging from 0.1 μm to 20 μm , silicon and silicon nitride micro-engineered membranes and ceramic membranes (Charcosset, 2009; Wagdare et al., 2010).

INFLUENCE OF PROCESS PARAMETERS

During membrane emulsification numerous parameters must be taken in consideration for effective emulsification regarding the final droplet size and stability of the produced emulsion. These parameters can be divided into three categories which are 1) continuous and dispersed phases formulations and parameters, 2) membrane parameters and 3) process parameters.

Among continuous and dispersed phases formulations and parameters we can cite the surfactant type and content and the viscosity of both phases. The selection of the adequate surfactant at the right concentration is crucial for the production of a stable emulsion as it can reduce water-oil interfacial tension, help with the detachment of the droplets from the pores, prevent from coalescences and has an effect on the determination of droplet size (Charcosset, 2009; Schröder et al., 1998; Van Der Graaf et al., 2004). Membrane parameters are the characteristics of the chosen membrane for emulsification such as porosity, pores geometry, pore size and distribution, activated pores and membranes wettability, in fact for the production of oil-in-water emulsions, the membrane must be hydrophilic, and must be hydrophobic in case of water-in-oil emulsion production. The membrane parameters highly affect the droplet size and distribution as well as stability of the emulsion (Abrahamse et al., 2002; Kukizaki, 2009; Nakashima et al., 2000). Regarding Process parameters, transmembrane pressure used to press the dispersed phase through the membrane pores, continuous phase flow rate and dispersed phase flux are essential parameters that should be taken in consideration in order to control the average droplet size and size distribution and to avoid droplets coalescence (Charcosset, 2009).

CONCLUSION

Membrane emulsification technique has been grabbing a lot of attention and has been significantly used for producing food grade emulsions. Among its advantages, membrane emulsification has been proven to be a low energy consuming technique, simple to manipulate and allows a smooth a control of droplet sizes. The challenge of choosing membrane emulsification as a technique for emulsion productions comes with the optimization and selection of phases formulations and parameters, membrane parameters and process parameters. In this context several studies have been conducted to investigate the effect of these parameters in the stability of the produced emulsion and the uniformity of the droplet size.

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QUANTIFICATION OF TOTAL BETALAINS, TOTAL PHENOLIC COMPOUNDS AND ANTIOXIDANT ACTIVITY OF BEETROOT (CORE) EXTRACTS

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Summary: *Evaluation of bioactive compounds from the different variety of fruits and vegetables with emerging technology and their application back into food industries has been noticed over a decade. In this research, the microwave-assisted extraction of bioactive compounds from the core of beetroot and spectrophotometric analysis of them were discussed. In addition, validation of correlations between total betalains, total phenolic compounds and their antioxidant activity was accomplished.*

Key words: *beetroot, total betalain compounds, total phenolic compounds, antioxidant activity, microwave-assisted extraction*

INTRODUCTION

Even though bioactive compounds are non-nutrients, they can modulate metabolic processes and demonstrate positive properties such as antioxidant, anti-inflammatory, antiviral and antimicrobial, anticarcinogenic, and anti cardiovascular effects (Teodoro, 2019). Their scavenging for practical application is limited due to their sensitiveness but can still be regarded as worthy owing to their abilities to improve the sensory properties of foods as well as preserving them.

Beetroots are rich in therapeutic property distributing micronutrients such as vitamin A, B, C, and trace minerals such as iron, copper, magnesium, and calcium (Székely et al., 2014). Beetroot has some benefits for skin and hair proliferation (Akbarhussain, 2018). The most important bioactive compounds in beetroots are betalains, ascorbic acid, carotenoids, phenolic acids, flavonoids (Clifford et al., 2015). Tonalities of yellow and red-violet depend upon the ratio of betaxanthin and betacyanin contents in the extract which in turn is influenced by their stability (Zin et al., 2020a). As both betalains and phenolics are plant secondary metabolites condensed with antiradical scavenging property, they can defend oxidative stresses, cardiovascular diseases, and neurodegenerative disorders like Parkinson and Alzheimer, chronic inflammation, hypertension, atherosclerosis, type 2 diabetes, dementia, chronic inflammation such as liver disease, arthritis and even cancer (Bucur et al., 2016; Clifford et al., 2015).

Utilization of emerging technology in the food industries with several purposes has been proved as highly efficient. Microwave-assisted extraction (MAE) is one of the well-known novel extraction technology with comparative advantages of reaction time reduction and lower or no solvent requirement compared to conventional extraction techniques. Besides,

among the non-ionizing radiations, microwave is longer in wavelength with lower frequency and so energy as per infrared, visible and ultraviolet which makes them compatible for use in food processing (Zin et al., 2020b). The major consideration for our work was how to examine the content of bioactive compounds from the beetroot with pure water which not only being greenest solvent is also the most efficient solvent for MAE.

MATERIAL AND METHODS

Microwave-Assisted Extraction (MAE)

Rhonda type beetroots (*Beta vulgaris L.*) were picked up from the local market. Primarily the beetroots were gently cleaned to remove foreign materials and the innermost part of the beetroot (core) was diced into small pieces after paring. Aqueous extraction was carried out with home-used microwave with pulse mode (15 s on 15 s off). The operation parameters were set up based on the central composite design of response surface methodology (Design - Expert 11 software) whereas the variables are microwave power (300 W – 600 W) and treatment time (1 min – 5 min).

Spectrophotometric Analysis

Spectronic Genesys 5 Spectrophotometer (MILTON ROY, U.S.A) was applied for the quantification of bioactive compounds. For colour compounds measurement, Nilson's method was used for the examination of total betalains colour compounds and calculation was based on Beer-Lambert law;

$$c = \frac{A \times DF}{\epsilon \times l}$$

Where; c is molar concentration, A is real absorbance, DF is the dilution factor, ϵ is molar attenuation coefficient ($E_{1\%}^{1cm} = 1120$ for betacyanin and $E_{1\%}^{1cm} = 750$ for betaxanthin), and l is the path length (1 cm). The total betalain contents (TBC) was calculated by the following equations (Costa et al., 2017);

$$X (\text{Betacyanin}) = 1.095 (A_{538} - A_{600})$$

$$Y (\text{Betaxanthin}) = A_{476} - A_{538} - X/3.1$$

$$Z (\text{Impurities}) = A_{538} - X$$

$$TBC = X + Y - Z$$

Analysis of total phenolic compounds (TPC) by Folin-Ciocalteu method and radical scavenging activity by ferric reducing antioxidant power (FRAP) method were performed in the same way as adopted by Zin et al. (2020c). DPPH (2,2-diphenyl-1-picryl-hydrazyl-hydrate) assay was performed according to the method of Ravichandran et al. (2013). In summary, extract sample 0.1 mL was mixed with 3.9 mL (6×10^{-5} M) of DPPH solution in

methanol and incubated in dark for 30 minutes. Optical density (OD) was then detected at 515 nm and (%) radical scavenging activity was predicted following the equation;

$$DPPH (\%) = \left[\frac{(A_{control} - A_{sample})}{A_{control}} \right] \times 100$$

All analysis were performed as triplicates.

RESULTS AND DISCUSSION

Correlation between the factors and the responses was investigated using randomized central composite design (CCD) of response surface methodology (RSM) modelled by a quadratic function. Table (1) represents the analysis of variance (ANOVA) for each response under the respective factor built by Design Expert Software version (11.0.3). Generally, the model is significant for all outputs (P -value < 0.0001) as shown in the table (1) and the high influence of independent variables on the variation of the target responses was proved by high adjusted R -squared values (Table 1). Furthermore, R -squared values of all responses which are over 0.9 indicate the good correlation between the independent and dependent variables. Considering the model's suggestion, ANOVA was built in the quadratic model for TBC analysis while 2FI model was fitted for the rest analysis. Herein, the operated regression equations by the model for each dependent variable in terms of coding units are as follows;

$$Sqrt(TBC) = + 3.29 + 0.6986 * A + 1.72 * B + 0.8512 * AB - 0.0435 * A^2 + 0.5897 * B^2$$

$$Sqrt(TPC) = + 4.06 + 1.19 * A + 2.23 * B + 0.8844 * AB$$

$$Sqrt(DPPH) = + 2.86 + 0.4887 * A + 1.04 * B + 0.5351 * AB$$

$$Sqrt(FRAP) = + 4.46 + 1.58 * A + 2.83 * B + 2.08 * AB$$

Whereas A means applied microwave power and B means operation time.

Table 1. ANOVA for corresponding responses

Model	TBC	TPC	DPPH	FRAP
P value	< 0.0001	< 0.0001	< 0.0001	< 0.0001
F value	62.41	43.07	34.85	64.71
Sum of squares	24.71	41.48	9.06	80.17
Df	5	3	3	3
Mean squares	4.94	13.83	3.02	26.72
R^2	0.9781	0.9349	0.9207	0.9557
Adjusted R^2	0.9624	0.9132	0.8943	0.9409
Predicted R^2	0.89	0.8499	0.8426	0.8689

Table 2. Operational parameters with coded variables (natural variables)

Run	Factor 1	Factor 2	Response 1	Response 2	Response 3	Response 4
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	A (Power)	B (Time)	TBC (mg/L)	TPC (mg/L)	AA (DPPH, %)	AA (FRAP, mg/L)
1	-1(300)	-1(1)	4.80	4.17	4.03	8.35
2	1(600)	0(3)	18.37	35.87	11.86	48.49
3	0(450)	1(5)	33.42	46.71	17.71	65.61
4	0(450)	0(3)	12.47	17.52	8.31	14.44
5	0(450)	0(3)	10.38	13.35	7.84	16.57
6	1(600)	1(5)	48.37	68.82	25.37	113.45
7	-1(300)	0(3)	7.00	9.18	6.45	7.96
8	1(600)	-1(1)	3.11	5.63	3.80	1.96
9	-1(300)	1(5)	15.81	19.60	8.72	14.67
10	0(450)	0(3)	10.33	13.76	6.20	18.56
11	0(450)	-1(1)	5.84	3.13	4.02	1.77
12	0(450)	0(3)	8.27	9.59	5.47	13.26
13	0(450)	0(3)	9.89	11.68	6.24	17.88

Table (2) represents the thirteen experimental runs with respective factors and responses. To set the process variables, the following equation written by the model in terms of the “usual” coded variables was applied;

$$A = \frac{Power - 450}{150}, \quad B = \frac{Time - 3}{2}$$

As depicted in the table (2), the range of minimum to maximum yields of the outcomes were as follow: 3.11 mg/L – 48.37 mg/L (TBC), 3.13 mg/L – 68.82 mg/L (TPC), 3.8 % – 25.37 % (DPPH), and 1.77 mg/L – 113.45 mg/L (FRAP), respectively. Based on the graphs of the RSM (Fig. 1), the yield of all investigated bioactive compounds increased according to the elevated microwave power and extraction time. The reason for that might be the proper time of swelling the plant matrix was needed for the penetration of microwave which improves the distribution of plant compounds into the solvent and so does its extraction yield (Zin et al., 2020c). Though, in accordance with Ravichandran et al. (2013), the degradation of the betalains occurred later by increasing the microwave power up to 1800 W. Cardoso-Ugarte et al. (2014) discussed that the rapid pigment diffusion occurs under quick contact of microwaves with the matrix followed by the breaking down of the cell membrane. In the same vein, antioxidant activity seemed to increase with longer extraction period at elevated power wattage. According to Casazza et al. (2010), the yield of total phenolic compounds from grape wastes increased with prolonged heating time. From the observation of Ballard et al. (2010), short irradiation time along with higher power nominal was the most effective way for phenolic compound extraction, however, the difference in applied wattage should be considered. Most importantly, the efficiency of power and time setting on the products might be varied with the type of raw material and the specific properties of the examined compounds as well. In the case of antioxidant activity analysis, the results of both FRAP and DPPH methods showed to be highest under the same process condition implying the good correlation between them. Moreover, the correlational tests among the other specific compounds revealed their high connection between each other as well (Fig. 2). Correlations

between the results predicted by the model and the experimental values were presented in Fig. 3. Fig. 4 depicts the desirabilities of the process conditions and their outcomes.

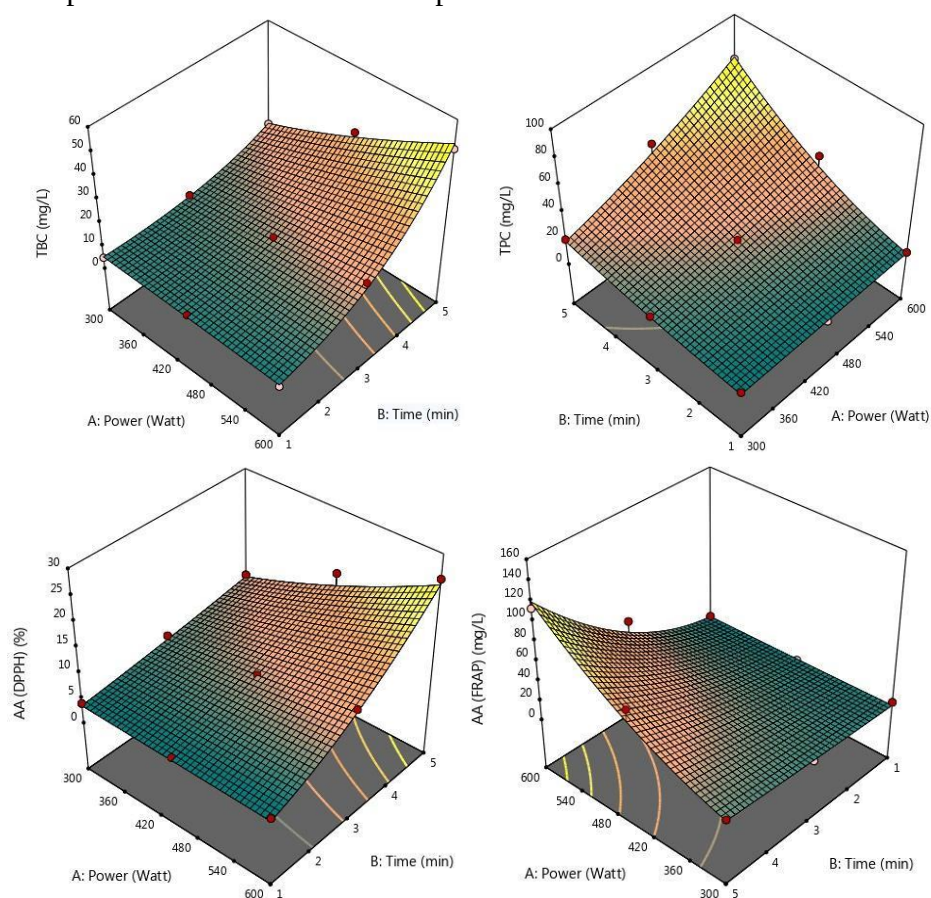


Fig. 1 RSM plots showing the effects of the process variables on their responses

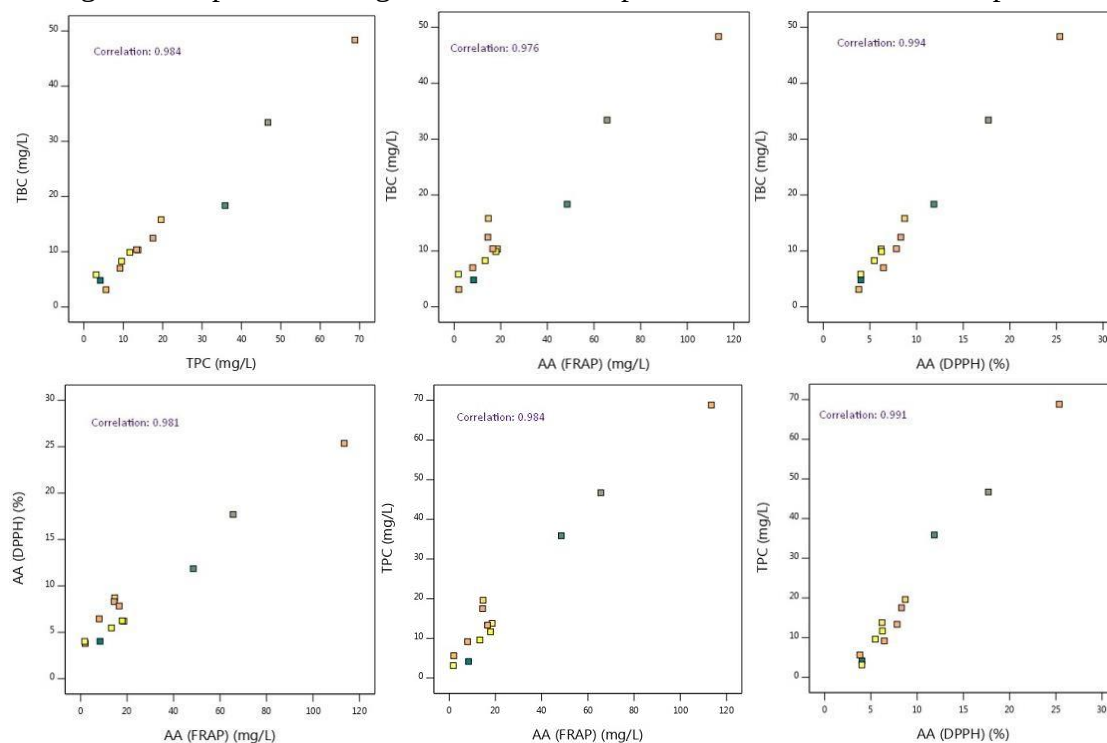


Fig. 2 Correlational tests among the corresponding bioactive compounds

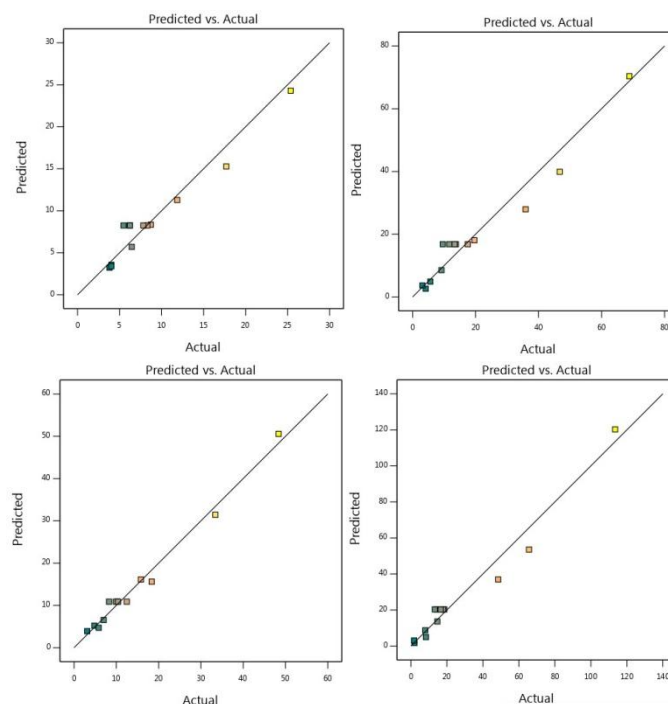


Fig. 3 Correlation between the predicted and actual values of the responses

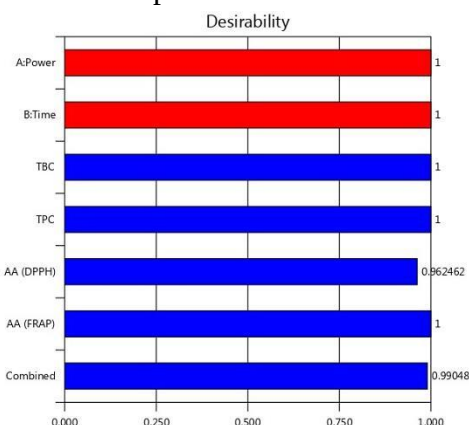


Fig. 4 Desirability test of independent and dependent variables

CONCLUSION

The maximum results of targeted bioactive compounds were achieved at microwave power 600 W after 5 minutes of irradiation time. Under this condition, the values of the recovered individual compounds predicted by the model were; TBC (50.49 mg/L), TPC (70.05 mg/L), DPPH (24.21 %), and FRAP (119.85 mg/L). Attention should be made in the selection of the power and time based on the focused compounds since the model can predict the yield of outcomes within the experimental range only. This work was a part of the examinations of the microwave efficiency in the extraction of TBC, TPC and AA from different parts of the beetroot. Further investigations for the optimization of the extraction of those valuable compounds from the waste parts of beetroots are in progress with different types of the solvent as well. It can be said that the amounts of the outcomes were satisfactory

from the perspective of energy consumption as it accomplished in a noticeably short extraction time. In conclusion, our study encouraged MAE as the novel extraction technology for the practical application in food processing with low energy consumption and time.

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ORIGANUM VULGARE L. AND SATUREYA MONTANA L. ESSENTIAL OILS AND MASTITIS PATHOGENS IN DAIRY COWS

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Mastitis continues to be the single most expensive health-related problem confronting the dairy industry worldwide. Conventional treatment of mastitis is based on intramammary infusion of antibiotics. Given the growing need of developing alternative strategies as a replacement for antibiotics, the objective of this study was to investigate antimicrobial efficacy of in vitro plant-derived molecules against Staphylococcus spp. and E. coli isolates of mastitis origin using data obtained from several studies. Both Origanum vulgare L. and Satureja montana L. essential oils and their components have been found to be effective against both gram-positive and gram-negative pathogens with a wide range of Minimum Inhibitory Concentration (MIC) and Minimum Bactericidal Concentration (MBC). Further clinical trials are needed to assess therapeutical effects, safety and possible milk withdrawal time for essential oils described in this article. In addition, further studies are needed with a larger number of bacterial strains and other types of essential oils in order to gain a clearer picture on the antimicrobial activity of tested essential oils, considering the number of essential oils, their components and causative agents included in this research

Key words: mastitis, essential oils, pathogens, dairy cows, veterinary

INTRODUCTION

Mastitis is defined as an inflammatory reaction of the mammary gland induced when pathogenic microorganisms in the udder produce toxins that are harmful to the mammary gland, being in clinical or subclinical form. Mastitis represents an important problem for farmers because of milk waste, loss of udder functionality, causing sometimes death of the animal [Fratini *et al.*, 2016]. Worldwide, economic losses due to mastitis have been estimated at \$35 billion [Wellenberg *et al.*, 2002]. The main causative agent of bovine mastitis is *Staphylococcus spp.*, with high prevalence in cases of clinical and subclinical manifestations. Regarding environmental mastitis *Staphylococcus spp.* and *Escherichia coli* are the main pathogens responsible for the inflammation and, together with coagulase-negative strains, are the most frequent pathogens causing this disease [Fratini *et al.*, 2016]. Conventional treatment of mastitis is based on intramammary infusion of antibiotics. A large number of commercial antibiotics cause drug resistance, super infections and alteration of enteric microbiota, and negative repercussions due to an increase of the chemoresistance of certain bacterial strains which all lead to serious consequences to public health [Fratini *et al.*, 2016,

Zafalon *et al.*, 2007]. Therefore, it is necessary to develop alternative natural and safe methods for controlling bacterial infections. Medicinal and aromatic plants (MAPs) are well known to have antibacterial activity against different pathogenic agents. Alternative treatments in bovine mastitis therapy were carried out with natural compounds from plants giving interesting results for new phytotherapeutic approaches. Essential oils (EOs) are defined by The International Organization for Standardization (ISO) as “products obtained from a natural raw material of plant origin, by steam distillation, by mechanical processes from the epicarp of citrus fruits, or by dry distillation, after separation of the aqueous phase- if any- by physical processes”. EOs and their constituent's antiseptic properties are well known and many scientific investigations were performed to test their antimicrobial activity in the last twenty years [Hammer *et al.*, 1999, Baydar *et al.*, 2004]. Moreover, plant-derived antimicrobials have been reported not to induce resistance in gram-positive and gram-negative bacteria after prolonged exposure [Domadia *et al.*, 2007]. Today the use of EOs and herbs in protecting livestock from infections mainly in organic farms is becoming a common practice.

Therefore, the aim of the present study was to present the current state of knowledge regarding alternative solution in prevention and treatment of mastitis. This includes *in vitro* antimicrobial activity of two different EOs and their components (*Origanum vulgare* L. and *Satureja montana* L.) against *Staphylococcus* spp. and *E. coli* bacterial strains involved in the pathogenesis of mastitis.

MATERIAL AND METHODS

Publications were searched through Google Scholar search engine and Kobson database, and relevant ones were selected and analyzed. *In vitro* studies that have been reviewed for this study are summarized in Table 1.

Table 1. *In vitro* studies reviewed for this study

Study	Country of work	Reference number in Table 2.
Baskaran et al., 2009.	USA	1
Choi et al., 2012.	Korea	2
Dal Pozzo et al., 2011.	Brazil	3
Norby and Halbert, 2016.	USA	4

Antibacterial activity of *Origanum vulgare* L. and *Satureja montana* L. essential oils and their components on *Staphylococcus* spp. and *E. coli* associated with mastitis in terms of Minimum Inhibitory Concentration (MIC) and Minimum Bactericidal Concentration (MBC) as observed in different *in vitro* studies is summarized in Table 2. The lowest concentration of the antimicrobial treatment that inhibited visible growth of the pathogen after incubation was

taken as the MIC of the treatment. The lowest concentration of the treatment that prevented growth of the organism after subculture following serial dilution and plating was taken as the MBC.

Table 2. Minimum inhibitory and minimum bactericidal concentrations of essential oils/ components as reported by different studies

Bovine isolate (clinical or subclinical)	EO/ component	MIC	MBC (mg/ml)	Reference number
<i>Staphylococcus</i> <i>spp.</i>	<i>Origanum</i> <i>vulgare</i> L., Carvacrol, Thymol	0.16>10 µl/ml	-	4
<i>Staphylococcus</i> <i>spp.</i>	Thymol, Carvacrol <i>Origanum</i> <i>vulgare</i> L.	0.2-0.8 mg/ml 0.2-1.6 mg/ml 0.8-3.2 mg/ml	0.4-6.4 0.2-3.2 1.6-6.4	3
<i>Staphylococcus</i> <i>aureus</i>	Carvacrol, Thymol	5 mg/ml, 6 mg/ml	12	1
<i>Staphylococcus</i> <i>aureus</i>	<i>Origanum</i> <i>vulgare</i> L.	1-4 mg/ml	-	2
<i>E. coli</i>	Carvacrol, Thymol	8 mg/ml	12, 15	1
<i>E. coli</i>	<i>Origanum</i> <i>vulgare</i> L.	0.13>10 µl/ml	-	4

RESULTS AND DISCUSSION

Mastitis continues to be the single most expensive health-related problem confronting the dairy industry worldwide [Baskaran et al., 2009]. Given the growing need of developing alternative strategies as a replacement for antibiotics, this study presents data obtained from several studies which indicate antimicrobial efficacy of plant-derived molecules against *Staphylococcus* spp. and *E. coli* isolates of mastitis origin. Both *Origanum vulgare* L. and *Satureja montana* L. essential oils and their components have been found to be effective against both gram-positive and gram-negative pathogens [Burt, 2004]. *Staphylococcus* spp. including *Staphylococcus aureus*, also *E. coli* have been chosen for this study regarding their importance as contagious and environmental mastitis pathogens, respectively.

As far as *Satureja montana* L. is concerned, only a few studies have addressed the antimicrobial potential of its essential oil, although its most abundant compounds such as carvacrol and thymol are used in large number of studies and are found also in *Origanum vulgare* L. [Serrano et al., 2011]. In a study conducted in the USA, Norby and Halbert (2016) reported that essential oils have excellent antibacterial properties with a range of MICs for the *E. coli* isolates from 0.13 to >10 µl/ml for oregano, 0.63 to 1 µl/ml for carvacrol and 0.63 to >1 µl/ml for thymol, respectively. For the *Staphylococcus* isolates, the MIC values were higher and ranged from 0.16 to >10 µl/ml for oregano, carvacrol, and thymol [Norby and Halbert, 2016]. Similar to these results, in another study from the USA, Baskaran et al. (2009) reported that *Staphylococcus aureus* required the highest MIC and MBC values of tested EOs, with carvacrol being slightly more efficient than thymol with MIC values 5 mg/ml and 6mg/ml, respectively. Also, it is pointed out that *Staphylococcus aureus* was more commonly associated with contagious mastitis than other causative agents [Baskaran et al., 2009].

In a study conducted in Brazil, Dal Pozzo et al. (2011) showed thymol and carvacrol to be more effective against *Staphylococcus* spp. isolates than *Origanum vulgare* L. with lower MIC values ranged from 0.2 mg/ml for thymol and carvacrol and from 0.8 mg/ml for *O. vulgare* L.

However, although EOs may be effective *in vitro*, their use as therapeutic agents is still being questioned. *In vitro* studies showed encouraging results but more tests, namely *in vivo*, are still critical [Gomes and Henriques, 2016]. In their study, Norby and Halbert have suggested that despite the low MICs of EOs, they would need to be applied at high concentrations hence their dilution in milk of the quarters. Such high concentrations may cause damage to udder tissue due to irritation [Norby and Halbert, 2016]. In addition, Venkitanarayanan pointed out in a report on an investigation of potential of natural antimicrobials for controlling bovine mastitis increased signs of discomfort after intramammary application of EOs [Venkitanarayanan, 2013]. Hence, clinical trials regarding intramammary tolerance test are needed first [European Medicines Agency, 2017].

CONCLUSION

This study reveals that *Origanum vulgare* L. and *Satureya montana* L. essential oils and their components against *Staphylococcus spp.* and *E. coli* isolates of mastitis origin have been found to be effective with a wide range of MIC and MBC, with *Origanum vulgare* having the lowest MICs. Considering the number of essential oils, their components and causative agents included in this research, further studies are needed with a larger number of bacterial strains and other types of essential oils in order to gain a clearer picture on the antimicrobial activity of tested essential oils. Especially EOs such as *Thymus vulgaris* L. and *Thymus serpyllum*, EOs with thymol and carvacrol as dominant compounds. Also, further clinical trials are needed to assess their therapeutical effects, safety and possible milk withdrawal time in order to optimize antimicrobial treatment in mastitis therapy.

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TOBACCO PRODUCTION IN 2019. ON FAMILY FARM "DIKŠIĆ"

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Summary: *The tobacco plant (*Nicotiana tabacum* L.) is used in industrial processing, and its main purpose is for smoking, snorting or chewing. This work analyses the tobacco production on Family farm «Dikšić» in 2019. The Family farm is located in Virovitica and the most important agricultural plant on farm is tobacco. Their cultivated area in 2019 was on 25 ha with average yield of 1.6 t/ha in year 2019. Plant of tobacco makes possible the seasonal employment, but year after year the workers are not satisfied with the payment, so more and more tobacco producers are making a transit to machine harvesting. The agroecological conditions in 2019 were not favorable because of hail, which negatively reflected to the leaf yield (t/ha). The tobacco was put into a hydropon on March 12 th 2019 at a temperature of 6 °C and it was replanted in the field on May 7 th 2019. Planting was performed with a semi-automatic four-row planter. Harvest began on July 16 th 2019 and was brought to an end on October 5 th 2019. The average repurchase value of tobacco was 9.52 kn/kg (cca 1.25 EUR/kg).*

Key words: *2019 year, tobacco, agrotechnic, yield, Family farm «Dikšić»*

INTRODUCTION

Tobacco (*Nicotiana tabacum* L.) is a plant used for industrial processing originating from South America. It has great economic significance in various branches. The goal of tobacco cultivation is a leaf which, after drying represent the main raw material for the production of cigarettes, cigars and other similar products intended for human consumption. Perhaps the most important use of tobacco in South American society is as a medicine its mild analgesic and antiseptic properties make it ideal for the treatment of minor ailments (Charlton, 2004; Cobb et al., 2010). Tobacco has a specific effect on the human central nervous system due to its leaf-containing ingredients, especially the toxic alkaloid nicotine, whose content is 0.6 to 3% of the total dry weight of dried industrial tobacco (Gadžo et al., 2011). Lovrinčević et al. (2015) state that the industry of production and distribution of tobacco, cigarettes and alcohol in the literature that studies its pronounced economic and sociological aspects is most often classified in the category of "Sin Industries". According to Cummins et al. (2008) the negative link between tobacco dependence and quality of life is most often confirmed. In the area of Podravina and Slavonia in the Republic of Croatia, American cigarette tobacco is grown, and it is divided into light (Virginia) and dark (Burley). After drying, Virginia in Croatia has an orange-yellow color, a slightly fuller leaf and belongs to the group of semi-aromatic tobacco (Gadžo et al., 2011).

Čogalj and Ivanković (2017) state that the most common in the EU is “Virginia” dried in warm air, followed by “Burley” dried in the shade and “Eastern” or sun-dried tobacco. The authors point out that the most common tobacco type is "Virginia" with about 50% representation, followed by "eastern" or sun-dried tobacco with approximately 35%, and "Burley" with 15%. Miceska and Dimitrieski (2017) point out that Macedonia is a major producer of small-leaf tobacco in the Pelagonia region, the Eastern region and the Strumica-Radovis region. The authors point out the presence of one domestic company Tututnski Kombinat - Prilep and foreign companies Socotab, Alliance One, Strumica Tabak and others, which buy and process oriental (small-leaf) tobacco.

For the production of seedlings in hydroponic system at 1m² come 6.4 trays, or 1.316 openings (Pospišil, 2013). The seeds are peeled, and sowing is done mechanically in the so-called. polyester plates. Sown trays are placed in the pool where the process of swelling, germination and germination begins (Hrgović, 2005). The substrate for the trays mixture is 60% white and 40% black peat. Filling the trays is performed mechanically, the machine evenly fills the trays with the substrate, compacts it as needed and makes holes in the sowing place in which it sows one piled seed (Butorac, 2009).

In the period from 2000 to 2009 in Republic of Croatia, there were on average 5.584 ha of tobacco in Croatia with a yield of 2.0 t/ha with a total production of 11.038 t of tobacco leaves. In the next ten-year period (2010-2019), the total production of tobacco leaf was slightly lower, ie. 9.243 t, due to the reduction of average cultivation areas to 4.671 ha (Bureau of Statistics, 2020).

The aim of this study was to analyze the production of tobacco on the family farm "Dikšić", implemented agrotechnics (sowing of seedlings, transplanting to the field, fertilization, crop protection, harvesting) and analyze the achieved yield and quality of tobacco leaves.

MATERIAL AND METHODS

This study was conducted on the family farm "Dikšić" for the production period in 2019 for the cultivation of light Virginia type tobacco. The family farm "Dikšić" was founded in 2000. There are 2 family members employed on the family farm. The farm have 30 ha of agricultural land, in addition to the production of tobacco grown on 25 ha, corn, oats and wheat are produced.

The seedlings were grown in hydroponics in tunnel 22.05 m long and 4.70 m wide. It is covered with polyethylene foil (nylon) 0.20 mm thick. The hydropon is placed on flat ground, sheltered from the wind, and exposed to the sun, so that it can be ventilated. The pool of the greenhouse is 22.05 m long and 3.48 m wide. A black foil 0.2 mm thick has been placed on the bottom of the pool and on the side of the pool. It is very important to leave a 0.5 m wide path around the pool for passability and various jobs in the greenhouse. At the family farm "Dikšić", the Virginia tobacco variety DH 17 and DH 36 were grown, which have been the most common Virginia tobacco hybrids for several years.

In March 2019, pools with a minimum depth of 20 cm are usually made of fir boards, filled with water to a height of 15 cm. The pool stays closed for a few days due to the heating of the

water, which is why black foil is placed for faster heating. To fill one pool requires 12.000 – 13.000 liters of water, and the water temperature should be above 7 °C. It is very important what kind of water the pool is filled with, the filling of the pool with water from canals, lakes, etc. should be avoided. Before placing the trays in the pool, the initial fertilization is performed with NPK fertilizers 20: 0:2 in the amount of 750 g/m³ of water and KAN in the amount of 150 g/m³ of water.

The fungicide Merpan 80 WDG was preventively added to the nutrient solution in a concentration of 0.3% and 300 g of agent in 1000 liters of water, respectively.

The trays were placed in the pool on March 12 at a temperature of 6 °C. The minimum temperature for placing trays in the greenhouse should not be lower than 5 °C, and during the whole growth the temperature should not be higher than 35 °C.

After placing the trays, the protective nylon is placed at about 50 cm due to the possibility of drops of water falling out of the trays. The protective nylon is left until the seedlings germinate, until the stage of 4 leaves when top-dressing with 50 mg/L KAN is performed.

As the danger of frost passes, the greenhouse is aerated daily in either sunny or cloudy weather, aeration affects the drying of the leaves and reduces the occurrence of diseases and algae in the solutions. Production in hydroponics lasts from 50 to 55 days.

Scything of tobacco seedlings on the family farm "Dikšić" is done 4 times starting on April 20 2019 in the 3-4 leaf stage. The cutter was disinfected with 10% sodium hypochlorite (NaClO) solution before use due to the possibility of disease transmission. This measure achieves more uniform seedlings that are more resistant to disease and ensures better and more uniform growth and development after transplanting into the field.

Previous crop was wheat. In the middle of the March, fertilization with mineral fertilizers P and K (PK 15:30) was done. Also, microelements boron and magnesium in the amount of 600 kg/ha was added. Immediately before planting, KAN fertilization is carried out in the amount of 90 kg/ha. Two weeks after transplanting, KAN was fertilized during cultivation and after 15 days after the first cultivation, the second cultivation was performed with KAN in the amount of 90 kg/ha.

The seedlings were taken out of the greenhouse 24 hours before planting and placed on a means of transport, and during these 24 hours the seedlings get used to the external conditions. Seedlings of better quality 12 - 15 cm, elastic with a well-developed root are used for planting. On the family farm "Dikšić", tobacco seedlings were planted in field on May 7, 2019.

RESULTS AND DISCUSSION

Agroecological conditions have a great influence on the morphological and technological properties of tobacco, on yield and quality of leaves. The average daily air temperatures and precipitation in the period from May to September - from planting tobacco in the field until the end of the harvest, are of the greatest importance for tobacco cultivation (Pospišil, 2013). Beljo et al. (1995) point out that the yield and chemical composition of tobacco leaves depends on climatic conditions, and in particular on the distribution of precipitation during

the year. In addition, a negative correlation was found between leaf yield and nicotine content ($r = - 0.695$). Figure 1. shows the range of average daily temperatures and total monthly precipitation in the period from May to September 2019 in the area of Virovitica.

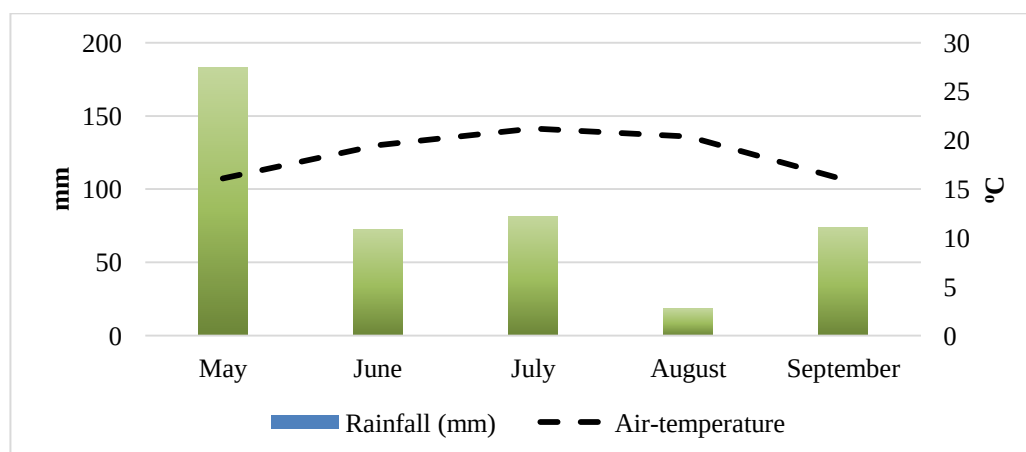


Fig. 1. Monthly rainfall (mm) and mean air temperatures (°C) in 2019 for Virovitica area (Croatian Meteorological and Hydrological service, 2020)

During vegetation period in the field, in addition to machine cultivation, weed control was performed with the herbicide KALIF 480 EC in amount of 0.5 l/ha. One hand hoeing was done, but if necessary, it is done several times during field growth.

After cutting of flowering, tobacco is treated with the contact physiotropic Antak and Itcan (Figure 14) at a dose of Antak of 9 l/ha, and a dose of Ethan of 6 l/ha. One treatment with a mixed dose of these agents is performed. The treatment is done in the afternoon, when the leaves are dry and ready for spraying. Spraying is done with a topper.

The tobacco harvest at the family farm "Dikšić" began on July 16, 2019 and was performed 3 times a week. A dryer with a capacity of 108 frames was picked on Tuesdays, a dryer with a capacity of 105 frames on Thursdays and a dryer with a capacity of 99 frames on Fridays. The harvest was attended by 10 pickers, 2 porters and 3 people who stack tobacco leaves in containers. Harvesting took place according to inserts of sub-selection, sub-selection, middle leaves, undergrowth and top. The harvest lasted much shorter than in previous years due to the occurrence of frequent ice showers. The average yield of dried leaf was about 1.6 t/ha.

In republic of Croatia is a huge problem of the tobacco harvesting workforce throughout Virovitica-Podravina County. Tobacco producers go to a place 50 km away from the field in search of labor. With the emigration of the population in recent years, this problem has become even greater. One of the solution can be tobacco harvesting machine. The advantage of machine harvesting is speed, but that is why manual harvesting gives a higher quality product. In manual harvesting, the picker only picks technologically mature leaves, while the machine does not differentiate between them. After machine harvesting, the tobacco must be rearranged so that it can be placed in frames and dried. The leaves are often broken after machine harvesting. The harvesting machine is adjusted to the height of the harvest and usually picks 3-4 leaves, and maybe more.

The family farm "Dikšić" has three dryers of 108, 105 and 99 frames. After drying, the tobacco is packed in boxes, which can hold 12-14 frames.

Table 1. Purchase price HRK / kg for virginia type tobacco (Source: owner of the family farm "Dikšić" - Mihajl Dikšić - production contract)

Class of tobacco leaf	Purchase price HRK and EUR/kg for tobacco
I	19,67 kn / 2.61 EUR
II	17,34 kn / 2.30 EUR
III	13,18 kn / 1.75 EUR
IV	9,52 kn / 1.26 EUR
V	8,07 kn / 1.07 EUR
VI	5,61 kn / 0.74 EUR

*calculated an 1 EUR = 7,539263 HRK (on 11 th December 2020) – Hrvatska narodna banka (Currency list on 11 December 2020 (<https://www.hnb.hr/>))

Tobacco is bought for purchase by picking sections from pick to top. On the family farm after drying the tobacco is packed in cardboard boxes, the boxes are of different weights from 80 - to 130 kg net dry leaf of tobacco. The technologist prints a travel order for the removal of tobacco. Tobacco is driven every week for redemption on a specific date. The purchasing technologist inspects all the boxes separately, and they are weighed and graded. At the end of the classification, labels are affixed to each box with the manufacturer's data, and the tobacco is paid on the basis of the net yield of the dry leaf and the corresponding class. In 2019, the incentive to grow tobacco was abolished.

The total income in 2019 for the area of 25 acres under tobacco amounted to 320.111,00 HRK (42.459,19 EUR) and the total costs was 232.174,14 HRK (30.795,33 EUR) an additional compensation for ice damage was paid in the amount of 75.513,85 HRK (10.016,08 EUR). A positive financial profit in the amount of 163,450.73 HRK (21.679,93 EUR) was generated from the said account. The highest production costs are set by the labor force, which costs 70.000 to 80.000 HRK (9.284,73 to 10.611,12 EUR) in the tobacco production season.

Price of tobacco depends of the class (Table 1). The average selling price of tobacco of Family farm „Dikšić“ in 2019 was 9.52 HRK/kg (1.26 EUR/kg) , the 4th class of tobacco prevailed, due to significant damage caused by ice.

CONCLUSION

The paper analyzes the production of tobacco on the family farm "Dikšić" in 2019. The study analyzes agrotechnics of tobacco production from cultivation on hydroponics to cultivation in the field, fertilization, harvesting, drying and purchase of tobacco. Agroecological conditions in 2019 were not favorable for tobacco due to hail. Tobacco production is becoming more demanding every year due to labor shortages and high costs. Finally, the economic analysis of

production and the profit from the insurance company due to ice is presented. Tobacco production on a family farm is the most important branch of the farm. It is grown on 25 acres with an average yield of 1.6 t/ha.

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EDN AND RESIDUAL PHYTOTOXICITY- A PECULIAR CASE OF THE EFFECT OF THE PROMISING FUMIGANT ON PLANT GROWTH

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Summary: *Many pathogens causing diseases in crops resulting in yield loss belong to the soil borne pathogen group. Methyl bromide had been known to be widely used in controlling the soil borne diseases. However, due to its environmental impact as an atmospheric ozone-depleting substance, the United Nations (Montreal protocol) made the decision to phase out its use globally (EIP-AGRI Focus Group, 2015). In such a scenario, Ethanedinitrile (EDN; C_2N_2) has been considered as a promising alternative to methyl bromide due to its environment safe properties. So far EDN has been proven to be effective against many soil borne pests and pathogens including nematodes as well as stored food product pests. However, there is a necessity to analyse any possible negative effects of its application in plants as well. Therefore, this study was aimed at analysing the effect of EDN on plant growth. Five different crops (winter rape, cabbage, cauliflower, kale and broccoli) were grown in soil treated with EDN at the concentration of 35 g/m³ for 12 hours and 24 hours, respectively, with 10 replicates per treatment. Similar numbers of plants were grown in untreated soil respectively as control. The plants were monitored closely during the growth period for any signs of phytotoxicity. After 6 weeks, the plants were harvested and a detailed analysis was carried out on the plant growth parameters comprising of measurement of diameter of hypocotyl, fresh weight and dry weight of roots and shoots, respectively. No sign of phytotoxicity was observed in the plants grown in the treated soil. Surprisingly, the growth of the plants under the treated set of conditions was significantly better than those grown under a controlled setting. Mostly, the plants grown in the soil treated for 12 hours exhibited better growth performance than the other set of treatment as well as the control. From this, it can be implied that EDN exhibits fertilization properties in addition to be effective against pathogens and pests.*

Keywords: EDN, fumigant, residual phytotoxicity plant growth

INTRODUCTION

Soil-borne pests and pathogens, known to be particularly difficult to control are the major yield-limiting factors in crop production. Plant parasitic nematodes alone, have been calculated to take away approximately 10% of the World's global agricultural output, causing economic losses valued at over \$125 billion each year (Chitwood, 2003). With limited applied knowledge on suppression techniques, Methyl bromide was one of the most widely used pesticides to control soilborne pests and pathogens. However, the United Nations under the Montreal protocol decided to phase out its use in 2010 in the developed countries and in 2015 all over the world due to its ozone depleting characteristics. Other soil disinfectants such as dichloropropene and methylisothiocyanate are also being banned in more and more European countries (EIP-AGRI Focus Group, 2015). In such a scenario, Ethanedinitrile (EDN; C₂N₂ molecular weight 52.04) has been considered as a promising alternative to methyl

bromide

due to its ozone depleting safe characteristics. Ethanedinitrile is a colourless, pungent gas with a boiling point of -21°C (Pranamornkith et al., 2014). So far EDN has been proven to be effective against not only soil borne pathogens but it has been highly effective against stored food product pests and many other pathogens and pests. Post harvest decay in apples caused by *Colletotrichum gloeosporioides* and *Penicillium expansum* was controlled by fumigation treatment with EDN in combination with gamma irradiation even in a lower dose of irradiation (Cheon et al., 2016). EDN was highly effective as a preplant fumigant in strawberry production against the fungus *Macrophomina phaseolina*, the causal agent of charcoal rot of strawberry as well as the weed *Cyperus rotundus* (Yu et al., 2020). EDN was completely effective in preventing post-treatment recovery of pinewood nematode in pine wood log samples and was much better performing in comparison to other fumigants Phosphine and Sulfuryl fluoride (Seabright et al., 2020). The efficacy of EDN against insect pests was confirmed with the burnt pine longhorn beetle *Arhopalus fesus* (Mulsant) (Pranamornkith et al., 2014). The high efficacy of EDN on the mortality of *Bursaphelenchus xylophilus* individuals was confirmed in field trials under varying physical conditions (Park et al. 2014, Lee et al. 2017). The full registration of EDN for wood fumigation in the European Union is currently in progress. However, temporary registration has been issued to mitigate the European spruce bark beetle (*Ips typographus*) disaster in the Czech Republic (Douda et al., 2020). Since EDN has been showing promising prospects in protection against various pathogens and pests, there is a necessity to analyse any possible negative effects of its application specially in commonly consumed crops. Therefore, the goal of this study was to analyse the effect of EDN application in plant growth and to detect any visible occurrence of phytotoxicity during the growth period of the plants.

MATERIALS AND METHODS

The soil for the experiment was of sandy texture and was collected from the experiment field of Czech University of Life Sciences. The soil was subjected to treatment by fumigation with Ethanedinitrile (EDN) in 2 batches at the concentration of 35 g/m³ of chamber space for 12 hours and 24 hours respectively. The fumigation treatment of the soil was carried out in

cooperation with Draslovka Holding a.s in their fumigation chamber facility located in the Czech Republic.

Pots of size 7x7x6 cm were prepared with 120 g of soil per pot for growing plants. The pots with soil were ventilated for 14 days after fumigation. Seeds of five different crops viz. winter oilseed rape, cauliflower, cabbage, kale and broccoli were used for sowing. All the treatments were carried out in replicates of 10 each. Another set of treatment with non fumigated soil was used as control. The plants were grown for 6 weeks and closely monitored for any visible signs of phytotoxicity. No fertilizer was applied in any of the treatments. After 6 weeks, the plants were harvested and cleaned for analysis. The diameter of the hypocotyl of the plants was measured with the help of a digital caliper. The roots were separated from the plants. Then the fresh weight of roots and the shoots were measured separately by a weighing scale.

The root and shoot samples were then further subjected to drying in an oven at a temperature of 105°C for 24 hours. After thorough drying, the weight of the roots and shoots were measured again separately for determining the dry mass of the samples.

RESULT AND DISCUSSION

Effect on the diameter of the hypocotyl

In oilseed rape, the diameter of hypocotyl was maximum in the plants treated with EDN for 12 hours followed by the controlled plants while it was least in the plants treated for 24 hours (Fig.1). Cabbage, kale and broccoli showed similar results with better performance in 12 hour treatment and the least in the control plants. The hypocotyl diameter in cauliflower plants was the most in 24 hour treatment, followed by control and least in 12 hour treatment which was unlikely the other results.

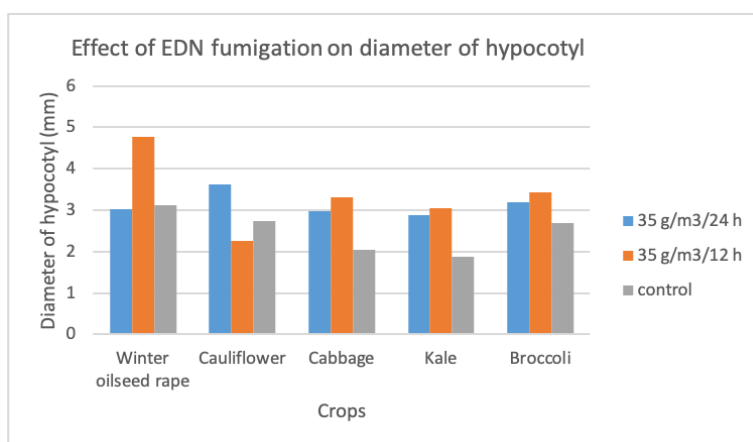


Fig 1. Effect of different treatment of EDN and control on the diameter of the hypocotyl in different crops

Effect on the fresh weight of root

Fumigation treatment by EDN showed varied results on the fresh weight of root in different crops (Fig. 2). Oilseed rape and kale plant roots showed maximum weight in the 12 hour EDN treatment set. However, the minimum weight for oilseed rape was in the 24 hour treatment

while that of kale was in the control set. For cauliflower, cabbage and broccoli roots, the weight was maximum in the 24 hour EDN treatment with the minimum weight in the control plants.

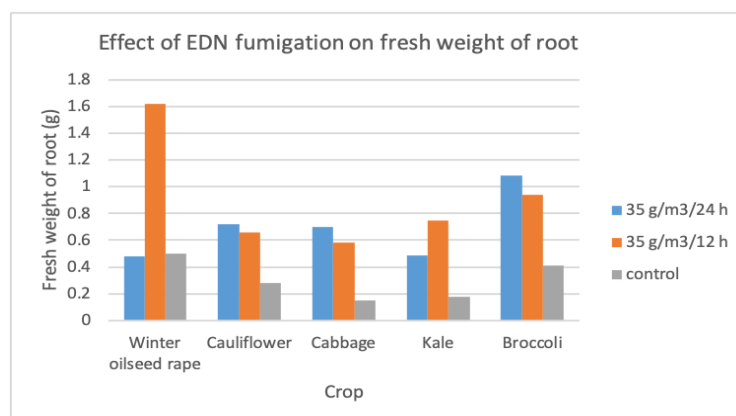


Fig 2. Effect of different treatment of EDN and control on the fresh weight of root in different crops

Effect on the fresh weight of shoot

All the plants except cauliflower showed the maximum weight under 12 hour EDN treatment condition while cauliflower plants showed maximum weight in the 24 hour EDN treatment (Fig. 3). On the other hand, all the plants showed minimum weight in the control set.

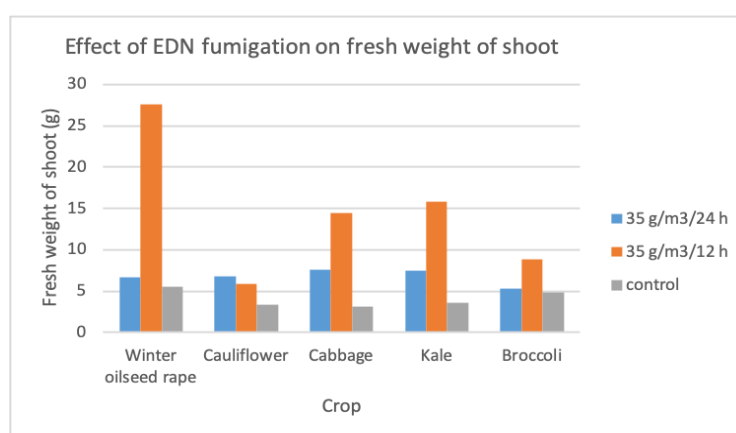


Fig 3. Effect of different treatment of EDN and control on the fresh weight of shoot in different crops

Effect on the dry weight of root

The dry weight of root for oilseed rape, cauliflower, cabbage and kale was maximum in the 12 hour treatment set while the maximum weight in case of broccoli was in the 24 hour treatment (Fig. 4). The minimum weight in all the plants except oilseed rape was in the controlled set. For oilseed rape, the minimum weight was in the 24 hour treatment.

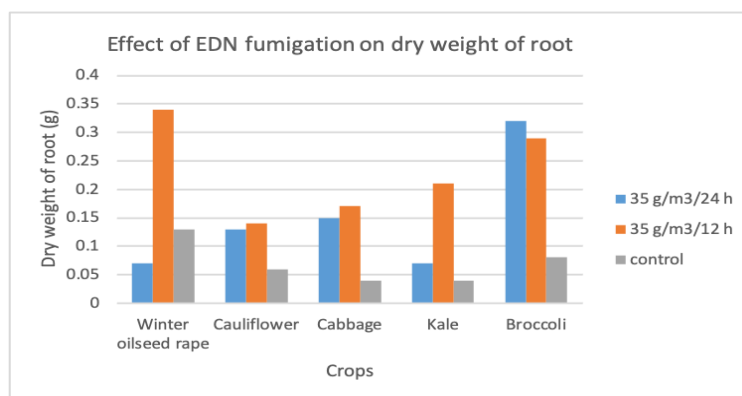


Fig 4. Effect of different treatment of EDN and control on the dry weight of root in different crops

Effect on the dry weight of shoot

All the plants except oilseed rape showed the maximum and minimum weight in the 12 hour treatment and control set, respectively. Although in oilseed rape, the maximum weight was in 12 hour treatment set, the minimum weight was evident in the 24 hour treatment set (Fig. 5).

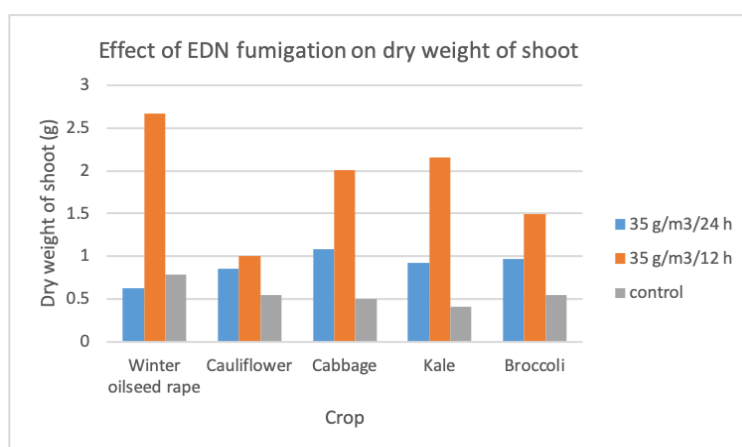


Fig 5. Effect of different treatment of EDN and control on the dry weight of shoot in different crops

In all the crops the treated ones showed better result than the control for all the parameters. It is hypothesised that after the decomposition of EDN, nitrogen is readily available to plants and therefore the plants exhibited better growth results.

Significantly, that there has been the record of phytotoxic activity by fumigants such as chloropicrin (CP) 1,3-dichloropropene (1,3-D), dimethyl disulfide (DMDS) and sulfuryl fluoride (SF) with signs of phytotoxicity like yellow leaves, smaller plant size and visible reduction and deformity of the leaf itself (Huang et al., 2016). However, such incidence has not been recorded in the current experiment. Moreover, other signs like leaf crinkling caused by chlorpropham (Franzaring et al., 2001) were not visible in the plants treated with EDN in either of application timing set. No signs of leaf damage (Kyung et al., 2019) were recorded either.

On comparison of the growth of plants based on the timing of exposure of EDN fumigant, the 12 h application period showed better results than the 24 h application period in all the crops

except cauliflower for almost all the parameters. Thus, according to our findings, exposure up to 12 h of EDN with the concentration of 35g/L-1 seems to be relatively safer in terms of phytotoxicity. This finding is analogous to the application timing of HCN used on garlic as fumigant against *Ditylenchus dipsaci* (Zouhar et al., 2016). On the other hand, it can be implied that in case of greater exposure timing of soil to EDN, there can be the residual activity of the fumigant resulting in the suppression of growth in plants. The exception in the results in case of cauliflower may be due to the higher nitrogen and organic matter requirement. Therefore, the application timing of EDN in this case may play a major role in the growth of the plants.

CONCLUSION

Based on the experiment conducted, it is evident that there are no visible phytotoxic effects on the plants due to the fumigation treatment of soil by EDN. On the contrary, all the plants performed better in the treated soil in comparison to the plants grown in the untreated soil. From the findings, it can be implied that EDN promotes the growth of the plants which can substitute the need for fertilizer application in the soil for plant growth. Majority of the plants showed enhanced growth in the EDN treated soil with 12 hour exposure limit. Therefore, it can be said that EDN treatment of soil with a 12 hour timing of exposure may be the optimum time limit of fumigation for better results in plant growth in addition to protection against pathogens.

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ANTIMICROBIAL AND ANTIOXIDANT ACTIVITY OF SALVIA OFFICINALIS ESSENTIAL OIL FOR ITS APPLICATION IN FOOD MODEL

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The aim of the study was to determine potential antimicrobial and antioxidant activity of S. officinalis essential oil. Antimicrobial activity was determined by disk diffusion method and on a food model by in situ gas phase analysis. Antioxidant activity was assessed by the DPPH method and the MIC was set by the agar microdilution method. The results showed the potential use of S. officinalis as an antioxidant. Also, antimicrobial activity and effect on a food model was demonstrated. This finding confirms that essential oil could have an application in the storage of fruit and vegetables.

Key words: *Salvia officinalis*, essential oil, antimicrobial, antioxidant

INTRODUCTION

The most frequently used method for inhibition of microorganism growth is application of antibiotics. Due to overuse of the antibiotics, the antibiotic resistance of the microorganisms rises (Chang et al., 2019). Development of the resistance leads the scientific society to investigation of natural alternatives with antimicrobial properties. One of the promising methods are essential oils which are becoming more and more attractive due to their safety and high efficiency (Cui et al., 2018). *Salvia officinalis* L. (family Lamiaceae) is a plant with a high content of bioactive compounds such as polyphenols and flavonoids, which are responsible for the antioxidant properties of this plant (Jasicka-Misiak et al., 2018). The sage is used for culinary and medicinal purposes, in the production of a wide range of phytopreparations, such as tea mixtures, nutritional supplements. *S. officinalis* extracts show significant antimicrobial and antifungal activity on various strains of bacteria and fungi (Armini et al., 2020). The aim of the study was to determine the antimicrobial potential of *S. officinalis* essential oil, its antioxidant activity and possible application for preservation of fruits and vegetables.

MATERIAL AND METHODS

Salvia officinalis essential oil was purchased from Hanus, s.r.o.

Microorganisms (*Staphylococcus aureus* CCM 2461, *Enterococcus faecalis* CCM 4224 and *Bacillus subtilis* subsp. *Spizizenii* CCM 1999, three Gram-negative strains *Pseudomonas aeruginosa* CCM 1960, *Salmonella enterica* subsp. *enterica* CCM 4420, *Yersinia enterocolitica* CCM 5671), and yeasts *Candida albicans* CCM 8186, *Candida glabrata* CCM 8270, *Candida krusei* CCM 8271 and *Candida tropicalis* CCM 8223) (Brno) for analyses was obtained from the Czech collection of microorganisms. The biofilm forming bacteria *Bacillus subtilis* and *Stenotrophomonas maltophilia* were obtained from dairy industry. They were identified by 16S rRNA sequencing and MALDI-TOF MS Biotyper.

Antioxidant activity of the *S. officinalis* essential oil was determined by method based on scavenging of free radical 2,2-diphenyl-1-picrylhydrazyl (DPPH) described in Kačániová et al. (2020).

The disc diffusion method was used for detection of antimicrobial activity. Bacteria were inoculated 24 hours aerobically on Tryptone soy agar (TSA) at 37 °C and the yeasts were inoculated 24 hours aerobically on Sabouraud dextrose broth (SBD) at 25 °C. The cell culture was diluted to optical density 0.5 according to McFarland standard. The disc diffusion method was performed on Mueller Hinton agar (MHA) for bacteria and Sabouraud dextrose agar (SDA) for yeasts. 100 µl of cell suspension was added on the Petri dish (PD) and smeared by a L-shaped cell spreader. Subsequently, the 6 mm paper discs were placed on the PD and 10 µl of *S. officinalis* essential oil was added. Prepared PD were incubated aerobically for 24 hours. Bacterial cultures were incubated at 37 °C and yeast cultures were incubated at 25 °C. After incubation, the inhibition zones were measured. The analyses were repeated in triplicate for each strain.

Minimal inhibitory concentration (MIC) was determined by agar microdilution method. Bacteria were inoculated aerobically for 24 hours in MHB at 37 °C and yeasts in SDB at 25 °C. Cell culture was diluted to optical density 0.5 according to McFarland standard. 50 µl of inoculum was transferred to each well of 96-well microplate and subsequently, 100 µl of MHB and SDB were added. Serial dilution of the *S. officinalis* essential oil was prepared and 100 µl of essential oil was added to final concentration in range 400 µl.ml⁻¹ to 0.1953125 µl.ml⁻¹ and the content in the wells were mixed. MHB, SDB and essential oil were used as the negative control. MHB, SDB and inoculum were used as a control of maximal growth. Absorbance of the samples was evaluated according to Kačániová et al. (2020).

Antimicrobial activity in gas phase was tested on biofilm-forming bacteria. The analysis was performed on a carrot, a potato, and an apple. All samples were cut into 5 mm slices. Warm MHA were poured to 60 mm PD and also to its lid. The potato, the carrot, and the

apple were placed to PD and the inoculum of bacteria was added onto fruit and vegetable. Essential oil was diluted to 500, 250, and 125 μ l and added to 55 mm sterile filter paper. The filter paper was dried for 1 minute to evaporate the ethyl acetate and placed to PD. The distance of 2 mm between the paper and the agar was ensured. Finally, the PD was hermetically enclosed with a lid containing solidified agar. PD were incubated at 37 °C for 7 days.

The disc diffusion method, MIC, and the antimicrobial activity in gas phase were conducted in triplicate and the statistical variability of the data was processed using software Microsoft TM Excel®. Results of the MIC value (concentration causing 50 % and 90 % reduction of bacterial growth) was determined by logarithm analysis. Statistical evaluation of the antioxidant activity of the obtained data was performed using GraphPad Prism 8.0.1 (GraphPad Software Incorporated, San Diego, California, USA). One-way analysis of variance (ANOVA) followed by Tukey's test was used for statistical analysis.

RESULTS AND DISCUSSION

The antioxidant activity of the *S. officinalis* essential oil was determined at 26.99 % $\pm$ 4.42 % that corresponds with a medium-high antioxidant activity. Jasicka-Misiak et al. (2017) determined the activity of *S. officinalis* essential oil by DPPH method at 41.25 % $\pm$ 3.10 %. Garcia et al. (2016) determined activity of radical scavenging at 73.00 % $\pm$ 3.00 %. Adrar et al. (2016) detect the level of antioxidant capacity at 20.93 % $\pm$ 3.45 % and Ghadermazi et al. (2017) determined the antioxidant activity of *S. officinalis* at 31.73 % $\pm$ 1.29 %. Pereira et al. (2018) state the antioxidant activity at 34.80 % $\pm$ 3.30 % and Pavič et al. (2019) determined the activity of *S. officinalis* essential oil at 26.90 % $\pm$ 0.91 %.

The results of the *S. officinalis* essential oil antimicrobial activity using the disc diffusion method are shown in Table 1. The results obtained from the experiments including statistical analyses and any comments should be presented graphically or as tables in this section.

Table 1. Zones of inhibition of the microorganisms by *S. officinalis* essential oil.

Microorganism	Inhibition zone (mm)
<i>P. aeruginosa</i>	10.00 ± 2.00
<i>S. aureus</i>	3.00 ± 1.00
<i>Y. enterocolitica</i>	3.33 ± 0.58
<i>B. subtilis</i>	4.00 ± 1.00
<i>E. faecalis</i>	1.33 ± 0.58
<i>S. enteritidis</i>	1.33 ± 0.58
<i>C. glabrata</i>	3.33 ± 0.58
<i>C. tropicalis</i>	2.33 ± 0.58
<i>C. albicans</i>	8.67 ± 0.58
<i>C. krusei</i>	5.33 ± 0.58
<i>S. maltophilia</i> (biofilm)	1.33 ± 0.58
<i>B. subtilis</i> (biofilm)	1.00 ± 0.58

Kačániová et al. (2018) tested the antimicrobial activity of the *S. officinalis* essential oil with the disc diffusion method on the microorganisms isolated from chicken breast. They determined the inhibition zones for *P. fluorescens* 4.33 ± 0.58 mm, *K. rhizophila* 4.67 ± 0.58 mm, *Y. enterocolitica* 12.33 ± 1.53 mm, *E. coli* 8.67 ± 0.58 mm, and others. Pop et al. (2016) determined the inhibition zones for *P. aeruginosa* 9.50 ± 0.08 mm and for *S. aureus* 10.20 ± 0.47 mm. Oliviera et al. (2018) measured inhibition zones for *C. albicans* 12.40 ± 0.31 mm, *C. tropicalis* 2.90 ± 0.50 mm, *C. glabrata* 4.20 ± 0.08 mm, and for *S. aureus* 14.10 ± 0.10 mm. Tafi et al. (2020) examined inhibitory effect of *S. officinalis* essential oil on *S. iniae* and detected the inhibition zone 15.61 ± 0.25 mm which was similar to tested antibiotic erythromycin 16.07 ± 0.58 mm. Ed-Dra et al. (2020) measured the inhibition zones for *S. enteritidis* 8.10 ± 0.10 mm.

The concentrations of *S. officinalis* essential oil that inhibit 50 % and 90 % growth of the microorganisms (MIC 50 and MIC 90 values), are shown in Table 2.

Table 2. Minimal inhibitory concentration of *S. officinalis* essential oil that caused 50 % and 90 % inhibition of microorganism growth.

Microorganism	MIC 50 ($\mu\text{g.ml}^{-1}$)	MIC 90 ($\mu\text{g.ml}^{-1}$)
<i>P. aeruginosa</i>	1.56	3.13
<i>S. aureus</i>	12.50	25.00
<i>Y. enterocolitica</i>	6.25	12.50
<i>B. subtilis</i>	6.25	12.50
<i>E. faecalis</i>	25.00	50.00
<i>S. enteritidis</i>	25.00	50.00
<i>C. glabrata</i>	6.25	12.50
<i>C. tropicalis</i>	12.50	25.00
<i>C. albicans</i>	1.56	3.13
<i>C. krusei</i>	3.13	6.25
<i>S. maltophilia</i> (biofilm)	100.00	200.00
<i>B. subtilis</i> (biofilm)	100.00	200.00

Kačániová et al. (2018) determined MIC 50 $6.25 \mu\text{g.ml}^{-1}$ and MIC 90 $12.50 \mu\text{g.ml}^{-1}$ for *Y. enterocolitica* equally at *S. officinalis* essential oil. Oliviera et al. (2018) determined MIC 50 and MIC 90 of *S. officinalis* essential oil for *C. albicans* at 6.25 and $12.50 \mu\text{g.ml}^{-1}$ respectively, for *C. tropicalis* 12.50 and $25 \mu\text{g.ml}^{-1}$ respectively, for *C. glabrata* 6.25 and $12.50 \mu\text{g.ml}^{-1}$ respectively, and for *S. aureus* 5.28 and $10.56 \mu\text{g.ml}^{-1}$ respectively. Tafi et al. (2020) determined minimal bactericidal concentration of *S. officinalis* essential oil at $5.185 \mu\text{g.ml}^{-1}$. Tadi et al. (2020) determined MIC 50 and MIC 90 for *S. aureus* and *S. typhimurium* 15.62 and $31.14 \mu\text{g.ml}^{-1}$, respectively 19.50 and $39.00 \mu\text{g.ml}^{-1}$. Pavič et al. (2019) MIC for *B. subtilis* $15.6 \mu\text{l.ml}^{-1}$, for *S. aureus* $62.50 \mu\text{l.ml}^{-1}$, and for *P. aeruginosa* $10.82 \pm 0.02 \mu\text{l.ml}^{-1}$.

The gas phase antimicrobial analysis on biofilm-forming bacteria *B. subtilis* and *S. maltophilia* was determined by the *in situ* method. *S. officinalis* essential oil showed an inhibitory effect on *B. subtilis* applied on carrot at concentration $250 \mu\text{g.ml}^{-1}$, on potato at $125 \mu\text{g.ml}^{-1}$, and on apple up to a concentration of $500 \mu\text{g.ml}^{-1}$ (Fig. 1). Inhibition against *S. maltophilia* was observed on carrot sample at concentration $250 \mu\text{g.ml}^{-1}$, on potato also at concentration $250 \mu\text{g.ml}^{-1}$, and on apple showed an effect at the lowest tested concentration $125 \mu\text{g.ml}^{-1}$ (Fig. 2).

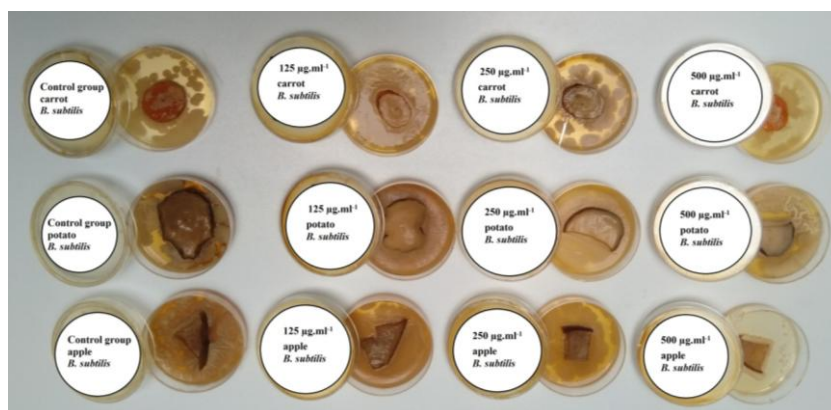


Fig. 2 *In situ* antimicrobial analyses of vegetables and fruit with *B. subtilis* in vapor phase with essential oil *S. officinalis*.

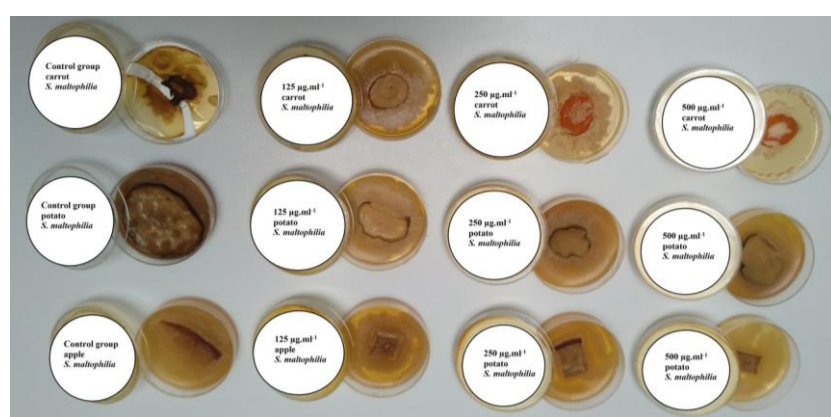


Fig. 3 *In situ* antimicrobial analyses of vegetables and fruit with *S. maltophilia* in vapor phase with essential oil *S. officinalis*.

CONCLUSION

The results of our study showed the potential use of *S. officinalis* as an antioxidant. The antioxidant activity was $26.99 \% \pm 4.42 \%$ which represents a medium-high antioxidant activity. That confirms its suitability as a source of natural antioxidants. An antimicrobial effect on various microorganisms has also been demonstrated, with the most significant effect recorded in *P. aeruginosa* and *C. albicans*. Their zones of inhibition were the largest and their MIC were the lowest among the tested microorganisms. The antimicrobial effect in the food model was significant with *S. maltophilia*, but inhibitory effect also occurred in *B. subtilis*. This finding confirms that *S. officinalis* essential oil could find an application for the preservation of fruit and vegetables in the future.

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DEVELOPMENT OF MUFFIN FORMED FLEUR PASTRY CAKE ENRICHED WITH BLOOD POWDER, AS A FUNCTIONAL FOOD

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Summary: *Animal blood is a valuable resource, which is not utilized in a value-added way by the industry like other animal by-products, however it has plenty benefit in terms of sustainability and human health especially protection against iron deficiency anemia. Animal blood is perfectly adequate for providing special functions, which are necessary for functional foods, based on scientific sources. In this paper animal blood products (whole blood powder, blood plasma powder and red blood cell powder) was substituted for egg powder in fleur pastry molded into muffin form. Sensory quality characteristics (texture, color) and dry matter content were measured for four days. Based on the results, powdered blood products are adequate for substituting for egg powder in fleur pastry cakes and develop functional foods.*

Key words: *animal blood, by-product, functional food, iron, iron deficiency anemia, sustainability*

INTRODUCTION

Animal blood is usually handled as hazardous waste and annihilated or in best case used for producing feed raw material. However, it may play a role in sustainability in value-added further processing and more sustainable products are preferred by consumers (Floros et al., 2010). Blood is a liquid connective tissue. Plasma is the intercellular part, in which RBC (red blood cell) fraction, or hemoglobin fraction is suspended. The RBC fraction becomes its name from red blood cells because they make the most part of this fraction, but it contains the other blood cells like leucocytes, lymphocytes and macrocytes as well as the most part of thrombocytes (Liu et al., 1996). Animal blood may be a perfect raw material for developing and producing common and functional foods from many aspects (Duarte et al., 1999; Hsieh, 2011; Ofori & Hsieh, 2012; Toldrá et al., 2012; Bah et al., 2013).

The iron content of the human body varies between 4-5 grams. Its most important property comes from its metal ion transition characteristic: the $\text{Fe}^{2+} \leftrightarrow \text{Fe}^{3+}$ transition by electron uptake-downtake has a key role. This property allows the catalysis of oxidoreductive processes as the cofactor of oxidoreductase enzymes. Most of the iron is in hemoglobin and myoglobin, but it also can be found in several proteins (Fe-S proteins) and enzymes (peroxidase, catalase, hydroxylases, flavin enzymes). It is also important in the transport of oxygen and carbon dioxide (Hegedűs & Stefanovitsné, 2012; Egyed Miklós, 2007). Table 1 shows the average daily iron requirement of people at different stages of life.

Table 1. Average daily iron requirement [mg] of people at different stages of life [months or years]

Age	Infant/Child (mg)	Men (mg)	Wome n (mg)	Pregna nt women (mg)	Breastfeedi ng women (mg)
7–12 months	11				
1-3 years	7				
4-8 years	10				
9-13 years		8	8		
14-18 years		11	15	27	10
19-50 years		8	18	27	9
51+ years		8	8		

Primarily in foods of animal origin (meat, liver, fish, poultry) about 40% of the iron is found in hem bonding which can thus be better absorbed than the non-hem bonding iron form in plants (Hegedűs & Stefanovitsné, 2012). The prevalence of anemia caused by iron deficiency is around 30% globally. Symptoms of anemia include: physical and mental decline and miscarriage in pregnant women. The most sensitive groups are children over six months of age, pregnant women, menstruating women and the elderly (EFSA, 2006). Currently, 1.6 billion people suffer from iron deficiency anemia. Among children alone, iron deficiency anemia affects 64.6% of children in Africa, 47.7% in Asia, 16.7% in Europe, 39.5% in South and Central America, 3.4% in North America and 28% in Oceania (Miller, 2013).

Hundred g of porcine whole blood powder is able to cover the daily essential amino acid needs of an average 70 kg adult male with the exception of methionine (Ockerman & Hansen, 2000), which can be supplemented with cereals. The iron content of porcine blood is 1490.14 mg/kg, in bovine blood the iron content is 2810.62 mg/kg expressed in terms of blood matter content (Sorapukdee & Narunatsopanon, 2017; USDA, 2018). In Russia and other post-Soviet states, blood chocolates and blood-colored candies are popular, especially for iron-deficient children. These are bought for them with a medical recommendation. Like in Russia, Hungary's soil conditions are not the best in terms of iron, so it would be expedient to help eradicate childhood iron deficiency in Hungary with similar products.

In the case of by-products, it should be pointed out that, given their nutrient composition, they can generally be considered as excellent sources of functional foods, either on their own or by separating into their fractions. However, due to our culture and trends, they are unlikely to become a major product within the foreseeable period of time, the use of by-products directly, in themselves, should not be forced. Functional foods are foods that contain more

than average quantities – often resulting from artificial enrichment – of components that have been demonstrated to have a positive health impact on the average normal consumption of the food concerned (e.g. vitamin-enriched foods); or foods that contain less than the expected quantity from a harmful component on the average normal consumption of the certain foods (e.g. reduced-fat products). A functional food may also be a food intended for a specific consumer group (e.g. people who suffer from a food allergy, bodybuilders). Innovations in the development of functional foods can be achieved primarily by using new materials, but it is possible to innovate through the introduction of new technologies or new packaging and innovative use of traditional materials and technologies.

For the technology to design the structure of a cake, we must first understand the structure of the foams. As a result of mechanical stress, the foaming agents create a foam with an airy structure, the volume of which increases rapidly in the first stage of the whipping, then this growth slows down, finally, the air bubbles grow and merge when exceeding the critical value and the foam collapses (Somogyi, 2017). The dispersion medium of the foams can be liquid or solid, while the dispersed phase can only be gaseous. In the case of sponge-cake dough, the proteins of the egg white (ovalbumin, conalbumin, ovomucoid, globulin, ovomucin) are usually responsible for the formation of the foam structure. Under strong mechanical action, gas bubbles are dispersed in the liquid egg, then the water-soluble proteins are denatured, adsorbed on the surface of the colloidal solution, and the bubbles are surrounded by the formed film layer. Foaming is affected by the following factors: 1.) pH, 2.) salinity. Flour and sugar are responsible for foam stability and foam hardness in bakery products. The discharged sponge flan is essentially a solid foam (Szücs, 2018).

In the present case, the egg was replaced with blood. Since albumin-type proteins typically make up the majority of proteins in the blood, especially in plasma, the blood is similarly capable of forming a colloidal structure. However, the big advantage over eggs is that they are not allergenic. Globally, 5-8% of children and 2-19% of adults are affected by food allergies or intolerances, thus the blood is also suitable for the development of functional foods for allergen replacement purposes (Csurka et al., 2019).

The purpose of this paper is to briefly summarize the importance of blood in the development of functional foods and to describe the product development of mixed dough pastries enriched with several types of blood powder and to quantify its properties (colour, texture, dry matter content) and techno-functional properties (pH, water activity). The functionality of the product obtained during the trials is demonstrated in 1.) the role of blood in the replacement of the allergic egg 2.) the effect of increasing the protein content and 3.) in the case of hemoglobin powder and total blood powder, the effect of increasing the iron content. The latter is particularly important because for children, who have a very high presence of iron deficiency anemia, a muffin is an easily acceptable product. It was also intended to carry out a storage experiment: From day 0, we examined the change in the quality of the muffins during storage at room temperature for four days, since the expected consumption also takes place within this period of time at the latest in the case of a fresh pastry.

In further experiments, we investigated multiple mixed dough pastries in several forms but this study will only deal with the results of the products prepared in the form of muffins. In the whole research, our aim is to compare the impact of the administration of egg powder, blood powder, blood plasma powder and hemoglobin powder into mixed dough pastries on quality-determining properties, so that we will be able to develop appropriate functional foods suitable for a well-defined consumer group using animal blood.

MATERIAL AND METHODS

Material

Muffin was prepared based on a common fleur pastry recipe, in which the protein content of egg was substituted by different powdered blood products. The fat content of egg was substituted by sunflower oil. Special ingredients, which were used during this investigation, were the followings:

- Hemoglobin powder 92B (Sonac Burgum B.V., Netherlands)
- Plasma powder 70B (Sonac Burgum B.V., Netherlands)
- Vepro 95 phf whole blood powder (Solvent Kereskedőház Zrt., Hungary)
- Egg powder (Capriovus Kft., Hungary)

Whole blood powder and powdered form of two main blood products (plasma powder, hemoglobin powder) were investigated for substitution of egg powder as well as for developing functional product features (iron content, protein content). Besides, we were curious about the change, which is caused by adding whole blood powder in ten times amount than required to substitute eggs according to the recipe. Recipes are shown in table 2.

Ingredients	Muffin with egg powder (control)	Muffin with whole blood powder	Muffin with whole blood powder (ten times the dose)	Muffin with hemoglobin powder	Muffin with plasma powder
Fine flour (g)	300	300	300	300	300
Margarin (g)	120	120	120	120	120
Powdered sugar (g)	250	250	250	250	250
Lea cocoa powder (g)	30	30	30	30	30
Dutch cocoa powder (g)	10	10	10	10	10
Baking powder (g)	12	12	12	12	12
Vanilla aroma (g)	2	2	2	2	2
Milk with 2.8% fat (ml)	100	100	100	100	100
Egg powder / Whole blood powder / Hemoglobin powder / Plasma powder (g)	28	14,16	141,6	13,8	18
Water (g)	88	92,06	92,06	92,66	88,8
Sunflower oil (g)	-	10,64	10,64	10,64	10,64

Table 2. Ingredients [g or ml] in recipes of different investigated products

Method

Determination of the recipe

The aim was to substitute egg protein by blood protein. Beside this the iron content, as another functional feature, was considered. At the same time fat content of egg has been substituted as well with sunflower oil. Water content of egg was replaced by water in case of each recipe. Thus the flour pastry matrix surrounding the proteins was closely the same as with egg. Raw material specifications and food nutrition database (USDA, 2018) were used for calculating the recipe. We calculated with average M sized eggs, which are 58 g of weight.

Muffin preparation

Ingredients were mixed well then a foamy mass was made from the mixture by a household mixer. Mass was filled into muffin forms. Muffins were baked at 180 °C for 20 min in a convection oven. Muffins were stored for four days, because a fresh pastry is expected to be stored for maximum four days before consumption. The storage conditions correspond to the average expected home storage of the pastries: at room temperature in imperfectly sealed (folded) plastic bag.

Water activity measurement

Water activity was measured by Novasina LabMaster-aw neo type instrument (Novasina AG, Switzerland) that requires very small sample amount and can fully control the temperature between 0 and 60 °C during the measurement. Measurements were performed at room temperature to control the integrity of samples for relevant data detection.

Dry matter content measurement

3-5 g sample measured by Kern ABJ-NM/ABS-N (Kern & Sohn GmbH, Germany) analytical balance was put into an open Petri dish. Then samples were dried at 120 °C until constant mass in a laboratory drying oven (Labor Műszeripari Művek, Hungary). Samples were cooled in a desiccator then their residual mass was measured by the analytical balance. Each sample was measured in triplicate.

Color measurement

Minolta CR-400 (Konica Minolta, INC., Japan) chroma meter was used for the reflection color measurement. The measurement is based on the fact that any color can be generated by the mixture of three ones defined by light wavelength. The ratio of these three different wavelength lights are plotted in a coordinate system called CIELAB color space. The color coordinates can be numbered making colors analysable. In case of blood there is a relation between the red color (and chroma) and the iron content hereby between the physical and chemical attributes.

The instrument was calibrated with a standard white etalon. Each sample was measured three times. Measured attributes are the followings: redness/greenness (a^*), yellowness/blueness (b^*) and brightness (L^*).

Texture measurement

Texture of muffins was measured by Stable Micro System (SMS) TA. XT Plus texture analyser (Stable Micro System, United Kingdom). A computer controlled the movement of the upper rod. Different probes are helping measurements with different shear, compressive and torsional stresses in different directions lasting different times while behavior of samples was investigated.

Muffins had upward widening cylindrical shape with 5 cm basis diameter and an average height of 10 cm. The probe was a p/75v steel cylinder plate, which started to press the sample with 0.002 N force. The extent of the load was 40% of sample height. The pre-test speed of the probe was 10 mm/s and test speed and post-test speed 0.5 mm/s. 'Hold until time' type compression measurement method was used with 30 s compressing time. Then the sample relaxed. Sampling frequency was 50/s. Each sample was measured three times.

Crust hardness was calculated from the force, which caused 40% deformation in the sample height, and the size of this deformation. Cake flexibility was calculated from the ratio of original height and relaxed height of muffins after 30 s 40% deformation.

Statistical analysis

Measurement results were evaluated by IBM SPSS statistic v25 (IBM Corp., Armonk, NY) and Microsoft Excel 365 version: 2010 (build: 13328.20356) software. To detect the effect of raw material and storage time on redness-greenness, yellowness-blueness, brightness

and dry matter content in the first analysis and on crust hardness and cake flexibility in the second analysis, we carried out multivariate analysis of variance (MANOVA), that can compare the means of different sample groups of related variables. According to Levene's test, the homogeneity of variances was slightly violated ($p < 0.05$). The normality of residuals was checked by Shapiro-Wilk test ($p > 0.05$). The value of the unexplained variance rate (Wilks's lambda) was evaluated. The homogeneous groups were separated by Tukey post hoc test.

RESULTS AND DISCUSSION

Results of water activity measurement were difficult to evaluate because of the large variance. Water activity results of samples at the day of baking and on the first day of storage were between 0.52 and 0.66 in case of all products. Only the water activity of muffin containing ten times the amount of blood powder, which can substitute the protein content of egg, was significantly different from the other samples. It was 0.76, which was probably caused by the water binding capacity of whole blood powder because the rehydrated proteins of blood powder could bind proportionally more water and thus less water could evaporate during baking.

As it was shown in figure 1, results of dry matter content were not significantly different from each other except samples containing ten times the amount of blood powder. Most of sample groups typically changed similarly. It is interesting that dry matter content increased in case of the samples with whole blood powder but the dry matter content of other sample groups decreased, these got wet. This observation may be connected with syneresis, the aging of colloid system. Dry matter content of sample groups with different raw material was not significantly different.

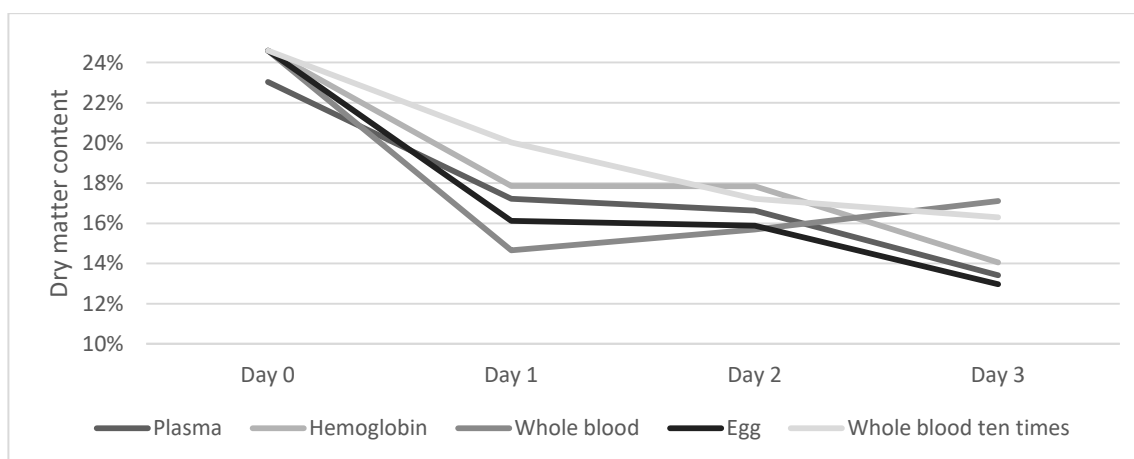


Figure 1. Dry matter content [%] of muffins with different raw materials and different storage time

However, cocoa could well mask the color change caused by blood, the color difference between samples containing hem-iron and samples without hem-iron was well perceptible by naked eye. At the same time this color difference was not disturbing, moreover, it seemed like it were a muffin with extra cocoa. The color of muffins with different raw materials and different storage time can be seen in Figure 2-4.

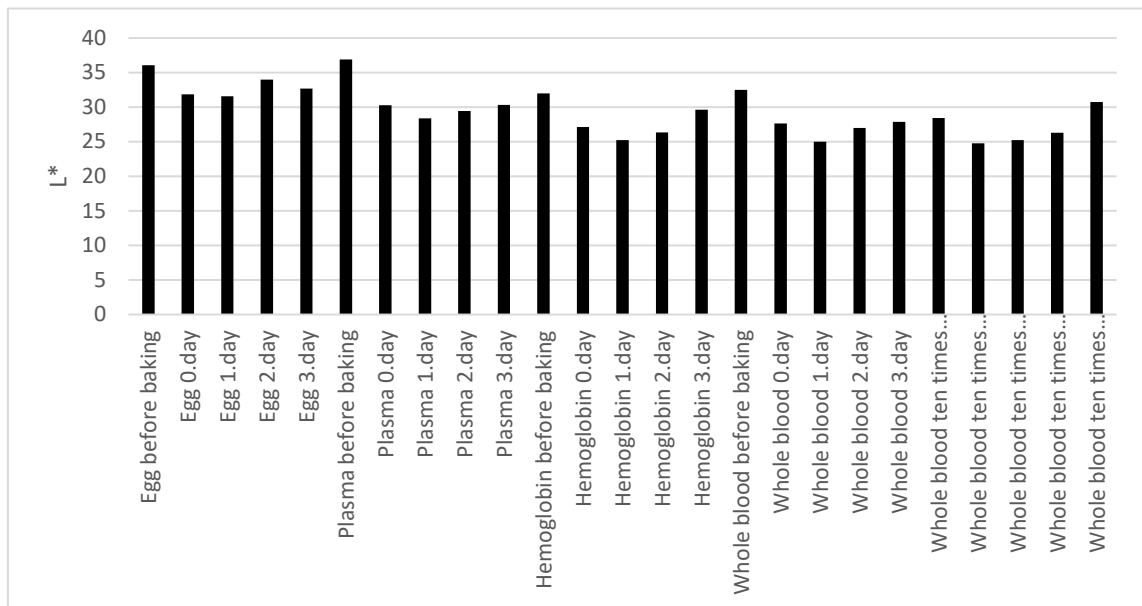


Figure 2. Brightness (L^*) of muffins with different raw materials and different storage time

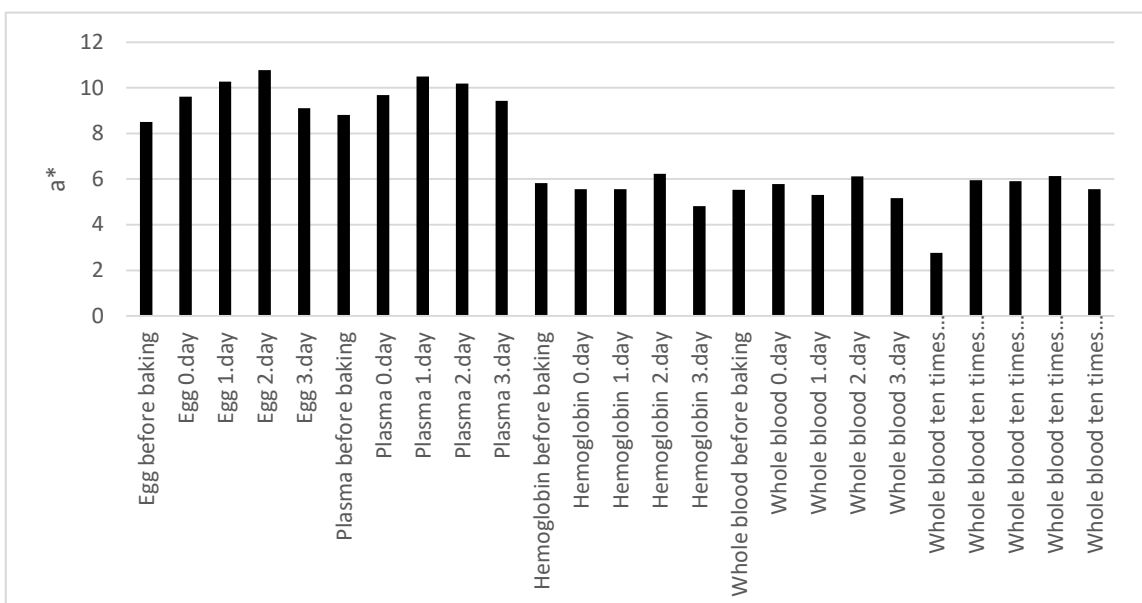


Figure 3. Redness-greenness (a^*) of muffins with different raw materials and different storage time

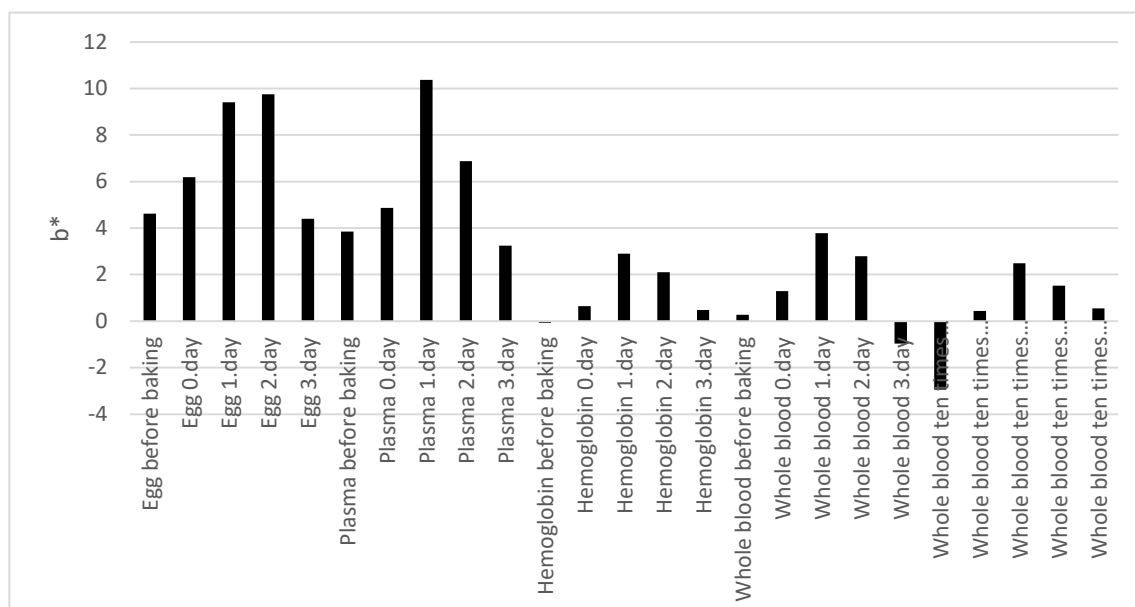


Figure 4. Yellowness-blueness (b^*) of muffins with different raw materials and different storage time

The first MANOVA investigated the effect of raw material and storage time on attributes of color and dry matter content. The overall MANOVA result was significant for both the raw materials and and storage time (Wilks's lambda: 0.004, 0.005 with $p < 0.001$, respectively). The two-way interaction was also significant (Wilks's lambda: 0.068 with $p < 0.001$). Tukey post hoc test (at 95% confidence interval) could separate the sample groups with different storage times based on color coordinates especially yellowness-blueness, but the change was not linear. The change in the dry matter content was well traceable. Tukey post hoc test could separate three groups: there were samples of the third and second day in the first group, there were samples of second and first day in the second group and there were samples of the day of baking in the third group. In the case of raw materials, muffins with egg powder and plasma powder were separated from each other and from the other samples based on brightness and yellowness-blueness by the Tukey post hoc test. Muffins with egg powder were the brightest, but muffins with plasma powder were also significantly brighter than the other muffins, which were similarly bright. Muffins containing egg powder and plasma powder were separated in one group from the other samples based on their redness-greenness. Muffins made with a raw material containing hem-iron were redder.

As it was shown in Figure 5-6, texture was similar in case of most recipes except the recipe containing ten times the amount of blood powder, which can substitute the protein content of egg. It was interesting to make the statistical evaluation with and without these sample groups.

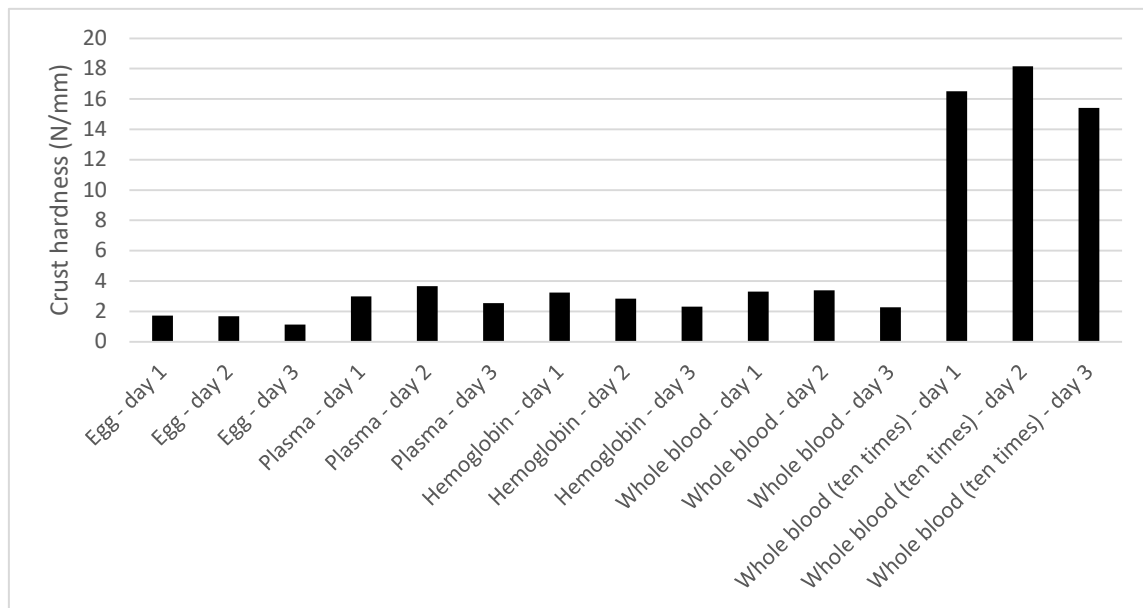


Figure 5. Crust hardness [N mm^{-1}] of muffins with different raw materials and different storage time

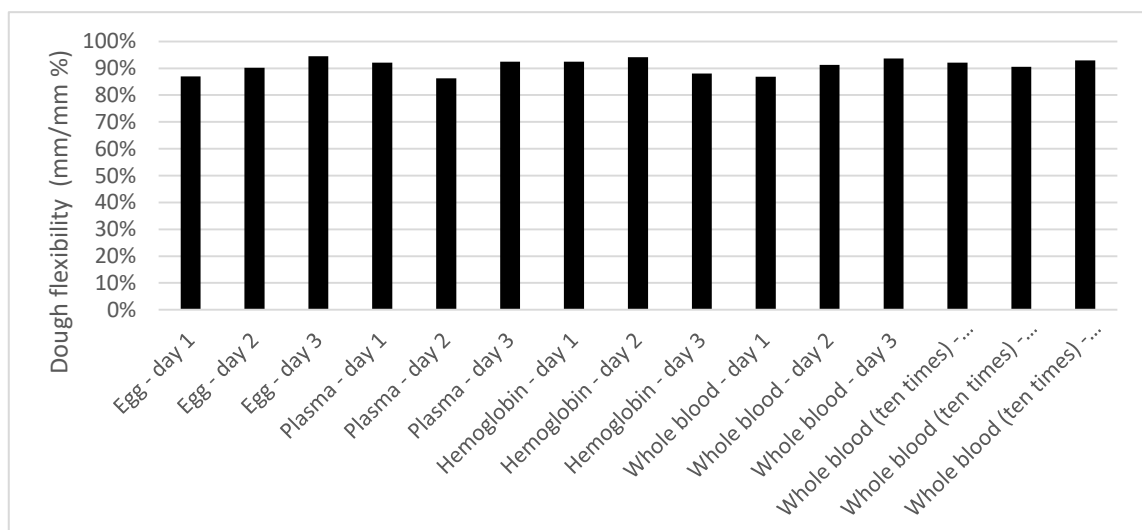


Figure 6. Cake flexibility [$\text{mm mm}^{-1} \%$] of muffins with different raw materials and different storage time

In the second MANOVA test we investigated the effect of raw material and storage time on attributes of crust hardness and cake flexibility. The overall MANOVA result was significant for raw material (Wilks's lambda: 0.02 with $p < 0.001$) when the muffin containing ten times the amount of blood powder was included in the analysis. The two-way interaction was not significant. The Tukey post hoc test (at 95% confidence interval) could separate only two groups based on crust hardness: one group contained the muffin containing ten times the amount of blood powder and the other group contained all the other sample groups, and for the rest of the measured characteristics all the samples were included in only one group. The overall MANOVA result was significant for raw material and storage time (Wilks's lambda: 0.38, 0.19 with $p < 0.03$) when the muffin containing ten times the amount of blood powder was not included in the analysis. The two-way interaction was not significant. Tukey post hoc test (at 95% confidence interval) could separate the sample group stored for three days from the sample groups with other storage time based on crust hardness, as well as could separate the samples that contained egg powder from the samples with other raw materials again based on the crust hardness. The difference between the samples of the third day and samples with other storage time can be explained by syneresis caused by drying, which had been significant at the third day. According to this test all other cases formed only one group.

CONCLUSION

Product development was successful. However, this topic needs further investigations. Texture analyses have to be carried out with more geometrical forms of flour pastry cakes as well as organoleptic quality has to be validated by a sensory analysis according to the GSP.

Based on the results of this paper powered blood products are adequate to substitute egg in flour pastry cakes. Although the difference is measurable and perceptible, some quality attributes of the product rather improved (more cocoa-like color, more crisp crust) as well as an allergic compound of cake is substituted. The quality of plasma powder fortified muffin is similar to common egg muffin. Hem origin iron content of whole blood powder and muffin fortified with hemoglobin powder can decrease the amount of childhood iron deficiency anemia in an acceptable form for children.

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DRUG QUALITY ASSESSMENT ON IVERMECTIN PREPARATIONS CIRCULATING IN ADDIS ABABA PHARMACEUTICAL MARKET

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Summary: *This was a cross sectional experimental study carried out in the Addis Ababa from November 2017 to April 2018 and all available Ivermectin brands, batches and preparations was randomly collected from veterinary shops, clinics and from dry port as consignment crosscheck. The samples were then subjected to appropriate qualitative and quantitative tests to determine the documentation fitness, dossier evaluation, identity and amount of active ingredient/API present in the product using HPLC. 19 samples formulated as generic (1), brand (14) products, (19) batches and (4) different preparations representing different drugs were collected and subjected to evaluation and analysis.*

Among the total samples collected during the survey, 19 (100%) passed primary drug quality test that is Physical characterization, packing information, labeling and registration status. 19 (100%) passed identity test, 3 (15.8 %) and 16 (84.2%) samples were failed and passed dossier evaluation test (weight variation and content uniformity test) respectively and 3 (15.8 %) and 16 (84.2%) samples were and passed HPLC % of API content determination in relation to label claim respectively. All of the three brands were below limit specified by the United States and British pharmacopeia, giving rise to a failure rate of 15.8% and all of which are from formal source and fulfill primary drug quality testes.

The finding indicates the presence of substandard veterinary pharmaceutical products among the investigated products in the central market of Ethiopian even slightly higher than the level estimated by WHO and various monitoring and surveillance reports. It is impossible to judge drug quality by physical appearance and registration status alone as all the failed drugs in this particular study fulfill primary drug quality test and registered. Thus it seems the quality compromising is due to sub standardization rather than counterfeiting.

As recommendation due attention should be paid by concerned bodies particularly federal Animal feed and veterinary drugs control and administration authority. So that it could reduce to the level below the current finding. In the addition the media must step up their efforts to raise awareness about the dangers of substandard and falsified medicines and work to stop such products from entering the supply chain. Furthermore laboratory testes has to be strengthened as primary physical test and registration status are not enough to assure drug quality, safety and efficacy. Quality control at the critical points (dry port while import and before market authorization for domestic pharmaceutical companies) has to be established.

Key words: *Drugs, quality control, Regulatory system, Counterfeit, Substandard, Pharmaceutical market, Ethiopia*

INTRODUCTION

Parasitic diseases are repeatedly challenging livestock productivity, due to these reasons producers use anthelmintic drug to boost animals reaching their full production potential. Several anthelmintic with different modes of action are available in the market for the control of helminthosis; however, intensive and indiscriminate use of these drugs to suppress infestation has resulted in rapid selection for resistance (FAO, 2004). To tackle such challenges as measure for disease constraint we need to have drugs which are effective, safe, fulfill quality standards set by official pharmacopeia and available for rural farmers throughout the year (WHO, 2018).

Ivermectin is a mixture of two chemically modified avermectins that contain at least 80% of 22, 23-dihydroivermectin- B1a and >20% 22, 23-dihydroivermectin-B1b. It is a highly lipophilic, but is practically insoluble in water (Steel, 1993), and it remains the leading worldwide antiparasitic agent for livestock. It has exceptional potency against endo and ectoparasites at extremely low doses; this accounts for its large margin of safety (Aranzazu *et.al*, 2009). The therapeutic interest in this compound justifies research to establish analytical methods for the determination of IVM in pharmaceutical preparations. It is highly active against a wide spectrum of nematode species, including most larvae and adult forms. All important gastrointestinal and lung nematodes are susceptible to the drug, including sensitive mites, ticks, biting flies, and parasitic dipteran larvae. In dogs, IVM is also active against developing larvae of *Dirofilaria immitis* and is used in heartworm prophylaxis (Aranzazu *et.al.*, 2009).

Ivermectin resistance in the canine heartworm, *Dirofilaria immitis* had been reported (Wolstenholme *et al.*, 2015). Concurrently, reports of reduced embryostatic effects of IVM on *Oesophagostomum volvulus* in Ghana and Cameroon have raised concerns that IVM resistance may evolve in human parasites (Nana Djeunga *et al.*, 2014).

In Ethiopia, various stakeholders forward accusations and counter accusations on the treatment failures in general (Morton *et al.*, 2007; Melaku *et al.*, 2017). It is attributed to presence of substandard veterinary drugs as smuggling and misuse of veterinary drugs involving anthelmintics is a wide spread practice in the country.

There are several studies conducted on efficacy evaluation and anthelmintic resistance using in vivo and in vitro assays (Egualle *et al.*, 2009). The assessment of the drug quality as a product using physicochemical methods is a missing gap. To the best of our knowledge there are no studies conducted so far to evaluate the quality, API content of veterinary medicinal products of Ivermectin in Ethiopia. Therefore this study was conducted with an aim to assess the physicochemical quality of selected brands of Ivermectin marketed in the country with the following Objective.

To carryout drug quality assessment of different brands, batches and preparations of Ivermectin preparations circulating in the legal pharmaceutical markets of Addis Ababa.

MATERIALS AND METHODS

Facilities, Chemicals and reagents: HPLC equipped with a UV-vis detector, Fem hood, and double distilled water, Volumetric flasks, graduating cylinders, beakers, Duran bottles, Erlenmeyer flasks, pipette of different volumes, spoon, spatula, piston and mortar, ultrasonicator, orbital shaker, vortex mixer, magnetic stirrer, syringe filters of 0.45 μ , disposable syringes, funnels, filtration units, sample vials of 1.5 ml capacity and column with ODS silica gel, whatman filter paper, labeling paper, parafim paper, methanol, Acetonitrile Acetic acid, IVM reference material 200mg and double distilled water were used in the analysis process.

Physicochemical Assessment of drug quality

Identification test;- Retention time (RT) is to compare the retention time (RT) of the IVM sample with that of the RT of the standard. (USP-NF, 2015).

High performance liquid chromatography method for Assay of API

Cleaning agent: (1:1 of Acetonitrile/organic and water/inorganic)

Mobile phase preparation: a filtered and degassed mixture of acetonitrile, methanol, and water at ratio of (106:55:39) was prepared (USP, 2015)

Standard preparation: 40 mg of USP IVM reference standard (RS) was accurately weighed twice, transferred to each 100 ml volumetric flask, added some methanol, sonicated for five minute, methanol was added to the volume to obtain 0.4 mg per ml, mixed on a vortex mixer to dissolve well and transferred to HPLC vial labeled (RF1 and RF2)

Sample preparation: (1) For Injectable preparations; Injectable 1% (five 50 ml Ivermectin vial was mixed into beaker using magnetic stirrer, 4 ml of sample IVM was taken to obtain 0.04 w/v based on the label claim, poured in three separate 100 ml volumetric flask labeled (S1, S2 and S3), added some methanol in each, sonicated for 5 minute, methanol was added to the volume mixed on a vortex mixer to dissolve well and transferred to HPLC vial). (2) For tablet preparations; Tablet 5 mg (weighed and finely powder of twenty (20) tablets. made average, Transferred an accurately weighed portion of the powder equivalent to 20 mg IVM (3.6761 g) to three separate a 50 ml volumetric flask. Added some methanol, sonicated for 10 minutes, diluted with methanol to the volume, mixed with vortex mixer and filtered with whatman filter paper putting on the funnel, and discarded the filtrate. (USP, 2015). Afterward, the assay preparation was taken in to a syringe, pass it through 0.45 μ syringe filter and filter it in to $\frac{3}{4}$ of 1.5 ml HPLC vial, and make it ready for assay determination (USP, 2015).

Chromatographic system:

Table 1.Chromatographic condition for IVM API determination (USP, 2015), AVICO – manufacturer Standard working Certificate (QWI 11/01/L Mec)

Set parameters	USP standard	Manufacturer/AVICO
Column	5 µm, 4.6 mm *25 cm	5 µm, 4.6 mm *25 cm/ C18
Detection	UV 245-nm	UV 245-nm
Flow	1.5 ml/min	1.2 ml/min
Column Temperature	not indicated	35°C
Injection Volume	20 µL	20 µL
Run Time	not indicated	15 min

Procedure: Separately injected equal volumes (about 20 µL) of the standard preparation and assay preparation into the chromatograph, recorded the chromatograms, and the responses for the component H2B1a plus component H2B1b were measured

Calculation: the % of label claim of ivermectin (H2B1a + H2B1b) in the portion of injection taken by the formula was calculated.

$$\text{Percentage Content (\%)} = \frac{\text{Peak of area of Sample}}{\text{Peak area of standard}} * 100$$

The ratio of contents, in percent, of H2B1a /H2B1a + H2B1b) in the portion of injection taken by the formula was calculated (USP, 2015).

Uniformity of weight test for tablets and filling to the volume for injectable preparations

The uniformity of mass between tablets of each brand was evaluated against Pharmacopoeial specification limit. For this purpose, twenty (20) tablets from a particular brand were released from the aluminum blister pack and weighed individually and collectively. The differences between the masses of the individual tablets and the mean from the 20 tablets were compared each other. The % deviations were also calculated (USP, 2015).

Data collection, Study design, sample and sampling technique

The study was conducted in Addis Ababa city, the capital city of Ethiopia starting from Oct, 2017 to April, 2018. It lies at 9.03° North latitude and 38.8° East longitudes with an average altitude of 2500 meters above sea level. It has an average annual temperature of 15.9°C. It also receives an annual rain fall of 1800 mm or 91 mm per month with 60.1% annual relative humidity which ranges from 49% in February to 82% in July (NMSA, 2007). The city is hub for distribution of veterinary pharmaceutical in the country.

The samples were collected from different veterinary pharmacy shops, clinics, and from dry port as consignment crosscheck and brought to the laboratory of veterinary drug, animal product and feed quality assessment center which is located in Akaki Kaliti sub-city, Addis Ababa. Different preparations of IVM i.e. injectable (1%), tablet (5mg/boli), oral drench/syrup (0.8mg/ml) and IVM + Clorsulon Combination (1% + 10%). Hence to ensure that each product or sample is obtained on the basis of chance and to run one sample five 50 ml IVM vial for injectable, twenty boli for tablet and 1 bottle for oral drench were used for each brands and batches.

Sample size and sampling: According to the report released by WHO's and pooled data from 100 literature reviews 10 % presence substandard pharmaceutical was found in developing countries (RAFS, 2017). Thus the rate of prevalence of substandard veterinary pharmaceutical product is expected to be 10 % and the confidence interval taken chosen as 95% and precision 5%. Thus, the sample size is calculated according to Thrusfield (2007). By substituting these values in the formula, the sample size become 138.

$$N = \frac{1.96^2 (p_{exp}) (1-p_{exp})}{d^2}$$

Where: N= required sample size, Pexp=Expected prevalence, d=Absolute precision (usually 0.05). But here the number of sample size required depends on availability of different brands of IVM on market during studies, hence sample size adjustment was employed as follow. i.e. $N = N_o * n / N_o + n$ (Sample size adjusted = population size * sample size required/ population size + sample size required) Where N = sample size adjusted, N_o = population size or max sample available and n = required sample size (Thrusfield, 2007).

$$N \text{ (sample size adjusted)} = \frac{\text{Population size} * \text{sample size required}}{\text{Population size} + \text{sample size required}}$$

Data management and analysis: The data were checked, coded and entered in to Microsoft excel work sheet and analyzed using descriptive statistics like %, mean and RSD to compare IVM API standard with different brands and batches of IVM Preparations.

RESULTS

Identity test: the identity test is performed to confirm the presence of IVM in each sample of IVM subjected for test. This is by considering the RT of the samples with that of the standard in the chromatographic response (Table 1.). That means the sample was retained with equivalent RT with that of the standard.

Table 1: Retention times of different brands of Ivermectin as comparison with the standard

Product	Preparation	RT of the sample		RT of the standard/USP		Difference		Remark
		H2B1a	H2B1b	H2B1a	H2B1b	H2B1a	H2B1b	
Brand 1	Tablet	8.572	7.175	8.573	7.174	0.001	0.001	Negligible
Brand 2	Injectable	8.573	7.176	8.572	7.177	0.001	0.001	Negligible
Brand 3	Injectable	8.573	7.176	8.571	7.173	0.002	0.003	Negligible
Brand 4	Injectable	8.574	7.176	8.576	7.178	0.002	0.002	Negligible
	Injectable	8.573	7.176	8.572	7.177	0.001	0.001	Negligible
Brand 5	Injectable	8.574	7.177	8.576	7.179	0.002	0.002	Negligible
Brand 6	Injectable	8.574	7.178	8.577	7.179	0.003	0.001	Negligible
Brand 7	Injectable	8.573	7.178	8.571	7.174	0.002	0.004	Negligible
Brand 8	Tablet	8.572	7.175	8.573	7.174	0.001	0.001	Negligible
Brand 9	Injectable	8.573	7.176	8.572	7.177	0.001	0.001	Negligible
Brand 10	Injectable	8.573	7.176	8.571	7.173	0.002	0.003	Negligible
Brand 11	Injectable	8.574	7.176	8.576	7.178	0.002	0.002	Negligible
	Injectable	8.573	7.178	8.571	7.174	0.002	0.004	Negligible
	Injectable	8.572	7.175	8.573	7.174	0.001	0.001	Negligible
	Injectable	8.573	7.176	8.572	7.177	0.001	0.001	Negligible
Brand 12	Injectable	8.573	7.176	8.571	7.173	0.002	0.003	Negligible
Brand 13	Oral drench	8.574	7.178	8.577	7.179	0.003	0.001	Negligible
	Oral drench	8.573	7.176	8.571	7.173	0.002	0.003	Negligible
Brand 14	Oral drench	8.573	7.178	8.571	7.174	0.002	0.004	Negligible

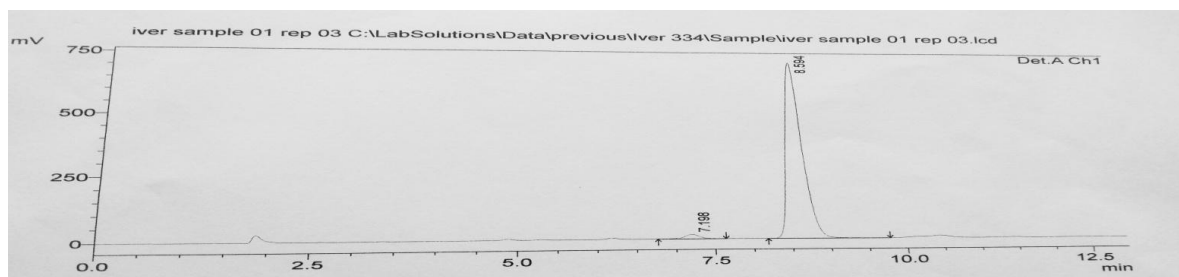


Figure 1: Chromatograph of one of the test sample (injectable) identified to have Ivermectin API. The peak area ratios of all the samples with that of the respective standard peak are were used to determine assay of the sample (Fig, 1).

Filling to the volume or dossier evaluation: the results of filling to the volume of injectable IVM preparation of each brand and batches are presented in Table 4 below. The results revealed that out of the 13 different brands of IVM injectable and oral drench preparations with 18 different sample, 3(23.1%) brands did not comply with the official Pharmacopoeial specification limit (USP, 2015).

Mass uniformity: the results of mass uniformity of Ivermectin tablet is presented in Table 4 below. Single brand and batch of tablet preparation of Ivermectin evaluated for mass uniformity revealed specification limit set by official Pharmacopoeial (USP, 2015).

Table 2. General information and mass uniformity results of tablet preparations of Ivermectin boli

Product	Strength (mg)	Blisters x Boli	Expiry date	Shelf Life	Weight (mg; mean $\pm$ %RSD) (n = 20)
Brand 14	5 mg	40 (10 x 4)	2021	4 year	919.05 $\pm$ 1 %

Assay of Product: Table 5 describes that, Single brand and batch of tablet preparation of Ivermectin is within the acceptance range and meet the expected requirements set by USP-NF 2015 official Pharmacopoeial (i.e. RSD not more than 2).

Table 3. Results of API-assay of the different products of Ivermectin brands and batches

Product	Preparations	Strength	Peak area (Triplicate inject.)		Assay (%LF; mean ± %RSD) (n ₁ =5 & n ₂ =20)	Content per blister or bottle (mg)
			H2B1a	H2B1b		
Brand 1	Tablet/bolus	5 mg/blister	10857740	322917	97.28 ± 1.78	4.864
Brand 2	Injectable	10 mg/ml	11276290	300033	101.03 ± 1.62	50515
Brand 3	Injectable	10 mg/ml	9633342	117835	86.31 ± 0.59 [#]	43155
Brand 4	Injectable	10 mg/ml	11065341	315695	99.14 ± 1.82	49570
	Injectable	10 mg/ml	11199277	248144	100.34 ± 1.09	50170
Brand 5	Injectable	10 mg/ml	11596620	239272	103.90 ± 1.84	51950
Brand 6	Injectable	10 mg/ml	10064170	258093	90.17 ± 1.28 [#]	45085
Brand 7	Injectable	10 mg/ml	11174722	213173	100.12 ± 0.71	50060
Brand 8	Injectable	10 mg/ml	10931405	257964	97.94 ± 1.45	48970
Brand 9	Injectable	10 mg/ml	10666573	234815	95.62 ± 0.52	47810
Brand 10	Injectable	10 mg/ml	10388964	314328	93.08 ± 0.38 [#]	46540
Brand 11	Injectable	10 mg/ml	11596620	207382	103.90 ± 0.38	51950
	Injectable	10 mg/ml	11518491	190617	103.20 ± 1.96	51600
	Injectable	10 mg/ml	11424735	188714	102.36 ± 0.49	51180
	Injectable	10 mg/ml	11554207	184256	103.52 ± 1.51	51760
Brand 12	Injectable	10 mg + 100	11335445	248392	101.56±0.79, 99.89±1.46 *	50780 plus 499451
	Iver + Clorsulon	mg/ml				
Brand 13	Oral drench	0.8 mg/ml	11040786	312184	98.92 ± 0.74	39568
	Oral drench	0.8 mg/ml	11422503	309587	102.34 ± 1.71	40936
Brand 14	Oral drench	0.8 mg/ml	11312006	237124	101.35 ± 0.98	40540

[#] Not within recommended range (the range is 90–110% for the tablet preparation and oral drench and 95–105% for injectable preparations as USP Specification). * Combination of Ivermectin and Colrsulon.

High performance liquid chromatography calculation: To calculate the content (percentage and milligram content) of the active pharmaceutical compound in each brand, the following formulas were used:

$$\text{Percentage Content of Brand 1 (300mg) (\%)} = \frac{\text{Peak of area of sample} \times 100}{\text{Peak area of standard}} =$$

$$\frac{10244117.7 \times 100\%}{10553968} \cong 97\%$$

$$\text{Milligram Content (mg)} = \frac{\text{Percentage content} \times 300\text{mg}}{100} = \frac{0.9764 \times 300}{100} \cong 292.92$$

DISCUSSIONS

According to USP (2015), Considering the average weight per tablet for each brands of drug, and twenty (20) tablets were selected at random, no two tablets should deviate by 5% and no single tablet by 10% of W/V. This is important to assure uniform distribution of API content of the drug. RSD should not be more than 2% (USP, 2007).

USP dosage uniformity method for weight uniformity test to the tablet and filling to the volume test for injectable as well as oral syrup preparations have shown single brand and batch of IVM tablet meet the specification for uniformity of dosage unit by W/V.

According to USP (2015), assay or active ingredient content for IVM bolus should be between 90% and 110%, and 95 % and 105% for injectable and oral drench preparations in relation to the label claim. If drugs are able to fall in the acceptance range with acceptable RSD, we can assure that the drugs can produce the required therapeutic effects, Thus out of the 14 brands of 19 different samples (batches) and four different preparations of IVM, 3(21.4%) of the brands has less than recommended API content that is 86.31, 90.17 and 93.08 which is below acceptance limit and 16 (84.2%) pass the requirement of percent API content with all RSDs were within acceptable range that is less than 2 which comply the specification of USP.

This assay determination by HPLC method revealed the presence of drug samples below specification limit for API content and this confirms the circulation of sub-therapeutic doses of veterinary drugs in the legal market of the country even slightly higher than the level estimated by WHO i.e. 10% (RAPS, 2017). (Geerts and Gryseels, 2000).

The implications of these can be; inability to achieve therapeutic goal which compromises the welfare of the animals and the productivity, obligatory repeated treatment incurs additional cost for the farmer/producer and it will result increasingly more resistant parasite population against treatment with IVM. All the aforementioned consequences of sub therapeutic dose drug circulating in the legal market will further aggravate the already existing anthelmintic resistance and poor efficacy scenarios reported from different parts of the country. The clinical significance of drug resistance is highly crucial in notorious parasites like *H. contortus* where a massive infection can kill a host (Kumsa and Wossene, 2006; Eguale *et al.*, 2009; Roos, 2009).

Laboratory experiments have shown that under-dosing contributes to the selection of resistance or tolerant strains. Moreover, variation in bioavailability in different host species also is crucial for making a decision about correct dose. Some indirect field evidence further support this conclusion (Shalaby, 2013).

There is no anthelmintic use policy in the country as a result misuse and smuggling of anthelmintics in many forms like illegal trading in open markets and irrational administration is a widespread practice in Ethiopia. In addition, no method is in use that can preserve and maintain the efficacy of anthelmintics or delay and prevent the emergence of anthelmintic resistance (Kumsa *et al.*, 2010).

CONCLUSION AND RECOMMENDATIONS

The failure to meet the specification could not always attributed to pharmaceutical companies and drug counterfeiting as case example failed drugs in this particular study all are registered, legal and produced by known manufacturers thus the quality compromision may not be only be as result of poor preparation techniques during formulation and subsequent manufacturing, incorrect weighing and inappropriate implementation of good manufacturing procedure/GMP , but also incorrect handling and storage conditions by different actors of market chain (agents/importers, distributors and retailers) after dispatching by manufacturers.

Based on the above conclusions the following recommendations are forwarded; There must have effective regulatory system due the fact Without strong regulatory capacity and systems in place we will not see a resolution to the problem, but we also need sound investment in safeguarding the manufacture, distribution and supply of medicinal products. We need to rule out the exact point where drug quality was compromised by regular inspection of manufacturers for implementation of GMP specification and evaluation of different actors who take part pharmaceutical market, and measures should be taken accordingly.

- ❖ There should be strong pre-registration evaluation mechanism before market authorization
- ❖ Laboratory analysis on consignment and post market Ivermectin samples for API determination should be strengthened
- ❖ Determination of API by HPLC method should be supported with other clinical and preclinical efficacy tests.
- ❖ Rapid reporting of suspected spurious products, risk-based inspections and post market surveillance, as well as access to laboratories and screening equipment.

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